



US012410159B2

(12) **United States Patent**
Lim et al.

(10) **Patent No.: US 12,410,159 B2**

(45) **Date of Patent: Sep. 9, 2025**

(54) **PREPARATION OF OXINDOLE
DERIVATIVES AS NOVEL
DIACYLGLYCERIDE
O-ACYLTRANSFERASE 2 INHIBITORS**

(71) Applicant: **Merck Sharp & Dohme LLC**,
Rahway, NJ (US)

(72) Inventors: **Yeon-Hee Lim**, South San Francisco,
CA (US); **Jianming Bao**, Princeton, NJ
(US); **James P. Roane**, Brisbane, CA
(US)

(73) Assignee: **Merck Sharp & Dohme LLC**,
Rahway, NJ (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

(21) Appl. No.: **17/494,896**

(22) Filed: **Oct. 6, 2021**

(65) **Prior Publication Data**
US 2022/0112181 A1 Apr. 14, 2022

Related U.S. Application Data

(60) Provisional application No. 63/089,063, filed on Oct.
8, 2020.

(51) **Int. Cl.**
C07D 409/14 (2006.01)

(52) **U.S. Cl.**
CPC **C07D 409/14** (2013.01)

(58) **Field of Classification Search**
CPC **C07D 409/14**
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,054,587	A	4/2000	Reddy
6,110,903	A	8/2000	Kasibhatla
6,284,748	B1	9/2001	Dang et al.
6,399,782	B1	6/2002	Kasibhatla et al.
6,489,476	B1	12/2002	Dang et al.
6,730,690	B2	5/2004	Olson et al.
2015/0259323	A1	9/2015	Cabral et al.
2018/0001628	A1	1/2018	Maxfield

FOREIGN PATENT DOCUMENTS

JP	2010513456	A	4/2010
WO	200208188	A1	1/2002
WO	2002060388	A1	1/2002
WO	2003053363	A2	7/2003
WO	2004019869	A2	3/2004
WO	2004020408	A1	3/2004
WO	2004020409	A1	3/2004
WO	2004066963	A2	8/2004
WO	2008084300	A1	7/2008
WO	2009001836	A1	12/2008
WO	2009042053	A2	4/2009
WO	2009000087	A1	12/2009
WO	2010077839	A1	7/2010

WO	2011150288	A1	12/2011
WO	2013150416	A1	10/2013
WO	2015077299	A1	5/2015
WO	2016036633	A1	3/2016
WO	2016036636	A1	3/2016
WO	2016036638	A1	3/2016
WO	2017011276	A1	1/2017
WO	2018033832	A1	2/2018
WO	2018093696	A1	5/2018
WO	2018093698	A1	5/2018
WO	2021064590	A1	4/2021
WO	2021127443	A1	6/2021
WO	2022074103	A1	4/2022
WO	2022076417	A1	4/2022
WO	2022076495	A1	4/2022

OTHER PUBLICATIONS

Registry No. 1234338-51-2, File Registry on STN, Jul. 29, 2010.*
Choi et al., Suppression of Diacylglycerol Acyltransferase-2 (DGAT2),
but Not DGAT1, with Antisense Oligonucleotides Reverses Diet-
induced Hepatic Steatosis and Insulin Resistance, Journal of Bio-
logical Chemistry, 2007, 22678-22688, 282(31).

Li, Chen et al., Roles of Acyl-CoA:Diacylglycerol Acyltransferases
1 and 2 in Triacylglycerol Synthesis and Secretion in Primary
Hepatocytes, Arterioscler. Thromb. Vase. Biol., 2015, 1080-1091,
35.

Liu et al., Knockdown of Acyl-CoA:diacylglycerol acyltransferase
2 with antisense oligonucleotide reduces VLDL TG and ApoB
secretion in mice, Biochimica et Biophysica Acta, Molecular and
Cell Biology of Lipids, 2008, 97-104, 1781 (3).

Yen, Chi-Liang Eric et al., DGAT enzymes and triacylglycerol
biosynthesis, Journal of Lipid Research, 2008, 2283-2301, 49.

Yu, Xing Xian et al., Antisense Oligonucleotide Reduction of
DGAT2 Expression Improves Hepatic Steatosis and Hyperlipidemia
in Obese Mice, Hepatology, 2005, 362-371, 42.

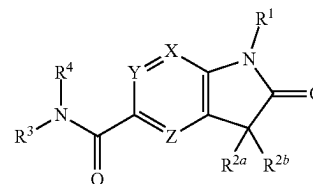
* cited by examiner

Primary Examiner — Rebecca L Anderson

(74) *Attorney, Agent, or Firm* — Kristi K. Harman;
Catherine D Fitch

(57) **ABSTRACT**

Invented are compounds of formula I



and the pharmaceutically acceptable salts, esters, and prod-
rugs thereof, which are DGAT2 inhibitors. Also provided are
methods of making compounds of Formula I, pharmaceuti-
cal compositions comprising compounds of Formula I, and
methods of using these compounds to treat hepatic steatosis,
nonalcoholic steatohepatitis (NASH), fibrosis, type-2 diabe-
tes mellitus, obesity, hyperlipidemia, hypercholesterolemia,
atherosclerosis, cognitive decline, dementia, cardiorenal dis-
eases such as chronic kidney diseases and heart failure and
related diseases and conditions, comprising administering a
compound of Formula I to a patient in need thereof.

48 Claims, No Drawings

1

**PREPARATION OF OXINDOLE
DERIVATIVES AS NOVEL
DIACYLGLYCERIDE
O-ACYLTRANSFERASE 2 INHIBITORS**

FIELD OF THE INVENTION

The present invention is directed to novel pharmaceutical compounds which inhibit diacylglyceride O-acyltransferase 2 ("DGAT2"), and may be useful for preventing, treating or acting as a reversing agent for hepatic steatosis, nonalcoholic steatohepatitis (NASH), fibrosis, type-2 diabetes mellitus, obesity, hyperlipidemia, hypercholesterolemia, atherosclerosis, cognitive decline, dementia, cardiorenal diseases such as chronic kidney diseases and heart failure, and related diseases and conditions, as well as methods of making such compounds and pharmaceutical compositions comprising such a compound and a pharmaceutical carrier.

BACKGROUND OF THE INVENTION

Triacylglycerols ("TGs") serve several functions in living organisms. One such function of TGs is in the storage of energy. TGs also play a role in the synthesis of membrane lipids. TG synthesis in cells may protect them from the potentially toxic effects of excess fatty acid ("FA"). In enterocytes and hepatocytes, TGs are synthesized for the assembly and secretion of lipoproteins which transport FA between tissues. TGs play a role in the skin's surface water barrier, and TGs in adipose tissue provide insulation for organisms.

The glycerol phosphate and the monoacylglycerol pathways are the major pathways for the biosynthesis of TG. However, the last step in the synthesis of TG involves the reaction of a fatty acyl-CoA and diacylglycerol ("DAG") to form TG. The reaction is catalyzed by acyl-CoA: diacylglycerol acyltransferase ("DGAT") enzymes. There have been identified two DGAT enzymes, DGAT1 and DGAT2. Although DGAT1 and DGAT2 catalyze the same reaction, they differ significantly at the level of DNA and protein sequences. DGAT2 can utilize endogenous fatty acid to synthesize TG in vitro assays, whereas DGAT1 appears to be more dependent on exogenous fatty acid (Yen et al., *J. Lipid Research*, 2008, 49, 2283). Inactivation of DGAT2 impaired cytosolic lipid droplet growth, whereas inactivation of DGAT1 exerts opposite effect. (Li et al., *Arterioscler. Thromb. Vasc. Biol.* 2015, 35, 1080).

DGAT2 is an integral membrane protein of the endoplasmic reticulum and is expressed strongly in adipose tissue and the liver. DGAT2 appears to be the dominant DGAT enzyme controlling TG homeostasis in vivo. DGAT2 deficient mice survive for only a few hours after birth. On the other hand, DGAT1 deficient mice are viable (Yen et al., *J. Lipid Research*, 2008, 49, 2283).

Despite this perinatal lethal phenotype, the metabolic role of DGAT2 has been mostly understood from effort exploiting anti-sense oligonucleotides (ASO) in rodents. In this setting, DGAT2 knockdown in ob/ob mice with a DGAT2 gene-specific ASO resulted in a dose dependent decrease in very low density lipoprotein ("VLDL") and a reduction in plasma TG, total cholesterol, and ApoB (Liu, et al., *Biochim. Biophys. Acta* 2008, 1781, 97). In the same study, DGAT2

2

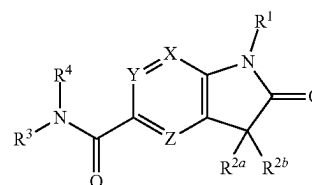
antisense oligonucleotide treatment of ob/ob mice showed a decrease in weight gain, adipose weight and hepatic TG content. Id. In another study, antisense treatment of ob/ob mice improved hepatic steatosis and hyperlipidemia (Yu, et al., *Hepatology*, 2005, 42, 362). Another study showed that diet-induced hepatic steatosis and insulin resistance was improved by knocking down DGAT2 in rats. These effects seem to be unique to inhibition of DGAT2, as ASO against DGAT1 did not lead to similar beneficial effects. Although the molecular mechanism behind these observations remains uncertain, the collective data suggest that suppression of DGAT2 is associated with reduced expression of lipogenic genes (SREBP1c, ACC1, SCD1, and mtGPAT) and increased expression of oxidative/thermogenic genes (CPT1, UCP2) (Choi et al., *J. Bio. Chem.*, 2007, 282, 22678).

In light of the above, inhibitors of DGAT2 are useful for treating disease related to the spectrum of metabolic syndrome such as hepatic steatosis, non-alcoholic steatohepatitis (NASH), fibrosis, type-2 diabetes mellitus, obesity, hyperlipidemia, hypercholesterolemia, atherosclerosis, cognitive decline, dementia, cardiorenal diseases such as chronic kidney diseases and heart failure and related diseases and conditions.

DGAT2 inhibitor compounds are described in WO2021064590, WO2016036633, WO2016036636, WO2016036638, WO2018093696, WO2018093698, WO2013150416, US20150259323, WO2015077299, WO2017011276, WO2018033832, US201801628, WO2003053363.

SUMMARY OF THE INVENTION

The present invention relates to compounds represented by Formula I:

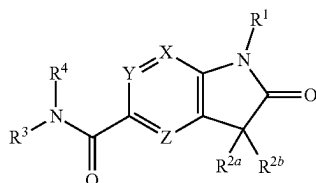


as well as pharmaceutically acceptable salts, esters, and prodrugs thereof, which are DGAT2 inhibitors. Also provided are methods of making compounds of Formula I, pharmaceutical compositions comprising compounds of Formula I, and methods of using these compounds to treat hepatic steatosis, nonalcoholic steatohepatitis (NASH), fibrosis, type-2 diabetes mellitus, obesity, hyperlipidemia, hypercholesterolemia, atherosclerosis, cognitive decline, dementia, cardiorenal diseases such as chronic kidney diseases and heart failure and related diseases and conditions, comprising administering a compound of formula I to a patient in need thereof.

3

DETAILED DESCRIPTION OF THE
INVENTION

The present disclosure is directed to compounds having structural Formula I:



or pharmaceutically acceptable salts thereof wherein:
X, Y, and Z are independently selected from N or C(R⁵);
R¹ is

- (1) unsubstituted or substituted with 1, 2, or 3 R⁶, or
- (2) 5- or 6-membered heteroaryl containing 1, 2, 3 or 4 heteroatoms independently selected from N, O, and S, wherein the heteroaryl is unsubstituted or substituted with 1, 2, or 3 R⁶, or
- (3) 8- to 10-membered fused heteroaryl containing 1, 2, 3 heteroatoms independently selected from N, O, and S, wherein the heteroaryl is unsubstituted or substituted with 1, 2, or 3 R⁶;

R^{2a} and R^{2b} are independently selected from

- (1) hydrogen,
- (2) halogen,
- (3) hydroxy,
- (4) (C₁₋₆)alkyl,
- (5) (C₁₋₆)haloalkyl,
- (6) R^{2a} and R^{2b} join to form spiro(C₃₋₈)cycloalkyl unsubstituted or optionally mono-substituted, or disubstituted with C₁₋₃alkyl, halogen, OH, or
- (7) R^{2a} and R^{2b} join to form spiro 4- to 8-membered heterocyclyl containing 1 or 2 heteroatoms independently selected from N, O, and S wherein the heterocyclyl is unsubstituted or substituted by 1, 2, or 3 R⁷;

R³ is

- (1) 4- to 7-membered heterocyclyl containing 1, 2 or 3 heteroatoms independently selected from N, O and S,
- (2) 5- or 6-membered heteroaryl containing 1, 2 or 3 heteroatoms independently selected from N, O, and S,
- (3) —(C₁₋₆)alkyl-heteroaryl, wherein the heteroaryl is a 5- or 6-membered heteroaryl containing 1, 2, or 3 heteroatoms independently selected from N, O and S,
- (4) —(C₁₋₆)alkyl-aryl,
- (5) —(C₁₋₆)alkyl-heterocyclyl, wherein the heterocyclyl is a 3- to 6-membered ring containing 1 or 2 heteroatoms independently selected from N, O and S,
- (6) —(C₁₋₆)alkyl,
- (7) —(C₃₋₆)cycloalkyl,
- (8) —(C₁₋₆) hydroxyalkyl,
- (9) —(C₁₋₆)alkyl-S(O)₂—NR^{8a}R^{8b}, or
- (10) —(C₁₋₆)alkyl-S(O)₂—(C₁₋₃)alkyl,

wherein each cycloalkyl, or heterocyclyl is unsubstituted or substituted with 1, 2, or 3 R⁹, and

wherein each alkyl, aryl or heteroaryl is unsubstituted or substituted with 1, 2, or 3 R¹⁰;

4

R⁴ is

- (1) hydrogen, or
- (2) (C₁₋₃)alkyl,

or R³ and R⁴ combine along with the nitrogen atom to which they are attached to form a mono- or bicyclic heterocyclyl ring containing 1 or 2 heteroatoms independently selected from N, O and S, wherein the heterocyclyl ring is unsubstituted or substituted by 1, 2, or 3 R¹¹,

when present, each R⁵ is selected from

- (1) hydrogen,
- (2) (C₁₋₃)alkyl,
- (3) (C₁₋₃)haloalkyl,
- (4) cyano,
- (5) halogen,

when present, each R⁶ is independently selected from

- (1) cyano,
- (2) halogen,
- (3) —OC₁₋₆alkyl,
- (4) (C₃₋₆)cycloalkyl optionally substituted with halogen or OH,
- (5) —(C=O)NH₂,
- (6) (C₃₋₆)cycloalkyloxy wherein the cycloalkyl is optionally substituted with halogen, or OH,
- (7) hydroxy,
- (8) —N(R¹¹)₂,
- (9) —NH(C=O)(C₁₋₆)alkyl,
- (10) (C₂₋₆) cyclic amine optionally substituted with one or two halogen substituents,
- (11) (C₁₋₆)haloalkyl-,
- (12) —O(C₁₋₆)haloalkyl optionally substituted with OH,
- (13) —O(C₀₋₃)alkyl-(C₃₋₆)cycloalkyl optionally substituted with one or two halogen substituents,
- (14) —SO₂(C₁₋₆)alkyl,
- (15) —SO₂NH(C₁₋₆)alkyl,
- (16) —SC₁₋₆alkyl,
- (17) —SC₁₋₆haloalkyl, or
- (18) (C₁₋₆)alkyl,

when present, each R⁷ is independently selected from

- (1) (C₁₋₃)alkyl,
- (2) halogen,
- (3) (C₁₋₃)alkoxy-,
- (4) (C₁₋₃)haloalkyl-, or
- (5) hydroxy,

when present, R^{8a} and R^{8b} are independently selected from

- (1) hydrogen,
- (2) (C₁₋₃)alkyl, or
- (3) (C₃₋₇)cycloalkyl;

when present, each R⁹ is independently selected from

- (1) (C₁₋₃)alkyl,
- (2) (C₁₋₃)haloalkyl-,
- (3) oxo,
- (4) (C₃₋₆)cycloalkyl,
- (5) N(R¹¹)₂,
- (6) hydroxy,
- (7) (C₁₋₃)alkoxy-,
- (8) cyano,
- (9) halogen,
- (10) —SO₂(C₁₋₆)alkyl,
- (11) —(C₁₋₆)alkyl SO₂(C₁₋₆)alkyl,
- (12) —C(O)(C₁₋₃)alkyl, or
- (13) O(C₁₋₃)alkyl,

when present, R¹⁰ is independently selected from

5

- (1) hydrogen,
- (2) (C₁₋₃)alkyl,
- (3) (C₁₋₃)alkoxy-,
- (4) hydroxy,
- (5) halogen,
- (6) (C₁₋₃)alkyl-S—,
- (7) (C₁₋₃)haloalkyl-, or
- (8) N(R¹¹)₂,

R¹¹, when present, is independently

- (1) hydrogen, or
- (2) (C₁₋₃)alkyl.

In Embodiment 1a of this disclosure are compounds of Formula I, or a pharmaceutically acceptable salt of any of the foregoing, wherein R¹ is

- a) phenyl optionally substituted with one to three substituents independently selected from halogen, hydroxy, CN, C₁₋₃alkyl, C₁₋₃haloalkyl, C₃₋₆cycloalkyl optionally substituted with halogen or OH, OC₁₋₃alkyl, —OC₁₋₃haloalkyl optionally substituted with OH, —OC₃₋₆cycloalkyl, —SC₁₋₃alkyl, —SC₁₋₃haloalkyl, S(O)₂C₁₋₃alkyl, —(NH(C=O))C₁₋₃alkyl, C(O)NH₂, S(O)₂NHC₁₋₃alkyl, N(R¹¹)₂, or azetidiny optionally substituted with one or two halogen substituents.
- b) a 6 membered heteroaryl containing one or two nitrogen atoms substituted with one or two substituents independently selected from: halogen, hydroxy, C₁₋₃alkyl, C₁₋₃haloalkyl, C₃₋₆cycloalkyl, —OC₁₋₃alkyl, —OC₁₋₃haloalkyl, OC₀₋₃alkyl-C₃₋₆cycloalkyl optionally substituted with 1-2 halogen substituents, azetidiny optionally substituted with 1-2 halogens, CN, —SC₁₋₃alkyl, or —SC₁₋₃haloalkyl;
- c) a 5 membered heteroaryl containing one to four nitrogen atoms or hetero atoms independently selected from N, O, and S optionally substituted with one to two substituents independently selected from with halogen, (C₁₋₃)alkyl, (C₃₋₆)cycloalkyl, (C₁₋₃)haloalkyl-, OH or OC₁₋₃alkyl, OC₀₋₃alkyl-C₃₋₆cycloalkyl optionally substituted with 1-2 halogens, O(C₁₋₃)haloalkyl-, O(C₁₋₃)haloalkyl-OH, azetidiny optionally substituted with 1-2 halogens; or
- d) 8- to 10-membered fused heteroaryl containing 1, 2, 3 heteroatoms independently selected from N, O, and S, wherein the heteroaryl is unsubstituted or substituted with halogen, (C₁₋₃)alkyl, (C₃₋₆)cycloalkyl, (C₁₋₃)haloalkyl-, OH or OC₁₋₃alkyl.

In Embodiment 1b of this disclosure are compounds of Formula I, or a pharmaceutically acceptable salt of any of the foregoing, wherein R¹ is

- a) phenyl optionally substituted with one to three substituents independently selected from halogen, hydroxy, CN, C₁₋₃alkyl, C₁₋₃haloalkyl, C₃₋₆cycloalkyl optionally substituted with halogen or OH, OC₁₋₃alkyl, —OC₁₋₃haloalkyl optionally substituted with OH, —OC₃₋₆cycloalkyl, —SC₁₋₃alkyl, —SC₁₋₃haloalkyl, S(O)₂C₁₋₃alkyl, —(NH(C=O))C₁₋₃alkyl.
- b) a 6 membered heteroaryl containing one or two nitrogen atoms substituted with one or two substituents independently selected from: halogen, hydroxy, C₁₋₃alkyl, C₁₋₃haloalkyl, C₃₋₆cycloalkyl, —OC₁₋₃alkyl, —OC₁₋₃haloalkyl, —O—C₃₋₆cycloalkyl, CN, —SC₁₋₃alkyl, or —SC₁₋₃haloalkyl;
- c) a 5 membered heteroaryl containing one to four nitrogen atoms or hetero atoms independently selected from N, O, and S optionally substituted with one to two substituents independently selected from with halogen, (C₁₋₃)alkyl, (C₃₋₆)cycloalkyl, (C₁₋₃)haloalkyl-, OH or OC₁₋₃alkyl; or

6

- d) 8- to 10-membered fused heteroaryl containing 1, 2, 3 heteroatoms independently selected from N, O, and S, wherein the heteroaryl is unsubstituted or substituted with halogen, (C₁₋₃)alkyl, ((C₃₋₆)cycloalkyl, (C₁₋₃)haloalkyl-, OH or OC₁₋₃alkyl.

In Embodiment 2 of this disclosure are compounds of Formula I, or of Embodiment 1, or a pharmaceutically acceptable salt of any of the foregoing, wherein R¹ is not alkyl.

In Embodiment 3a of this disclosure are compounds of Formula I, or any one of Embodiments 1-2, or a pharmaceutically acceptable salt of any of the foregoing, wherein R¹ is

- a) phenyl substituted with one to three substituents independently selected from halogen, hydroxy, CN, C₁₋₃alkyl, C₁₋₃haloalkyl, C₃₋₆cycloalkyl optionally substituted with halogen or OH, —OC₁₋₃alkyl, —OC₁₋₃haloalkyl optionally substituted with OH, —OC₃₋₆cycloalkyl, —SC₁₋₃alkyl, —SC₁₋₃haloalkyl, S(O)₂C₁₋₆alkyl, or —NH(C=O)C₁₋₃alkyl, C(O)NH₂, S(O)₂NHC₁₋₃alkyl, N(R¹¹)₂, azetidiny optionally substituted with one or two halogen substituents;
- b) a 6 membered heteroaryl containing one or two nitrogen atoms substituted with one to two substituent selected from: halogen, hydroxy, C₁₋₃alkyl, C₁₋₃haloalkyl, C₃₋₆cycloalkyl, —OC₁₋₃alkyl, —OC₁₋₃haloalkyl, OC₀₋₃alkyl-C₃₋₆cycloalkyl optionally substituted with 1-2 halogens, azetidiny optionally substituted with 1-2 halogens, CN, —SC₁₋₃alkyl, —SC₁₋₃haloalkyl;
- c) a 5 membered heteroaryl containing one to four nitrogen atoms or heteroatoms independently selected from N, O, and S optionally substituted with one to two substituents independently selected from with halogen, (C₁₋₃)alkyl, (C₃₋₆)cycloalkyl, (C₁₋₃)haloalkyl-, OH, OC₁₋₃alkyl, OC₀₋₃alkyl-C₃₋₆cycloalkyl optionally substituted with 1-2 halogens, O(C₁₋₃)haloalkyl-, O(C₁₋₃)haloalkyl-OH, azetidiny optionally substituted with 1-2 halogens; or
- d) 8- to 10-membered fused heteroaryl containing at least one nitrogen and optionally containing one oxygen.

In Embodiment 3b of this disclosure are compounds of Formula I, or any one of Embodiments 1-2, or a pharmaceutically acceptable salt of any of the foregoing, wherein R¹ is

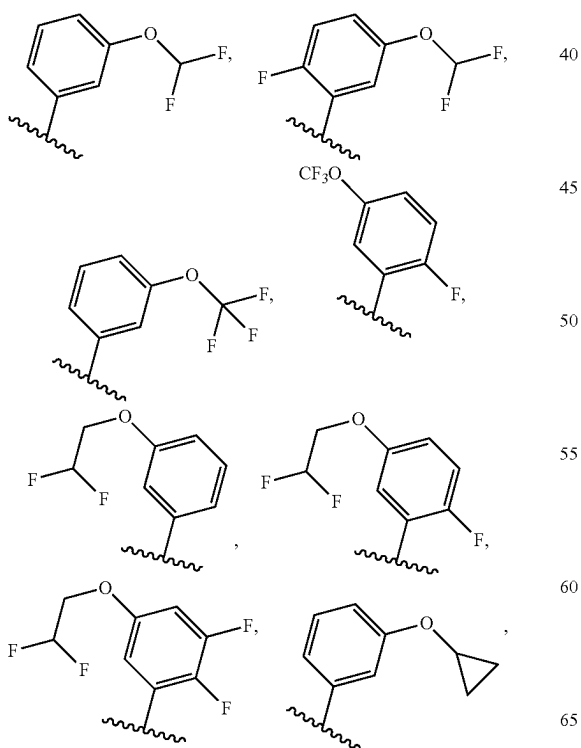
- a) phenyl substituted with one to three substituents independently selected from halogen, hydroxy, CN, C₁₋₃alkyl, C₁₋₃haloalkyl, C₃₋₆cycloalkyl optionally substituted with halogen or OH, —OC₁₋₃alkyl, —OC₁₋₃haloalkyl optionally substituted with OH, —OC₃₋₆cycloalkyl, —SC₁₋₃alkyl, —SC₁₋₃haloalkyl, S(O)₂C₁₋₆alkyl, or —NH(C=O)C₁₋₃alkyl, and wherein the cycloalkyl is optionally substituted with halogen, or OH;
- b) a 6 membered heteroaryl containing one or two nitrogen atoms substituted with one to two substituent selected from: halogen, hydroxy, C₁₋₃alkyl, C₁₋₃haloalkyl, C₃₋₆cycloalkyl, —OC₁₋₃alkyl, —OC₁₋₃haloalkyl, —O—C₃₋₆cycloalkyl, CN, —SC₁₋₃alkyl, —SC₁₋₃haloalkyl;
- c) a 5 membered heteroaryl containing one to four nitrogen atoms or heteroatoms independently selected from N, O, and S optionally substituted with one to two substituents independently selected from with halogen, (C₁₋₃)alkyl, (C₃₋₆)cycloalkyl, (C₁₋₃)haloalkyl-, OH or OC₁₋₃alkyl, or
- d) 8- to 10-membered fused heteroaryl containing at least one nitrogen and optionally containing one oxygen.

In Embodiment 4 of this disclosure are compounds of Formula I, or any one of Embodiments 1-3, or a pharmaceutically acceptable salt of any of the foregoing, wherein R¹ is

7

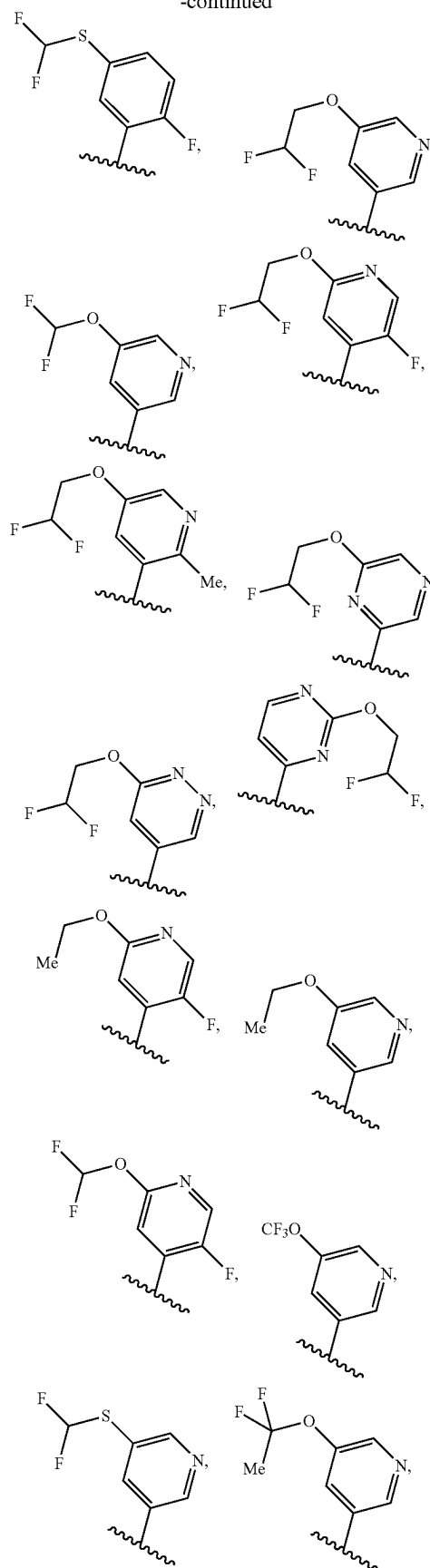
- a) phenyl substituted with a substituent selected from: halogen, hydroxy, C_{1-3} alkyl, C_{1-3} haloalkyl, $-OC_{1-3}$ alkyl, $-OC_{1-3}$ haloalkyl, $-O$ -cyclopropyl optionally substituted with halogen or hydroxy, $-SC_{1-3}$ alkyl, $-SC_{1-3}$ haloalkyl, $S(O)_2NHCH_3$, $S(O)_2C_{1-3}$ alkyl, $-NH(C=O)C_{1-3}$ alkyl, CN, azetidiny optionally substituted with one or two fluoro substituents, $C(O)NH_2$, $N(CH_3)_2$, and wherein the phenyl is optionally further substituted with 1 or 2 fluorine atoms;
- b) a 6 membered heteroaryl containing one or two nitrogen atoms substituted with one or two substituents selected from: C_{1-3} alkyl, C_{1-3} haloalkyl, $-OC_{1-3}$ alkyl, $-OC_{1-3}$ haloalkyl optionally substituted with hydroxy, $-O$ -cyclopropyl optionally substituted with halogen or hydroxy, $-O-C_{1-3}$ alkyl- C_{3-6} cycloalkyl, $-SC_{1-3}$ alkyl, $-SC_{1-3}$ haloalkyl, azetidiny optionally substituted with one or two fluoro substituents, halogen, $-OC_{1-3}$ alkyl-OH, $-OC_{1-3}$ haloalkyl-OH, and wherein the 6-membered heteroaryl is optionally further substituted with 1 or 2 fluorine atoms or CH_3 ;
- c) 5 membered heteroaryl containing one to four nitrogen atoms or heteroatoms independently selected from N, O, and S optionally substituted with one to two substituents independently selected from with halogen, (C_{1-3}) alkyl, (C_{3-6}) cycloalkyl, (C_{1-6}) haloalkyl-, $-OH$, OC_{1-6} alkyl, OC_{0-3} alkyl- C_{3-6} cycloalkyl optionally substituted with 1-2 halogens, $O(C_{1-3})$ haloalkyl-, $O(C_{1-3})$ haloalkyl-OH, azetidiny optionally substituted with 1-2 fluoro substituents; or
- d) 9 or 10-membered fused heteroaryl containing one or two heteroatoms independently selected from nitrogen or oxygen, optionally substituted with methyl.

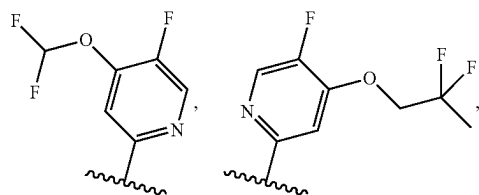
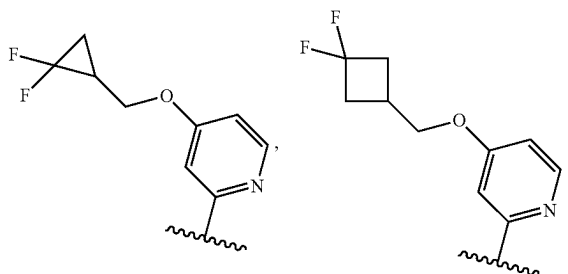
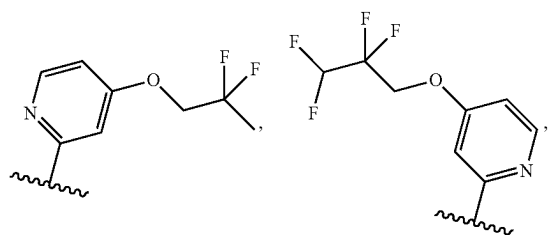
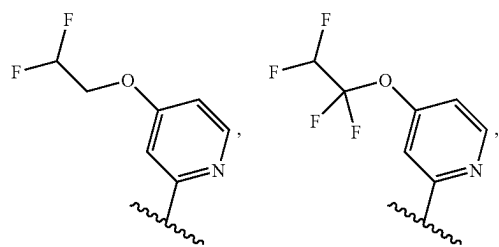
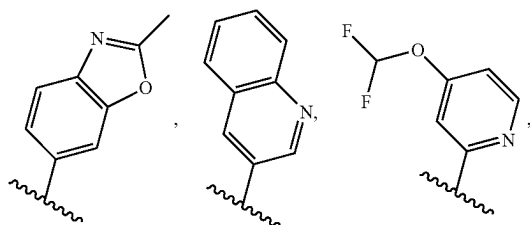
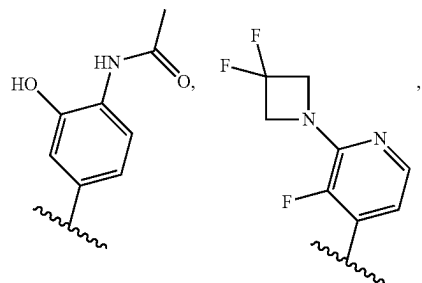
In Embodiment 5 of this disclosure are compounds of Formula I, or any one of Embodiments 1-4, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^1 is



8

-continued



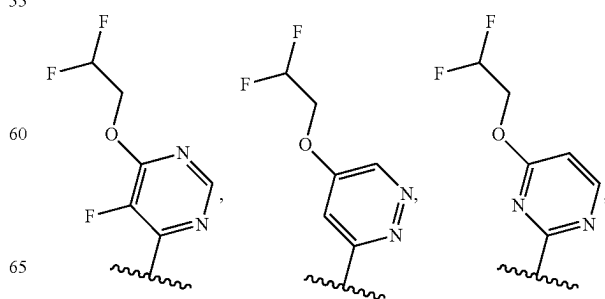
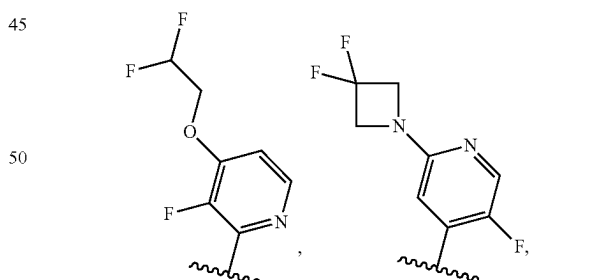
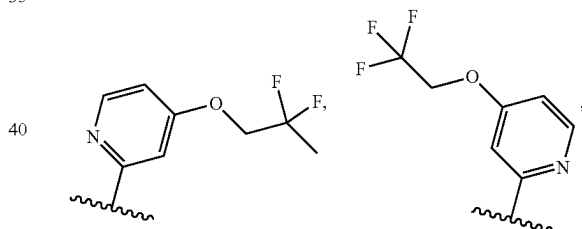
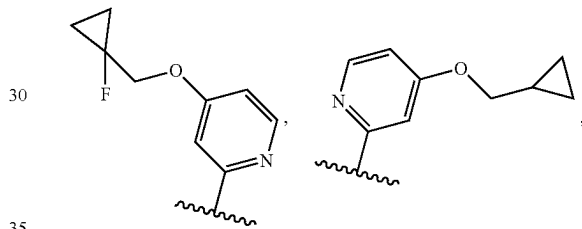
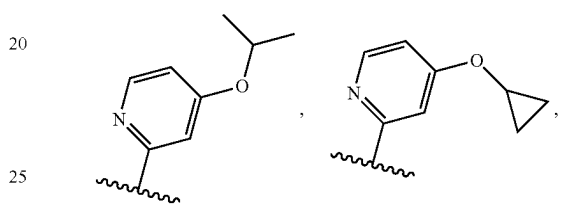
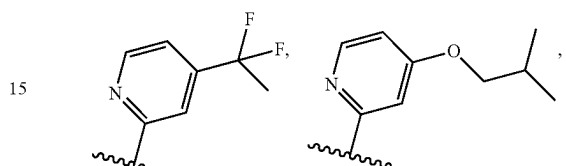


10

5

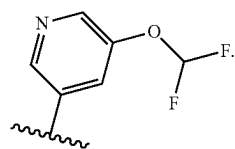
10

Chemical structures 1 and 2 are shown. Structure 1 is 2-(2-(2-fluoroethoxy)-5-fluoropyridin-3-yl)pyridine. Structure 2 is 2-(2-(2,2,2-trifluoroethoxy)-5-fluoropyridin-3-yl)pyridine.

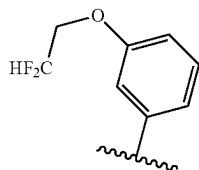


In Embodiment 12 of this disclosure are compounds of 65 Formula I, or any one of Embodiments 1-5, or 7, or a pharmaceutically acceptable salt of any of the foregoing, wherein R¹ is

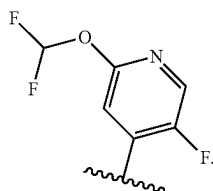
13



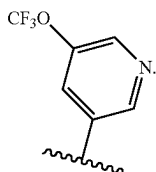
In Embodiment 13 of this disclosure are compounds of Formula I, or any one of Embodiments 1-6, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^1 is



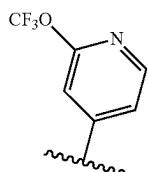
In Embodiment 14 of this disclosure are compounds of Formula I, or any one of Embodiments 1-5, or 7, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^1 is



In Embodiment 15 of this disclosure are compounds of Formula I, or any one of Embodiments 1-5, or 7, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^1 is

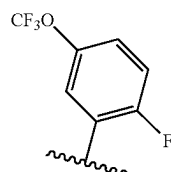


In Embodiment 16 of this disclosure are compounds of Formula I, or any one of Embodiments 1-5, or 7, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^1 is

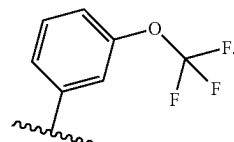


In Embodiment 17 of this disclosure are compounds of Formula I, or any one of Embodiments 1-6, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^1 is

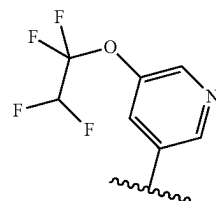
14



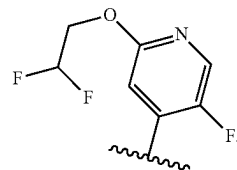
In Embodiment 18 of this disclosure are compounds of Formula I, or any one of Embodiments 1-6, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^1 is



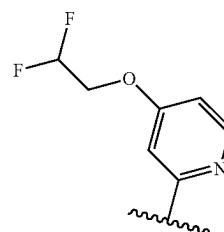
In Embodiment 19 of this disclosure are compounds of Formula I, or any one of Embodiments 1-5, or 7, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^1 is



In Embodiment 20 of this disclosure are compounds of Formula I, or any one of Embodiments 1-5, or 7, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^1 is

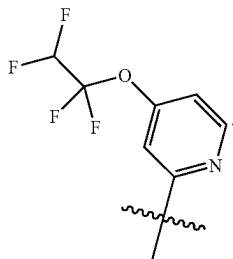


In Embodiment 21 of this disclosure are compounds of Formula I, or any one of Embodiments 1-5, or 7, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^1 is



15

is In Embodiment 22 of this disclosure are compounds of Formula I, or any one of Embodiments 1-5, or 7, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^1 is



In Embodiment 23 of this disclosure are compounds of Formula I, or Embodiments 1-22, or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^{2a} and R^{2b} are selected from one of the following:

- (1) R^{2a} and R^{2b} are hydrogen,
- (2) R^{2a} is hydrogen and R^{2b} is (C_{1-6}) alkyl, or R^{2a} is (C_{1-6}) alkyl and R^{2b} is hydrogen,
- (3) R^{2a} is halogen and R^{2b} is (C_{1-6}) alkyl, or R^{2a} is (C_{1-6}) alkyl and R^{2b} is halogen,
- (4) R^{2a} is hydroxy and R^{2b} is (C_{1-6}) alkyl, or R^{2a} is (C_{1-6}) alkyl and R^{2b} is hydroxy,
- (5) R^{2a} is hydroxy and R^{2b} is (C_{1-6}) haloalkyl, or R^{2a} is (C_{1-6}) haloalkyl and R^{2b} is hydroxy,
- (6) R^{2a} and R^{2b} are each independently selected from (C_{1-6}) alkyl,
- (7) R^{2a} and R^{2b} join to form spiro 4- to 8-membered heterocyclyl containing 1 or 2 heteroatoms independently selected from N, O and S wherein the heterocyclyl is unsubstituted or substituted by 1, 2, or 3 R^7 , or
- (8) R^{2a} and R^{2b} join to form spiro (C_{3-8}) cycloalkyl unsubstituted or optionally mono-substituted, or disubstituted with C_{1-6} alkyl, halogen, or OH.

In Embodiment 24 of this disclosure are compounds of Formula I, or Embodiments 1-23, or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^{2a} and R^{2b} are independently selected from H, and C_{1-3} alkyl, or together to form a spiro C_{1-6} cycloalkyl.

In Embodiment 25a of this disclosure are compounds of Formula I, or Embodiments 1-23, or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^{2a} and R^{2b} are independently selected from hydrogen, CH_3 , OH, CF_3 , $CH(CH_3)_2$, or CH_2CH_3 .

In Embodiment 25b of this disclosure are compounds of Formula I, or Embodiments 1-23, or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^{2a} and R^{2b} are independently selected from hydrogen, CH_3 , $CH(CH_3)_2$, or CH_2CH_3 .

In Embodiment 26 of this disclosure are compounds of Formula I, or Embodiments 1-23, or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^{2a} and R^{2b} are joined at the carbon to which they both attach to form cyclobutyl, cyclopropyl, or cyclopentyl.

In Embodiment 27a of this disclosure are compounds of Formula I, or any one of Embodiments 1-22, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^3 is

- (1) 4- to 7-membered heterocyclyl containing 1, 2 or 3 heteroatoms independently selected from N, O and S,

16

- (2) 5- or 6-membered heteroaryl containing 1, 2 or 3 heteroatoms independently selected from N, O, and S,
- (3) $-(C_{1-6})$ alkyl-heteroaryl, wherein the heteroaryl is a 5- or 6-membered heteroaryl containing 1, 2 or 3 heteroatoms independently selected from N, O and S,
- (4) $-(C_{1-6})$ alkyl-aryl,
- (5) $-(C_{1-6})$ alkyl-heterocyclyl, wherein the heterocyclyl is a 3- to 6-membered ring containing 1 or 2 heteroatoms independently selected from N, O and S,
- (6) $-(C_{1-6})$ alkyl,
- (7) $-(C_{3-6})$ cycloalkyl,
- (8) (C_{1-6}) hydroxyalkyl,
- (9) $-(C_{1-6})$ alkyl-S(O)₂-NR^{8a}R^{8b}, or
- (10) $-(C_{1-6})$ alkyl-S(O)₂-(C₁₋₃)alkyl,

wherein each aryl, heteroaryl, cycloalkyl, or heterocyclyl is unsubstituted or substituted with 1, 2, or 3 R^9 , and wherein each alkyl, aryl or heteroaryl is unsubstituted or substituted with 1, 2, or 3 R^{10} .

In Embodiment 27b of this disclosure are compounds of Formula I, or any one of Embodiments 1-26, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^3 is

- (1) 4- to 7-membered heterocyclyl containing 1, 2 or 3 heteroatoms independently selected from N, O and S,
- (2) 5- or 6-membered heteroaryl containing 1, 2 or 3 heteroatoms independently selected from N, O, and S,
- (3) $-(C_{1-6})$ alkyl-heteroaryl, wherein the heteroaryl is a 5- or 6-membered heteroaryl containing 1, 2 or 3 heteroatoms independently selected from N, O and S,
- (4) $-(C_{1-6})$ alkyl-aryl,
- (5) $-(C_{1-6})$ alkyl-heterocyclyl, wherein the heterocyclyl is a 3- to 6-membered ring containing 1 or 2 heteroatoms independently selected from N, O and S,
- (6) $-(C_{1-6})$ alkyl,
- (7) $-(C_{3-6})$ cycloalkyl,
- (8) (C_{1-6}) hydroxyalkyl,
- (9) $-(C_{1-6})$ alkyl-S(O)₂-NR^{8a}R^{8b}, or
- (10) $-(C_{1-6})$ alkyl-S(O)₂-(C₁₋₃)alkyl,

wherein each aryl, heteroaryl, cycloalkyl, or heterocyclyl is unsubstituted or substituted with 1, 2, or 3 R^9 , and wherein each alkyl, aryl or heteroaryl is unsubstituted or substituted with 1, 2, or 3 R^{10} .

In Embodiment 28 of this disclosure are compounds of Formula I, or any one of Embodiments 1-26, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^3 is

- (1) 4- to 6-membered heterocyclyl containing 1, 2 or 3 heteroatoms independently selected from N, O and S,
- (2) $-(C_{3-6})$ cycloalkyl,
- (3) $-(C_{1-6})$ hydroxyalkyl,
- (4) $-(C_{1-6})$ alkyl-S(O)₂-NR^{8a}R^{8b}, or
- (5) $-(C_{1-6})$ alkyl-S(O)₂-(C₁₋₃)alkyl,

wherein each cycloalkyl, or heterocyclyl is unsubstituted or substituted with 1, 2, or 3 R^9 , and wherein each alkyl, is unsubstituted or substituted with 1, 2, or 3 R^{10} .

In Embodiment 29 of this disclosure are compounds of Formula I, or any one of Embodiments 1-26, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^3 is

- (1) 4- to 6-membered heterocyclyl containing 1, 2, or 3 heteroatoms independently selected from N, O and S,
- (2) $-(C_{3-6})$ cycloalkyl,
- (3) $-(C_{1-6})$ hydroxyalkyl,
- (4) $-(C_{1-6})$ alkyl-S(O)₂-NR^{8a}R^{8b},
- (5) $-(C_{1-6})$ alkyl-S(O)₂-(C₁₋₃)alkyl,
- (6) (C_{1-6}) alkyl-heterocyclyl, wherein the heterocyclyl is a 5-membered heterocyclyl containing 1 O heteroatom,

17

- (7) 5- to 6-membered heteroaryl containing 1, 2, or 3 heteroatoms independently selected from N, O and S, or
 (8) (C₁₋₆)alkyl-heteroaryl, wherein the heteroaryl is a 5-membered heteroaryl containing 2 N heteroatoms, wherein each aryl, heteroaryl, cycloalkyl, or heterocyclyl is unsubstituted or substituted with 1, 2, or 3 R⁹, and wherein each alkyl is unsubstituted or substituted with 1, 2, or 3 R¹⁰.

In Embodiment 30 of this disclosure are compounds of Formula I, or any one of Embodiments 1-26, or a pharmaceutically acceptable salt of any of the foregoing, wherein R³ is

- (1) 4 to 6 membered cycloalkyl optionally substituted with halogen, hydroxy, C₁₋₆alkyl, C₁₋₆haloalkyl, OC₁₋₃haloalkyl, S(O)₂C₁₋₃alkyl, C₁₋₃alkylS(O)₂C₁₋₃alkyl, OC₁₋₃alkyl, or C₁₋₆alkoxy,
- (2) 4, 5, or 6 membered heterocyclyl containing 1 sulfur atom or 1 oxygen atom optionally mono-substituted, disubstituted, or trisubstituted with C₁₋₆alkyl, C₁₋₃alkyl (OH), OH, C₁₋₃haloalkyl, or oxo,
- (3) C₁₋₆alkyl wherein the alkyl is optionally mono-substituted, disubstituted, or trisubstituted with halogen, OH, CF₃, S(O)₂CH₃, S(O)₂NH-cyclohexyl or C₁₋₃alkyl,
- (4) 6 membered heterocyclyl containing 1 nitrogen atom optionally mono-substituted, disubstituted, or trisubstituted with C₁₋₆alkyl, C₁₋₆haloalkyl, C(O)C₁₋₃alkyl, S(O)₂C₁₋₃alkyl,
- (5) (C₁₋₆)alkyl-heterocyclyl, wherein the heterocyclyl is a 5-membered heterocyclyl containing 1 O heteroatom optionally mono-substituted, disubstituted, trisubstituted with OH,
- (6) 5-membered heteroaryl containing 1 S atom and 2 N heteroatoms optionally mono-substituted, disubstituted, trisubstituted with C₁₋₆alkyl, or
- (7) (C₁₋₆)alkyl-heteroaryl, wherein the heteroaryl is a 5-membered heteroaryl containing 2 N heteroatoms optionally mono-substituted, disubstituted, trisubstituted with C₁₋₆alkyl.

In Embodiment 31 of this disclosure are compounds of Formula I, or any one of Embodiments 1-26 or a pharmaceutically acceptable salt of any of the foregoing, wherein R³ is

- (1) 4 to 6 membered cycloalkyl optionally substituted with F, OH, S(O)₂CH₃, CH₂S(O)₂CH₃, CF₃, OCH₃, CH₃,
- (2) 4 to 6 membered heterocyclyl containing 1 sulfur atom optionally mono-substituted, disubstituted, or trisubstituted with oxo, CHF₂, CH₂CH₃, or CH₃,
- (3) 5 or 6 membered heterocyclyl containing 1 oxygen atom optionally mono-substituted, disubstituted, or trisubstituted with CH₃, OH, or CH₂OH,
- (4) C₁₋₆alkyl, wherein the alkyl is optionally mono-substituted, disubstituted, trisubstituted with halogen, OH, CF₃, S(O)₂CH₃, S(O)₂NH-cyclohexyl, or CH₃,
- (5) 6 membered heterocyclyl containing 1 nitrogen atom optionally N-substituted, disubstituted, or trisubstituted with CH₃, C(O)CH₃, CH₂CHF₂, S(O)₂CH₃,
- (6) (C₁₋₆)alkyl-heterocyclyl, wherein the heterocyclyl is a 5-membered heterocyclyl containing 1 O heteroatom optionally mono-substituted, disubstituted, trisubstituted with OH,
- (7) 5-membered heteroaryl containing 1 S atom and 2 N heteroatoms optionally mono-substituted, disubstituted, trisubstituted with CH₂CH₃, or

18

- (8) CH₃-heteroaryl, wherein the heteroaryl is a 5-membered heteroaryl containing 2 N heteroatoms optionally mono-substituted, disubstituted, trisubstituted with CH₃.

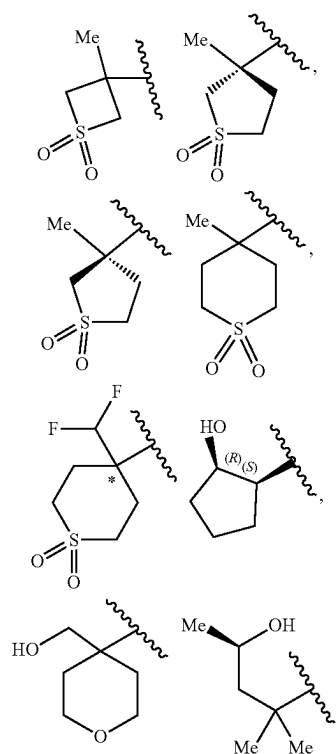
In Embodiment 32 of this disclosure are compounds of Formula I, or any one of Embodiments 1-26, or a pharmaceutically acceptable salt of any of the foregoing, wherein R³ is

- (1) 4 to 6 membered cycloalkyl optionally substituted with halogen, hydroxy, C₁₋₆alkyl, C₁₋₆haloalkyl, or C₁₋₆alkoxy,
- (2) 4, 5, or 6 membered heterocyclyl containing 1 sulfur atom or 1 oxygen atom optionally mono-substituted, disubstituted, or trisubstituted with C₁₋₆alkyl, C₁₋₃alkyl (OH), OH, or oxo,
- (3) C₁₋₆alkyl (OH) wherein the alkyl is optionally mono-substituted, disubstituted, or trisubstituted with halogen, CF₃, or C₁₋₃alkyl.

In Embodiment 33 of this disclosure are compounds of Formula I, or any one of Embodiments 1-26 or a pharmaceutically acceptable salt of any of the foregoing, wherein R³ is

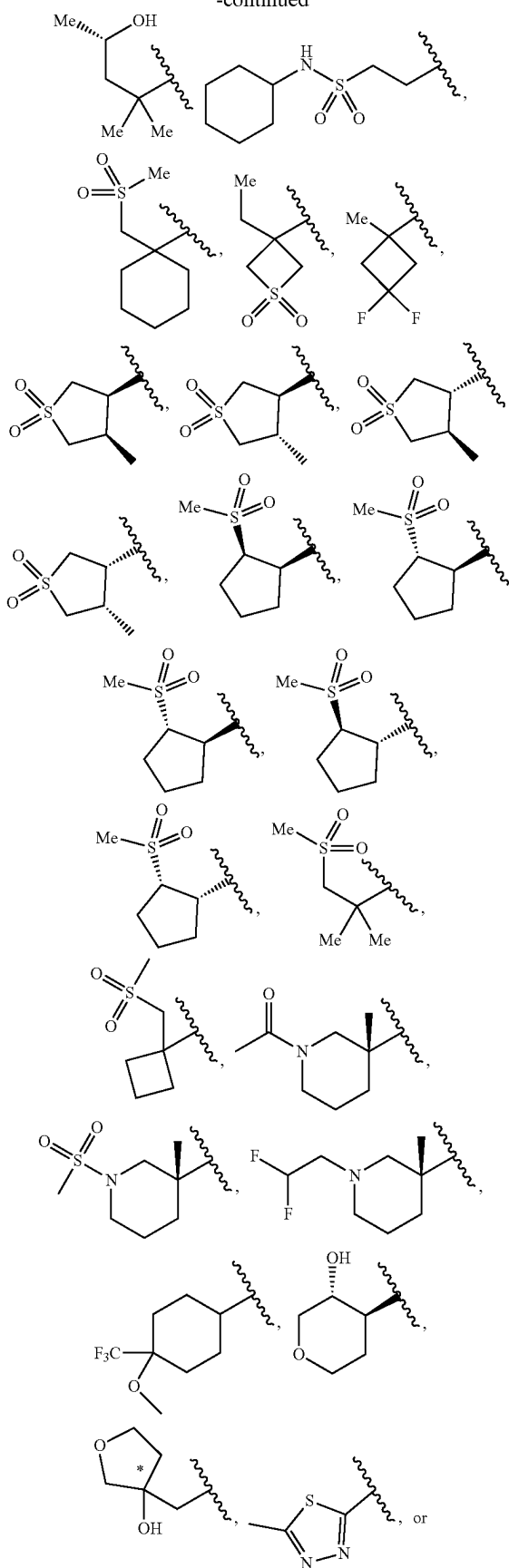
- (1) 4 to 6 membered cycloalkyl optionally substituted with F, OH, S(O)₂CH₃, CH₂S(O)₂CH₃, CF₃, OCH₃,
- (2) 4 to 6 membered heterocyclyl containing 1 sulfur atom optionally mono-substituted, disubstituted, or trisubstituted with oxo, or CH₃;
- (3) 5 or 6 membered heterocyclyl containing 1 oxygen atom optionally mono-substituted, disubstituted, or trisubstituted with CH₃, or CH₂OH; or
- (4) C₁₋₆alkyl, wherein the alkyl is optionally mono-substituted, disubstituted, or trisubstituted with halogen, OH, CF₃, S(O)₂CH₃, S(O)₂NR^{8a}R^{8b}, or CH₃.

In Embodiment 34 of this disclosure are compounds of Formula I, or any one of Embodiments 1-26, or a pharmaceutically acceptable salt of any of the foregoing, wherein R³ is



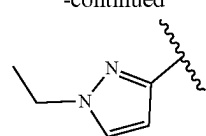
19

-continued

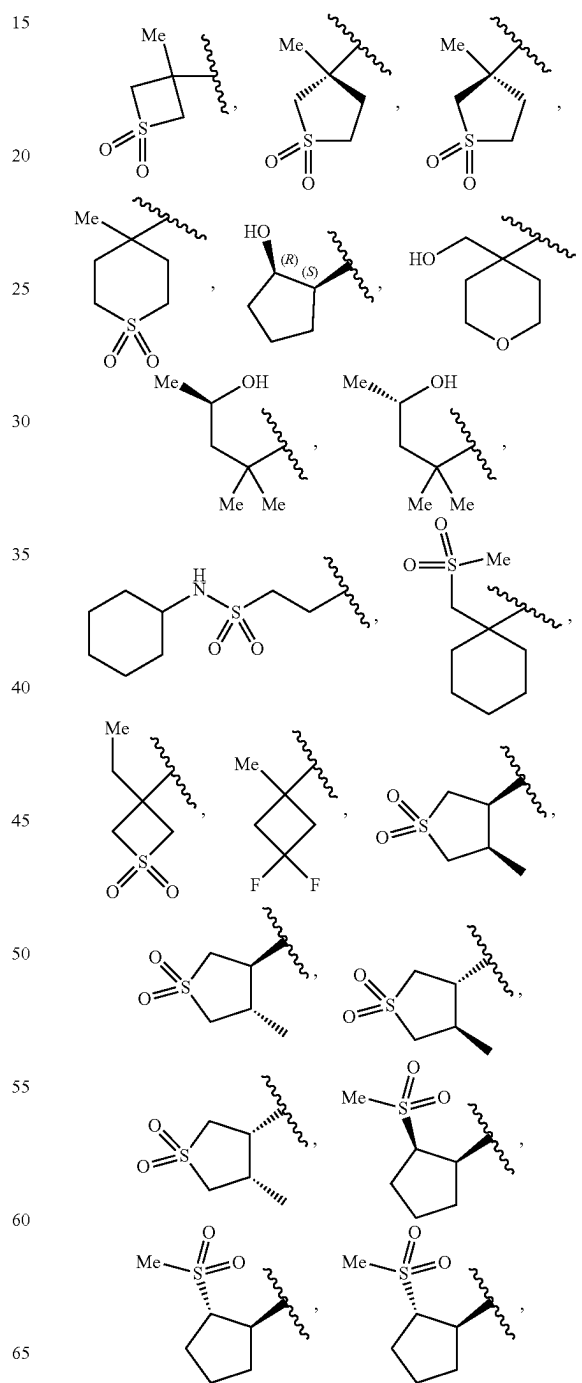


20

-continued

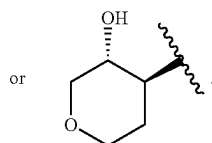
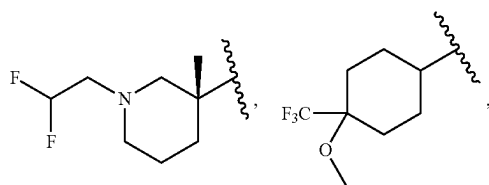
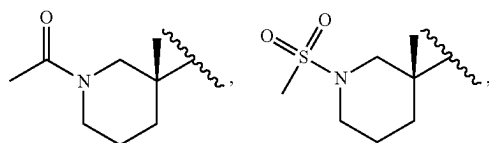
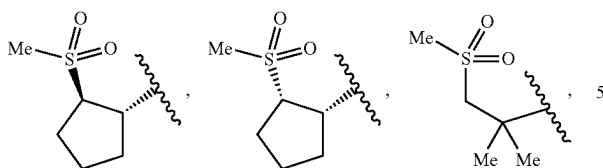


In Embodiment 35 of this disclosure are compounds of
 10 Formula I, or any one of Embodiments 1-26, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^3 is

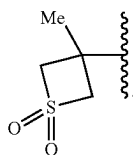


21

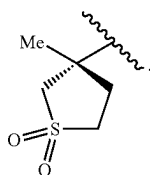
-continued



In Embodiment 36 of this disclosure are compounds of Formula I, or any one of Embodiments 1-26, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^3 is

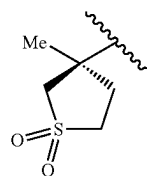


In Embodiment 37 of this disclosure are compounds of Formula I, or any one of Embodiments 1-26, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^3 is

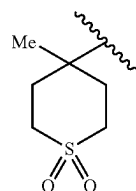


In Embodiment 38 of this disclosure are compounds of Formula I, or any one of Embodiments 1-26, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^3 is

22



In Embodiment 39 of this disclosure are compounds of Formula I, or any one of Embodiments 1-26, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^3 is



In Embodiment 40 of this disclosure are compounds of Formula I, or Embodiments 1-39, or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^4 is H.

In Embodiment 41 of this disclosure are compounds of Formula I, or Embodiments 1-40, or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^5 is H, halogen, or CN.

In Embodiment 42 of this disclosure are compounds of Formula I, or Embodiments 1-41, or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^5 is H, F, Cl, or CN.

In Embodiment 43 of this disclosure are compounds of Formula I, or Embodiments 23-42, or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^6 is cyano, halogen, $-\text{OC}_{1-6}\text{alkyl}$, $(\text{C}_{3-6})\text{cycloalkyl}$ optionally substituted with halogen or OH, $-(\text{C}=\text{O})\text{NH}_2$,

$(\text{C}_{3-6})\text{cycloalkyloxy}$ wherein the cycloalkyl is optionally substituted with halogen or OH, hydroxy, $-\text{N}(\text{R}^{11})_2$, $-\text{NH}(\text{C}=\text{O})(\text{C}_{1-6})\text{alkyl}$, (C_{2-6}) cyclic amine optionally substituted with one or two halogen substituents, $(\text{C}_{1-6})\text{haloalkyl}$, $-\text{O}(\text{C}_{1-6})\text{haloalkyl}$ optionally substituted with OH, $-\text{O}(\text{C}_{0-3})\text{alkyl}-(\text{C}_{3-6})\text{cycloalkyl}$ optionally substituted with one or two halogen, $-\text{SO}_2(\text{C}_{1-6})\text{alkyl}$, $-\text{SO}_2\text{NH}(\text{C}_{1-6})\text{alkyl}$, $-\text{SC}_{1-6}\text{alkyl}$, $-\text{SC}_{1-6}\text{haloalkyl}$, or $(\text{C}_{1-6})\text{alkyl}$.

In Embodiment 44 of this disclosure are compounds of Formula I, or Embodiments 23-43, or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^6 is cyano, halogen, $-\text{OC}_{1-6}\text{alkyl}$, $(\text{C}_{3-6})\text{cycloalkyl}$ optionally substituted with halogen or OH, $-(\text{C}=\text{O})\text{NH}_2$, $(\text{C}_{3-6})\text{cycloalkyloxy}$ wherein the cycloalkyl is optionally substituted with halogen or OH, hydroxy,

$-\text{N}(\text{R}^{11})_2$, $-\text{NH}(\text{C}=\text{O})(\text{C}_{1-6})\text{alkyl}$, (C_{2-6}) cyclic amine optionally substituted with one or two halogen substituents, $(\text{C}_{1-6})\text{haloalkyl}$, $-\text{O}(\text{C}_{1-6})\text{haloalkyl}$, $-\text{SO}_2(\text{C}_{1-6})\text{alkyl}$, $-\text{SO}_2\text{NH}(\text{C}_{1-6})\text{alkyl}$, $-\text{SC}_{1-6}\text{alkyl}$, $-\text{SC}_{1-6}\text{haloalkyl}$.

In Embodiment 45 of this disclosure are compounds of Formula I, or Embodiments 23-43, or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^6 is halogen, hydroxy, CN, $-\text{C}_{1-6}\text{alkyl}$, $-\text{C}_{1-6}\text{haloalkyl}$, $-\text{C}_{3-6}\text{cycloalkyl}$ optionally substituted with halogen or OH, $-\text{OC}_{1-6}\text{alkyl}$, $-\text{OC}_{1-6}\text{haloalkyl}$ optionally

23

substituted with OH, $-\text{O}(\text{C}_{1-3})\text{alkyl}-(\text{C}_{3-6})\text{cycloalkyl}$ optionally substituted with one or two halogen, $-\text{OC}_{3-6}\text{cycloalkyl}$, $-\text{SO}_2\text{NH}(\text{C}_{1-6})\text{alkyl}$, $-\text{S}(\text{O})_2\text{C}_{1-6}\text{alkyl}$, $-\text{S}-\text{C}_{1-6}\text{alkyl}$, $-\text{SC}_{1-6}\text{haloalkyl}$, or $\text{NH}(\text{C}=\text{O})\text{C}_{1-3}\text{alkyl}$, (C_{2-6}) cyclic amine optionally substituted with OH, $\text{N}(\text{C}_{1-3})\text{alkyl}$, $-\text{O}(\text{C}_{3-6})\text{cycloalkyl}$.

In Embodiment 46 of this disclosure are compounds of Formula I, or Embodiments 23-43, and 44 or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^6 is halogen, hydroxy, CN, $-\text{C}_{1-6}\text{alkyl}$, $-\text{C}_{1-6}\text{haloalkyl}$, $-\text{C}_{3-6}\text{cycloalkyl}$ optionally substituted with halogen or OH, $-\text{OC}_{1-6}\text{alkyl}$, $-\text{OC}_{1-6}\text{haloalkyl}$, $-\text{OC}_{3-6}\text{cycloalkyl}$, $-\text{S}(\text{O})_2\text{C}_{1-6}\text{alkyl}$, $-\text{S}-\text{C}_{1-6}\text{alkyl}$, $-\text{SC}_{1-6}\text{haloalkyl}$, or $\text{NH}(\text{C}=\text{O})\text{C}_{1-3}\text{alkyl}$, and wherein the cycloalkyl is optionally substituted with halogen, CN, or OH.

In Embodiment 47 of this disclosure are compounds of Formula I, or Embodiments 23-43, and 44 or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^6 is $-\text{F}$, OH, $\text{C}_{1-6}\text{alkyl}$, $-\text{C}_{1-6}\text{haloalkyl}$, $\text{O}-\text{C}_{1-6}\text{cycloalkyl}$, $-\text{OC}_{1-6}\text{alkyl}$, $-\text{OC}_{1-6}\text{haloalkyl}$, $-\text{S}-\text{C}_{1-6}\text{alkyl}$, $-\text{SC}_{1-6}\text{haloalkyl}$, or $-\text{NH}(\text{C}=\text{O})\text{C}_{1-3}\text{alkyl}$, wherein the cycloalkyl is additionally optionally substituted with 1-3 F.

In Embodiment 48 of this disclosure are compounds of Formula I, or Embodiments 23-43, and 45, or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^6 is F, CN, OH, $-\text{CH}_3$, $-\text{CF}_2\text{CH}_3$, $-\text{OCF}_2\text{CH}_3$, $-\text{OCHF}_2$, $-\text{OCF}_3$, OCH_2CF_3 , $-\text{OCH}_2\text{CH}_3$, $\text{OCH}(\text{CH}_3)_2$, $-\text{OCF}_2\text{CHF}_2$, $-\text{OCH}_2\text{CHF}_2$, $-\text{OCHF}_2\text{CH}_3$, $\text{OCH}_2\text{CF}_2\text{CH}_3$, $\text{OCH}_2\text{CH}(\text{CH}_3)_2$, $\text{OCH}_2\text{CF}_2\text{CHF}_2$, $-\text{cyclopropyl}$, $-\text{O-cyclopropyl}$, $\text{OCH}_2\text{-cyclopropyl}$ optionally substituted with 1 or 2 F, $\text{OCH}_2\text{-cyclobutyl}$ optionally substituted with 1 or 2 F, azetidine optionally substituted with 1 or 2 F, $-\text{SCHF}_2$, $\text{S}(\text{O})_2\text{CH}_3$, $\text{S}(\text{O})_2\text{NHCH}_3$, $\text{C}(\text{O})\text{NH}_2$, $\text{N}(\text{CH}_3)_2$, $\text{OCF}_2\text{CH}_2\text{OH}$, or $-\text{NH}(\text{C}=\text{O})\text{CH}_3$.

In Embodiment 49 of this disclosure are compounds of Formula I, or Embodiments 23-48, or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^6 is F, OH, $-\text{CH}_3$, $-\text{CF}_2\text{CH}_3$, $-\text{OCHF}_2$, $-\text{OCF}_3$, $-\text{OCH}_2\text{CH}_3$, $\text{OCH}(\text{CH}_3)_2$, $-\text{OCF}_2\text{CHF}_2$, $-\text{OCH}_2\text{CHF}_2$, $-\text{OCHF}_2\text{CH}_3$, $-\text{cyclopropyl}$, $-\text{O-cyclopropyl}$, $-\text{SCHF}_2$, or $-\text{NH}(\text{C}=\text{O})\text{CH}_3$.

In Embodiment 50 of this disclosure are compounds of Formula I, or Embodiments 1-49, or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^7 is halogen, $\text{C}_{1-6}\text{haloalkyl}$, or OH.

In Embodiment 51 of this disclosure are compounds of Formula I, or Embodiments 1-50, or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^7 is F, CF_3 , or OH.

In Embodiment 52 of this disclosure are compounds of Formula I, or Embodiments 1-51, or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^{8a} is H.

In Embodiment 53 of this disclosure are compounds of Formula I, or Embodiments 1-52, or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^{8b} is H.

In Embodiment 54 of this disclosure are compounds of Formula I, or Embodiments 1-51 or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^{8a} and R^{8b} are independently selected from hydrogen and cyclohexyl.

In Embodiment 55a of this disclosure are compounds of Formula I, or Embodiments 1-54, or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing,

24

wherein R^9 is $=\text{O}$, $-\text{OH}$, halogen, $\text{OC}_{1-3}\text{alkyl}$, $\text{C}_{1-3}\text{alkyl}$, $\text{C}_{1-3}\text{haloalkyl}$, $\text{C}_{1-3}\text{alkylOH}$, $-\text{SO}_2(\text{C}_{1-3})\text{alkyl}$, or $-\text{C}(\text{C}_{1-3})\text{alkylSO}_2(\text{C}_{1-3})\text{alkyl}$, $\text{C}(\text{O})\text{C}_{1-3}\text{alkyl}$.

In Embodiment 55b of this disclosure are compounds of Formula I, or Embodiments 1-55, or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^9 is $=\text{O}$, $-\text{OH}$, $\text{C}_{1-3}\text{alkyl}$, $\text{C}_{1-3}\text{haloalkyl}$, $\text{C}_{1-3}\text{alkylOH}$, $-\text{SO}_2(\text{C}_{1-3})\text{alkyl}$, or $-\text{C}(\text{C}_{1-3})\text{alkylSO}_2(\text{C}_{1-3})\text{alkyl}$.

In Embodiment 56 of this disclosure are compounds of Formula I, or Embodiments 1-56, or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^9 is $=\text{O}$, OH, CH_3 , CH_2CH_3 , F, CF_3 , CH_2OH , OCH_3 , SO_2CH_3 , or $\text{CH}_2\text{SO}_2\text{CH}_3$.

In Embodiment 57 of this disclosure are compounds of Formula I, or Embodiments 1-56, or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^{10} is hydrogen, OH, $\text{C}_{1-3}\text{alkyl}$, $\text{C}_{1-3}\text{haloalkyl}$, $\text{C}_{1-3}\text{alkylOH}$, or $-\text{C}(\text{C}_{1-6})\text{alkylSO}_2(\text{C}_{1-6})\text{alkyl}$.

In Embodiment 58 of this disclosure are compounds of Formula I, or Embodiments 1-57, or a class thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^{10} is hydrogen, OH, CH_3 , CH_2CH_3 , F, CF_3 , CH_2OH or $\text{CH}_2\text{SO}_2\text{CH}_3$.

In Embodiment 59 of this disclosure are compounds of Formula I, or any one of Embodiments 1-58, or a pharmaceutically acceptable salt of any of the foregoing, wherein X is N, Y is $\text{C}(\text{R}^5)$, and Z is $\text{C}(\text{R}^5)$.

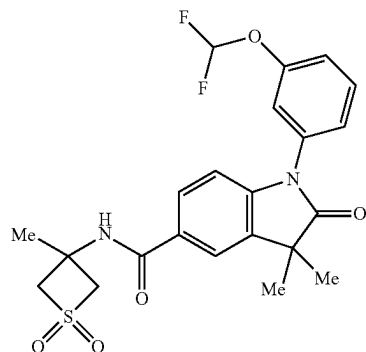
In Embodiment 60 of this disclosure are compounds of Formula I, or any one of Embodiments 1-58, or a pharmaceutically acceptable salt of any of the foregoing, wherein X is $\text{C}(\text{R}^5)$, Y is N, and Z is $\text{C}(\text{R}^5)$.

In Embodiment 61 of this disclosure are compounds of Formula I, or any one of Embodiments 1-58, or a pharmaceutically acceptable salt of any of the foregoing, wherein X, Y, and Z are $\text{C}(\text{R}^5)$.

In Embodiment 62 of this disclosure are compounds of Formula I, or any one of Embodiments 1-58, or a pharmaceutically acceptable salt of any of the foregoing, wherein X is N, Y is N and Z is $\text{C}(\text{R}^5)$.

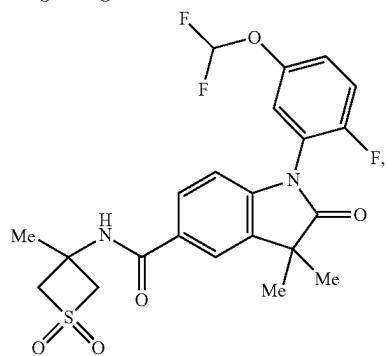
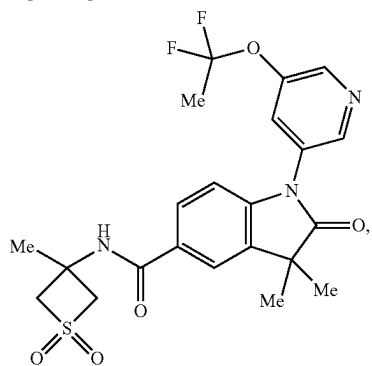
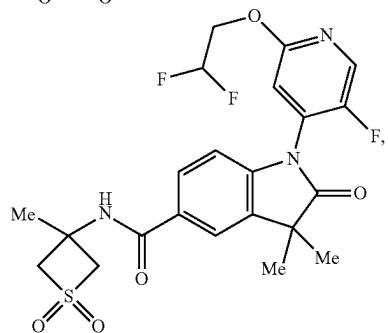
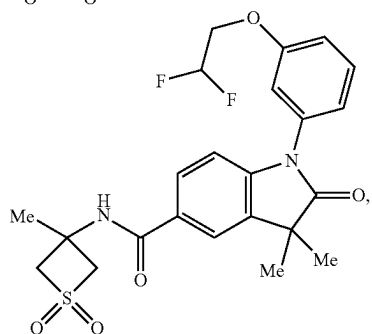
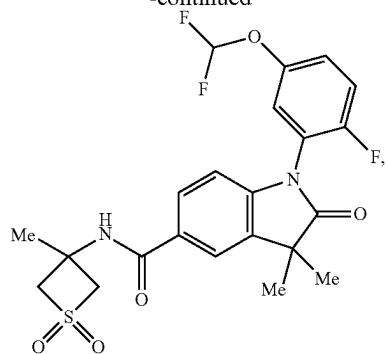
In Embodiment 63 of the present invention provides a compound as described in any one of Examples 1-159 as set forth below, or a pharmaceutically acceptable salt thereof.

In Embodiment 64, the present invention is a compound selected from:



25

-continued

**26**

-continued

5

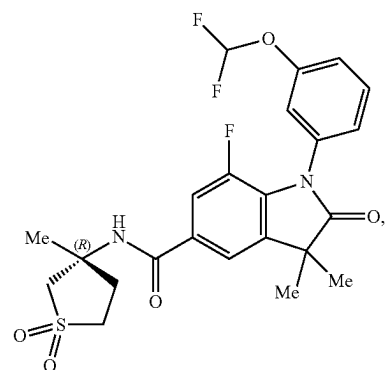
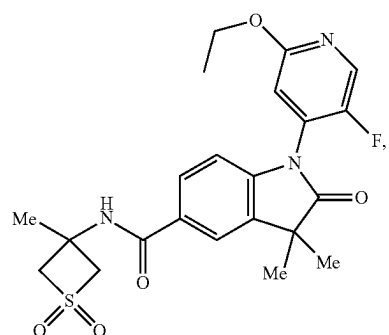
10

15

20

25

30



or a pharmaceutically acceptable salt thereof.

In Embodiment 65, the present invention is a compound selected from:

35

40

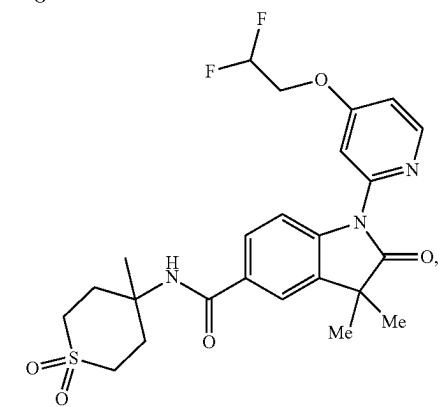
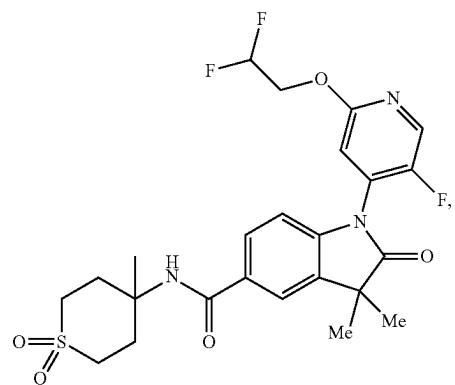
45

50

55

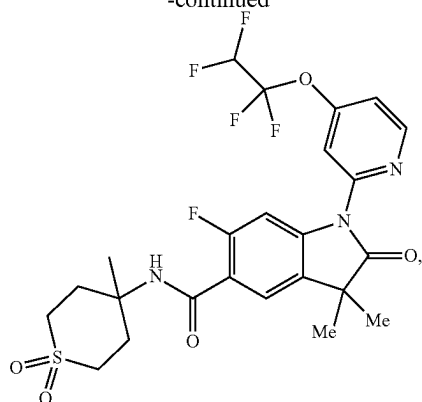
60

65



27

-continued

**28**

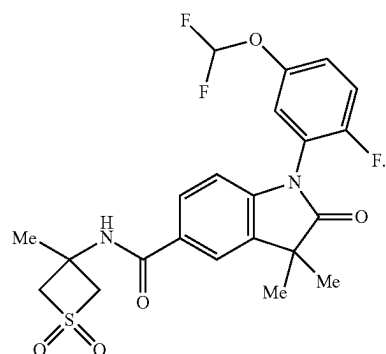
In Embodiment 67, the present invention is a compound,
or a pharmaceutically acceptable salt thereof, which is

5

10

15

20



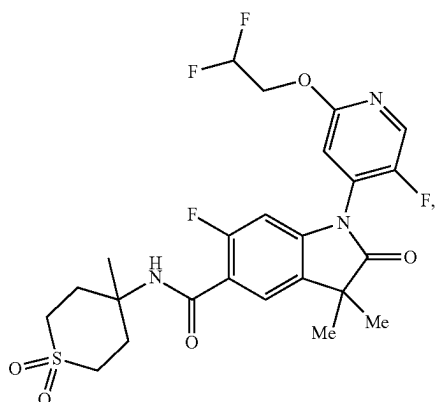
In Embodiment 68, the present invention is a compound,
or a pharmaceutically acceptable salt thereof, which is

30

35

40

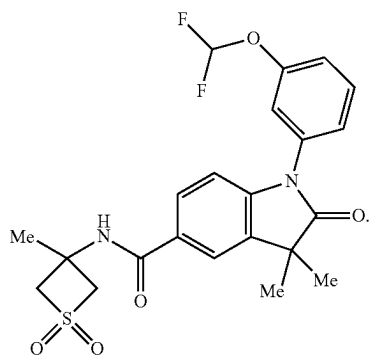
45



or a pharmaceutically acceptable salt thereof.

In Embodiment 66, the present invention is a compound,
or a pharmaceutically acceptable salt thereof, which is

50

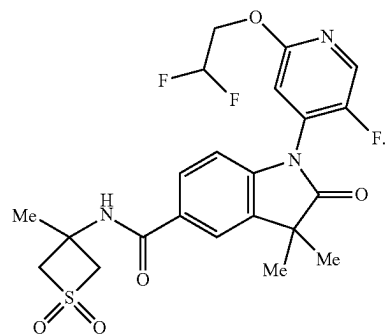


In Embodiment 69, the present invention is a compound,
or a pharmaceutically acceptable salt thereof, which is

55

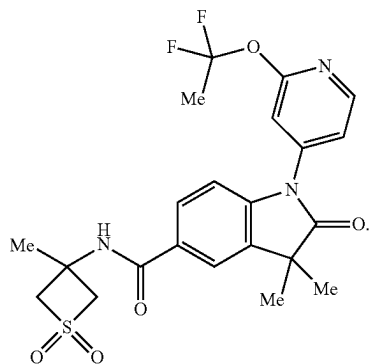
60

65

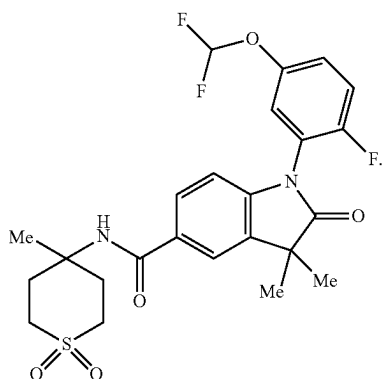


29

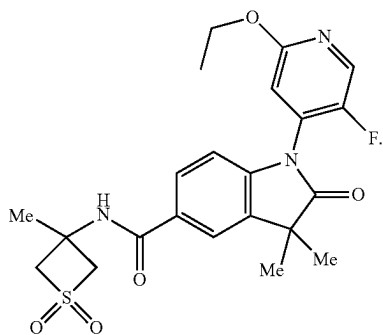
In Embodiment 70, the present invention is a compound, or a pharmaceutically acceptable salt thereof, which is



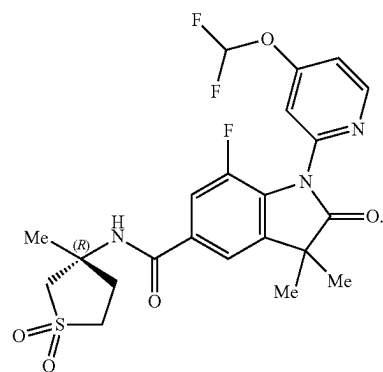
In Embodiment 71, the present invention is a compound, or a pharmaceutically acceptable salt thereof, which is



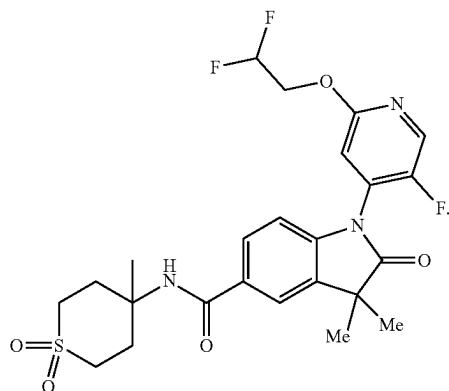
In Embodiment 72, the present invention is a compound, or a pharmaceutically acceptable salt thereof, which is

**30**

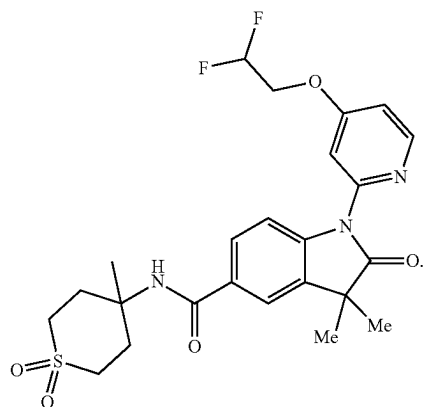
In Embodiment 73, the present invention is a compound, or a pharmaceutically acceptable salt thereof, which is



In Embodiment 74, the present invention is a compound, or a pharmaceutically acceptable salt thereof, which is

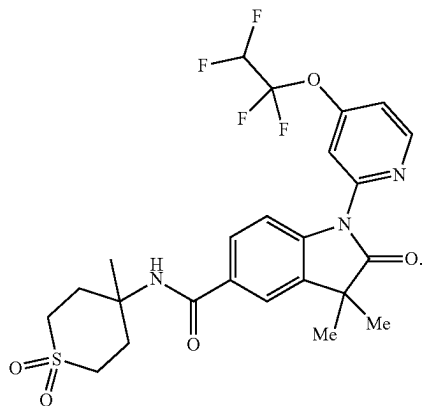


In Embodiment 75, the present invention is a compound, or a pharmaceutically acceptable salt thereof, which is

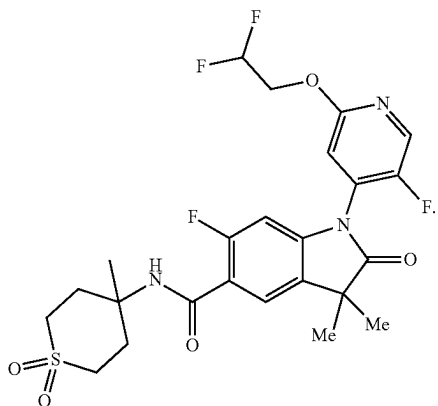


31

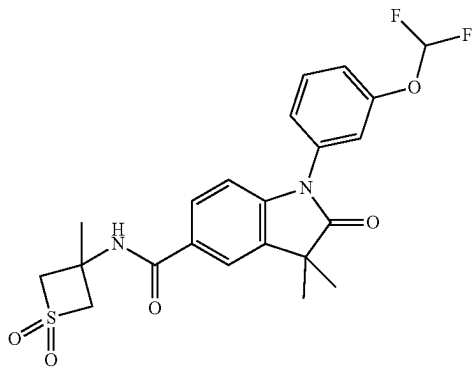
In Embodiment 76, the present invention is a compound, or a pharmaceutically acceptable salt thereof, which is



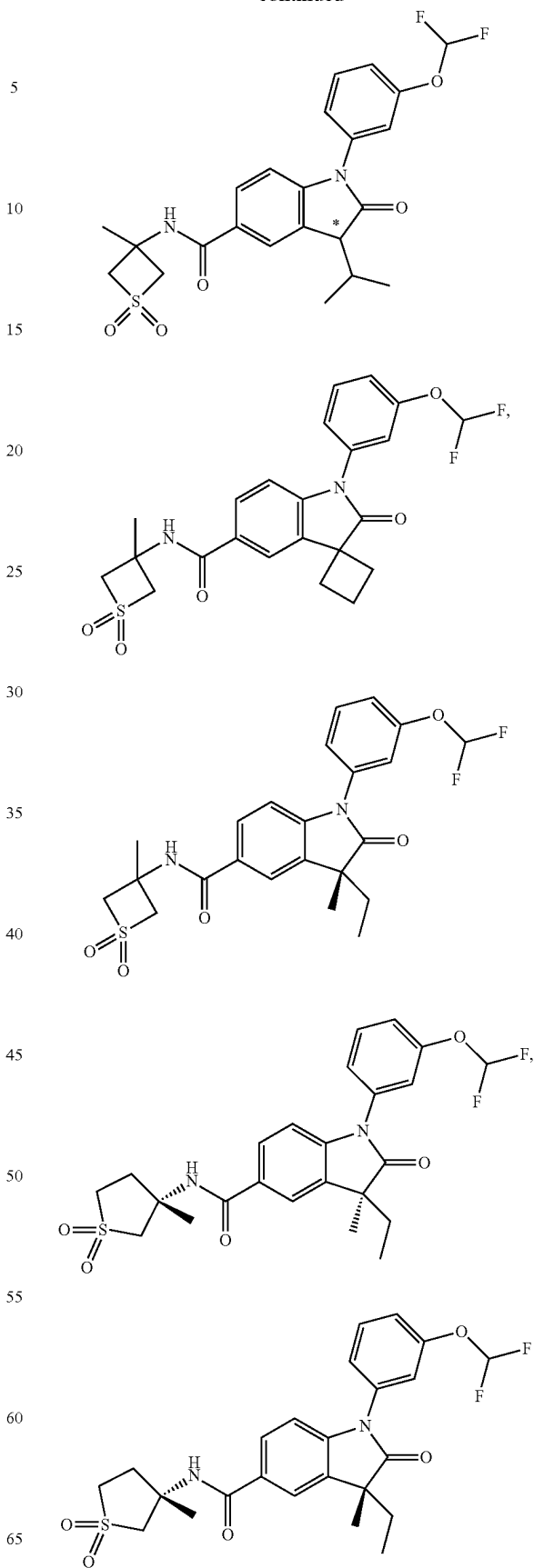
In Embodiment 77, the present invention is a compound, or a pharmaceutically acceptable salt thereof, which is



In Embodiment 78 of the invention, the compound of formula I, or a pharmaceutically acceptable salt thereof, is:

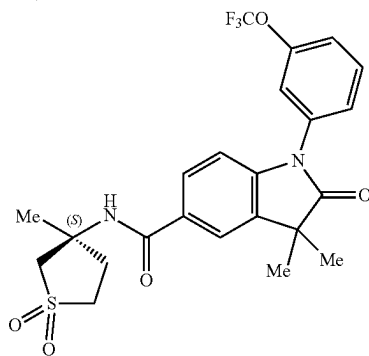
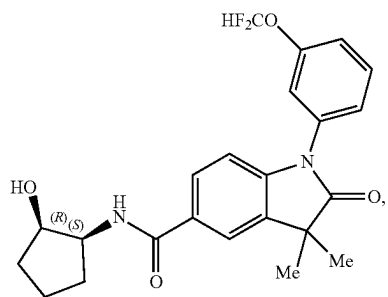
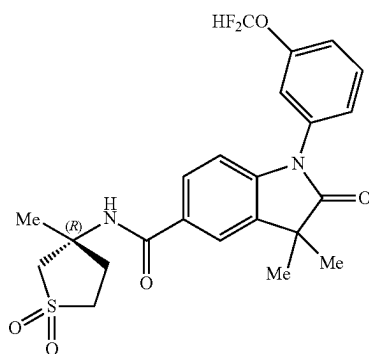
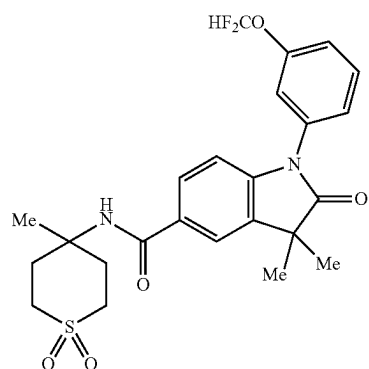
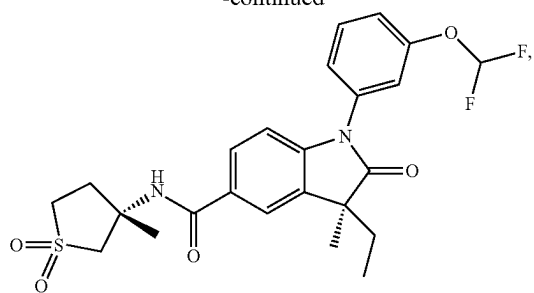
**32**

-continued

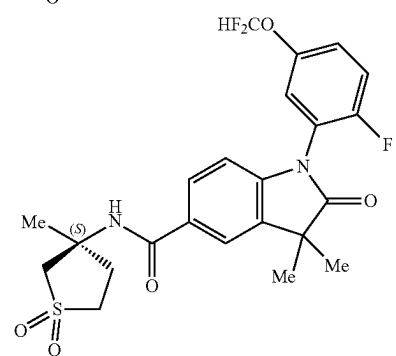
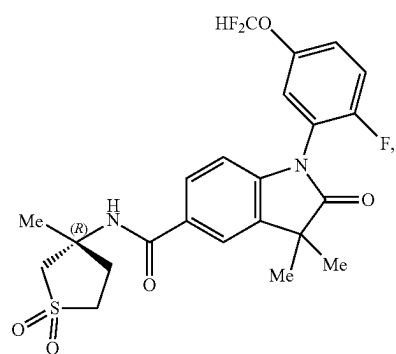
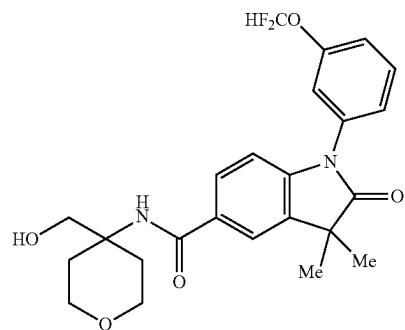
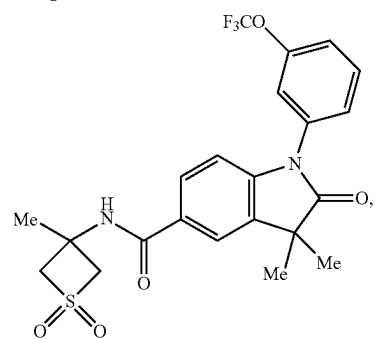
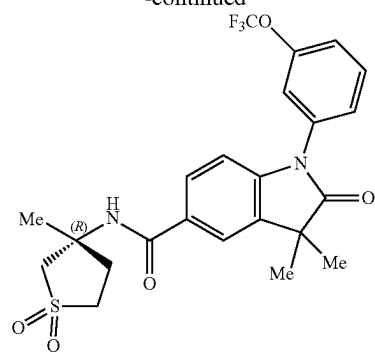


33

-continued

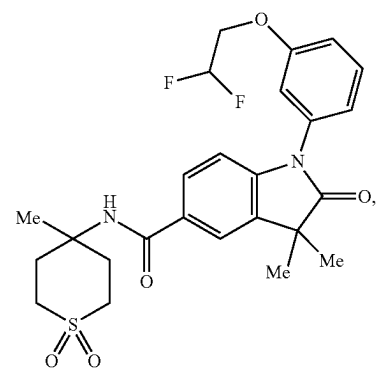
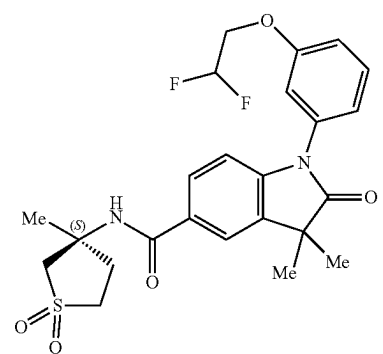
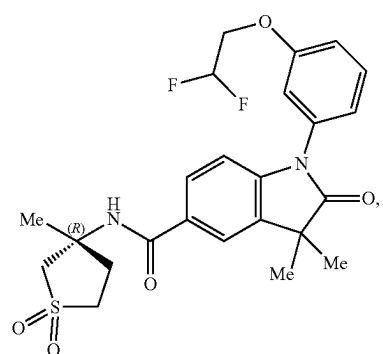
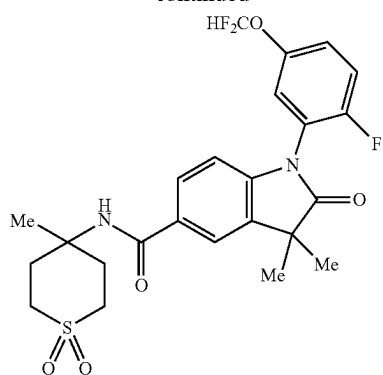
**34**

-continued

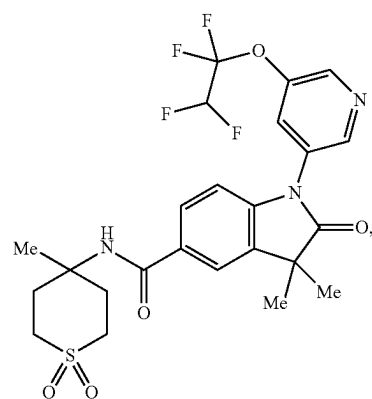
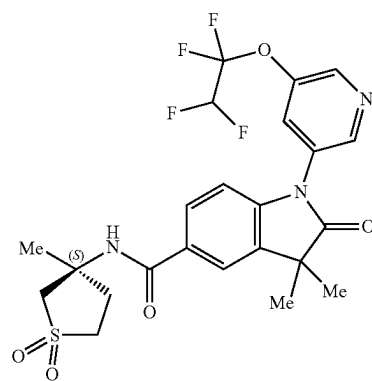
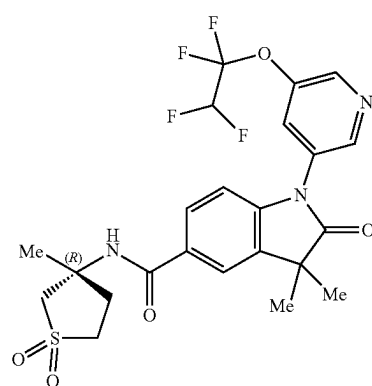
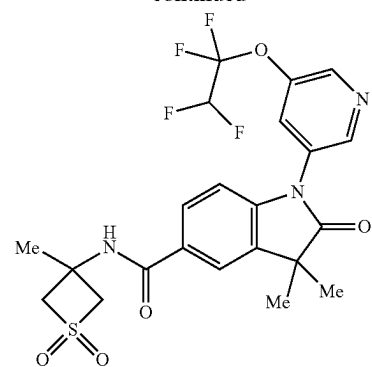


35

-continued

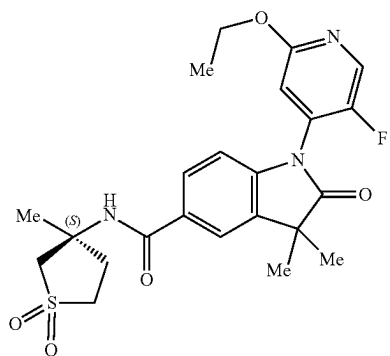
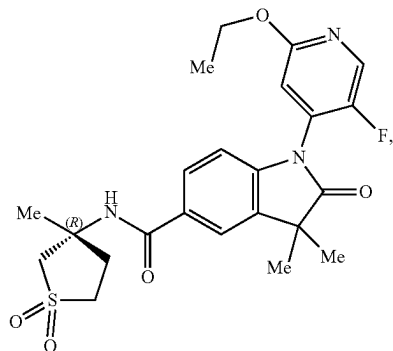
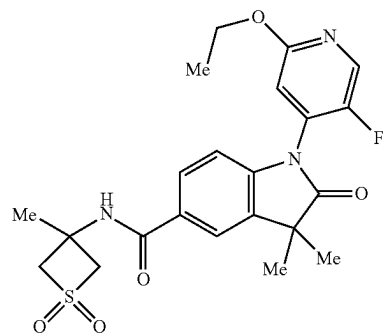
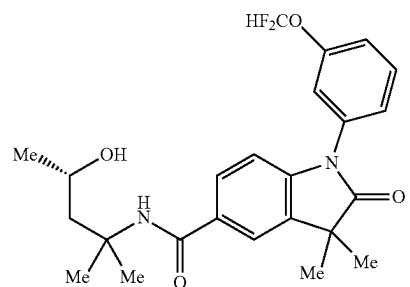
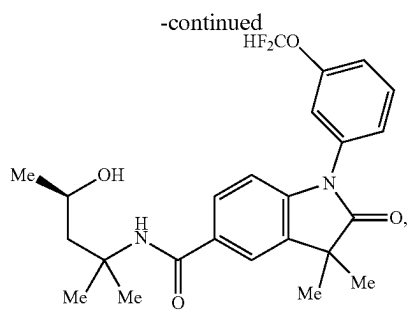
**36**

-continued



37

-continued

**38**

-continued

5

10

15

20

25

30

35

40

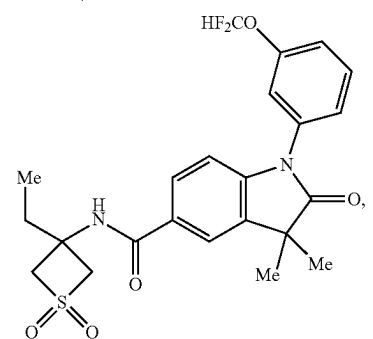
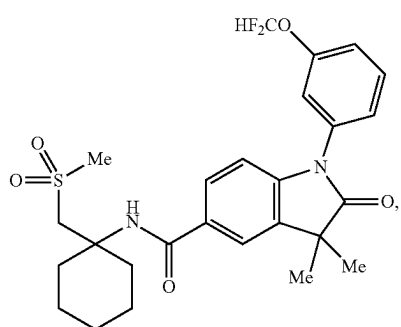
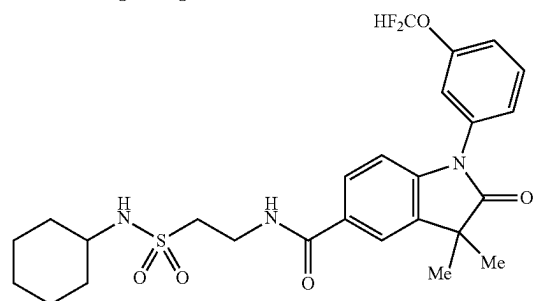
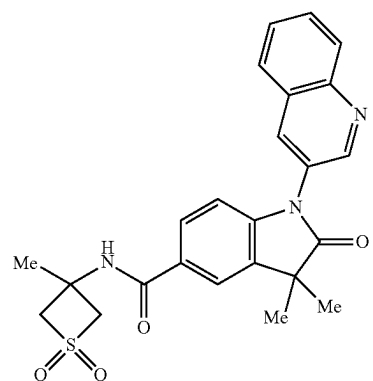
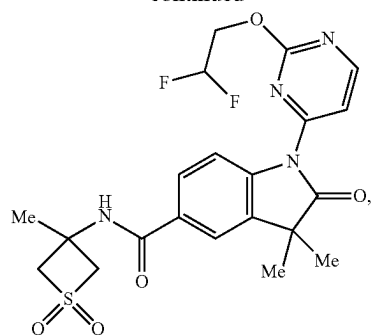
45

50

55

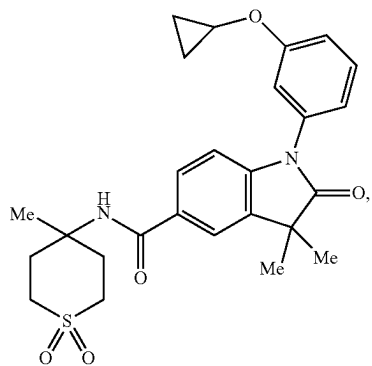
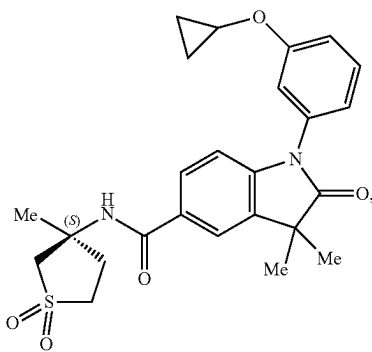
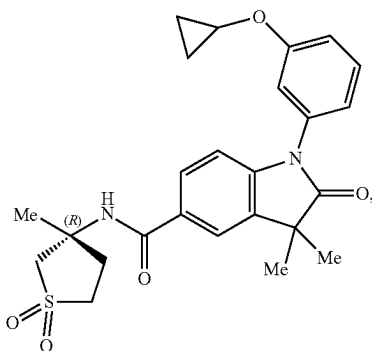
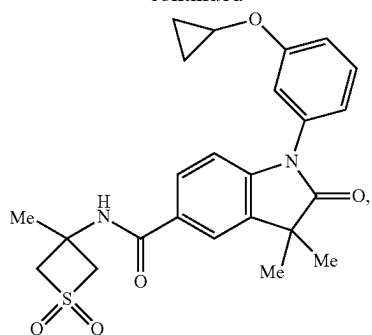
60

65

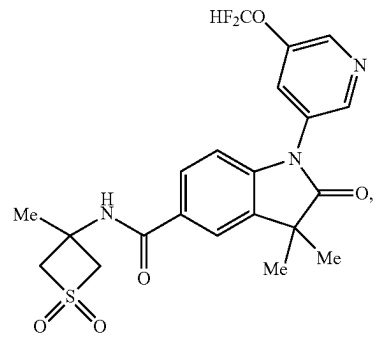
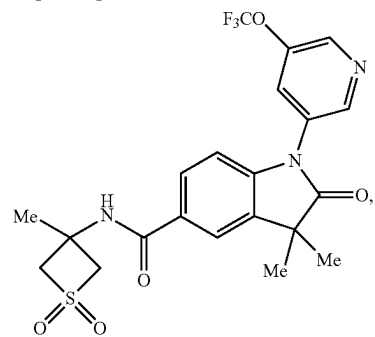
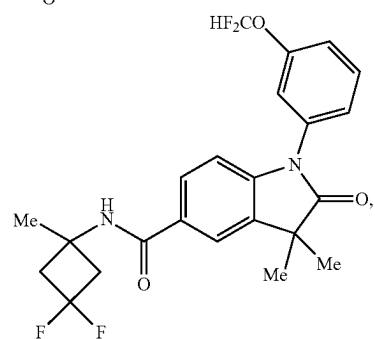
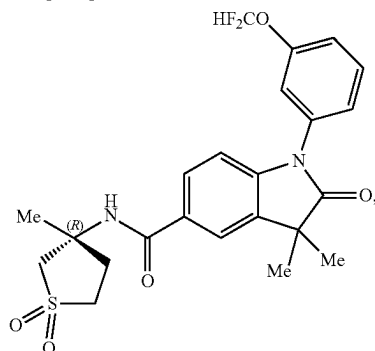
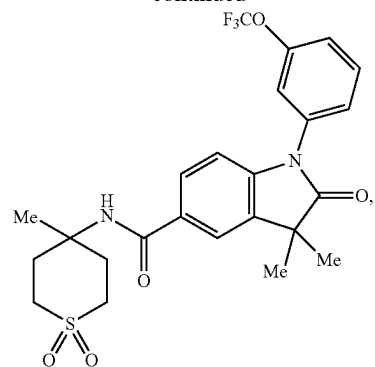


39

-continued

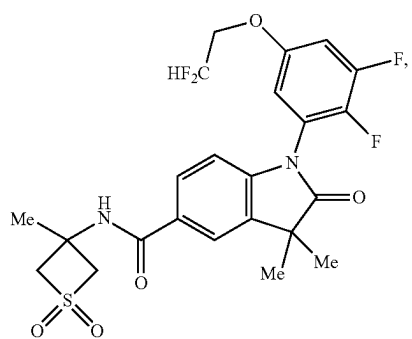
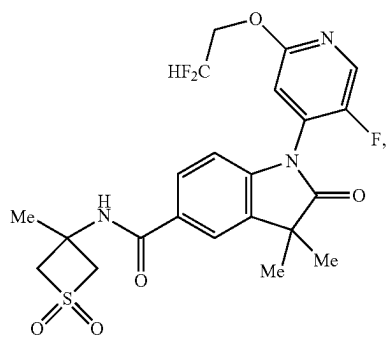
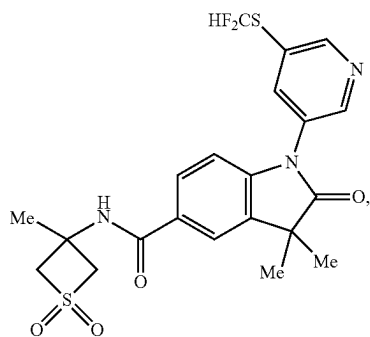
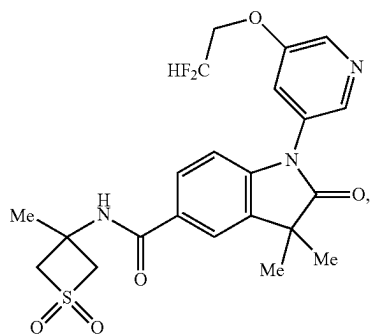
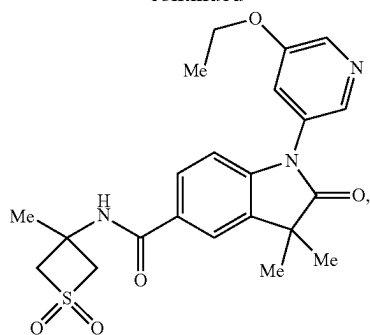
**40**

-continued

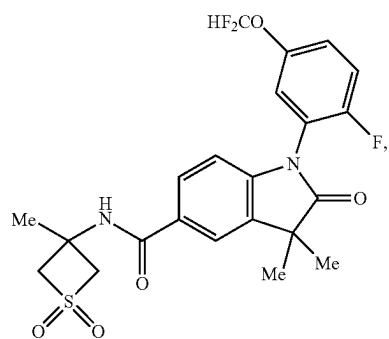
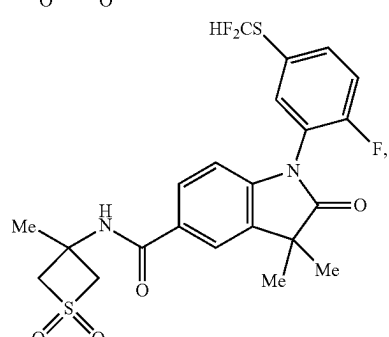
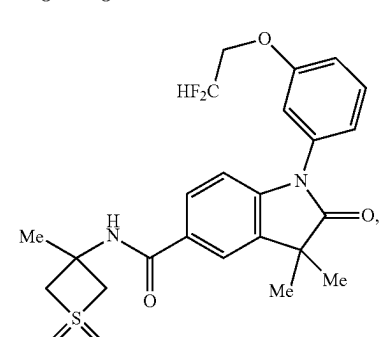
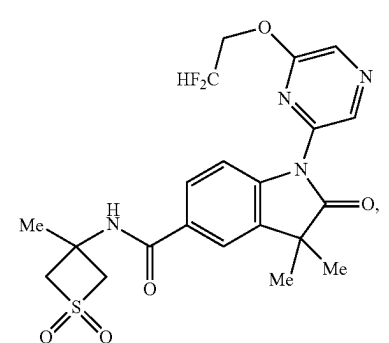
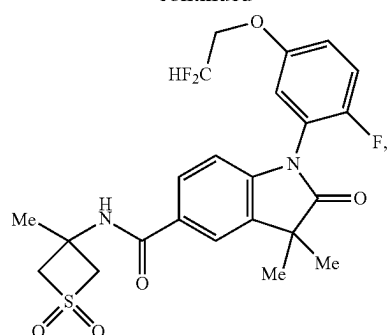


41

-continued

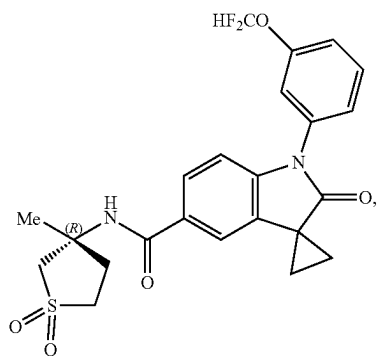
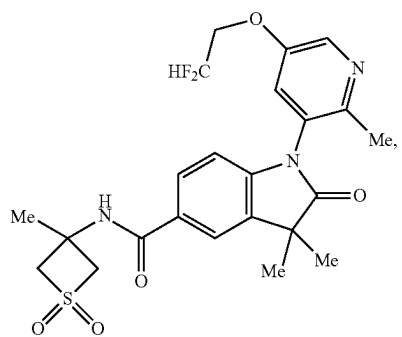
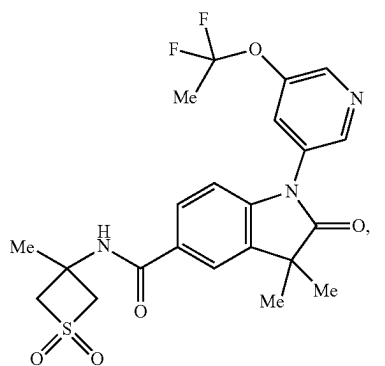
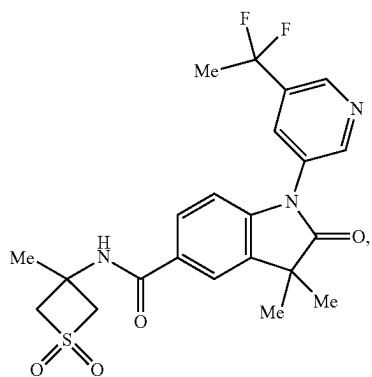
**42**

-continued



43

-continued

**44**

-continued

5

10

15

20

25

30

35

40

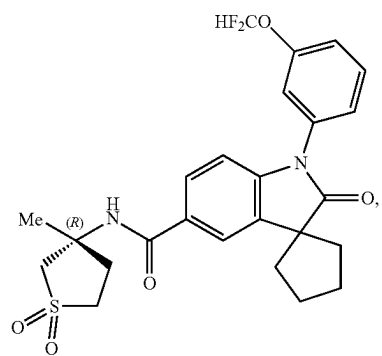
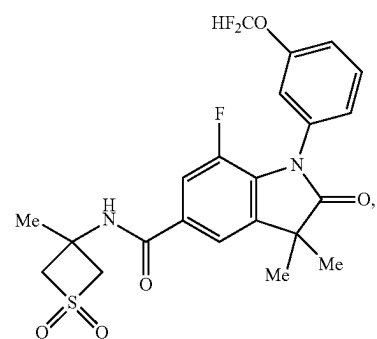
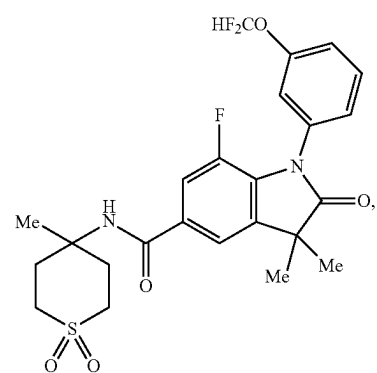
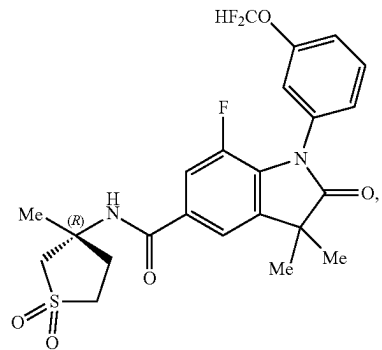
45

50

55

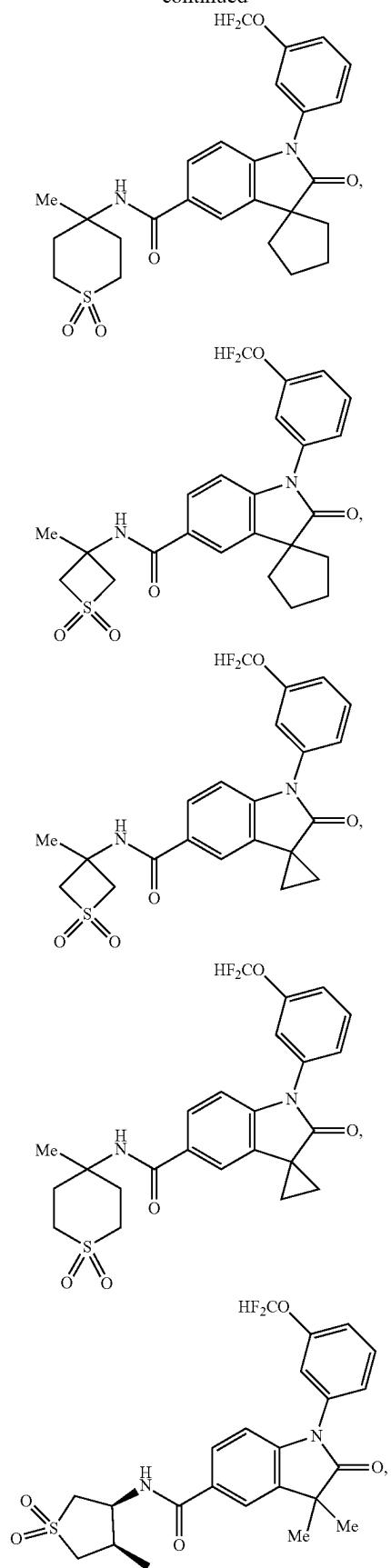
60

65

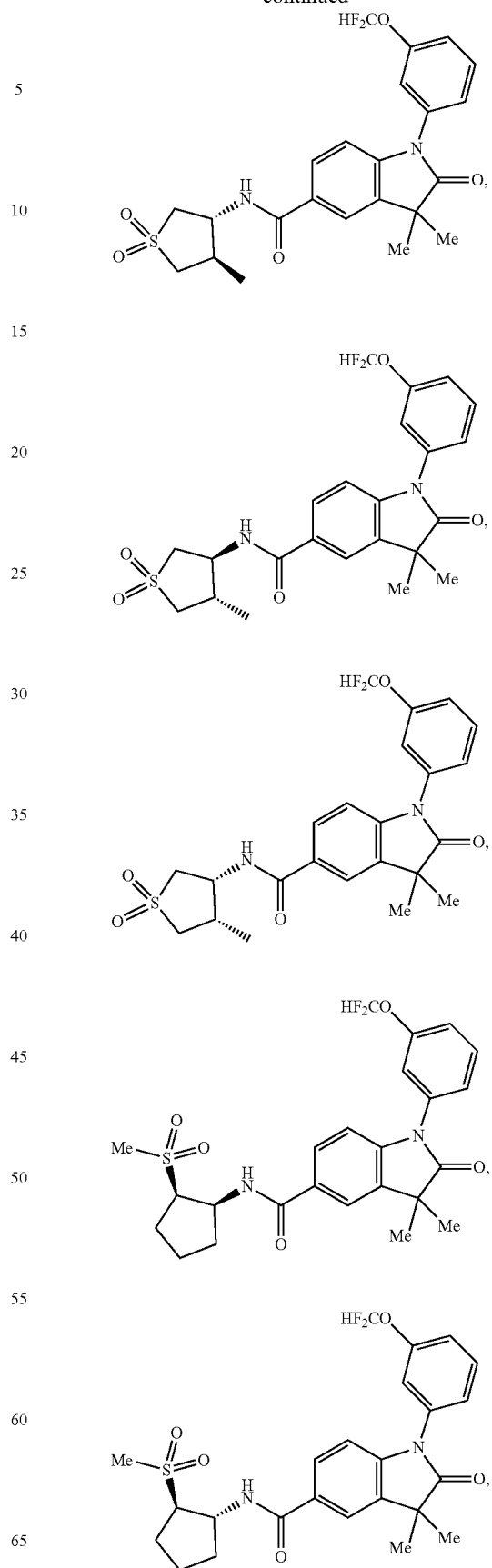


45

-continued

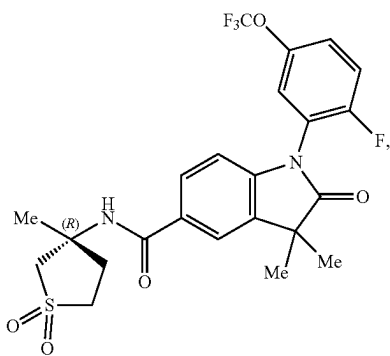
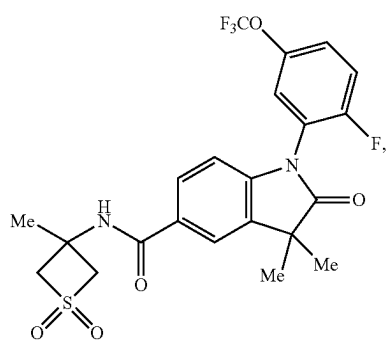
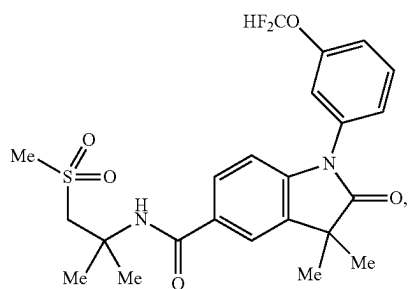
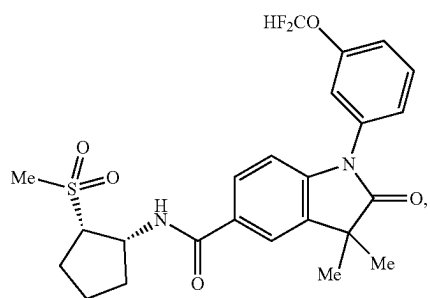
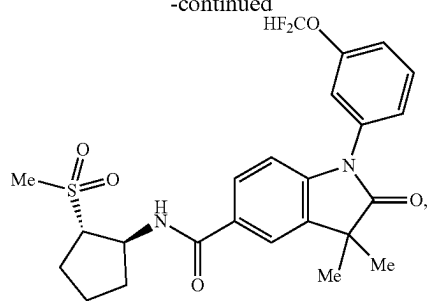
**46**

-continued

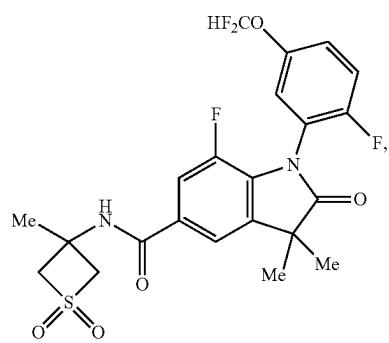
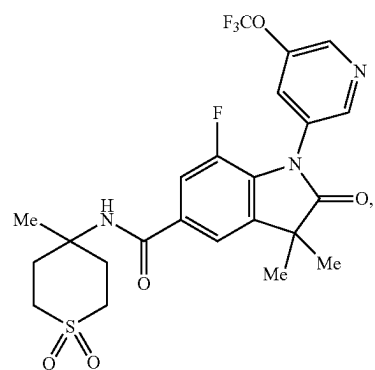
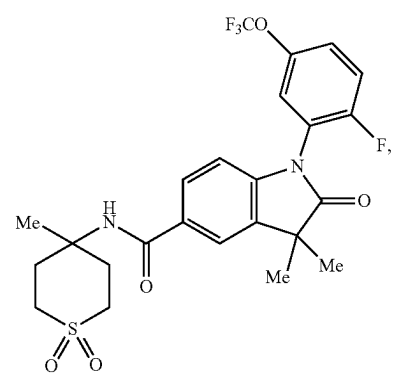
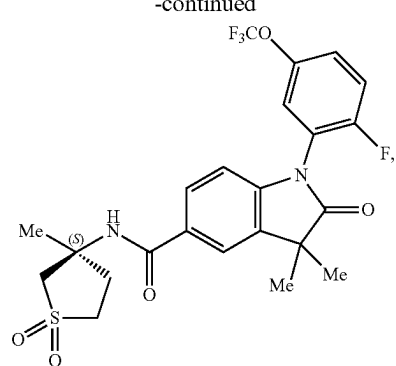


47

-continued

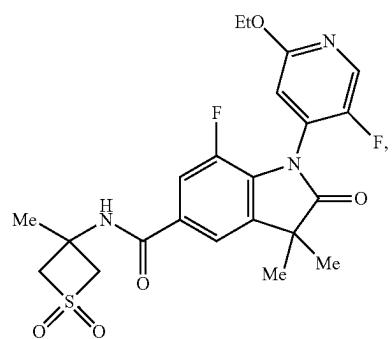
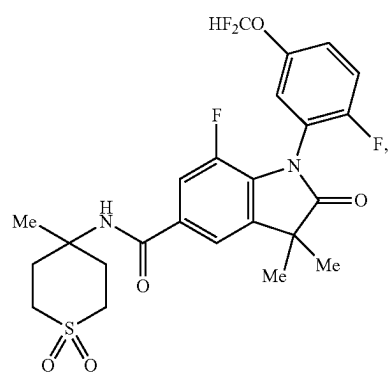
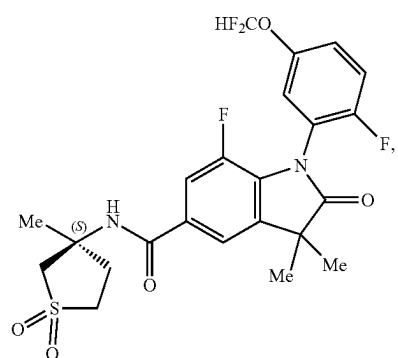
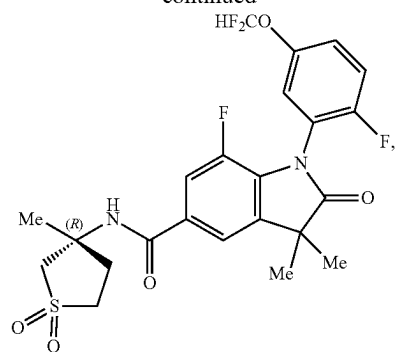
**48**

-continued

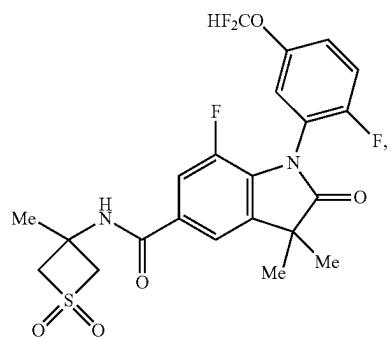
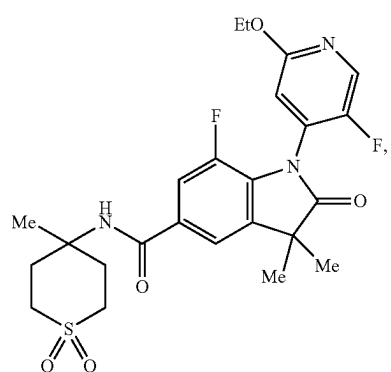
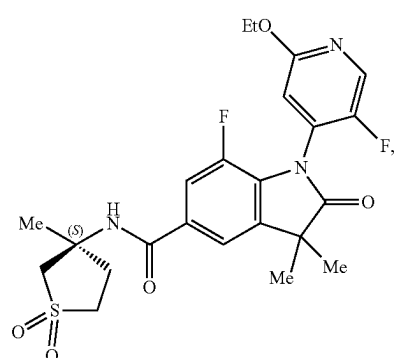
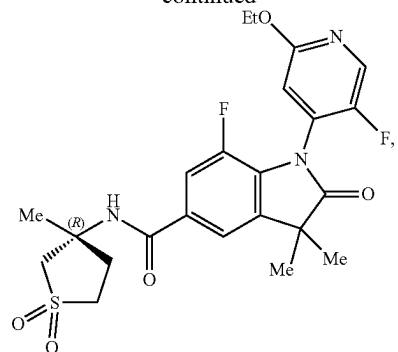


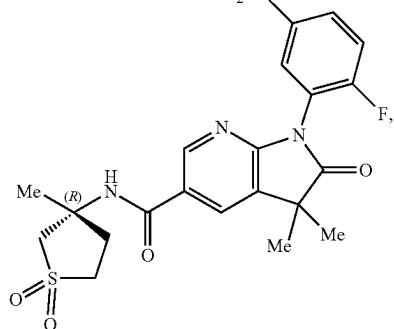
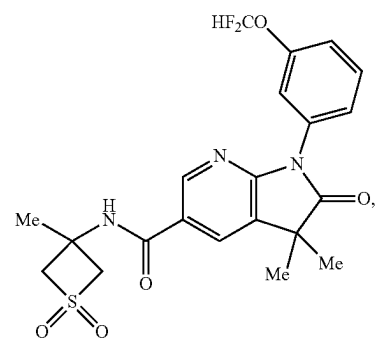
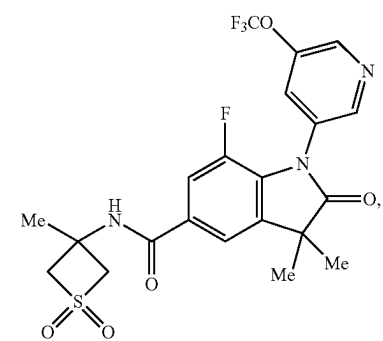
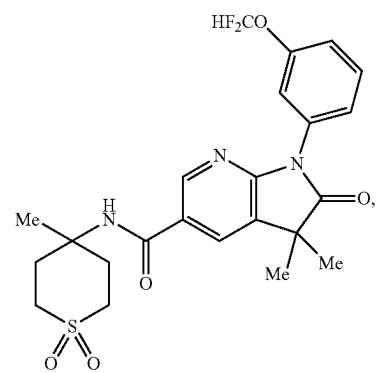
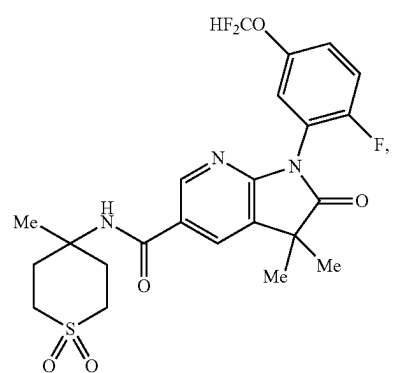
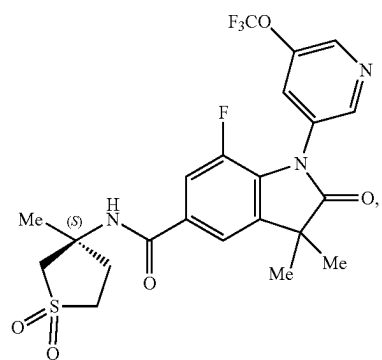
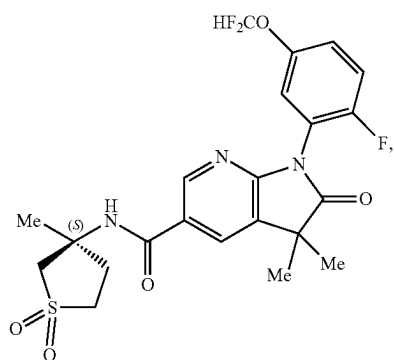
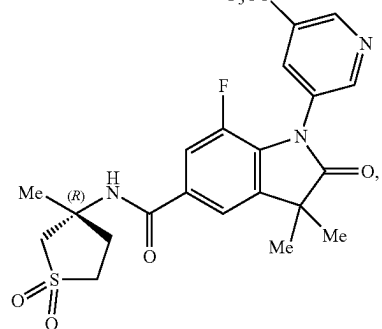
49

-continued

**50**

-continued



51-continued
HF₂CO**52**-continued
F₃CO

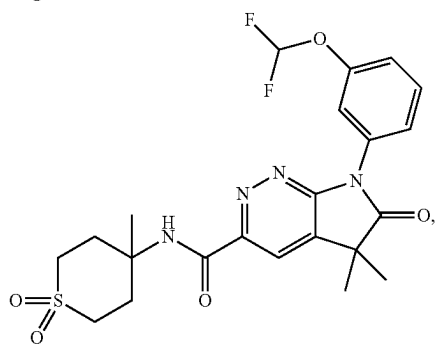
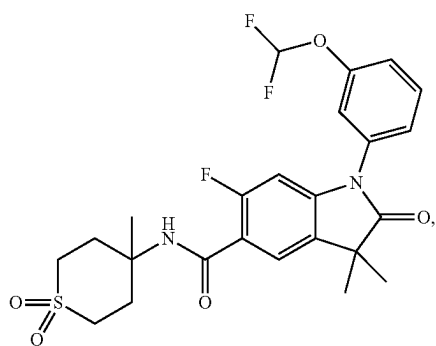
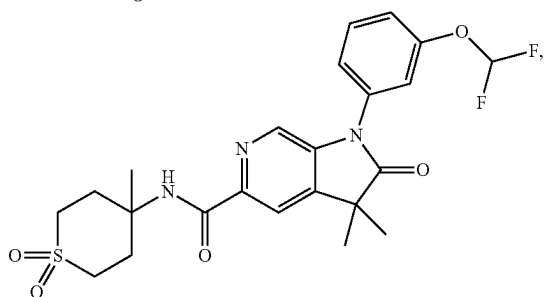
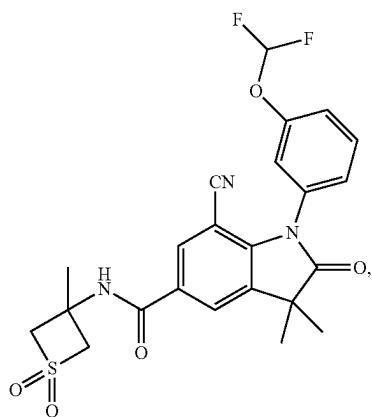
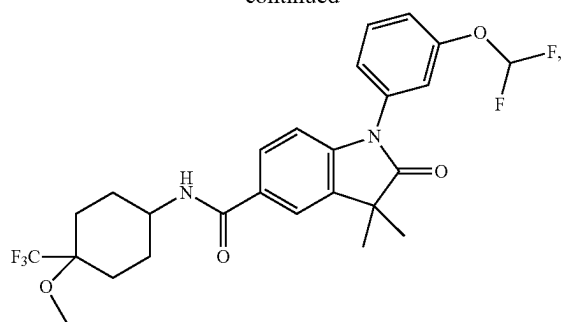
54

-continued

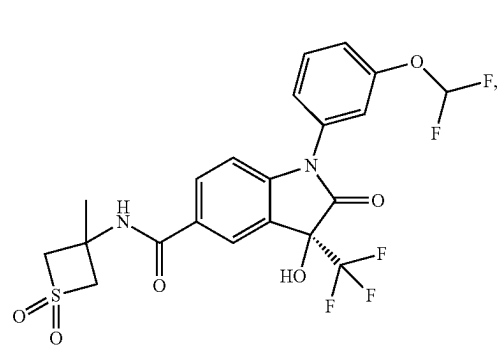
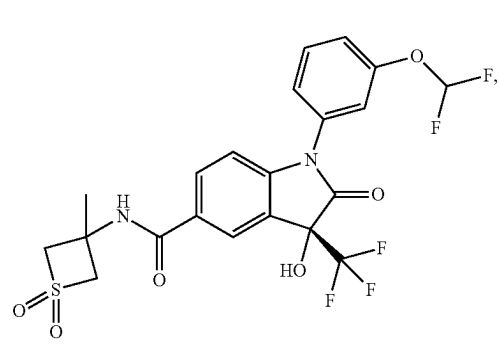
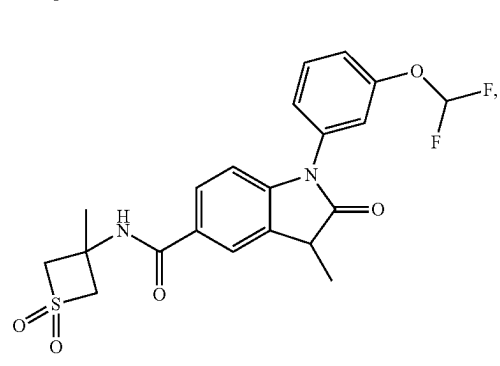
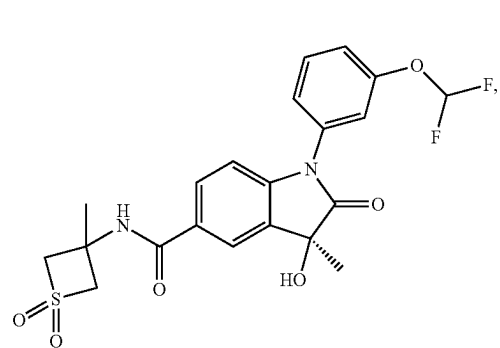
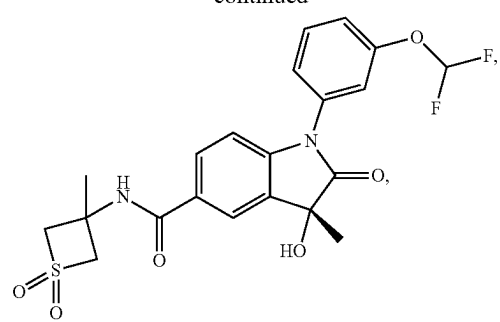


55

-continued

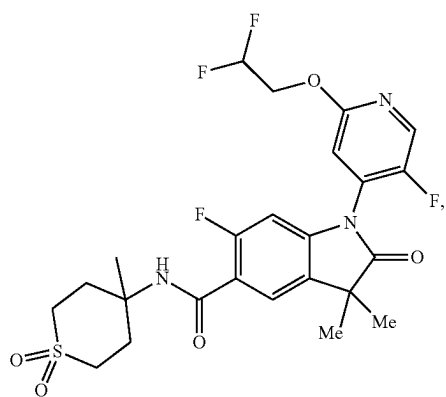
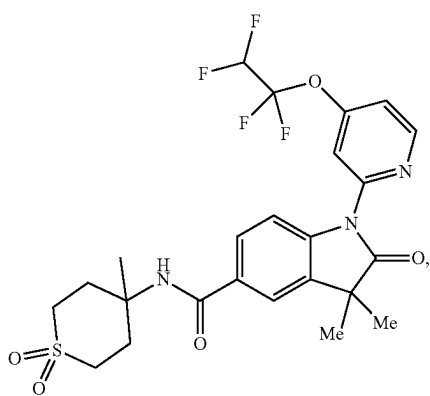
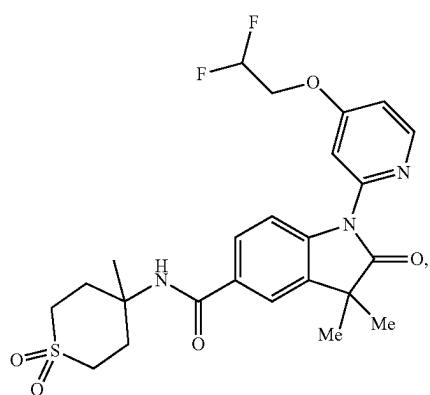
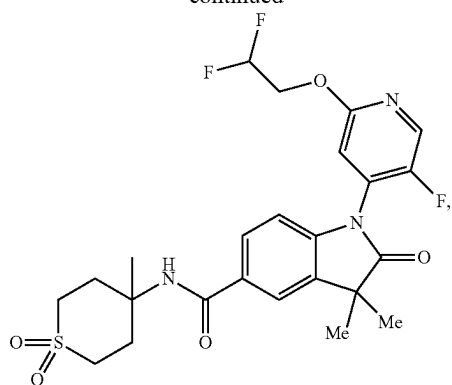
**56**

-continued



57

-continued

**58**

-continued

5

10

15

20

25

30

35

40

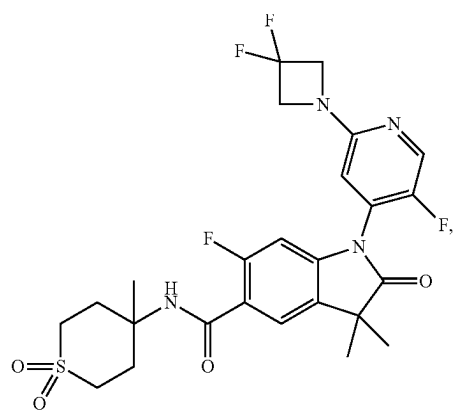
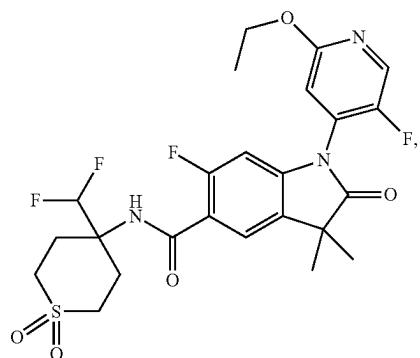
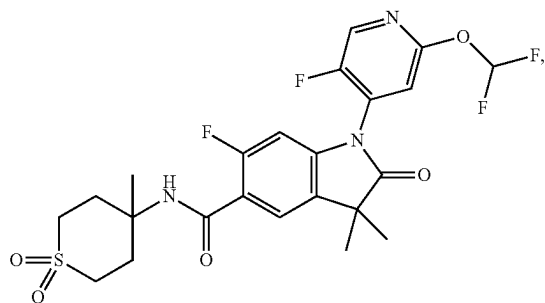
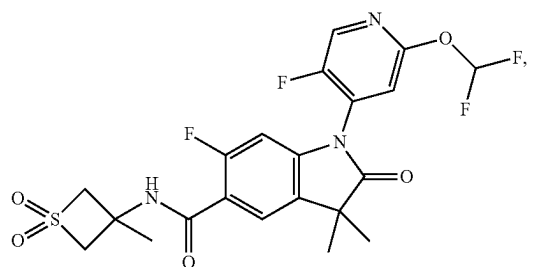
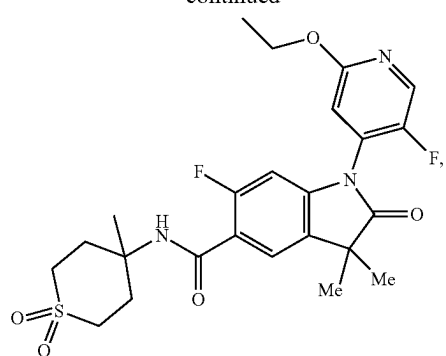
45

50

55

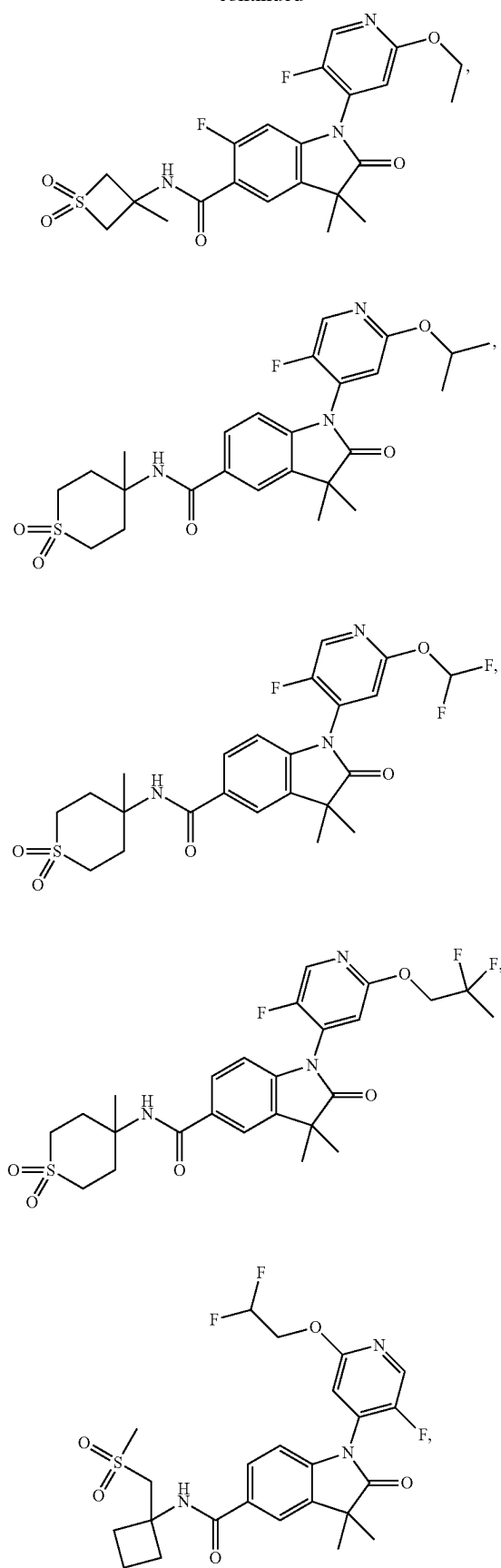
60

65

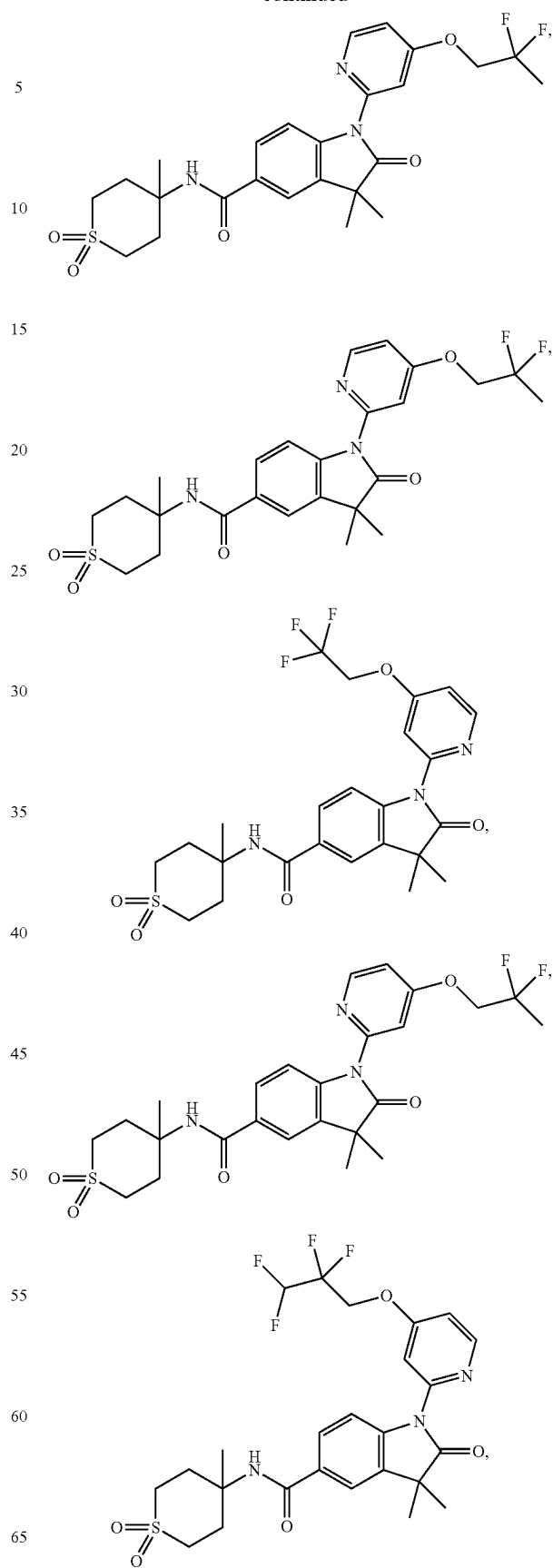


59

-continued

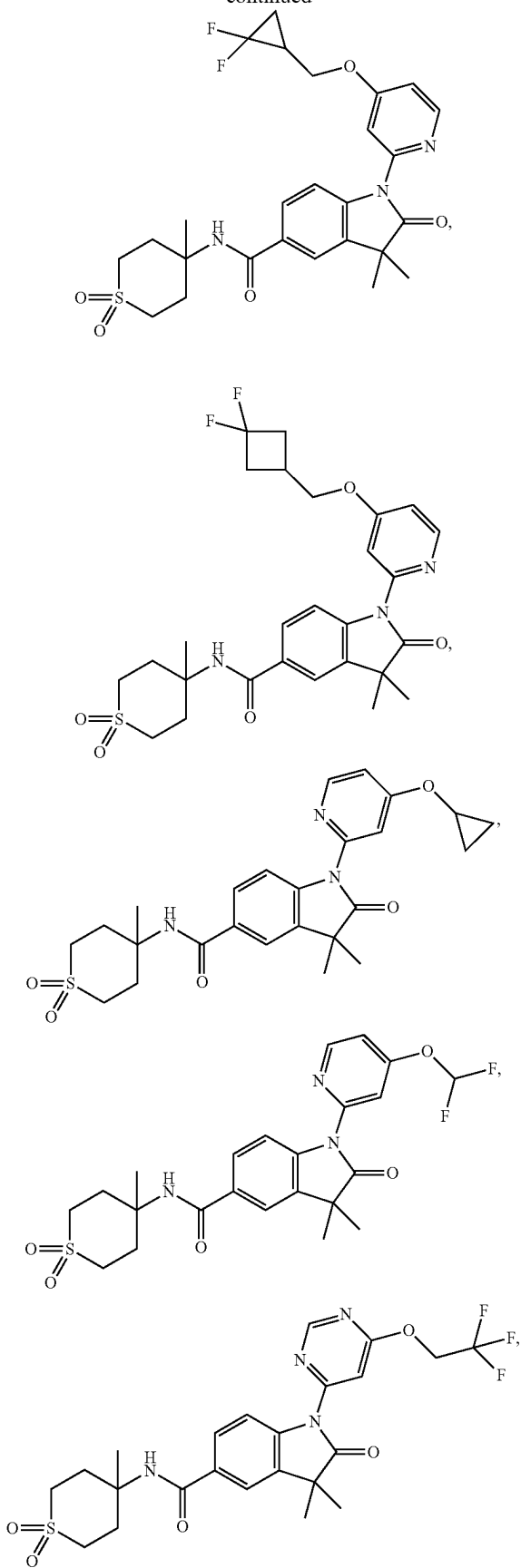
**60**

-continued

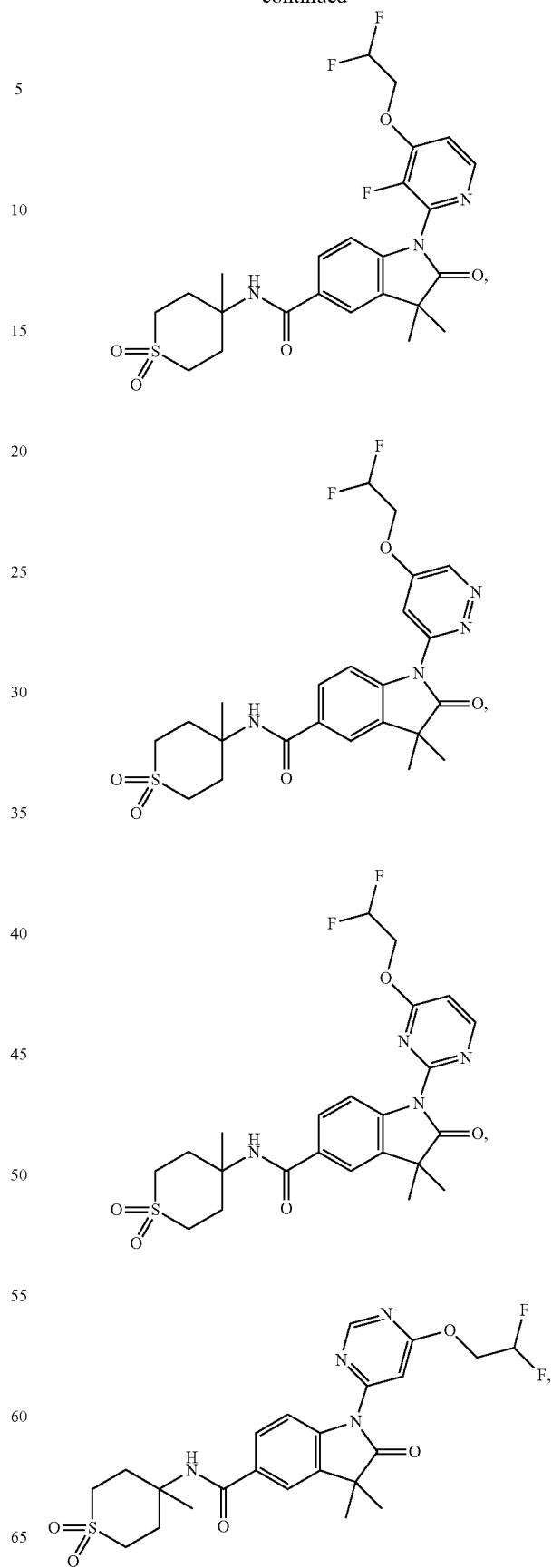


61

-continued

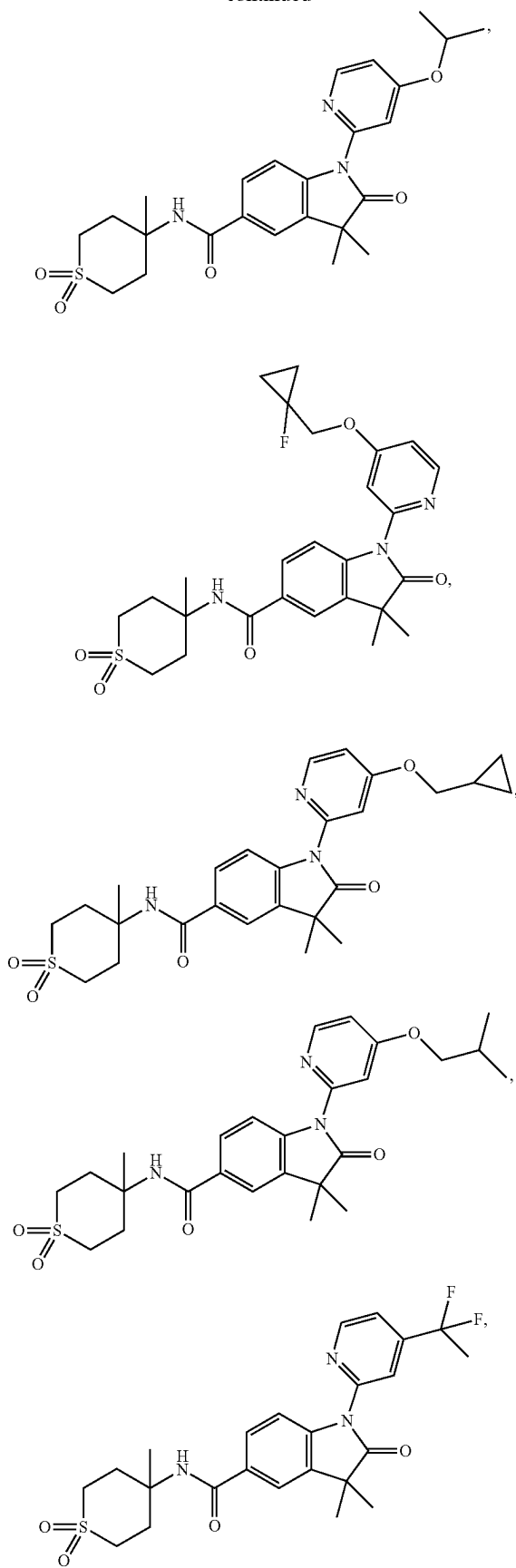
**62**

-continued

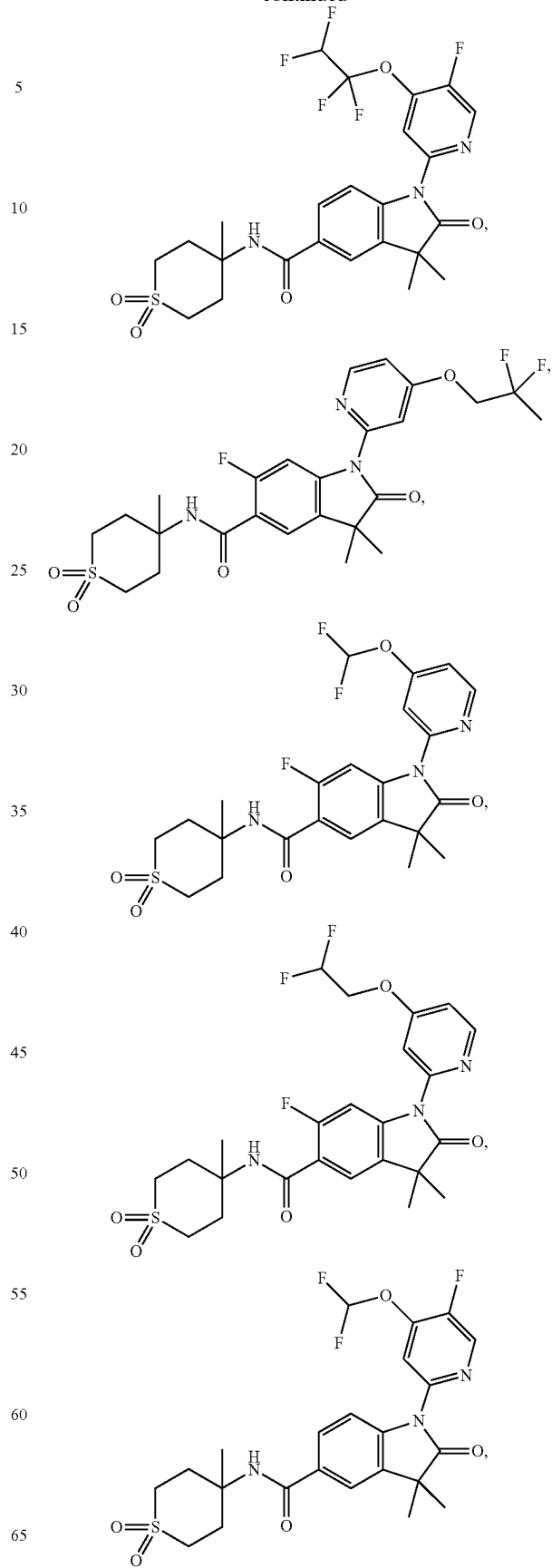


63

-continued

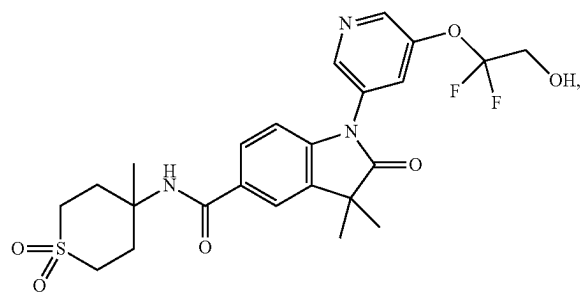
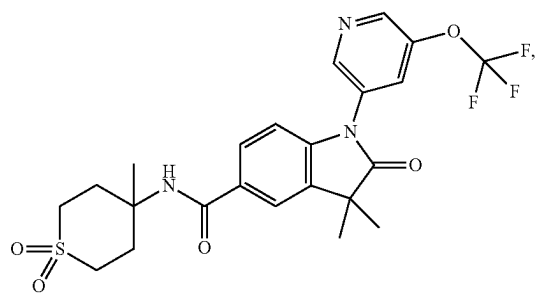
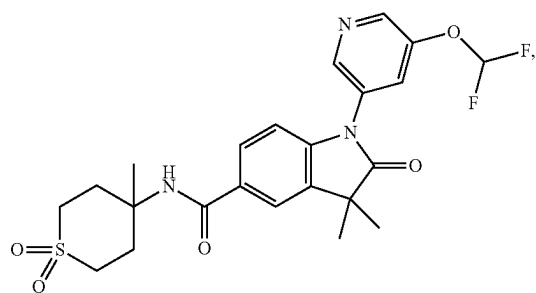
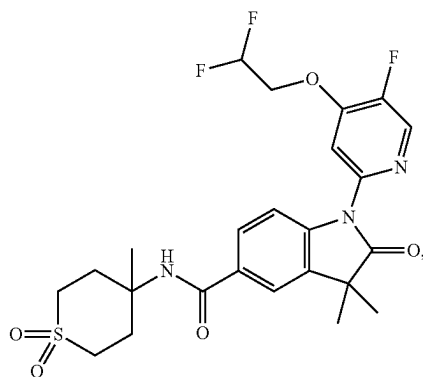
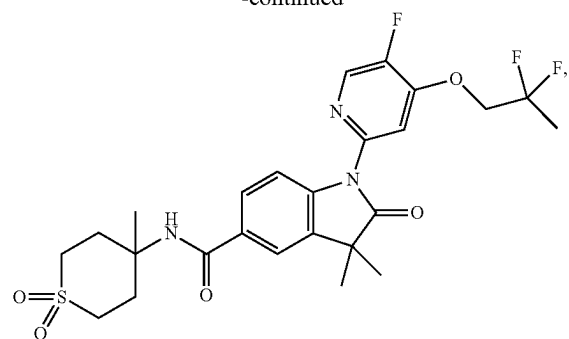
**64**

-continued

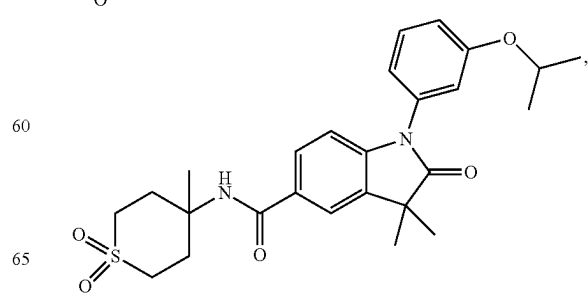
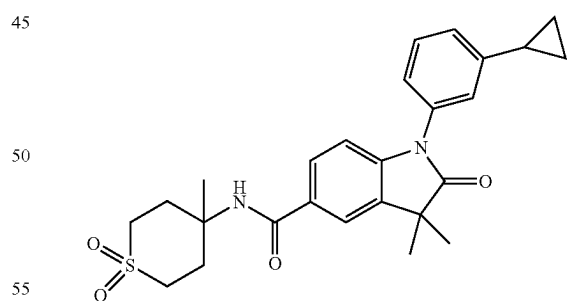
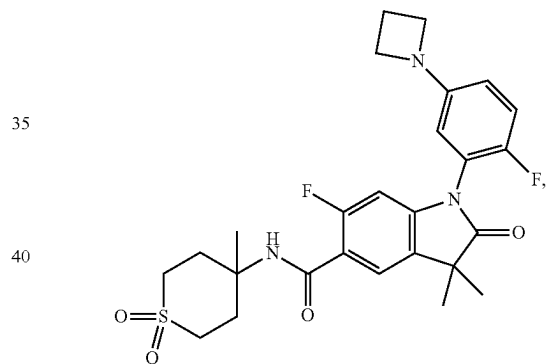
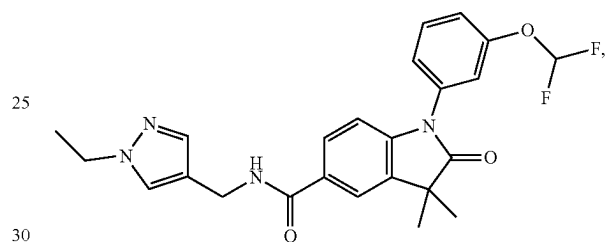
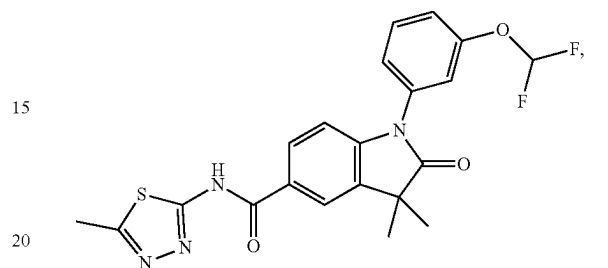
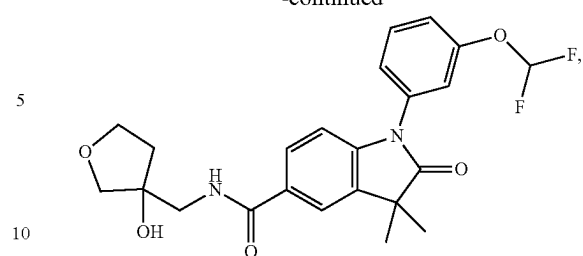


65

-continued

**66**

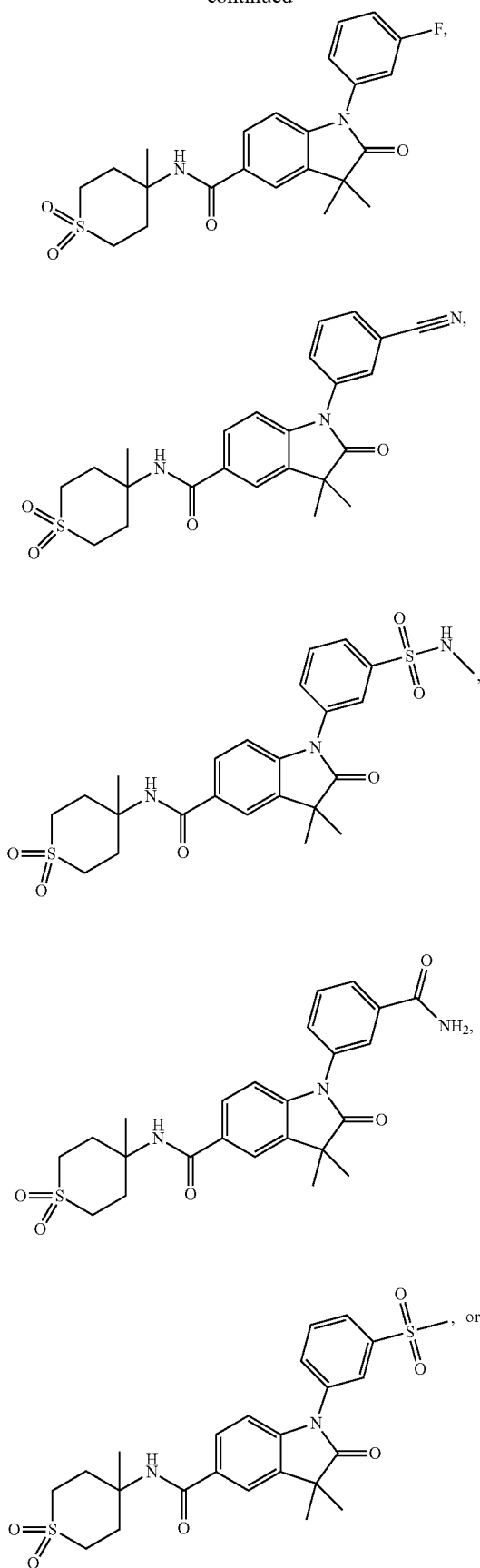
-continued



65

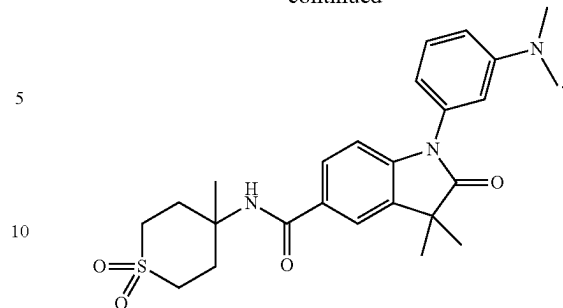
67

-continued



68

-continued



- 15 In Embodiment 79 of the invention, the compound of formula I, or a pharmaceutically acceptable salt thereof, is:
- 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide,
- 1-(3-(difluoromethoxy)phenyl)-3-isopropyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide,
- 1'-(3-(difluoromethoxy)phenyl)-N-(3-methyl-1,1-dioxidothietan-3-yl)-2'-oxospiro[cyclobutane-1,3'-indoline]-5'-carboxamide,
- (R)-1-(3-(difluoromethoxy)phenyl)-3-ethyl-3-methyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide,
- (S)-1-(3-(difluoromethoxy)phenyl)-3-ethyl-3-methyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide,
- (R)-1-(3-(difluoromethoxy)phenyl)-3-ethyl-3-methyl-N-((S)-3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide,
- (S)-1-(3-(difluoromethoxy)phenyl)-3-ethyl-3-methyl-N-((S)-3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide,
- 1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-indoline-5-carboxamide,
- 1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-[(3R)-(3-methyl-1,1-dioxo-thiolan-3-yl)]-2-oxo-indoline-5-carboxamide,
- 1-[3-(difluoromethoxy)phenyl]-N-[(1S,2R)-2-hydroxycyclopentyl]-3,3-dimethyl-2-oxo-indoline-5-carboxamide,
- 3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-1-[3-(trifluoromethoxy)phenyl]indoline-5-carboxamide,
- 3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-1-[3-(trifluoromethoxy)phenyl]indoline-5-carboxamide,
- 3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-1-[3-(trifluoromethoxy)phenyl]indoline-5-carboxamide,
- 1-[3-(difluoromethoxy)phenyl]-N-[4-(hydroxymethyl)tetrahydropyran-4-yl]-3,3-dimethyl-2-oxo-indoline-5-carboxamide,
- 1-[5-(difluoromethoxy)-2-fluoro-phenyl]-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide,
- 1-[5-(difluoromethoxy)-2-fluoro-phenyl]-3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide,
- 1-[5-(difluoromethoxy)-2-fluoro-phenyl]-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-indoline-5-carboxamide,
- 1-[3-(2,2-difluoroethoxy)phenyl]-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide,

1-[3-(2,2-difluoroethoxy)phenyl]-3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide,
 1-[3-(2,2-difluoroethoxy)phenyl]-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-indoline-5-carboxamide,
 3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-1-[5-(1,1,2,2-tetrafluoroethoxy)-3-pyridyl]indoline-5-carboxamide,
 3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-1-[5-(1,1,2,2-tetrafluoroethoxy)-3-pyridyl]indoline-5-carboxamide,
 3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-1-[5-(1,1,2,2-tetrafluoroethoxy)-3-pyridyl]indoline-5-carboxamide,
 3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-1-[5-(1,1,2,2-tetrafluoroethoxy)-3-pyridyl]indoline-5-carboxamide,
 (R)-1-(3-(difluoromethoxy)phenyl)-N-(4-hydroxy-2-methylpentan-2-yl)-3,3-dimethyl-2-oxoindoline-5-carboxamide,
 (S)-1-(3-(difluoromethoxy)phenyl)-N-(4-hydroxy-2-methylpentan-2-yl)-3,3-dimethyl-2-oxoindoline-5-carboxamide,
 1-(2-ethoxy-5-fluoro-4-pyridyl)-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
 1-(2-ethoxy-5-fluoro-4-pyridyl)-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide,
 1-(2-ethoxy-5-fluoro-4-pyridyl)-3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide,
 1-[2-(2,2-difluoroethoxy)pyrimidin-4-yl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
 3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-1-(3-quinolyl) indoline-5-carboxamide, N-[2-(cyclohexylsulfamoyl)ethyl]-1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-2-oxo-indoline-5-carboxamide,
 1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-[1-(methylsulfonylmethyl)cyclohexyl]-2-oxo-indoline-5-carboxamide,
 1-[3-(difluoromethoxy)phenyl]-N-(3-ethyl-1,1-dioxo-thietan-3-yl)-3,3-dimethyl-2-oxo-indoline-5-carboxamide,
 1-[3-(cyclopropoxy)phenyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
 1-[3-(cyclopropoxy)phenyl]-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide,
 1-[3-(cyclopropoxy)phenyl]-3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide,
 1-[3-(cyclopropoxy)phenyl]-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-indoline-5-carboxamide,
 3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-1-[3-(trifluoromethoxy)phenyl]indoline-5-carboxamide,
 1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-[(3S)-(3-methyl-1,1-dioxo-thiolan-3-yl)]-2-oxo-indoline-5-carboxamide,
 1-[3-(difluoromethoxy)phenyl]-N-(3,3-difluoro-1-methylcyclobutyl)-3,3-dimethyl-2-oxo-indoline-5-carboxamide,
 3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-1-[5-(trifluoromethoxy)-3-pyridyl]indoline-5-carboxamide,
 1-[5-(difluoromethoxy)-3-pyridyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,

1-(5-ethoxy-3-pyridyl)-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
 1-[5-(2,2-difluoroethoxy)-3-pyridyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
 1-[5-(difluoromethylsulfanyl)-3-pyridyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
 1-[2-(2,2-difluoroethoxy)-5-fluoro-4-pyridyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
 1-[5-(2,2-difluoroethoxy)-2,3-difluoro-phenyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
 1-[5-(2,2-difluoroethoxy)-2-fluoro-phenyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
 1-[6-(2,2-difluoroethoxy) pyrazin-2-yl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
 1-[3-(2,2-difluoroethoxy)phenyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
 1-[5-(difluoromethylsulfanyl)-2-fluoro-phenyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
 1-[5-(difluoromethoxy)-2-fluoro-phenyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
 1-[5-(1,1-difluoroethyl)-3-pyridyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
 1-[5-(1,1-difluoroethoxy)-3-pyridyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
 1-[5-(2,2-difluoroethoxy)-2-methyl-3-pyridyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
 (R)-1'-(3-(difluoromethoxy)phenyl)-N-(3-methyl-1,1-dioxodotetrahydrothiophen-3-yl)-2'-oxospiro[cyclopropane-1,3'-indoline]-5'-carboxamide,
 1-[3-(difluoromethoxy)phenyl]-7-fluoro-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide,
 1-[3-(difluoromethoxy)phenyl]-7-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-indoline-5-carboxamide,
 1-[3-(difluoromethoxy)phenyl]-7-fluoro-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
 1'-[3-(difluoromethoxy)phenyl]-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2'-oxo-spiro[cyclopentane-1,3'-indoline]-5'-carboxamide,
 1'-[3-(difluoromethoxy)phenyl]-N-(4-methyl-1,1-dioxo-thian-4-yl)-2'-oxo-spiro[cyclopentane-1,3'-indoline]-5'-carboxamide,
 1'-[3-(difluoromethoxy)phenyl]-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2'-oxo-spiro[cyclopentane-1,3'-indoline]-5'-carboxamide,
 1'-[3-(difluoromethoxy)phenyl]-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2'-oxo-spiro[cyclopropane-1,3'-indoline]-5'-carboxamide,
 1'-[3-(difluoromethoxy)phenyl]-N-(4-methyl-1,1-dioxo-thian-4-yl)-2'-oxo-spiro[cyclopropane-1,3'-indoline]-5'-carboxamide.

1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-((3S,4R)-4-methyl-1,1-dioxidotetrahydrothiophen-3-yl)-2-oxoindoline-5-carboxamide,
 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-((3R,4R)-4-methyl-1,1-dioxidotetrahydrothiophen-3-yl)-2-oxoindoline-5-carboxamide,
 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-((3S,4S)-4-methyl-1,1-dioxidotetrahydrothiophen-3-yl)-2-oxoindoline-5-carboxamide,
 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-((3R,4S)-4-methyl-1,1-dioxidotetrahydrothiophen-3-yl)-2-oxoindoline-5-carboxamide,
 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-((3S,5R)-5-methyl-1,1-dioxidotetrahydrothiophen-3-yl)-2-oxoindoline-5-carboxamide,
 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-((3R,5R)-5-methyl-1,1-dioxidotetrahydrothiophen-3-yl)-2-oxoindoline-5-carboxamide,
 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-((3S,5S)-5-methyl-1,1-dioxidotetrahydrothiophen-3-yl)-2-oxoindoline-5-carboxamide,
 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-((3R,5S)-5-methyl-1,1-dioxidotetrahydrothiophen-3-yl)-2-oxoindoline-5-carboxamide,
 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-((1S,2R)-2-(methylsulfonyl)cyclopentyl)-2-oxoindoline-5-carboxamide,
 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-((1R,2R)-2-(methylsulfonyl)cyclopentyl)-2-oxoindoline-5-carboxamide,
 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-((1S,2S)-2-(methylsulfonyl)cyclopentyl)-2-oxoindoline-5-carboxamide,
 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-((1R,2S)-2-(methylsulfonyl)cyclopentyl)-2-oxoindoline-5-carboxamide,
 1-[3-(difluoromethoxy)phenyl]-N-(1,1-dimethyl-2-methylsulfonyl-ethyl)-3,3-dimethyl-2-oxoindoline-5-carboxamide,
 1-[2-fluoro-5-(trifluoromethoxy)phenyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxoindoline-5-carboxamide,
 1-[2-fluoro-5-(trifluoromethoxy)phenyl]-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxoindoline-5-carboxamide,
 1-[2-fluoro-5-(trifluoromethoxy)phenyl]-3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxoindoline-5-carboxamide,
 1-[2-fluoro-5-(trifluoromethoxy)phenyl]-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxoindoline-5-carboxamide,
 7-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-1-[5-(trifluoromethoxy)-3-pyridyl]indoline-5-carboxamide,
 1-[5-(difluoromethoxy)-2-fluoro-phenyl]-7-fluoro-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxoindoline-5-carboxamide,
 1-[5-(difluoromethoxy)-2-fluoro-phenyl]-7-fluoro-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxoindoline-5-carboxamide,
 1-[5-(difluoromethoxy)-2-fluoro-phenyl]-7-fluoro-3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxoindoline-5-carboxamide,
 1-[5-(difluoromethoxy)-2-fluoro-phenyl]-7-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxoindoline-5-carboxamide,

1-(2-ethoxy-5-fluoro-4-pyridyl)-7-fluoro-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxoindoline-5-carboxamide,
 1-(2-ethoxy-5-fluoro-4-pyridyl)-7-fluoro-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxoindoline-5-carboxamide,
 1-(2-ethoxy-5-fluoro-4-pyridyl)-7-fluoro-3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxoindoline-5-carboxamide,
 1-(2-ethoxy-5-fluoro-4-pyridyl)-7-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxoindoline-5-carboxamide,
 1-[5-(difluoromethoxy)-2-fluoro-phenyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-pyrrolo[2,3-b]pyridine-5-carboxamide,
 1-[5-(difluoromethoxy)-2-fluoro-phenyl]-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-pyrrolo[2,3-b]pyridine-5-carboxamide,
 1-[5-(difluoromethoxy)-2-fluoro-phenyl]-3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-pyrrolo[2,3-b]pyridine-5-carboxamide,
 1-[5-(difluoromethoxy)-2-fluoro-phenyl]-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-pyrrolo[2,3-b]pyridine-5-carboxamide,
 7-fluoro-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-1-[5-(trifluoromethoxy)-3-pyridyl]indoline-5-carboxamide,
 7-fluoro-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-1-[5-(trifluoromethoxy)-3-pyridyl]indoline-5-carboxamide,
 7-fluoro-3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-1-[5-(trifluoromethoxy)-3-pyridyl]indoline-5-carboxamide,
 1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-pyrrolo[2,3-b]pyridine-5-carboxamide,
 1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-pyrrolo[2,3-b]pyridine-5-carboxamide,
 1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-pyrrolo[2,3-b]pyridine-5-carboxamide,
 1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-pyrrolo[2,3-b]pyridine-5-carboxamide,
 3,3-dimethyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-1-(2-methylbenzo[d]oxazol-6-yl)-2-oxoindoline-5-carboxamide,
 1-[6-(2,2-difluoroethoxy)pyridazin-4-yl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxoindoline-5-carboxamide,
 1-(2-(3,3-difluoroazetidin-1-yl)-3-fluoropyridin-4-yl)-3,3-dimethyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide,
 1-(4-acetamido-3-hydroxyphenyl)-3,3-dimethyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide,
 (R)-N-(1-(2,2-difluoroethyl)-3-methylpiperidin-3-yl)-1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-2-oxoindoline-5-carboxamide,
 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-(3-methyl-1-(methylsulfonyl)piperidin-3-yl)-2-oxoindoline-5-carboxamide,
 (R)-N-(1-acetyl-3-methylpiperidin-3-yl)-1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-2-oxoindoline-5-carboxamide,

1-(3-(difluoromethoxy)phenyl)-N-(4-methoxy-4-(trifluoromethyl)cyclohexyl)-3,3-dimethyl-2-oxoindoline-5-carboxamide,
 7-cyano-1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide,
 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxo-2,3-dihydro-1H-pyrrolo[2,3-c]pyridine-5-carboxamide,
 1-(3-(difluoromethoxy)phenyl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 7-(3-(difluoromethoxy)phenyl)-5,5-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-6-oxo-6,7-dihydro-5H-pyrrolo[2,3-c]pyridazine-3-carboxamide,
 1-(2-(2,2-difluoroethoxy)-5-fluoropyridin-4-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-(2,2-difluoroethoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxo-1-(4-(1,1,2,2-tetrafluoroethoxy)pyridin-2-yl) indoline-5-carboxamide,
 1-(2-(2,2-difluoroethoxy)-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(2-ethoxy-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(2-ethoxy-5-fluoropyridin-4-yl)-3,3-dimethyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide,
 3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxo-1-(4-(2,2,2-trifluoroethoxy)pyridin-2-yl) indoline-5-carboxamide,
 1-(4-(2,2-difluoropropoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxo-1-(4-(2,2,3,3-tetrafluoropropoxy)pyridin-2-yl) indoline-5-carboxamide,
 1-(4-((2,2-difluorocyclopropyl) methoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-((3,3-difluorocyclobutyl) methoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-cyclopropoxypyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-(difluoromethoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(2-(difluoromethoxy)-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxo-1-(6-(2,2,2-trifluoroethoxy)pyrimidin-4-yl) indoline-5-carboxamide,
 1-(5-(difluoromethoxy)pyridin-3-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxo-1-(5-(trifluoromethoxy)pyridin-3-yl) indoline-5-carboxamide,

1-(2-(difluoromethoxy)-5-fluoropyridin-4-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(2-(difluoromethoxy)-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide,
 1-(2-(2,2-difluoroethoxy)-5-fluoropyridin-4-yl)-3,3-dimethyl-N-(1-((methylsulfonyl)methyl)cyclobutyl)-2-oxoindoline-5-carboxamide,
 1-(4-(2,2-difluoroethoxy)-3-fluoropyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(2-(2,2-difluoropropoxy)-5-fluoropyridin-4-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(5-(2,2-difluoroethoxy)pyridazin-3-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-(2,2-difluoroethoxy)pyrimidin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(6-(2,2-difluoroethoxy)pyrimidin-4-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(5-(azetidin-1-yl)-2-fluorophenyl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-isopropoxypyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-((1-fluorocyclopropyl) methoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-(cyclopropylmethoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-isobutoxypyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-(1,1-difluoroethyl) pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(5-fluoro-4-(1,1,2,2-tetrafluoroethoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(5-(1,1-difluoro-2-hydroxyethoxy)pyridin-3-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(2-(3,3-difluoroazetidin-1-yl)-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-(2,2-difluoropropoxy)pyridin-2-yl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-(difluoromethoxy)pyridin-2-yl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-(2,2-difluoroethoxy)pyridin-2-yl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-(difluoromethoxy)-5-fluoropyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-(2,2-difluoropropoxy)-5-fluoropyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,

1-(4-(2,2-difluoroethoxy)-5-fluoropyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 N-(4-(difluoromethyl)-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-1-(2-ethoxy-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-2-oxoindoline-5-carboxamide,
 1-(5-fluoro-2-isopropoxypyridin-4-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(3-(difluoromethoxy)phenyl)-N-((3-hydroxytetrahydrofuran-3-yl)methyl)-3,3-dimethyl-2-oxoindoline-5-carboxamide,
 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-(5-methyl-1,3,4-thiadiazol-2-yl)-2-oxoindoline-5-carboxamide,
 1-(3-(difluoromethoxy)phenyl)-N-((1-ethyl-1H-pyrazol-4-yl)methyl)-3,3-dimethyl-2-oxoindoline-5-carboxamide,
 1-(3-cyclopropylphenyl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(3-isopropoxyphenyl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(3-fluorophenyl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(3-cyanophenyl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 (R)-1-(3-(difluoromethoxy)phenyl)-3-hydroxy-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxo-3-(trifluoromethyl) indoline-5-carboxamide,
 (S)-1-(3-(difluoromethoxy)phenyl)-3-hydroxy-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxo-3-(trifluoromethyl) indoline-5-carboxamide,
 (R)-1-(3-(difluoromethoxy)phenyl)-3-hydroxy-3-methyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide,
 (S)-1-(3-(difluoromethoxy)phenyl)-3-hydroxy-3-methyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide,
 1-(3-(difluoromethoxy)phenyl)-3-methyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide,
 3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-1-(3-(N-methylsulfamoyl)phenyl)-2-oxoindoline-5-carboxamide,
 1-(3-carbamoylphenyl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-1-(3-(methylsulfonyl)phenyl)-2-oxoindoline-5-carboxamide, or
 1-(3-(dimethylamino)phenyl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide.

The present invention includes the pharmaceutically acceptable salts of the compounds defined therein.

In one embodiment, the present invention is a composition comprising an effective amount of at least one compound of formula I, or a pharmaceutically acceptable salt thereof.

The invention also provides a pharmaceutical composition comprising an effective amount of at least one compound of formula I, or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable carrier.

The invention also provides a pharmaceutical composition comprising an effective amount of at least one compound of formula I, or a pharmaceutically acceptable salt

thereof, and an effective amount of at least one other pharmaceutically active ingredient (such as, for example, a chemotherapeutic agent).

The invention also provides a pharmaceutical composition comprising an effective amount of at least one compound of formula I, or a pharmaceutically acceptable salt thereof, and an effective amount of at least one other pharmaceutically active ingredient (such as, for example, a chemotherapeutic agent), and a pharmaceutically acceptable carrier.

In one embodiment, the present invention provides a composition for treating hepatic steatosis, nonalcoholic steatohepatitis (NASH), fibrosis, type-2 diabetes mellitus, obesity, hyperlipidemia, hypercholesterolemia, atherosclerosis, cognitive decline, dementia, cardiorenal diseases such as chronic kidney diseases or heart failure comprising an acceptable carrier and a compound of formula I or a pharmaceutically acceptable salt thereof.

In one embodiment, the present invention provides a composition for treating hepatic steatosis, nonalcoholic steatohepatitis (NASH), fibrosis, type-2 diabetes mellitus, obesity, hyperlipidemia, hypercholesterolemia, atherosclerosis, cognitive decline, dementia, cardiorenal diseases such as chronic kidney diseases or heart failure, comprising a compound of formula I, or a pharmaceutically acceptable salt thereof.

In one embodiment, the present invention provides a composition for treating hepatic steatosis, nonalcoholic steatohepatitis (NASH), fibrosis, type-2 diabetes mellitus, obesity, hyperlipidemia, hypercholesterolemia, atherosclerosis, cognitive decline, dementia, cardiorenal diseases such as chronic kidney diseases or heart failure, comprising a compound of formula I, or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable carrier.

In one embodiment, the present invention provides a method of treating hepatic steatosis, nonalcoholic steatohepatitis (NASH), fibrosis, type-2 diabetes mellitus, obesity, hyperlipidemia, hypercholesterolemia, atherosclerosis, cognitive decline, dementia, cardiorenal diseases such as chronic kidney diseases or heart failure in a subject in need of such treatment, comprising administering to said subject a therapeutically effective amount of at least one compound of formula I, or a pharmaceutically acceptable salt thereof.

In one embodiment, the present invention provides a method of treating hepatic steatosis, nonalcoholic steatohepatitis (NASH), fibrosis, type-2 diabetes mellitus, obesity, hyperlipidemia, hypercholesterolemia, atherosclerosis, cognitive decline, dementia, cardiorenal diseases such as chronic kidney diseases or heart failure in a patient in need thereof, comprising administering to said patient a therapeutically effective amount of at least one compound of formula I, or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable carrier.

The methods of the invention include the administration of a pharmaceutical composition comprising at least one compound of the invention, or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable carrier.

In another embodiment, the present invention includes a method of treating NASH and/or fibrosis, comprising administering to a patient in need thereof a compound of formula I, or a pharmaceutically acceptable salt thereof.

In another embodiment, the present invention includes a method of treating NASH and/or fibrosis, comprising administering to a patient in need thereof a compound of formula I, or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable carrier.

In another embodiment, the present invention includes a method of treating NASH and/or fibrosis, comprising administering to a patient in need thereof a composition comprising a compound of formula I, or a pharmaceutically acceptable salt thereof.

In another embodiment, the present invention includes a method of treating NASH and/or fibrosis, comprising administering to a patient in need thereof a composition comprising a compound of formula I, or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable carrier.

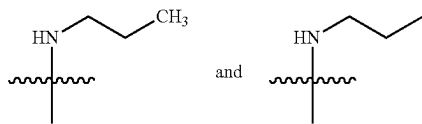
In another embodiment, the present invention provides for the use of a compound of formula I, or a pharmaceutically acceptable salt thereof, in the manufacture of a medicament for treating NASH and/or fibrosis,

In another embodiment, the present invention includes the use of a compound of formula I, or a pharmaceutically acceptable salt thereof, for the preparation of a medicament for the treatment of NASH and/or fibrosis,

“Alkyl” means branched- and straight-chain saturated aliphatic hydrocarbon groups having the specified number of carbon atoms when noted. If no number is specified, 1-6 carbon atoms are intended for linear and 3-7 carbon atoms for branched alkyl groups. Examples of alkyl groups include methyl, ethyl, propyl, isopropyl, cyclopropyl, butyl, sec- and tert-butyl, pentyl, hexyl, octyl, nonyl, and the like. For example, the term “C₁₋₆alkyl” includes all of “C₁₋₄alkyl” defined as follows, plus the linear or branched chain alkyl groups, including all possible isomers, having 5 or 6 carbon atoms. “C₁₋₆alkyl” means linear or branched chain alkyl groups, including all possible isomers, having 1, 2, 3, 4, 5 or 6 carbon atoms, and includes each of the alkyl groups within C₁₋₆alkyl including each of the hexyl and pentyl isomers as well as n-, iso-, sec- and tert-butyl (butyl, i-butyl, s-butyl, t-butyl, collectively “C₄alkyl”; Bu=butyl), n- and i-propyl (propyl, i-propyl, collectively “C₃alkyl”; Pr=propyl), ethyl (Et) and methyl (Me). Commonly used abbreviations for alkyl groups are used throughout the specification, e.g., methyl may be represented by conventional abbreviations including “Me” or CH₃ or a symbol that is an extended bond as the terminal group, e.g.,



ethyl may be represented by “Et” or CH₂CH₃, propyl may be represented by “Pr” or CH₂CH₂CH₃, butyl may be represented by “Bu” or CH₂CH₂CH₂CH₃, etc. For example, the structures



have equivalent meanings. C₁₋₆ alkyl includes n-, iso, sec- and t-butyl, n- and isopropyl, ethyl and methyl. If no number is specified, 1-6 carbon atoms are intended for linear or branched alkyl groups.

“Cyclic amine” refers to a cyclic ring comprising one nitrogen atom.

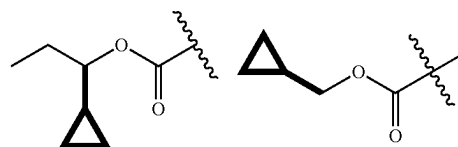
“Alkoxy” refers to an alkyl group linked to oxygen. Examples of alkoxy groups include methoxy, ethoxy, propoxy and the like.

“Aryl” refers to an aromatic monocyclic or multicyclic ring moiety comprising 6 to 14 ring carbon atoms. In one embodiment, an aryl group contains from about 6 to 10 ring carbon atoms. Monocyclic aryl rings include, but are not limited to, phenyl. Multicyclic rings include, but are not limited to, naphthyl and bicyclic rings wherein phenyl is fused to a C₅₋₇cycloalkyl or C₅₋₇cycloalkenyl ring. Aryl groups may be optionally substituted with one or more substituents as defined herein. Bonding can be through any of the carbon atoms of any ring.

“Fused Aryl” refers to an aryl ring fused with heterocyclyl or cycloalkyl.

“Halogen” or “Halo” includes fluorine, chlorine, bromine and iodine.

“Cycloalkyl” refers to a non-aromatic mono- or multicyclic ring system comprising about 3 to 10 ring carbon atoms. If no number of atoms is specified, 3-10 carbon atoms are intended. Cycloalkyl may also be fused, forming 1-3 carbocyclic rings. Non-limiting examples of monocyclic cycloalkyl include cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, and cyclooctyl. The term C₁₋₆cycloalkyl” refers to a cycloalkyl group having 1 to 6 ring carbon atoms. The term C₃₋₆cycloalkyl” refers to a cycloalkyl group having 3 to 6 ring carbon atoms. Thus, for example, “C₃₋₆ cycloalkyl” includes each of cyclopropyl, cyclobutyl, cyclopentyl, and cyclohexyl. A cycloalkyl group is unsubstituted or substituted with one or more ring system substituents which may be the same or different, and are as defined within. When cycloalkyl is a substituent on an alkyl group, the cycloalkyl substituent can be bonded to any available carbon in the alkyl group. The following are illustrations of —C₃₋₆cycloalkyl substituents on an alkyl group wherein the substituent is cyclopropyl in bold:



“Haloalkyl” refers to an alkyl group as defined within, wherein one or more of the alkyl group’s hydrogen atoms has been replaced with a halogen. In one embodiment, a haloalkyl group has from 1 to 6 carbon atoms. Non-limiting examples of haloalkyl groups include CH₂F, CHF₂, CF₃, CH₂CF₃, CH₂CHF₂, CF₂CF₃, CF₂CHF₂, CH₂Cl and CCl₃. The term “C₁₋₆haloalkyl” or “haloC₁₋₆alkyl” refers to a haloalkyl group having from 1 to 6 carbons.

“Haloalkoxy,” “haloalkyl-O” and derivatives such as “halo(C₁₋₆)alkoxy” are used interchangeably and refer to halo substituted alkyl groups linked through the oxygen atom. Haloalkoxy include mono-substituted as well as multiple halo substituted alkoxy groups. For example, trifluoromethoxy, chloromethoxy, and bromomethoxy are included as well as OCH₂CF₃, OCH₂CHF₂, OCF₂CF₃, and OCF₂CHF₂.

“Heterocyclyl,” “heterocycle” or “heterocyclic” refers to monocyclic ring structures in which one or more atoms in the ring, the heteroatom(s), is an element other than carbon. Heteroatoms are typically O, S or N atoms. A heterocycle containing more than one heteroatom may contain different heteroatoms. Bicyclic ring moieties include fused, spirocyclic

clic and bridged bicyclic rings and may comprise one or more heteroatoms in either of the rings. The ring attached to the remainder of the molecule may or may not contain a heteroatom. Either ring of a bicyclic heterocycle may be saturated, partially unsaturated or unsaturated. The heterocycle may be attached to the rest of the molecule via a ring carbon atom, a ring oxygen atom or a ring nitrogen atom. Examples of heterocyclyl groups include: piperidine, piperazine, morpholine, pyrrolidine, tetrahydrofuran, azetidine, oxirane, or aziridine, and the like.

"Bicyclic heterocyclyl," "bicyclic heterocycle" or "bicyclic heterocyclic" refers to a heterocyclic ring fused to another ring system. The fusion may be bridged or unbridged.

Except where noted, the term "heteroaryl", as used herein, represents a stable monocyclic, bicyclic or tricyclic ring of up to 10 atoms in each ring, wherein at least one ring is aromatic and contains from 1 to 4 heteroatoms selected from the group consisting of O, N and S. Heteroaryl groups within the scope of this definition include but are not limited to: benzoimidazolyl, benzofuranyl, benzofurazanyl, benzopyrazolyl, benzotriazolyl, benzothiophenyl, benzoxazolyl, carbazolyl, carbolinyl, cinnolinyl, furanyl, imidazolyl, indolinyl, indolyl, indolaziny, indazolyl, isobenzofuranyl, isoindolyl, isoquinolyl, isothiazolyl, isoxazolyl, naphthopyridinyl, oxadiazolyl, oxazolyl, oxazoline, isoxazoline, pyranyl, pyrazinyl, pyrazolyl, pyridazinyl, pyridopyridinyl, pyridyl, pyrimidinyl, pyrrolyl, quinazolinyl, quinolyl, quinoxalinyl, tetrazolyl, tetrazolopyridyl, thiadiazolyl, thiazolyl, thienyl, triazolyl, dihydrobenzoimidazolyl, dihydrobenzofuranyl, dihydrobenzothiophenyl, dihydrobenzoxazolyl, dihydroindolyl, dihydroquinolinyl, methylenedioxybenzene, benzothiazolyl, benzothienyl, quinolinyl, isoquinolinyl, oxazolyl, and tetra-hydroquinoline.

"Fused heteroaryl" is heteroaryl fused with an aryl or heteroaryl

"Oxo" means an oxygen linked to an atom by a double bond. An example of an oxo group is a doubly bonded oxygen in a ketone, sulfoxide, sulfone and sulfate.

"Hydroxyalkyl" or "-hydroxy (C₁₋₃)alkyl" means an alkyl group having one or more hydrogen atoms replaced by hydroxyl (—OH) groups

"Cyanoalkyl" means an alkyl group having one or more hydrogen atoms replaced by cyano (—CN) groups.

"Hydroxyhaloalkyl" means an alkyl group having one or more hydrogen atoms replaced by hydroxyl (—OH) groups, and one or more hydrogen atoms replaced by a halogen.

"Hydroxycycloalkyl" means a cyclic alkyl group having one or more hydrogen atoms replaced by hydroxyl (—OH) groups.

The term "composition" is intended to encompass a product comprising the specified ingredients in the specified amounts, as well as any product which results, directly or indirectly, from combination of the specified ingredients in the specified amounts.

The term "at least one" means one or more than one. The meaning of "at least one" with reference to the number of compounds of the invention is independent of the meaning with reference to the number of chemotherapeutic agents.

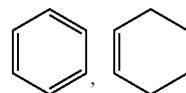
The term "chemotherapeutic agent" means a drug (medicament or pharmaceutically active ingredient) for treating cancer (i.e., an antineoplastic agent).

The term "effective amount" means a "therapeutically effective amount". The term "therapeutically effective amount" means that amount of active compound or pharmaceutical agent that elicits the biological or medicinal

response in a tissue, system, animal or human that is being sought by a researcher, veterinarian, medical doctor or other clinician.

The term "treating cancer" or "treatment of cancer" refers to administration to a mammal afflicted with a cancerous condition and refers to an effect that alleviates the cancerous condition by killing the cancerous cells, and also refers to an effect that results in the inhibition of growth and/or metastasis of the cancer.

Except where noted herein, the term "carbocycle" (and variations thereof such as "carbocyclic" or "carbocyclyl") as used herein, unless otherwise indicated, refers to a C₃ to C₆ monocyclic ring, e.g., C₃₋₆ monocyclic carbocycle. The carbocycle may be attached to the rest of the molecule at any carbon atom which results in a stable compound. Saturated carbocyclic rings include, for example, "cycloalkyl" rings, e.g., cyclopropyl, cyclobutyl, etc. Unsaturated carbocyclic rings include, for example



Except where noted, the term "saturated heterocycle" refers to a stable 4- to 7-membered mono-cyclic heteroatom-containing ring system, and which consists of carbon atoms and from one to four heteroatoms independently selected from the group consisting of N, O and S, and wherein the nitrogen and sulfur heteroatoms may optionally be oxidized, and the nitrogen heteroatom may optionally be quaternized. Especially useful are rings containing one oxygen or sulfur, one to four nitrogen atoms, or one oxygen or sulfur combined with one or two nitrogen atoms. The heterocyclic ring may be attached at any heteroatom or carbon atom which results in the creation of a stable structure. Representative examples include azetidine, oxetane, thietane, diazetidine, dioxetane, dithietane, pyrrolidine, tetrahydrofuran, thiolane, imidazolidine, pyrazolidine, oxazolidine, isoxazolidine, thiazolidine, isothiazolidine, dioxolane, dithiolane, piperidine, oxane, thiane, piperazine, morpholine, thiomorpholine, dioxane, dithiane, trioxane, trithiane, azepane, oxepane, thiepane and homopiperazine. For example, a "saturated 6 membered heterocycle" is a stable 6-membered mono-cyclic heteroatom-containing ring system. An example of a saturated 6 membered heterocycle is piperidine. Likewise, but not limiting, a "saturated 4 membered heterocycle" is a stable 4-membered mono-cyclic heteroatom-containing ring system.

Except where noted herein, the term "unsaturated heterocycle" refers to a monocyclic unsaturated heterocycle having a specified number of atom members (e.g., 5 membered), including a specified number of heteroatoms (e.g., 1, 2, 3 or 4 heteroatoms independently selected from N, O or S), e.g., 5-membered rings containing one nitrogen (pyrrole), one oxygen (furan) or one sulfur (thiophene) atom, 5-membered rings containing one nitrogen and one sulfur (thiazole) atom, 5-membered rings containing one nitrogen and one oxygen (oxazole or isoxazole) atom, 5-membered rings containing two nitrogen (imidazole or pyrazole) atoms, five-membered aromatic rings containing three nitrogen (triazole) atoms, five-membered aromatic rings containing one oxygen, one nitrogen or one sulfur atom, five-membered aromatic rings containing two heteroatoms independently selected from oxygen, nitrogen and sulfur (e.g., oxazole). Additional examples are thiophene, imidazole, isothiazole, oxadiazole,

81

and isoxazole. For example, a “unsaturated 6 membered heterocycle” is a 6 membered ring containing 6 atom members including at least one heteroatom. Likewise, an “unsaturated 5 membered heterocycle” is a 5 membered ring containing 5 atom members including at least one heteroatom.

A “stable” compound is a compound which can be prepared and isolated and whose structure and properties remain or can be caused to remain essentially unchanged for a period of time sufficient to allow use of the compound for the purposes described herein (e.g., therapeutic or prophylactic administration to a subject).

The compounds of the present disclosure are limited to stable compounds embraced by Formula I and its embodiments. For example, certain moieties as defined in Formula I, may be unsubstituted or substituted, and the latter is intended to encompass substitution patterns (i.e., number and kind of substituents) that are chemically possible for the moiety and that result in a stable compound.

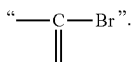
The term “substituted” means that one or more hydrogens on the designated atom is replaced with a selected from the indicated group, provided that the designated atom’s normal valency under the existing circumstances is not exceeded, and that the substitution results in a stable compound. Where multiple substituent moieties are disclosed or claimed, the substituted compound can be independently substituted by one or more of the disclosed or claimed substituent moieties, singly or plurally. By independently substituted, it is meant that the (two or more) substituents can be the same or different. Combinations of substituents and/or variables are permissible only if such combinations result in stable compounds. If a substituent is itself substituted with more than one group, it is understood that these multiple groups may be on the same carbon or on different carbons, so long as a stable structure result. By optionally substituted, it is meant that compounds containing the specified optional substituent(s) as well as compounds that do not contain the optional substituent(s).

The wavy line $\sim$, as used herein, indicates a point of attachment to the rest of the compound.

Where ring atoms are represented by variables such as “X”, e.g.,



the variables are defined by indicating the atom located at the variable ring position without depicting the ring bonds associated with the atom. For example, when X in the above ring is nitrogen, the definition will show “N” and will not depict the bonds associated with it, e.g., will not show “=N—”. Likewise, when X is a carbon atom that is substituted with bromide, the definition will show “C—Br” and will not depict the bonds associated with it, e.g., will not show



The invention also includes derivatives of the compound of formula I, acting as prodrugs and solvates. Any pharmaceutically acceptable pro-drug modification of a compound

82

of the invention which results in conversion in vivo to a compound within the scope of the invention is also within the scope of the invention. Prodrugs, following administration to the patient, are converted in the body by normal metabolic or chemical processes, such as through hydrolysis in the blood, to the compound of formula I. Such prodrugs include those that demonstrate enhanced bioavailability, tissue specificity, and/or cellular delivery, to improve drug absorption of the compound of I. The effect of such prodrugs may result from modification of physicochemical properties such as lipophilicity, molecular weight, charge, and other physicochemical properties that determine the permeation properties of the drug.

For example, esters can optionally be made by esterification of an available carboxylic acid group or by formation of an ester on an available hydroxy group in a compound. Similarly, labile amides can be made. Pharmaceutically acceptable esters or amides of the compounds of the invention may be prepared to act as pro-drugs which can be hydrolyzed back to an acid (or —COO^- depending on the pH of the fluid or tissue where conversion takes place) or hydroxy form particularly in vivo and as such are encompassed within the scope of the invention. Included are those esters and acyl groups known in the art for modifying the solubility or hydrolysis characteristics for use as sustained-release or prodrug formulations. Examples of pharmaceutically acceptable pro-drug modifications include, but are not limited to, $\text{—C}_{1-6}\text{alkyl}$ esters and $\text{—C}_{1-6}\text{alkyl}$ substituted with phenyl esters.

“Celite®” (Fluka) diatomite is diatomaceous earth, and can be referred to as “celite”.

When any variable (e.g., R^1 etc.) occurs more than one time in any constituent or in Formulas I or other generic Formula herein, its definition on each occurrence is independent of its definition at every other occurrence. Combinations of substituents and/or variables are permissible only if such combinations result in stable compounds. In choosing compounds of the present invention, one of ordinary skill in the art will recognize that the various substituents, i.e., R^1 etc., are to be chosen in conformity with well-known principles of chemical structure connectivity and stability. Unless expressly stated to the contrary, substitution by a named substituent is permitted on any atom in a ring (e.g., aryl, a heteroaryl ring, or a saturated heterocyclic ring) provided such ring substitution is chemically allowed and results in a stable compound.

It should be noted that, if a discrepancy between the chemical name and structure exists, the structure is understood to dominate.

Compounds of structural Formulas I may contain one or more asymmetric centers and can thus occur as racemates and racemic mixtures, single enantiomers, diastereoisomeric mixtures and individual diastereoisomers. Centers of asymmetry that are present in the compounds of Formulas I can all independently of one another have S configuration or R configuration. When bonds to the chiral carbon are depicted as straight lines in the structural Formulas of the invention, it is understood that both the (R) and (S) configurations of the chiral carbon, and hence both enantiomers and mixtures thereof, are embraced within the Formulas. Similarly, when a compound name is recited without a chiral designation for a chiral carbon, it is understood that both the (R) and (S) configurations of the chiral carbon, and hence individual enantiomers and mixtures thereof, are embraced by the name. The production of specific stereoisomers or mixtures thereof may be identified in the Examples where such stereoisomers or mixtures were obtained, but this in no way

limits the inclusion of all stereoisomers and mixtures thereof from being within the scope of the invention.

The compounds of this invention include all possible enantiomers and diastereomers and mixtures of two or more stereoisomers, for example mixtures of enantiomers and/or diastereomers, in all ratios. Thus, enantiomers are a subject of the invention in enantiomerically pure form, both as levorotatory and as dextrorotatory antipodes, in the form of racemates and in the form of mixtures of the two enantiomers in all ratios. In the case of a cis/trans isomerism the invention includes both the cis form and the trans form as well as mixtures of these forms in all ratios. The present invention is meant to comprehend all such stereo-isomeric forms of the compounds of structural Formulas I.

Compounds of structural Formulas I may be separated into their individual diastereoisomers by, for example, fractional crystallization from a suitable solvent, for example MeOH or EtOAc or a mixture thereof, or via chiral chromatography using an optically active stationary phase. Optionally a derivatization can be carried out before a separation of stereoisomers. The separation of a mixture of stereoisomers can be carried out at an intermediate step during the synthesis of a compound of formula I, or it can be done on a final racemic product. Absolute stereochemistry may be determined by X-ray crystallography of crystalline products or crystalline intermediates which are derivatized, if necessary, with a reagent containing an asymmetric center of known absolute configuration. Alternatively, any stereoisomer or isomers of a compound of Formulas I may be obtained by stereospecific synthesis using optically pure starting materials or reagents of known absolute configuration. The present invention includes all such isomers, as well as salts, solvates (including hydrates) and solvated salts of such racemates, enantiomers, diastereomers and tautomers and mixtures thereof.

If desired, racemic mixtures of the compounds may be separated so that the individual enantiomers are isolated. The separation can be carried out by methods well known in the art, such as the coupling of a racemic mixture of compounds to an enantiomerically pure compound to form a diastereomeric mixture, followed by separation of the individual diastereoisomers by standard methods, such as fractional crystallization or chromatography. The coupling reaction is often the formation of salts using an enantiomerically pure acid or base. The diastereomeric derivatives may then be converted to the pure enantiomers by cleavage of the added chiral residue. The racemic mixture of the compounds can also be separated directly by chromatographic methods utilizing chiral stationary phases, which methods are well known in the art.

For compounds of Formula I described herein which contain olefinic double bonds, unless specified otherwise, they are meant to include both E and Z, geometric isomers.

Some of the compounds described herein may exist as tautomers which have different points of attachment of hydrogen accompanied by one or more double bond shifts. For example, a ketone and its enol form are keto-enol tautomers. The individual tautomers as well as mixtures thereof are encompassed with compounds of Formula I of the present invention.

In the compounds of structural Formula I, the atoms may exhibit their natural isotopic abundances, or one or more of the atoms may be artificially enriched in a particular isotope having the same atomic number, but an atomic mass or mass number different from the atomic mass or mass number predominately found in nature. The present invention as described and claimed herein is meant to include all suitable

isotopic variations of the compounds of structural Formula I and embodiments thereof. For example, different isotopic forms of hydrogen (H) include protium (1H) and deuterium (2H, also denoted herein as D). Protium is the predominant hydrogen isotope found in nature. Enriching for deuterium may afford certain therapeutic advantages, such as increasing in vivo half-life or reducing dosage requirements, or may provide a compound useful as a standard for characterization of biological samples. Isotopically-enriched compounds within structural Formula I, can be prepared without undue experimentation by conventional techniques well known to those skilled in the art or by processes analogous to those described in the Schemes and Examples herein using appropriate isotopically-enriched reagents and/or intermediates.

It will be understood that the compounds of structural Formula I may be prepared as pharmaceutically acceptable salts or as salts that are not pharmaceutically acceptable when they are used as precursors to the free compounds or their pharmaceutically acceptable salts or in other synthetic manipulations. The compounds of the present invention, including the compounds of the Examples, may also include all salts of the compounds of Formula I which, owing to low physiological compatibility, are not directly suitable for use in pharmaceuticals but which can be used, for example, as intermediates for chemical reactions or for the preparation of physiologically acceptable salts.

The compounds of the present invention may be administered in the form of a pharmaceutically acceptable salt. The term "pharmaceutically acceptable salt" refers to salts prepared from pharmaceutically acceptable non-toxic bases or acids including inorganic or organic bases and inorganic or organic acids.

Salts of basic compounds encompassed within the term "pharmaceutically acceptable salt" refer to non-toxic salts of the compounds of this invention which are generally prepared by reacting the free base with a suitable organic or inorganic acid. Representative salts of basic compounds of the present invention include, but are not limited to, the following: acetate, ascorbate, benzenesulfonate, benzoate, bicarbonate, bisulfate, bitartrate, borate, bromide, butyrate, camphorate, camphorsulfonate, camsylate, carbonate, chloride, clavulanate, citrate, dihydrochloride, edetate, edisylate, estolate, esylate, fumarate, gluceptate, gluconate, glutamate, glycollylsarbanilate, hexylresorcinatate, hydrabamine, hydrobromide, hydrochloride, hydroxynaphthoate, iodide, isothionate, lactate, lactobionate, laurate, malate, maleate, mandelate, mesylate, methylbromide, methylnitrate, methylsulfate, methanesulfonate, mucate, napsylate, nitrate, N-methylglucamine ammonium salt, oleate, oxalate, pamoate (embonate), palmitate, pantothenate, phosphate/diphosphate, polygalacturonate, propionate, salicylate, stearate, sulfate, subacetate, succinate, tannate, tartrate, teoclate, thiocyanate, tosylate, triethiodide, valerate and the like. Furthermore, where the compounds of the invention carry an acidic moiety, suitable pharmaceutically acceptable salts thereof include, but are not limited to, salts derived from inorganic bases including aluminum, ammonium, calcium, copper, ferric, ferrous, lithium, magnesium, manganic, mangamous, potassium, sodium, zinc, and the like. In one embodiment, the salts of acidic compounds are as follows, the ammonium, calcium, magnesium, potassium, and sodium salts.

With basic reagents such as hydroxides, carbonates, hydrogencarbonates, alkoxides and ammonia, organic bases or alternatively basic amino acids the compounds of the formula I, form stable alkali metal, alkaline earth metal or optionally substituted ammonium salts.

Salts derived from pharmaceutically acceptable organic non-toxic bases include salts of primary, secondary, and tertiary amines, cyclic amines, dicyclohexyl amines and basic ion-exchange resins, such as arginine, betaine, caffeine, choline, N,N-dibenzylethylenediamine, diethylamine, 2-diethylaminoethanol, 2-dimethylaminoethanol, ethanolamine, ethylenediamine, N-ethylmorpholine, N-ethylpiperidine, glucamine, glucosamine, histidine, hydrabamine, isopropylamine, lysine, methylglucamine, morpholine, piperazine, piperidine, polyamine resins, procaine, purines, theobromine, triethylamine, trimethylamine, tripropylamine, tromethamine, and the like. Also, included are the basic nitrogen-containing groups may be quaternized with such agents as lower alkyl halides, such as methyl, ethyl, propyl, and butyl chloride, bromides and iodides; dialkyl sulfates like dimethyl, diethyl, dibutyl; and diamyl sulfates, long chain halides such as decyl, lauryl, myristyl and stearyl chlorides, bromides and iodides, aralkyl halides like benzyl and phenethyl bromides and others.

The preparation of pharmacologically acceptable salts from compounds of the formula I, capable of salt formation, including their stereoisomeric forms is carried out known methods, for example, by mixing a compound of the present invention with an equivalent amount and a solution containing a desired acid, base, or the like, and then collecting the desired salt by filtering the salt or distilling off the solvent. The compounds of the present invention and salts thereof may form solvates with a solvent such as water, ethanol, or glycerol. The compounds of the present invention may form an acid addition salt and a salt with a base at the same time according to the type of substituent of the side chain.

If the compounds of Formula I simultaneously contain acidic and basic groups in the molecule the invention also includes, in addition to the salt forms mentioned, inner salts or betaines (zwitterions). Salts can be obtained from the compounds of Formula I by customary methods which are known to the person skilled in the art, for example by combination with an organic or inorganic acid or base in a solvent or dispersant, or by anion exchange or cation exchange from other salts.

The present invention includes compounds of structural formula I, as well as salts thereof, particularly pharmaceutically acceptable salts, solvates of such compounds and solvated salt forms thereof, where such forms are possible unless specified otherwise.

Furthermore, compounds of the present invention may exist in amorphous form and/or one or more crystalline forms, and as such all amorphous and crystalline forms and mixtures thereof of the compounds of Formula I, including the Examples, are intended to be included within the scope of the present invention. In addition, some of the compounds of the instant invention may form solvates with water (i.e., a hydrate) or common organic solvents such as but not limited to EtOAc. Such solvates and hydrates, particularly the pharmaceutically acceptable solvates and hydrates, of the instant compounds are likewise encompassed within the scope of this invention, along with un-solvated and anhydrous forms.

Accordingly, the compounds within the generic structural formulas, embodiments and specific compounds described in the Examples and claimed herein encompass salts, all possible stereoisomers and tautomers, physical forms (e.g., amorphous and crystalline forms), solvate and hydrate forms thereof and any combination of these forms, as well as the salts, pro-drug forms thereof, and salts of pro-drug forms thereof, where such forms are possible unless specified otherwise,

The invention also relates to medicaments containing at least one compound of the Formula I and/or of a pharmaceutically acceptable salt of the compound of the Formula I and/or an optionally stereoisomeric form of the compound of the Formulas I or a pharmaceutically acceptable salt of the stereoisomeric form of the compound of Formulas I, together with a pharmaceutically acceptable vehicle, carrier, additive and/or other active substances and auxiliaries.

The medicaments according to the invention can be administered by oral, inhalative, rectal or transdermal administration or by subcutaneous, intraarticular, intraperitoneal or intravenous injection. Oral administration is preferred. Coating of stents with compounds of the Formula I and other surfaces which come into contact with blood in the body is possible.

The invention also relates to a process for the production of a medicament, which comprises bringing at least one compound of the Formula I into a suitable administration form using a pharmaceutically acceptable carrier and optionally further suitable active substances, additives or auxiliaries.

The present invention also relates to processes for the preparation of the compounds of Formula I which are described in the following and by which the compounds of the invention are obtainable.

The terms "therapeutically effective (or efficacious) amount" and similar descriptions such as "an amount efficacious for treatment" are intended to mean that amount of a pharmaceutical drug that will alleviate the symptoms of the disorder, condition or disease being treated (i.e., disorder, condition or disease associated with DGAT2 activity) in an animal or human. The terms "prophylactically effective (or efficacious) amount" and similar descriptions such as "an amount efficacious for prevention" are intended to mean that amount of a pharmaceutical drug that will prevent or reduce the symptoms or occurrence of the disorder, condition or disease being treated (i.e., disorder, condition or disease associated with DGAT2 activity) in an animal or human. The dosage regimen utilizing a compound of the instant invention is selected in accordance with a variety of factors including type, species, age, weight, sex and medical condition of the patient; the severity of the condition to be treated; the potency of the compound chosen to be administered; the route of administration; and the renal and hepatic function of the patient. A consideration of these factors is well within the purview of the ordinarily skilled clinician for the purpose of determining the therapeutically effective or prophylactically effective dosage amount needed to prevent, counter, or arrest the progress of the condition. It is understood that a specific daily dosage amount can simultaneously be both a therapeutically effective amount, e.g., for treatment of hepatic steatosis, diabetes mellitus, obesity, hyperlipidemia, hypercholesterolemia, and a prophylactically effective amount, e.g., for treatment of NASH.

Disorders, conditions and diseases which can be treated or prevented by inhibiting DGAT2 by using the compounds of Formula I are, for example, diseases such as non-alcoholic steatohepatitis (NASH), fibrosis, hyperlipidemia, type I diabetes, type II diabetes mellitus, cognitive decline, dementia, coronary heart disease, ischemic stroke, restenosis, peripheral vascular disease, intermittent claudication, myocardial infarction, dyslipidemia, post-prandial lipemia, obesity, osteoporosis, hypertension, congestive heart failure, left ventricular hypertrophy, peripheral arterial disease, diabetic retinopathy, diabetic nephropathy, glomerulosclerosis, chronic renal failure, diabetic neuropathy, metabolic syndrome, syndrome X, coronary heart disease, angina pectoris,

thrombosis, atherosclerosis, myocardial infarction, transient ischemic attacks, stroke, hyperglycemia, hyperinsulinemia, hypertriglyceridemia, hypertriglyceridemia, insulin resistance, impaired glucose tolerance, erectile dysfunction, skin and connective tissue disorders, hyper-apo B lipoproteinemia, non-alcoholic fatty liver disease, cardiorenal diseases such as chronic kidney diseases and heart failure, and related diseases and conditions.

The compounds of Formula I and their pharmaceutically acceptable salts can be administered to animals, preferably to mammals, and in particular to humans, as pharmaceuticals by themselves, in mixtures with one another or in the form of pharmaceutical preparations. The term "patient" includes animals, preferably mammals and especially humans, who use the instant active agents for the prevention or treatment of a medical condition. Administering of the drug to the patient includes both self-administration and administration to the patient by another person. The patient may need, or desire, treatment for an existing disease or medical condition, or may be in need of or desire prophylactic treatment to prevent or reduce the risk of occurrence of said disease or medical condition. As used herein, a patient "in need" of treatment of an existing condition or of prophylactic treatment encompasses both a determination of need by a medical professional as well as the desire of a patient for such treatment.

Furthermore, a subject of the present invention are pharmaceutical preparations (or pharmaceutical compositions) which comprise as active component a therapeutically effective dose of at least one compound of Formula I and/or a pharmaceutically acceptable salt thereof and a customary pharmaceutically acceptable carrier, i.e., one or more pharmaceutically acceptable carrier substances and/or additives.

Thus, a subject of the invention is, for example, said compound and its pharmaceutically acceptable salts for use as a pharmaceutical, pharmaceutical preparations which comprise as active component a therapeutically effective dose of said compound and/or a pharmaceutically acceptable salt thereof and a customary pharmaceutically acceptable carrier, and the uses of said compound and/or a pharmaceutically acceptable salt thereof in the therapy or prophylaxis of the above mentioned syndromes as well as their use for preparing medicaments for these purposes.

The pharmaceuticals according to the invention can be administered orally, for example in the form of pills, tablets, lacquered tablets, sugar-coated tablets, granules, hard and soft gelatin capsules, aqueous, alcoholic or oily solutions, syrups, emulsions or suspensions, or rectally, for example in the form of suppositories. Administration can also be carried out parenterally, for example subcutaneously, intramuscularly or intravenously in the form of solutions for injection or infusion. Other suitable administration forms are, for example, percutaneous or topical administration, for example in the form of ointments, tinctures, sprays or transdermal therapeutic systems, or the inhalative administration in the form of nasal sprays or aerosol mixtures, or, for example, microcapsules, implants or rods. The preferred administration form depends, for example, on the disease to be treated and on its severity.

For the production of pills, tablets, sugar-coated tablets and hard gelatin capsules it is possible to use, for example, lactose, starch, for example maize starch, or starch derivatives, talc, stearic acid or its salts, etc. Carriers for soft gelatin capsules and suppositories are, for example, fats, waxes, semisolid and liquid polyols, natural or hardened oils, etc. Suitable carriers for the preparation of solutions, for example of solutions for injection, or of emulsions or syrups

are, for example, water, physiologically sodium chloride solution, alcohols such as ethanol, glycerol, polyols, sucrose, invert sugar, glucose, mannitol, vegetable oils, etc. It is also possible to lyophilize the compounds of Formula I and their pharmaceutically acceptable salts and to use the resulting lyophilisates, for example, for preparing preparations for injection or infusion. Suitable carriers for microcapsules, implants or rods are, for example, copolymers of glycolic acid and lactic acid.

Suitable solid or galenical preparation forms are, for example, granules, powders, coated tablets, tablets, (micro) capsules, suppositories, syrups, juices, suspensions, emulsions, drops or injectable solutions and preparations having prolonged release of active substance, in whose preparation customary excipients such as vehicles, disintegrants, binders, coating agents, swelling agents, glidants or lubricants, flavorings, sweeteners and solubilizers are used. Frequently used auxiliaries which may be mentioned are magnesium carbonate, titanium dioxide, lactose, mannitol and other sugars, talc, lactose, gelatin, starch, cellulose and its derivatives, animal and plant oils such as cod liver oil, sunflower, peanut or sesame oil, polyethylene glycol and solvents such as, for example, sterile water and mono- or polyhydric alcohols such as glycerol.

Besides the active compounds and carriers, the pharmaceutical preparations can also contain customary additives, for example fillers, disintegrants, binders, lubricants, wetting agents, stabilizers, emulsifiers, dispersants, preservatives, sweeteners, colorants, flavorings, aromatizers, thickeners, diluents, buffer substances, solvents, solubilizers, agents for achieving a depot effect, salts for altering the osmotic pressure, coating agents or antioxidants.

The dosage of the active compound of Formula I and/or of a pharmaceutically acceptable salt thereof to be administered depends on the individual case and is, as is customary, to be adapted to the individual circumstances to achieve an optimum effect. Thus, it depends on the nature and the severity of the disorder, condition or disease to be treated, and also on the sex, age, weight and individual responsiveness of the human or animal to be treated, on the efficacy and duration of action of the compounds used, on whether the therapy is acute or chronic or prophylactic, or on whether other active compounds are administered in addition to compounds of Formula I.

Combination Agents

The compounds of the present invention can be administered alone or in combination with one or more additional therapeutic agents disclosed herein or other suitable agents, depending on the condition being treated. Hence, in some embodiments the one or more compounds of the invention will be co-administered with other agents as described above. When used in combination therapy, the compounds described herein are administered with the second agent simultaneously or separately. This administration in combination can include simultaneous administration of the two agents in the same dosage form, simultaneous administration in separate dosage forms, and separate administration. That is, a compound of Formula (I) and any of the agents described above can be formulated together in the same dosage form and administered simultaneously. Alternatively, a compound of Formula (I) and any of the agents described above can be simultaneously administered, wherein both the agents are present in separate formulations. In another alternative, a compound of Formula (I) can be administered just followed by and any of the agents described above, or

vice versa. In some embodiments of the separate administration protocol, a compound of Formula (I) and any of the agents described above are administered a few minutes apart, or a few hours apart, or a few days apart.

As one aspect of the present invention contemplates the treatment of the disease/conditions with a combination of pharmaceutically active compounds that may be administered separately, the invention further relates to combining separate pharmaceutical compositions in kit form. The kit comprises two separate pharmaceutical compositions: a compound of Formula (I), and a second pharmaceutical compound. The kit comprises a container for containing the separate compositions such as a divided bottle or a divided foil packet. Additional examples of containers include syringes, boxes, and bags. In some embodiments, the kit comprises directions for the use of the separate components. The kit form is particularly advantageous when the separate components are preferably administered in different dosage forms (e.g., oral, parenteral; IV, transdermal and subcutaneous), are administered at different dosage intervals, or when titration of the individual components of the combination is desired by the prescribing health care professional.

One or more additional pharmacologically active agents may be administered in combination with a compound of Formula I. An additional active agent (or agents) is intended to mean a pharmaceutically active agent (or agents) that is active in the body, including pro-drugs that convert to pharmaceutically active form after administration, which are different from the compound of Formula I, and also includes free-acid, free-base and pharmaceutically acceptable salts of said additional active agents. Generally, any suitable additional active agent or agents, including but not limited to anti-hypertensive agents, anti-obetic, anti-inflammatory, anti-fibrotic, and anti-atherosclerotic agents such as a lipid modifying compound, anti-diabetic agents and/or anti-obesity agents may be used in any combination with the compound of Formula I in a single dosage formulation (a fixed dose drug combination), or may be administered to the patient in one or more separate dosage formulations which allows for concurrent or sequential administration of the active agents (co-administration of the separate active agents).

Examples of additional active agents which may be employed include but are not limited to angiotensin converting enzyme inhibitors (e.g., alacepril, benazepril, captopril, ceronapril, cilazapril, delapril, enalapril, enalaprilat, fosinopril, imidapril, lisinopril, moveltipril, perindopril, quinapril, ramipril, spirapril, temocapril, ortrandolapril), angiotensin II receptor antagonists (e.g., losartan i.e., COZAAR®, valsartan, candesartan, olmesartan, telmesartan and any of these drugs used in combination with hydrochlorothiazide such as HYZAAR®); neutral endopeptidase inhibitors (e.g., thiorphan and phosphoramidon), aldosterone antagonists, aldosterone synthase inhibitors, renin inhibitors (e.g., urea derivatives of di- and tri-peptides, amino acids and derivatives, amino acid chains linked by non-peptidic bonds, di- and tri-peptide derivatives, peptidyl amino diols and peptidyl beta-aminoacyl aminodiol carbamates; also, and small molecule renin inhibitors including diol sulfonamides and, N-morpholino derivatives, N-heterocyclic alcohols and pyrrolimidazolones; also, pepstatin derivatives and fluoro- and chloro-derivatives of statone-containing peptides, enalkrein, RO 42-5892, A 65317, CP 80794, ES 1005, ES 8891, SQ 34017, aliskiren (2(S),4(S),5(S),7(S)-N-(2-carbamoyl-2-methylpropyl)-5-amino-4-hydroxy-2,7-diisopropyl-8-[4-methoxy-3-(3-methoxypropoxy)-phenyl]-octanamid hemifumarate) SPP600, SPP630 and

SPP635), endothelin receptor antagonists, phosphodiesterase-5 inhibitors (e.g., sildenafil, tadalafil and vardenafil), vasodilators, calcium channel blockers (e.g., amlodipine, nifedipine, verapamil, diltiazem, gallopamil, niludipine, nimodipine, nicardipine), potassium channel activators (e.g., nicorandil, pinacidil, cromakalim, minoxidil, aprilkalim, loprazolam), diuretics (e.g., hydrochlorothiazide), sympatholitics, beta-adrenergic blocking drugs (e.g., propranolol, atenolol, bisoprolol, carvedilol, metoprolol, or metoprolol tartrate), alpha adrenergic blocking drugs (e.g., doxazosin, prazosin or alpha methyl dopa) central alpha adrenergic agonists, peripheral vasodilators (e.g., hydralazine); lipid lowering agents e.g., HMG-CoA reductase inhibitors such as simvastatin and lovastatin which are marketed as ZOCOR® and MEVACOR® in lactone pro-drug form and function as inhibitors after administration, and pharmaceutically acceptable salts of dihydroxy open ring acid HMG-CoA reductase inhibitors such as atorvastatin (particularly the calcium salt sold in LIPITOR®), rosuvastatin (particularly the calcium salt sold in CRESTOR®), pravastatin (particularly the sodium salt sold in PRAVACHOL®), fluvastatin (particularly the sodium salt sold in LESCOL®), cerivastatin, and pitavastatin; a cholesterol absorption inhibitor such as ezetimibe (ZETIA®) and ezetimibe in combination with any other lipid lowering agents such as the HMG-CoA reductase inhibitors noted above and particularly with simvastatin (VYTORIN®) or with atorvastatin calcium; niacin in immediate-release or controlled release forms, and/or with an HMG-CoA reductase inhibitor; niacin receptor agonists such as acipimox and acifran, as well as niacin receptor partial agonists; anti-cholesterol agents such as PCSK9 inhibitors (alirocumab, evolocumab), Nexletol™ (bempedoic acid, ACL inhibitor), and Vascepa® (Icosapent ethyl); metabolic altering agents including insulin and insulin mimetics (e.g., insulin degludec, insulin glargine, insulin lispro), dipeptidyl peptidase-IV (DPP-4) inhibitors (e.g., sitagliptin, alogliptin, omarigliptin, linagliptin, vildagliptin); insulin sensitizers, including (i) β -klotho/FGFR1 activating monoclonal antibody (e.g., MK-3655), pan FGFR1-4/KLB modulators, FGF19 analogue (e.g., Aldafermin) (ii) PPAR γ agonists, such as the glitazones (e.g., pioglitazone, AMG 131, CHS 131, MBX2044, mitoglitazone, lobeglitazone, IDR-105, rosiglitazone, and balaglitazone), and other PPAR ligands, including (1) PPAR α/γ dual agonists (e.g., ZYH2, ZYH1, GFT505, chiglitazar, muraglitazar, aleglitazar, sodelglitazar, and naveglitazar); (2) PPAR α agonists such as fenofibric acid derivatives (e.g., gemfibrozil, clofibrate, ciprofibrate, fenofibrate, bezafibrate), (3) selective PPAR γ modulators (SPPAR γ M's), (e.g., such as those disclosed in WO 02/060388, WO 02/08188, WO 2004/019869, WO 2004/020409, WO 2004/020408, and WO 2004/066963); (4) PPAR γ partial agonists, (5) PPAR $\alpha/8$ dual agonists (e.g., Elafibranor); (iii) biguanides, such as metformin and its pharmaceutically acceptable salts, in particular, metformin hydrochloride, and extended-release formulations thereof, such as Glumetza™, Fortamet™, and GlucophageXR™; and (iv) protein tyrosine phosphatase-1B (PTP-1B) inhibitors (e.g., ISIS-113715 and TTP814); insulin or insulin analogs (e.g., insulin detemir, insulin glulisine, insulin degludec, insulin glargine, insulin lispro and inhalable formulations of each); leptin and leptin derivatives and agonists; amylin and amylin analogs (e.g., pramlintide);

sulfonylurea and non-sulfonylurea insulin secretagogues (e.g., tolbutamide, glyburide, glipizide, glimepiride, mitiglinide, meglitinides, nateglinide and repaglinide); α -glucosidase inhibitors (e.g., acarbose, voglibose and

migliitol); glucagon receptor antagonists (e.g., MK-3577, MK-0893, LY-2409021 and KT6-971); incretin mimetics, such as GLP-1, GLP-1 analogs, derivatives, and mimetics; and GLP-1 receptor agonists (e.g., dulaglutide, semaglutide, albiglutide, exenatide, liraglutide, lixisenatide, taspoglutide, CJC-1131, and BIM-51077, including intranasal, transdermal, and once-weekly formulations thereof), bile acid sequestering agents (e.g., colestilan, colestimide, colessevalam hydrochloride, colestipol, cholestyramine, and dialkylaminoalkyl derivatives of a cross-linked dextran), acyl CoA: cholesterol acyltransferase inhibitors, (e.g., avasimibe); antiobesity compounds; agents intended for use in inflammatory conditions, such as aspirin, non-steroidal anti-inflammatory drugs or NSAIDs, glucocorticoids, and selective cyclooxygenase-2 or COX-2 inhibitors; glucokinase activators (GKAs) (e.g., AZD6370); inhibitors of 11 β -hydroxysteroid dehydrogenase type 1, (e.g., such as those disclosed in U.S. Pat. No. 6,730,690, and LY-2523199); CETP inhibitors (e.g., anacetrapib, torcetrapib, and evacetrapib); inhibitors of fructose 1,6-bisphosphatase, (e.g., such as those disclosed in U.S. Pat. Nos. 6,054,587; 6,110,903; 6,284,748; 6,399,782; and 6,489,476); inhibitors of acetyl CoA carboxylase-1 or 2 (ACC1 or ACC2); AMP-activated Protein Kinase (AMPK) activators; other agonists of the G-protein-coupled receptors: (i) GPR-109, (ii) GPR-119 (e.g., MBX2982 and PSN821), and (iii) GPR-40 (e.g., TAK875); SSTR3 antagonists (e.g., such as those disclosed in WO 2009/001836); neuromedin U receptor agonists (e.g., such as those disclosed in WO 2009/042053, including, but not limited to, neuromedin S (NMS)); SCD modulators (e.g., Aramchol); GPR-105 antagonists (e.g., such as those disclosed in WO 2009/000087); SGLT inhibitors (e.g., ASP1941, SGLT-3, SGLT-2 such as empagliflozin, dapagliflozin, canagliflozin, and ertugliflozin, BI-10773, remogliflozin, TS-071, tofogliflozin, ipragliflozin, and LX-4211); inhibitors of acyl coenzyme A carboxylase (ACC, MK-4074); inhibitors of diacylglycerol acyltransferase 1 and 2 (DGAT-1 and DGAT-2); inhibitors of fatty acid synthase; inhibitors of acyl coenzyme A: monoacylglycerol acyltransferase 1 and 2 (MGAT-1 and MGAT-2); agonists of the TGR5 receptor (also known as GPBAR1, BG37, GPCR19, GPR131, and M-BAR); ileal bile acid transporter inhibitors; bile acid modulators; PACAP, PACAP mimetics, and PACAP receptor 3 agonists; IL-1b antibodies, (e.g., XOMA052 and canakinumab); anti-fibrotic and/or anti-inflammatory agents (CCR2/CCR5 dual receptor antagonist (e.g., cenicriviroc); galectin 3 inhibitor (e.g., belapectin, GB-1107, GB-1211), siRNA against HSP 47 (e.g., BMS-986263); NSAID derived from pifenidone (e.g., hydronidone), A3AR agonist (e.g., namodenoson, FM101); TGFTX4 (e.g., nitazoxanide); 5-lipoxygenase inhibitor (e.g., tiplelukast), Bifunctional urate inhibitor (e.g., ACQT1127), adiponectin receptor agonist (e.g., ALY688), TNF receptor antagonist (e.g., atosimab), Autotaxin inhibitor (e.g., BLD-0409, TJC 0265, TJC 0316), CCL24 blocking monoclonal antibody (e.g., CM101), IL-11 inhibitor (e.g., ENx 108A), LPA1 receptor antagonist (e.g., EPGN 696), Dual JAK1/2 inhibitor (e.g., EX 76545), GPR antagonist (e.g., GPR91 antagonist), Integrin α v β 1, α v β 3 and α v β 6 inhibitor (e.g., IDL 2965), NLRP3 antagonist (e.g., IFM-514), inflammasome inhibitors (e.g., JT194, JT349), Cell membrane permeability inhibitor (e.g., Larazotide), CCR5 antagonist (e.g., Ieronlimab), TNF inhibitor (e.g., LIVNate), integrin α v β 6 inhibitor (e.g., MORF beta6), NLRP inflammasome antagonists, siRNA (e.g., OLX 701), dual TGF β /Hedgehog inhibitor (e.g., Oxy 200), GPR40 agonist/GPR84 antagonist (e.g., PBI-4547), neutrophil elastase inhibitor

(e.g., PHP-303), integrin inhibitor (e.g., PLN-1474), TGF β 1 modulator (e.g., PRM-151), CCK receptor antagonist (e.g., proglumide), LOXL2 inhibitor (e.g., PXS-5338K, PXS-5382A), IL-11 inhibitors, MPYS protein inhibitor (e.g., cGAS/STING antagonists), kinase inhibiting RNase, membrane protein mAbs, tumor necrosis factor inhibitor, NRF2 activator (e.g., SCO 116), SSAO inhibitor (e.g., TERN 201), TRAIL2 agonist (e.g., TLY012), IL-6 receptor antagonist (e.g., TZLS 501), AOC3 inhibitor (e.g., UD-014), SSAO/VAP-1 inhibitor, TREM2; anti-oxidant (e.g., vitamin E); anti-inflammatory agents (e.g., norfloxacin, ciprofloxacin, ceftriaxone); coagulation modifiers (e.g., anti-coagulants, anti-platelet agents, pentoxifylline, vitamin K, DDAVP); dual GIP and GLP-1 receptor agonist (e.g., tirzepatide); dual GLP-1/GRA (e.g., cotadutide, ALT-801, DD 01, G49, PB-718); dual GLP-1 (e.g., CT 868); GLP-1/GRA/GIP triple agonist (e.g., HM15211); GRP120 stimulant/inflammasome modulator/PPAR γ dual agonist (e.g., KDT501); GLP-1/FGF21 (e.g., YH25724); GLP-1 agonist (e.g., Ozempic (semaglutide sc), XW 003); selective thyroid hormone receptor- β agonist (e.g., resmetirom); apoptosis modulators (JNK-1 inhibitor (e.g., CC-90001), Peroxidase inhibitor (e.g., AZM198), ASK-1 inhibitor (e.g., CS-17919, SRT 015)); erythropoietin-stimulating agents (erythropoietin receptor agonist (e.g., cibinetide)); glucose pathway modulators (SGLT-2 inhibitor (e.g., Forxiga, Farxiga (dapagliflozin)); dual SGLT-1/2 inhibitor (e.g., licogliflozin), Glucose-6-P dehydrogenase inhibitor (e.g., fluasterone) LAPS glucagon combo (e.g., HM14320), SGLT-1 inhibitor (e.g., SGL5213)); immune modulators (TLR4 inhibitor (e.g., GBK-233), immunomodulatory polyclonal antibody (e.g., IMM-124E), TLR4 antagonist (e.g., JKB-122), CD3 monoclonal antibody (e.g., foralumab), TLR4 antagonist (e.g., JKB 133), TLR4 inhibitor (e.g., mosedipimod), Macrophage inhibitor via CD206 targeting (e.g., MT2002), TLR2/4 antagonist (e.g., VB-201, VB-703), immunomodulatory polyclonal antibody (e.g., IMM-124E)); incretin-based therapies (GLP-1 agonist (e.g., Ozempic (semaglutide sc), XW 003), GLP-1/glucagon dual receptor agonist (e.g., HM12525A), prandial insulin (e.g., ORMD 0801)); lipid modulators (AMPK Activator/Glutathione transferase (e.g., oltipraz), THR-beta agonist (e.g., resmetirom, VK2809, MGL-3745, ALG-009, ASC41, CNPT-101101, TERN 501), IBAT inhibitor (e.g., elobixibat, CJ 14199), omega-6-fatty acid (e.g., epeleton), FASN inhibitor (e.g., TVB2640, FT 4101, FT 8225), ANGPTL3 inhibitor (e.g., vupanorsen), PNPLA3 inhibitor (e.g., AZD2693), RAS domain kinase inhibitor (e.g., BioE1115), NTCP inhibitor (e.g., bulevirtide), P2Y13 receptor agonist (e.g., CER-209), omega-3 fatty acid, HSD17 β 13 inhibitor; metabolism modulators (FXR agonist (e.g., Ocaliva (obeticholic acid), IOT022), recombinant variant of FGF19 (e.g., aldafermin), bi-specific FGFR1/KLB antibody (e.g., BFKB8488A), mTOT modulator (e.g., MSDC-0602K), pegylated analog of FGF21 (e.g., pegbelfermin, BMS-986171), non-bile FXR agonist (e.g., cilofexor, EDP-305, EYP 001, tropifexor, MET409, AGN-242256, AGN-242266, EDP 297, HPG 1860, MET642, RDX023, TERN 101), ACC inhibitor (e.g., firsocostat, PF-05221304), ketohexokinase inhibitor (e.g., PF-06835919), AMPK activator (e.g., PXL770, MSTM 101, O304), bile acid modulator (e.g., Albiro), FGF21 analog (e.g., BIO89-100), MOTSc analog (e.g., CB4211), cyclophilin inhibitor (e.g., CRV 431), FGF19 (e.g., DEL 30), mitochondrial uncoupler (e.g., GEN 3026), FXR/GPCR dual agonist (e.g., INT-767), Cysteamine derivative (e.g., KB-GE-001), dual amylin and calcitonin receptor agonist (e.g., KBP-089), transient FXR agonist (e.g., M 1217),

anti-beta-klotho (KLB)-FGFR1c receptor complex mAb (e.g., MK3655), GDF15 analog (e.g., NGM395), cyclophilin inhibitor (e.g., NV556), LXR modulator (e.g., PX 329, PX 655, PX 788), LXR inverse agonist (e.g., PX016), deuterated obeticholic acid (e.g., ZG 5216)); PPAR modulators (dual PPAR α/γ agonist (e.g., clafibranor), PPAR pan agonist (e.g., lanifibranor), PPAR α agonists (e.g., Parmodia), PPAR γ agonist (e.g., CHS 131), MPC inhibitor (e.g., PXL065), PPAR 8/Y agonist (e.g. T3D 959)); RAAS mIM-Modulators (mineralocorticoid receptor antagonist (e.g., apararenone, eplerenone, spironolactone), angiotensin receptor blocker (e.g., losartan potassium)); neurotransmitter modulators (cannabinoid receptor modulator, CB1 receptor antagonist (e.g., CRB-4001, IM-102, nimacimab), TPH1 inhibitor (e.g., CU 02), GPR120 agonist (e.g., KBR2001), combination of cannabinoid and botanical anti-inflammatory compound (e.g., SCN 002)); PDE Modulator (PDE4 inhibitor (e.g., ART 648)); CYP2E1 inhibitor (e.g., SNP-610); cell therapies (e.g., HepaStem) and bromocriptine mesylate and rapid-release formulations thereof; or with other drugs beneficial for the prevention or the treatment of the above-mentioned diseases including nitroprusside and diazoxide the free-acid, free-base, and pharmaceutically acceptable salt forms of the above active agents where chemically possible.

The present invention includes the pharmaceutically acceptable salts of the compounds defined herein, including the pharmaceutically acceptable salts of all structural formulas, embodiments and classes defined herein. Reference to the compounds of structural Formula (I) includes the compounds of other generic structural Formulas, such as Formulas and embodiments that fall within the scope of Formula (I).

Dosages of the Compounds of Formula (I)

If the patient is responding, or is stable, after completion of the therapy cycle, the therapy cycle can be repeated according to the judgment of the skilled clinician. Upon completion of the therapy cycles, the patient can be continued on the compounds of the invention at the same dose that was administered in the treatment protocol. This maintenance dose can be continued until the patient progresses or can no longer tolerate the dose (in which case the dose can be reduced and the patient can be continued on the reduced dose).

Those skilled in the art will recognize that the actual dosages and protocols for administration employed in the methods of the invention may be varied according to the judgment of the skilled clinician. The actual dosage employed may be varied depending upon the requirements of the patient and the severity of the condition being treated. Determination of the proper dosage for a particular situation is within the skill of the art. A determination to vary the dosages and protocols for administration may be made after the skilled clinician considers such factors as the patient's age, condition and size, as well as the severity of the condition being treated and the response of the patient to the treatment.

The dosage regimen utilizing a compound of the instant invention is selected in accordance with a variety of factors including type, species, age, weight, sex and medical condition of the patient; the severity of the condition to be treated; the potency of the compound chosen to be administered; the route of administration; and the renal and hepatic function of the patient. A consideration of these factors is well within the purview of the ordinarily skilled clinician for the purpose of determining the therapeutically effective or prophylactically effective dosage amount needed to prevent,

counter, or arrest the progress of the condition. It is understood that a specific daily dosage amount can simultaneously be both a therapeutically effective amount, e.g., for treatment of an oncological condition, and a prophylactically effective amount, e.g., for prevention of an oncological condition.

While individual needs vary, determination of optimal ranges of effective amounts of the compound of the invention is within the skill of the art. For administration to a human in the curative or prophylactic treatment of the conditions and disorders identified herein, for example, typical dosages of the compounds of the present invention can be about 0.05 mg/kg/day to about 50 mg/kg/day, for example at least 0.05 mg/kg, at least 0.08 mg/kg, at least 0.1 mg/kg, at least 0.2 mg/kg, at least 0.3 mg/kg, at least 0.4 mg/kg, or at least 0.5 mg/kg, and preferably 50 mg/kg or less, 40 mg/kg or less, 30 mg/kg or less, 20 mg/kg or less, or 10 mg/kg or less, which can be about 2.5 mg/day (0.5 mg/kg $\times$ 5 kg) to about 5000 mg/day (50 mg/kg $\times$ 100 kg), for example. For example, dosages of the compounds can be about 0.1 mg/kg/day to about 50 mg/kg/day, about 0.05 mg/kg/day to about 10 mg/kg/day, about 0.05 mg/kg/day to about 5 mg/kg/day, about 0.05 mg/kg/day to about 3 mg/kg/day, about 0.07 mg/kg/day to about 3 mg/kg/day, about 0.09 mg/kg/day to about 3 mg/kg/day, about 0.05 mg/kg/day to about 0.1 mg/kg/day, about 0.1 mg/kg/day to about 1 mg/kg/day, about 1 mg/kg/day to about 10 mg/kg/day, about 1 mg/kg/day to about 5 mg/kg/day, about 1 mg/kg/day to about 3 mg/kg/day, about 3 mg/day to about 500 mg/day, about 5 mg/day to about 250 mg/day, about 10 mg/day to about 100 mg/day, about 3 mg/day to about 10 mg/day, or about 100 mg/day to about 250 mg/day. Such doses may be administered in a single dose or may be divided into multiple doses.

Pharmaceutical Compositions

The compounds of Formula I and their pharmaceutically acceptable salts can be administered to animals, preferably to mammals, and in particular to humans, as pharmaceuticals by themselves, in mixtures with one another or in the form of pharmaceutical compositions. The term "subject" or "patient" includes animals, preferably mammals and especially humans, who use the instant active agents for the prevention or treatment of a medical condition. Administering of the compound of Formula I to the subject includes both self-administration and administration to the patient by another person. The subject may need, or desire, treatment for an existing disease or medical condition, or may be in need of or desire prophylactic treatment to prevent or reduce the risk of occurrence of said disease or medical condition. As used herein, a subject "in need" of treatment of an existing condition or of prophylactic treatment encompasses both a determination of need by a medical professional as well as the desire of a patient for such treatment.

Methods for the safe and effective administration of most of these agents are known to those skilled in the art. In addition, their administration is described in the standard literature.

If the patient is responding, or is stable, after completion of the therapy cycle, the therapy cycle can be repeated according to the judgment of the skilled clinician. Upon completion of the therapy cycles, the patient can be continued on the compounds of the invention at the same dose that was administered in the treatment protocol. This maintenance dose can be continued until the patient progresses or can no longer tolerate the dose (in which case the dose can be reduced and the patient can be continued on the reduced dose).

Those skilled in the art will recognize that the actual dosages and protocols for administration employed in the methods of the invention may be varied according to the judgment of the skilled clinician. The actual dosage employed may be varied depending upon the requirements of the patient and the severity of the condition being treated. Determination of the proper dosage for a particular situation is within the skill of the art. A determination to vary the dosages and protocols for administration may be made after the skilled clinician takes into account such factors as the patient's age, condition and size, as well as the severity of the condition being treated and the response of the patient to the treatment.

The amount and frequency of administration of the compound of formula I, and any additional agents will be regulated according to the judgment of the attending clinician (physician) considering such factors as age, condition and size of the patient as well as severity of the condition being treated.

The compounds of the invention are also useful in preparing a medicament that is useful in treating NASH and fibrosis.

The instant compounds are also useful in combination with therapeutic, chemotherapeutic and anti-cancer agents for the treatment of hepatic cellular carcinoma. Combinations of the presently disclosed compounds with therapeutic, chemotherapeutic and anti-cancer agents are within the scope of the invention. Examples of such agents can be found in *Cancer Principles and Practice of Oncology* by V. T. Devita and S. Hellman (editors), 9th edition (May 16, 2011), Lippincott Williams & Wilkins Publishers. A person of ordinary skill in the art would be able to discern which combinations of agents would be useful based on the particular characteristics of the drugs and the cancer involved. Such agents include the following: estrogen receptor modulators, programmed cell death protein 1 (PD-1) inhibitors, programmed death-ligand 1 (PD-L1) inhibitors, androgen receptor modulators, retinoid receptor modulators, cytotoxic/cytostatic agents, antiproliferative agents, prenyl-protein transferase inhibitors, HMG-CoA reductase inhibitors and other angiogenesis inhibitors, HIV protease inhibitors, reverse transcriptase inhibitors, inhibitors of cell proliferation and survival signaling, bisphosphonates, aromatase inhibitors, siRNA therapeutics, γ -secretase inhibitors, agents that interfere with receptor tyrosine kinases (RTKs) and agents that interfere with cell cycle checkpoints.

The chemotherapeutic agent can be administered according to therapeutic protocols well known in the art. It will be apparent to those skilled in the art that the administration of the chemotherapeutic agent can be varied depending on the cancer being treated and the known effects of the chemotherapeutic agent on that disease. Also, in accordance with the knowledge of the skilled clinician, the therapeutic protocols (e.g., dosage amounts and times of administration) can be varied in view of the observed effects of the administered therapeutic agents on the patient, and in view of the observed responses of the cancer to the administered therapeutic agents. The particular choice of chemotherapeutic agent will depend upon the diagnosis of the attending physicians and their judgment of the condition of the patient and the appropriate treatment protocol.

The initial administration can be made according to established protocols known in the art, and then, based upon the observed effects, the dosage, modes of administration and times of administration can be modified by the skilled clinician.

The determination of the order of administration, and the number of repetitions of administration of the chemotherapeutic agent during a treatment protocol, is well within the knowledge of the skilled physician after evaluation of the condition being treated and the condition of the patient.

Thus, in accordance with experience and knowledge, the practicing physician can modify each protocol for the administration of a chemotherapeutic agent according to the individual patient's needs, as the treatment proceeds. All such modifications are within the scope of the present invention.

The agent can be administered according to therapeutic protocols well known in the art. It will be apparent to those skilled in the art that the administration of the anti-cancer agent can be varied depending on the cancer being treated and the known effects of the anti-cancer agent on that disease.

The initial administration can be made according to established protocols known in the art, and then, based upon the observed effects, the dosage, modes of administration and times of administration can be modified by the skilled clinician.

The particular choice of agent will depend upon the diagnosis of the attending physicians and their judgment of the condition of the patient and the appropriate treatment protocol.

The determination of the order of administration, and the number of repetitions of administration of the agent during a treatment protocol, is well within the knowledge of the skilled physician after evaluation of the cancer being treated and the condition of the patient.

Thus, in accordance with experience and knowledge, the practicing physician can modify each protocol for the administration of an anti-cancer agent according to the individual patient's needs, as the treatment proceeds. All such modifications are within the scope of the present invention.

The attending clinician, in judging whether treatment is effective at the dosage administered, will consider the general well-being of the patient as well as more definite signs such as relief of cancer-related symptoms (e.g., pain), inhibition of tumor growth, actual shrinkage of the tumor, or inhibition of metastasis. Size of the tumor can be measured by standard methods such as radiological studies, e.g., CAT or MRI scan, and successive measurements can be used to judge whether or not growth of the tumor has been retarded or even reversed. Relief of disease-related symptoms such as pain, and improvement in overall condition can also be used to help judge effectiveness of treatment.

The compounds, compositions and methods provided herein are useful for the treatment of cancer. Cancers that may be treated by the compounds, compositions and methods disclosed herein include, but are not limited to: Liver: hepatoma (hepatocellular carcinoma), cholangiocarcinoma, hepatoblastoma, angiosarcoma, hepatocellular adenoma, hemangioma.

PD-1 inhibitors include pembrolizumab (lambrolizumab), nivolumab and MPDL3280A. PDL-inhibitors include atezolizumab, avelumab, and durvalumab.

The invention further relates to a method of treating hepatic cellular carcinoma in a human patient comprising administration of a compound of the invention (i.e., a compound of Formula I) and a PD-1 antagonist to the patient. The compound of the invention and the PD-1 antagonist may be administered concurrently or sequentially.

In particular embodiments, the PD-1 antagonist is an anti-PD-1 antibody, or antigen binding fragment thereof. In alternative embodiments, the PD-1 antagonist is an anti-PD-L1 antibody, or antigen binding fragment thereof. In some embodiments, the PD-1 antagonist is pembrolizumab (KEYTRUDA™, Merck & Co., Inc., Kenilworth, NJ, USA), nivolumab (OPDIVO™, Bristol-Myers Squibb Company, Princeton, NJ, USA), cemiplimab (LIBTAYO™, Regeneron Pharmaceuticals, Inc., Tarrytown, NY, USA), atezolizumab (TECENTRIQ™, Genentech, San Francisco, CA, USA), durvalumab (IMFINZI™, AstraZeneca Pharmaceuticals LP, Wilmington, DE), or avelumab (BAVENCIO™, Merck KGaA, Darmstadt, Germany).

In some embodiments, the PD-1 antagonist is pembrolizumab. In particular sub-embodiments, the method comprises administering 200 mg of pembrolizumab to the patient about every three weeks. In other sub-embodiments, the method comprises administering 400 mg of pembrolizumab to the patient about every six weeks.

In further sub-embodiments, the method comprises administering 2 mg/kg of pembrolizumab to the patient about every three weeks. In particular sub-embodiments, the patient is a pediatric patient.

In some embodiments, the PD-1 antagonist is nivolumab. In particular sub-embodiments, the method comprises administering 240 mg of nivolumab to the patient about every two weeks. In other sub-embodiments, the method comprises administering 480 mg of nivolumab to the patient about every four weeks.

In some embodiments, the PD-1 antagonist is cemiplimab. In particular embodiments, the method comprises administering 350 mg of cemiplimab to the patient about every 3 weeks.

In some embodiments, the PD-1 antagonist is atezolizumab. In particular sub-embodiments, the method comprises administering 1200 mg of atezolizumab to the patient about every three weeks.

In some embodiments, the PD-1 antagonist is durvalumab. In particular sub-embodiments, the method comprises administering 10 mg/kg of durvalumab to the patient about every two weeks.

In some embodiments, the PD-1 antagonist is avelumab. In particular sub-embodiments, the method comprises administering 800 mg of avelumab to the patient about every two weeks.

A compound of the instant invention, or a pharmaceutically acceptable salt thereof, may also be useful for treating cancer in combination with the following therapeutic agents: pembrolizumab (Keytruda®), abarelix (Plenaxis Depot®); aldesleukin (Prokine®); Aldesleukin (Proleukin®); Alemtuzumab (Campath®); alitretinoin (Panretin®); allopurinol (Zyloprim®); altretamine (Hexalen®); amifostine (Ethyol®); anastrozole (Arimidex®); arsenic trioxide (Trisenox®); asparaginase (Elspar®); azacitidine (Vidaza®); bevacizumab (Avastin®); bexarotene capsules (Targretin®); bexarotene gel (Targretin®); bleomycin (Blenoxane®); bortezomib (Velcade®); busulfan intravenous (Busulfex®); busulfan oral (Myleran®); calusterone (Methosarb®); capecitabine (Xeloda®); carboplatin (Paraplatin®); carmustine (BCNU®, BICNUR); carmustine (Gliadel®); carmustine with Polifeprosan 20 Implant (Gliadel Wafer®); celecoxib (Celebrex®); cetuximab (Erbix®); chlorambucil (Leukeran®); cisplatin (Platinol®); cladribine (Leustatin®; 2-CdAR); clofarabine (Clolar®); cyclophosphamide (Cytosan®, Neosar®); cyclophosphamide (Cytosan Injection®); cyclophosphamide (Cytosan Tablet®); cytarabine (Cytosar-U®); cytarabine liposomal (DepoCyt®); dac-

arbazine (DTIC-Dome®); dactinomycin, actinomycin D (Cosmegen®); Darbepoetin alfa (Aranesp®); daunorubicin liposomal (DanuoXome®); daunorubicin, daunomycin (Daunorubicin®); daunorubicin, daunomycin (Cerubidine®); Denileukin diftitox (Ontak®); dexrazoxane (Zincard®); docetaxel (Taxotere®); doxorubicin (Adriamycin PFS®); doxorubicin (Adriamycin®, Rubex®); doxorubicin (Adriamycin PFS Injection®); doxorubicin liposomal (Doxil®); dromostanolone propionate (Dromostanolone®); dromostanolone propionate (Masterone Injection®); Elliott's B Solution (Elliott's B Solution®); epirubicin (Elevance®); Epoetin alfa (Epogen®); erlotinib (Tarceva®); estramustine (Emcyt®); etoposide phosphate (Etopophos®); ctoposide, VP-16 (Vepesid®); exemestane (Aromasin®); Filgrastim (Neupogen®); floxuridine (intraarterial) (FUDR®); fludarabine (Fludara®); fluorouracil, 5-FU (Adrucil®); fulvestrant (Faslodex®); gefitinib (Iressa®); gemcitabine (Gemzar®); gemtuzumab ozogamicin (Mylotarg®); goserelin acetate (Zoladex Implant®); goserelin acetate (Zoladex®); histrelin acetate (Histrelin Implant®); hydroxyurea (Hydrea®); Ibritumomab Tiuxetan (Zevalin®); idarubicin (Idamycin®); ifosfamide (IFEX®); imatinib mesylate (Gleevec®); interferon alfa 2a (Roferon A®); Interferon alfa-2b (Intron AR); irinotecan (Camptosar®); lenalidomide (Revlimid®); letrozole (Femara®); leucovorin (Wellcovorin®, Leucovorin®); Leuprolide Acetate (Eliard®); levamisole (Ergamisol®); lomustine, CCNU (CeeBU®); meclizethamine, nitrogen mustard (Mustargen®); megestrol acetate (Megace®); melphalan, L-PAM (Alkeran®); mercaptopurine, 6-MP (Purinethol®); mesna (Mesnex®); mesna (Mesnex Tabs®); methotrexate (Methotrexate®); methoxsalen (Uvadex®); mitomycin C (Mutamycin®); mitotane (Lysodren®); mitoxantrone (Novantrone®); nandrolone phenpropionate (Durabolin-50®); nelarabine (Arranon®); Nofetumomab (Verluma®); Oprelvekin (Neumega®); oxaliplatin (Eloxatin®); paclitaxel (Paxene®); paclitaxel (Taxol®); paclitaxel protein-bound particles (Abraxane®); palifermin (Kepivance®); pamidronate (Aredia®); pegademase (Adagen (Pegademase Bovine®); pegaspargase (Oncaspar®); Pegfilgrastim (Neulasta®); pemetrexed disodium (Alimta®); pentostatin (Nipent®); pipobroman (Vercyte®); plicamycin, mithramycin (Mithracin®); porfimer sodium (Photofrin®); procarbazine (Matulane®); quinacrine (Atabrine®); Rasburicase (Elitek®); Rituximab (Rituxan®); Ridaforolimus; sargramostim (Leukine®); Sargramostim (Prokine®); sorafenib (Nexavar®); streptozocin (Zanosar®); sunitinib maleate (Sutent®); talc (Sclerosol®); tamoxifen (Nolvadex®); temozolomide (Temodar®); teniposide, VM-26 (Vumon®); testolactone (Teslac®); thioguanine, 6-TG (Thioguanine®); thiotepa (Thioplex®); topotecan (Hycamtin®); toremifene (Fareston®); Tositumomab (Bexxar®); Tositumomab/I-131 tositumomab (Bexxar®); Trastuzumab (Herceptin®); tretinoin, ATRA (Vesanoid®); Uracil Mustard (Uracil Mustard Capsules®); valrubicin (Valstar®); vinblastine (Velban®); vincristine (Oncovin®); vinorelbine (Navelbine®); vorinostat (Zolinza®) and zoledronate (Zometa®), or a pharmaceutically acceptable salt thereof.

Methods for Making the Compounds of Present Invention

The following examples are provided so that the invention might be more fully understood. Unless otherwise indicated, the starting materials are commercially available. They should not be construed as limiting the invention in any way.

Several methods for preparing the compounds of this invention are described in the following Schemes and Examples. Starting materials and intermediates are purchased, made from known procedures, or as otherwise illustrated. Some frequently applied routes to the compounds of Formula I are also described by the Schemes as follows. In some cases, the order of carrying out the steps of reaction schemes may be varied to facilitate the reaction or to avoid unwanted reaction products. For stereoisomers, enantiomer A refers to the faster/earlier eluting enantiomer and enantiomer B refers to the slower/late eluting enantiomer at the point of separation and this nomenclature is maintained through the remainder of a synthetic sequence for a given enantiomeric series regardless of the possibility that subsequent intermediates and final compounds may have the same or opposite orders of elution.

Throughout the synthetic schemes and examples, abbreviations and acronyms may be used with the following meanings unless otherwise indicated:

ACN=acetonitrile
 AcOH=acetic acid
 CDI=1,1'-Carbonyldiimidazole
 Cu(OAc)₂=copper acetate
 DAST=Diethylaminosulfur trifluoride
 DIEA=diisopropyl ethylamine
 DMF=dimethylformamide
 DMS=dimethyl sulfide
 DMSO=dimethyl sulfoxide
 DCE=1,2-dichloroethane
 DCM=dichloromethane
 DIPEA=N,N-Diisopropylethylamine
 DPPA=Diphenylphosphoryl azide
 dppf=1,1'-bis(diphenylphosphino)ferrocene
 Et=ethyl
 EtOAc=ethyl acetate
 EDC=N-(3-Dimethylaminopropyl)-N'-ethylcarbodiimide hydrochloride
 RP HPLC=Reverse Phase High Pressure Liquid Chromatography
 h or hrs=hour or hours
 HATU=1-[bis(dimethylamino)methylene]-1H-1,2,3-triazolo[4,5-b]pyridinium 3-oxid-hexafluorophosphate
 Hex=hexanes
 HOAc=acetic acid
 HOBT=Hydroxybenzotriazole
 IPA=iso-Propanol
 Me=methyl
 mCPBA=meta-chloroperoxybenzoic acid
 MgSO₄=magnesium sulfate
 MP-cyanoborohydride=macroporous polymer-supported cyanoborohydride
 rt or RT=room temperature
 nBu₄LI=Tetra-n-butylammonium iodide
 NBS=N-bromosuccinimide
 NCS=N-Chlorosuccinimide
 NIS=N-Iodosuccinimide
 PdCl₂ (dppf)=bis(diphenylphosphino)ferrocene]dichloropalladium (II)
 SFC=supercritical fluid chromatography
 SM=Starting material
 t-Bu-OH=tert-butyl alcohol
 tBuBrettPhos=2-(Di-tert-butylphosphino)-2',4',6'-triisopropyl-3,6-dimethoxy-1,1'-biphenyl
 THF=tetrahydrofuran
 TFA=Trifluoroacetic acid
 Xphos=2-Dicyclohexylphosphino-2',4',6'-triisopropylbiphenyl

XPhos Pd G2=chloro(2-dicyclohexylphosphino-2',4',6'-triisopropyl-1,1'-biphenyl) [2-(2-amino-1,1'-biphenyl)]palladium (II)

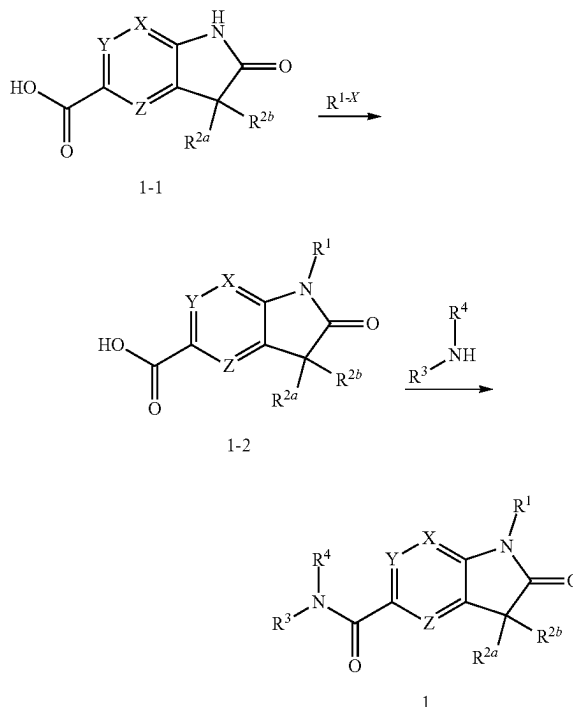
Also, aq. is aqueous, TLC is thin layer chromatography; Ts is tosyl; UV is ultraviolet; W is watts; wt. % is percentage by weight; xg is times gravity; [α]_D is the specific rotation of polarized light at 589 nm; ° C. is degrees Celsius; % w/v is percentage in weight of the former agent relative to the volume of the latter agent.

LCMS conditions: column: SUPELCO Ascentis Express C18 3×100 mm, 2.7 μ m. Solvent system: A—0.05% TFA in water and B—0.05% TFA in Acetonitrile. Gradient condition: 10-99% B in 3.5 min.

GENERAL SYNTHETIC SCHEMES

While the present invention has been described in conjunction with the specific examples set forth above, many alternatives, modifications and variations thereof will be apparent to those of ordinary skill in the art. In some cases, the order of carrying out the steps of the reaction schemes may be varied to facilitate the reaction or to avoid unwanted reaction products. All such alternatives, modifications and variations are intended to fall within the spirit and scope of the present invention.

[General Scheme]

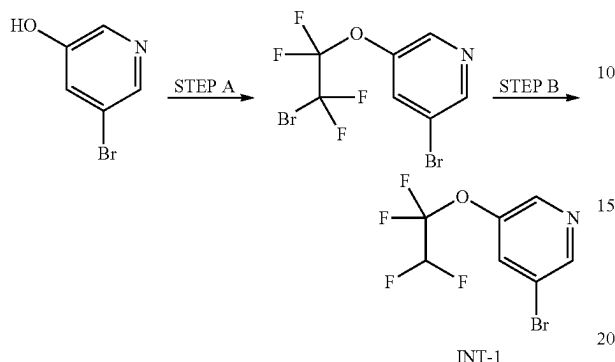


Compounds of formula I were prepared from 1-1 with R¹—X via copper-mediated N-functionalization reactions using either Chan-Lam or Goldberg-type systems to afford 1-2. Amide coupling of 1-2 with the appropriate amines provided compounds of formula (I) as described by the general scheme. The order of steps for some examples may be varied to facilitate the syntheses.

101

INTERMEDIATES

Intermediate-1:
3-bromo-5-(1,1,2,2-tetrafluoroethoxy)pyridine



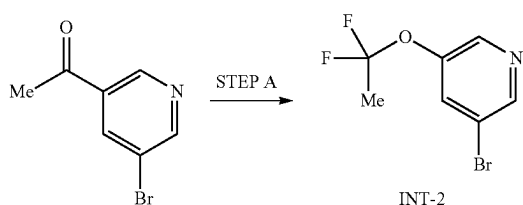
Step A: 3-bromo-5-(2-bromo-1,1,2,2-tetrafluoroethoxy)pyridine

To a stirred solution of 5-bromopyridin-3-ol (0.85 g, 4.90 mmol) and Cs_2CO_3 (3.19 g, 9.79 mmol) in DMF (9.8 ml) was added 1,2-dibromo-1,1,2,2-tetrafluoroethane (0.58 ml, 4.90 mmol) via syringe, and the reaction mixture allowed to stir at RT for 16 hours. The reaction mixture was diluted with water and extracted with Et_2O . The combined organic fractions were washed with water and NaCl (sat. aq. soln.), dried over MgSO_4 (s), filtered, and concentrated in vacuo. The crude residue was subjected to flash silica gel column chromatography using a 0-30% EtOAc/hexanes gradient to afford the title compound. LC/MS=351 [M+1].

Step B:
3-bromo-5-(1,1,2,2-tetrafluoroethoxy)pyridine

To a stirred solution of 3-bromo-5-(2-bromo-1,1,2,2-tetrafluoroethoxy)pyridine (1.42 g, 4.01 mmol) in acetic acid (4.0 ml) at 50°C . was added zinc dust (0.79 g, 12.04 mmol) portion wise. The reaction mixture was stirred at 50°C . for 1 hour. The reaction mixture was diluted with water, neutralized with NaHCO_3 (sat. aq. soln.) and extracted with DCM. The combined organic fractions were dried over MgSO_4 (s), filtered, and concentrated in vacuo. The crude residue was purified via flash silica gel column chromatography using a 0-40% EtOAc/hexanes gradient to afford the title compound. LC MS=275 [M+1].

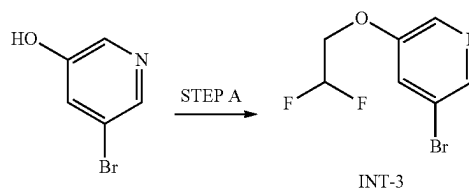
Intermediate-2:
3-bromo-5-(1,1-difluoroethoxy)pyridine



102

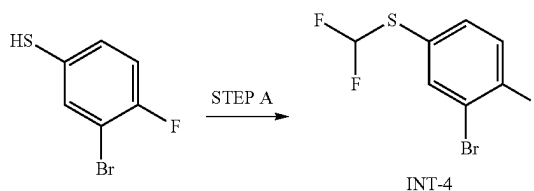
A solution of xenon difluoride (1.69 g, 10.00 mmol) and 1-(5-bromopyridin-3-yl) ethan-1-one (1.00 g, 5.00 mmol) in DCM (16.66 ml) was added HF-pyridine (14.86 g, 150 mmol) and the reaction mixture was allowed to stir at RT for 18 hours. The reaction mixture was slowly poured into a stirred mixture of NaHCO_3 (sat. aq. soln.) and ice. The aqueous phase was extracted with DCM. The combined organic fractions were washed with NaCl (sat. aq. soln.), dried over MgSO_4 (s), filtered, and concentrated in vacuo. The crude residue was subjected to flash silica gel column chromatography using a 0-40% EtOAc/hexanes gradient to afford the title compound. LC/MS=238 [M+1].

Intermediate-3:
3-bromo-5-(2,2-difluoroethoxy)pyridine



To a stirred solution of 5-bromopyridin-3-ol (0.50 g, 2.87 mmol) and Cs_2CO_3 (1.87 g, 5.75 mmol) in DMF (14.4 ml) was added 2,2-difluoroethyl trifluoromethanesulfonate (380 μl , 2.87 mmol) via syringe, and the reaction mixture allowed to stir at RT for 16 hours. The reaction mixture was diluted with water and extracted with Et_2O . The combined organic fractions were washed with water and NaCl (sat. aq. soln.), dried over MgSO_4 (s), filtered, and concentrated in vacuo. The crude residue was subjected to flash silica gel column chromatography using a 0-40% EtOAc/hexanes gradient to afford the title compound. LCMS=238 [M+1].

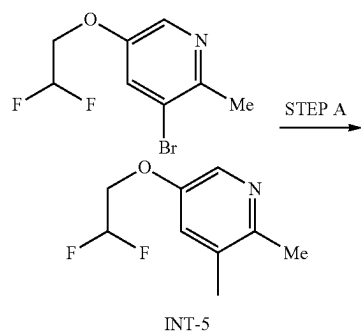
Intermediate-4: (3-bromo-4-fluorophenyl)
(difluoromethyl) sulfane



To a stirred solution of 3-bromo-4-fluorobenzenethiol (1.00 g, 4.83 mmol) and KOH (5.42 g, 97 mmol) in a mixture of acetonitrile (24.2 ml) and water (24.2 ml) was added diethyl (bromodifluoromethyl)phosphonate (1.72 ml, 9.66 mmol) dropwise via syringe at 0°C . The reaction mixture was allowed to warm to 20°C . naturally over 16 hours. The reaction mixture was diluted with Et_2O , washed with water and NaCl (sat. aq. soln.), dried over MgSO_4 (s), filtered, and concentrated in vacuo. The crude residue was subjected to flash silica gel column chromatography using 100% hexanes to afford the title compound.

103

Intermediate-5:3-iodo-5-(2,2-difluoroethoxy)-2-methylpyridine

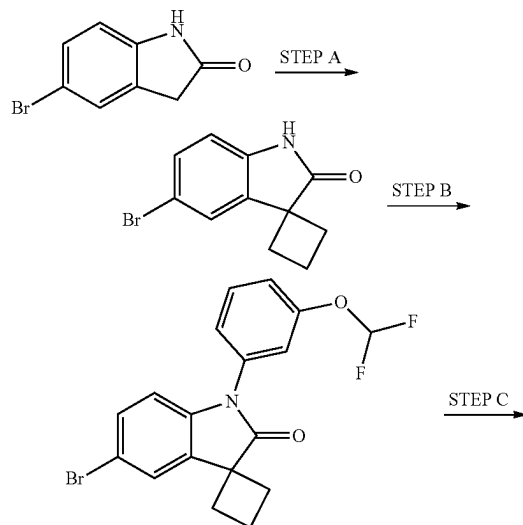


To a stirred mixture of sodium iodide (1.40 g, 9.33 mmol) and CuI (0.044 g, 0.23 mmol) was added 3-bromo-5-(2,2-difluoroethoxy)-2-methylpyridine (1.18 g, 4.67 mmol) and (1R,2R)-N1,N2-dimethylcyclohexane-1,2-diamine (0.066 g, 0.47 mmol) as a solution in dioxane (9.3 ml). The reaction mixture was heated to 110° C. for 16 hours. The reaction mixture was diluted with EtOAc, filtered through a plug of SiO₂, and concentrated in vacuo. The crude residue was subjected to flash silica gel column chromatography using a 0-30% EtOAc hexanes gradient to afford the title compound.

EXAMPLES

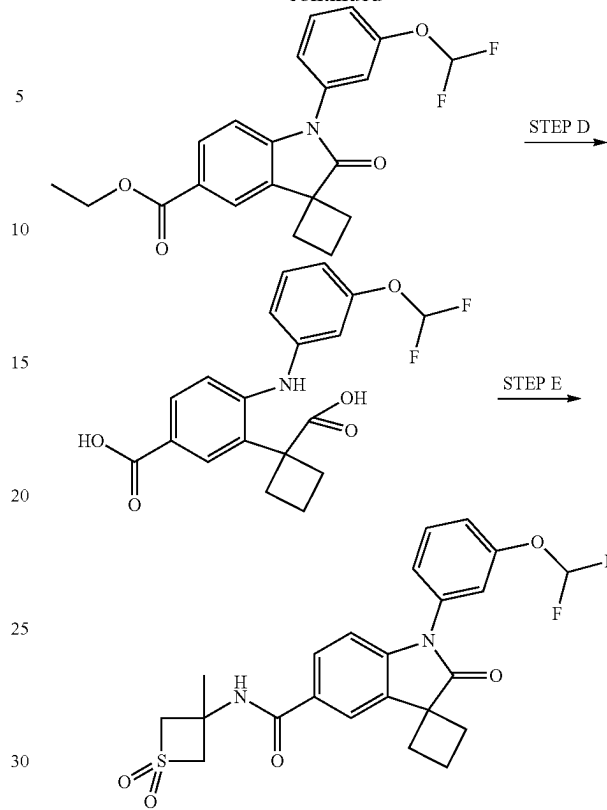
The following experimental procedures detail the preparation of specific examples of the instant disclosure. The examples are for illustrative purposes only and are not intended to limit the scope of the instant disclosure in any way.

Example 1: 1'-(3-(difluoromethoxy)phenyl)-N-(3-methyl-1,1-dioxidothietan-3-yl)-2'-oxospiro[cyclobutane-1,3'-indoline]-5'-carboxamide



104

-continued



Step A: 5'-bromospiro[cyclobutane-1,3'-indolin]-2'-one

To a solution of 5-bromoindolin-2-one (400 mg, 1.89 mmol) in THF (40 mL) was added n-butyllithium (2.5 M in hexane, 2.57 mL, 6.41 mmol) at -10° C., and the reaction mixture was stirred at -10° C. for 0.5 h under N₂. Then, the N,N,N',N'-tetramethylethylenediamine (745 mg, 6.41 mmol) and 11,3-diiodopropane (0.214 mL, 1.868 mmol) were added to the mixture at -10° C. under N₂. The reaction mixture was slowly warmed to 25° C. and stirred for 3 h under N₂. The reaction mixture was quenched with saturated aqueous NH₄Cl and extracted with DCM (x3). The combined organic layers were washed with brine, dried over Na₂SO₄, filtered and concentrated under reduced pressure. The residue was purified by flash silica gel column chromatography (0-100% EtOAc/hexanes) to afford the title compound. LC/MS=252 and 254 [M+1].

Step B: 5'-bromo-1'-(3-(difluoromethoxy)phenyl)spiro[cyclobutane-1,3'-indolin]-2'-one

To a solution of 5'-bromospiro[cyclobutane-1,3'-indolin]-2'-one (120 mg, 0.476 mmol) in acetonitrile (2.5 mL) was added K₂CO₃ (132 mg, 0.952 mmol), N,N'-dimethylethylenediamine (16.78 mg, 0.190 mmol), 1-bromo-3-(difluoromethoxy)benzene (0.401 mL, 2.86 mmol) and copper (I) iodide (18.13 mg, 0.095 mmol) in a glove box. The reaction mixture was stirred at 90° C. for 14 h, then quenched with water and extracted with DCM (x3). The combined organic layers were washed with brine, dried over Na₂SO₄, then filtered and concentrated under reduced pressure. The resi-

105

due was purified by flash silica gel column chromatography (0-100% EtOAc/hexanes) to afford the title compound. LC/MS=394 and 396 [M+1].

Step C: ethyl 1'-(3-(difluoromethoxy)phenyl)-2'-oxospiro[cyclobutane-1,3'-indoline]-5'-carboxylate

To a solution of 5'-bromo-1'-(3-(difluoromethoxy)phenyl)spiro[cyclobutane-1,3'-indoline]-2'-one (125 mg, 0.317 mmol) in EtOH (5 mL) were added potassium acetate (93 mg, 0.951 mmol) and PdCl₂ (dppf) (23.20 mg, 0.032 mmol). The mixture was stirred at 80° C. for 14 h under a CO balloon. The mixture was poured into H₂O and extracted with DCM (x3). The combined organic layers were washed with brine, dried over Na₂SO₄, filtered and concentrated under reduced pressure. The residue was purified by flash silica gel column chromatography (0-100% EtOAc/hexanes) to afford the title compound. LC/MS=388 [M+1].

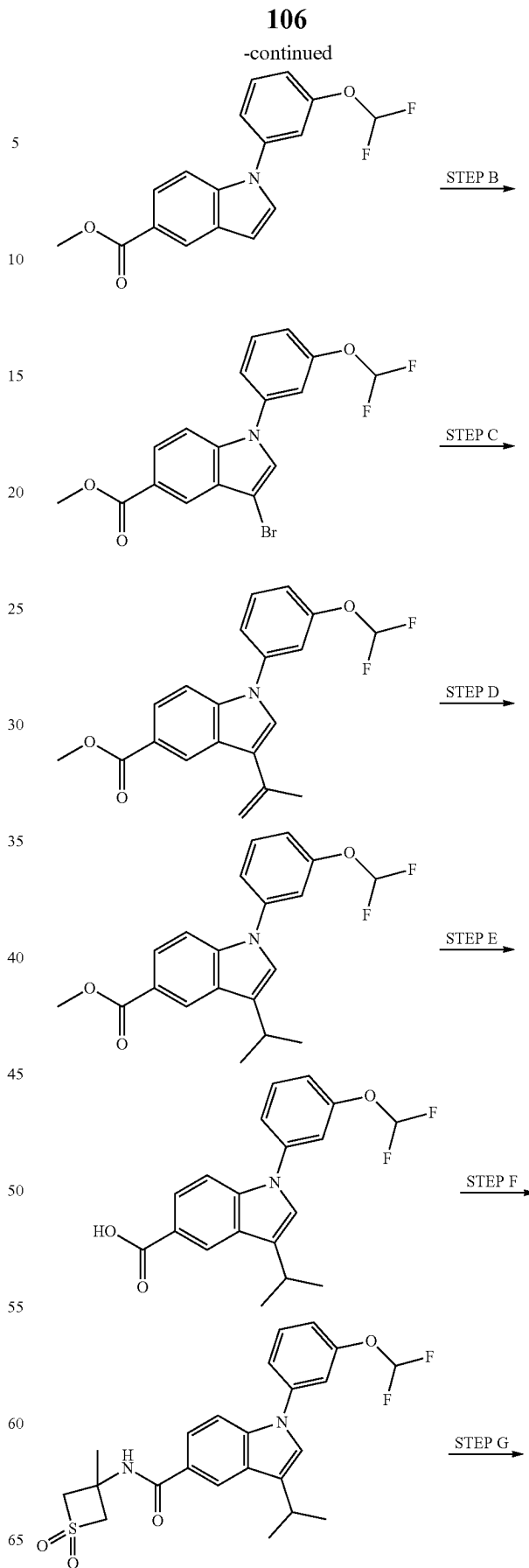
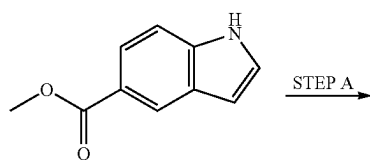
Step D: 3-(1-carboxycyclobutyl)-4-((3-(difluoromethoxy)phenyl)amino)benzoic acid

To a solution of ethyl 1'-(3-(difluoromethoxy)phenyl)-2'-oxospiro[cyclobutane-1,3'-indoline]-5'-carboxylate (80 mg, 0.207 mmol) in MeOH (4 mL) and water (0.4 mL) was added lithium hydroxide monohydrate (17.33 mg, 0.413 mmol). The mixture was stirred at 25° C. for 12 h. The mixture was concentrated under reduced pressure and was dissolved in H₂O. HCl (1N in water) was added to the mixture until pH=4. Then, the mixture was extracted with DCM (x3). The combined organic layers were washed with brine, dried over Na₂SO₄, filtered and concentrated under reduced pressure to afford the title compound. LC/MS=378 [M+1].

Step E: 1'-(3-(difluoromethoxy)phenyl)-N-(3-methyl-1,1-dioxidothietan-3-yl)-2'-oxospiro[cyclobutane-1,3'-indoline]-5'-carboxamide

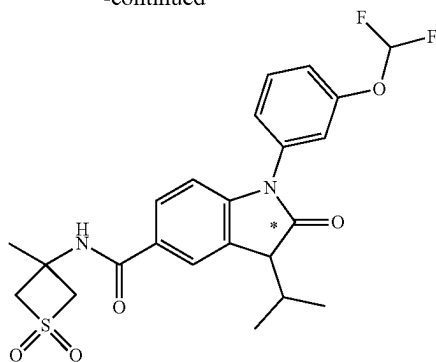
To a solution of 3-(1-carboxycyclobutyl)-4-((3-(difluoromethoxy)phenyl)amino)benzoic acid (20 mg, 0.053 mmol) in D₂O (2 mL) was added DIEA (0.046 mL, 0.265 mmol) and HATU (30.2 mg, 0.080 mmol). The mixture was stirred at 25° C. for 0.5 h. Then, the 3-amino-3-methylthietane 1,1-dioxide (9.31 mg, 0.069 mmol) was added to the mixture. The mixture was stirred at 25° C. for 1 h. The crude mixture was purified by mass triggered reverse phase HPLC (ACN/water with 0.1% TFA modifier) to afford the title compound. LC/MS=477 [M+1]. ¹H NMR (400 MHz, CDCl₃-d₁) δ 8.05 (d, J=1.6 Hz, 1H), 7.49-7.62 (m, 2H), 7.26-7.31 (m, 1H), 7.20 (dd, J=1.6, 4.0 Hz, 2H), 6.36-6.91 (m, 3H), 4.62 (d, J=14.4 Hz, 2H), 4.12-4.29 (m, 2H), 2.73-2.85 (m, 2H), 2.32-2.52 (m, 4H), 1.94 (s, 3H). Human DGAT2 IC₅₀=65.4 nM

Example 2: 1-(3-(difluoromethoxy)phenyl)-3-isopropyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxindoline-5-carboxamide



107

-continued



EX-2

Step A: methyl 1-(3-(difluoromethoxy)phenyl)-1H-indole-5-carboxylate

To a solution of 1-bromo-3-(difluoromethoxy)benzene (50 mg, 0.224 mmol) in toluene (1.5 mL) was added methyl 1H-indole-5-carboxylate (47.1 mg, 0.269 mmol), potassium phosphate tribasic (143 mg, 0.673 mmol), copper (I) iodide (4.27 mg, 0.022 mmol) and N,N'-dimethylethylenediamine (3.95 mg, 0.045 mmol) at 28° C. under N₂ atmosphere. The mixture was stirred at 100° C. for 12 h, then poured into H₂O and extracted with DCM. The combined organic layers were washed with brine, dried over Na₂SO₄ (s), filtered and concentrated under reduced pressure. The residue was purified by flash silica gel column chromatography (0-100% EtOAc/hexanes) to afford the title compound. LC/MS=318 [M+1].

Step B: methyl 3-bromo-1-(3-(difluoromethoxy)phenyl)-1H-indole-5-carboxylate

To a solution of methyl 1-(3-(difluoromethoxy)phenyl)-1H-indole-5-carboxylate (70 mg, 0.221 mmol) in DMF (1.5 mL) was added NBS (43.2 mg, 0.243 mmol) at 28° C. The mixture was stirred at 28° C. for 1 h, then poured into H₂O, and extracted with DCM (×3). The combined organic layers were washed with brine, dried over Na₂SO₄, filtered and concentrated under reduced pressure. The residue was purified by flash silica gel column chromatography (0-100% EtOAc/hexanes) to afford the title compound. LC/MS=396 [M+1].

Step C: methyl 1-(3-(difluoromethoxy)phenyl)-3-(prop-1-en-2-yl)-1H-indole-5-carboxylate

To a solution of methyl 3-bromo-1-(3-(difluoromethoxy)phenyl)-1H-indole-5-carboxylate (70 mg, 0.177 mmol) in dioxane (1.5 mL) and water (0.3 mL) was added K₂CO₃ (73.3 mg, 0.530 mmol), potassium isopropenyltrifluoroborate (39.2 mg, 0.265 mmol) and PdCl₂ (dppf) (12.93 mg, 0.018 mmol) at 28° C. under N₂ atmosphere. The mixture was stirred at 80° C. for 1 h. The mixture was poured into H₂O, the precipitates were filtered, then the precipitates were dissolved in ethyl acetate and concentrated under reduced pressure to afford the title compound LC/MS=358 [M+1].

Step D: methyl 1-(3-(difluoromethoxy)phenyl)-3-isopropyl-1H-indole-5-carboxylate

A mixture of methyl 1-(3-(difluoromethoxy)phenyl)-3-(prop-1-en-2-yl)-1H-indole-5-carboxylate (53 mg, 0.148

108

mmol) and wet Pd/C (15.78 mg, 0.148 mmol, 10%) in MeOH (5 mL) was stirred at 28° C. under 15 psi of H₂ for 1 h. The mixture was filtered through CELITE, and the solvent was evaporated under reduced pressure to give the crude product. The residue was purified by flash silica gel column chromatography (0-100% EtOAc/hexanes) to afford the title compound. LC/MS=360 [M-1].

Step E: 1-(3-(difluoromethoxy)phenyl)-3-isopropyl-1H-indole-5-carboxylic acid

To a solution of methyl 1-(3-(difluoromethoxy)phenyl)-3-isopropyl-1H-indole-5-carboxylate (35 mg, 0.097 mmol) in THF (1.5 mL) was added potassium trimethylsilanolate (37.5 mg, 0.292 mmol) at 28° C. The mixture was stirred at 60° C. for 5 h then concentrated under reduced pressure to afford the title compound. LC/MS=346 [M+1].

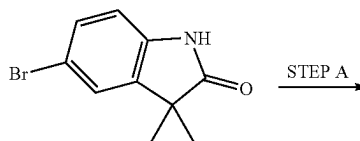
Step F: 1-(3-(difluoromethoxy)phenyl)-3-isopropyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-1H-indole-5-carboxamide

To a solution of 1-(3-(difluoromethoxy)phenyl)-3-isopropyl-1H-indole-5-carboxylic acid (20 mg, 0.058 mmol) in DMF (1 mL) were added DIEA (0.030 mL, 0.174 mmol) and HATU (26.4 mg, 0.069 mmol) at 28° C. The mixture was stirred at 28° C. for 5 minutes, then 3-amino-3-methylthietane 1,1-dioxide (9.39 mg, 0.069 mmol) was added. The mixture was stirred at 28° C. for another 10 minutes, then poured into H₂O, and extracted with DCM (×3). The combined organic layers were washed with brine, dried over Na₂SO₄, filtered and concentrated under reduced pressure. The residue was purified by flash silica gel column chromatography (0-100% EtOAc/hexanes) to afford the title compound. LC/MS=463 [M+1].

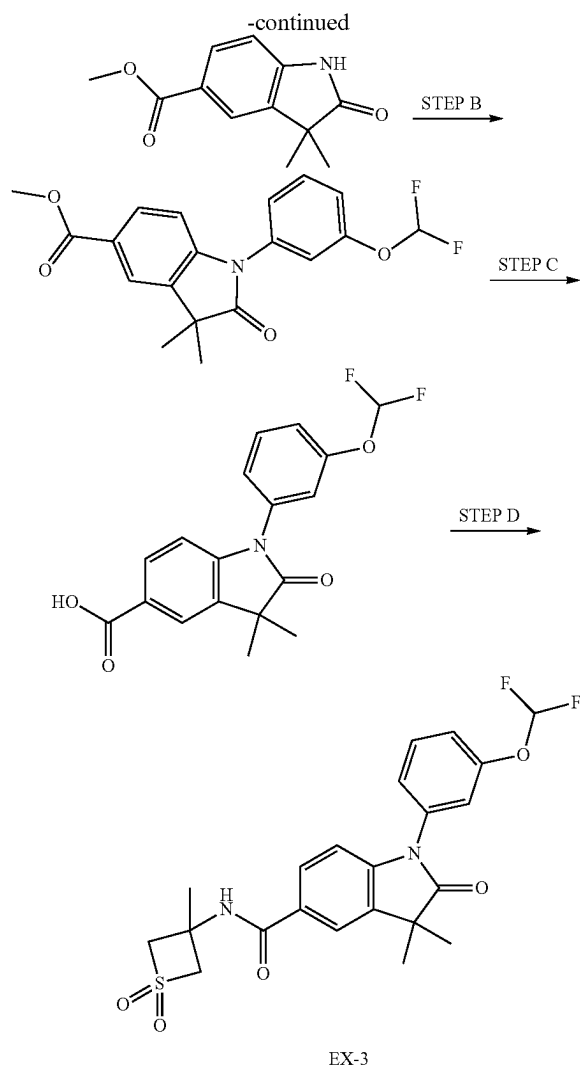
Step G: 1-(3-(difluoromethoxy)phenyl)-3-isopropyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxindoline-5-carboxamide

To a solution of 1-(3-(difluoromethoxy)phenyl)-3-isopropyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-1H-indole-5-carboxamide (15 mg, 0.032 mmol) in CH₂Cl₂ (1 mL) was added NCS (4.76 mg, 0.036 mmol) at 26° C. The mixture was stirred at 26° C. for 1 h. The mixture was concentrated, and the residue was dissolved in HOAc (1 mL) and H₃PO₄ (0.25 mL). The mixture was stirred at 100° C. for 1.5 h, then concentrated to give the crude product. The crude mixture was purified by mass triggered reverse phase HPLC (ACN/water with 0.1% TFA modifier) to afford the title compound. LC/MS=479 [M+1]. ¹H NMR (500 MHz, CHLOROFORM-d₁) δ 7.89 (s, 1H), 7.78 (dd, J₁=8.3 Hz, J₂=1.0 Hz, 1H), 7.60 (t, J=8.2 Hz, 1H), 7.26 (dd, J₁=8.2 Hz, J₂=2.1 Hz, 2H), 7.18 (s, 1H), 6.70-7.12 (m, 2H), 4.56 (br d, J=14.4 Hz, 2H), 4.20 (br d, J=14.9 Hz, 2H), 3.70 (d, J=3.2 Hz, 1H), 2.46-2.65 (m, 1H), 1.81 (s, 3H), 1.11 (d, J=6.8 Hz, 3H), 1.03 (d, J=6.8 Hz, 3H). Human DGAT2 IC₅₀=1093 nM

Example 3: 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxindoline-5-carboxamide



109



Step A: methyl
3,3-dimethyl-2-oxoindoline-5-carboxylate

To a solution of 5-bromo-3,3-dimethylindolin-2-one (700 mg, 2.92 mmol) in MeOH (30 mL) were added TEA (1.219 mL, 8.75 mmol) and PdCl₂ (dppf) (320 mg, 0.437 mmol) under argon atmosphere at 25° C. The mixture was stirred at 80° C. under CO atmosphere (50 psi) for 15 h. LCMS showed that the desired product was formed. The mixture was poured into H₂O, and extracted with DCM. The combined organic layers were washed with brine, dried over Na₂SO₄ (s), filtered and concentrated under reduced pressure. The residue was purified by flash silica gel column chromatography (0-100% EtOAc/hexanes) to afford the title compound. LC/MS=220 [M+1].

Step B: methyl 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-2-oxoindoline-5-carboxylate

To a solution of methyl 3,3-dimethyl-2-oxoindoline-5-carboxylate (40 mg, 0.182 mmol) in dioxane (1 mL) were added (1R,2R)-N1,N2-dimethylcyclohexane-1,2-diamine (5.19 mg, 0.036 mmol), potassium carbonate (76 mg, 0.547 mmol), copper (I) iodide (3.47 mg, 0.018 mmol) and

110

1-bromo-3-(difluoromethoxy)benzene (40.7 mg, 0.182 mmol) under N₂ atmosphere. The mixture was stirred at 100° C. for 15 h, then poured into H₂O and extracted with DCM (×3). The combined organic layers were washed with brine, dried over Na₂SO₄ (s), filtered and concentrated under reduced pressure. The residue was purified by flash silica gel column chromatography (0-100% EtOAc/hexanes) to afford the title compound. LC/MS=362 [M+1].

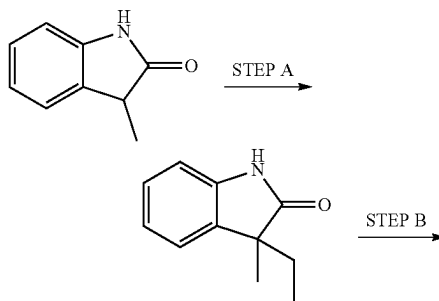
Step C: 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-2-oxoindoline-5-carboxylic acid

To a solution of methyl 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-2-oxoindoline-5-carboxylate (90 mg, 0.249 mmol) in MeOH (2 mL)/water (0.5 mL) was added LiOH·H₂O (23.86 mg, 0.996 mmol). The mixture was stirred at 30° C. for 2 h, then concentrated under reduced pressure. HCl (1N in water) was added until pH=4. Then, the mixture was extracted with DCM. The combined organic layers were washed with brine, dried over Na₂SO₄ (s), filtered and concentrated under reduced pressure to afford the title compound. LC/MS=348 [M+1].

Step D: 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide

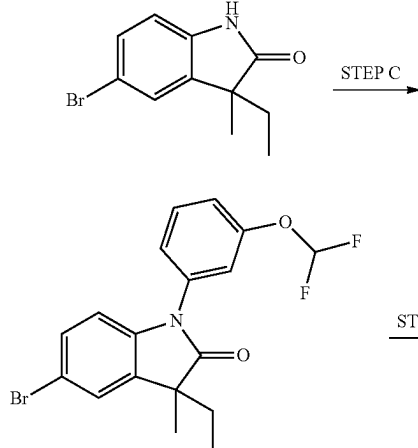
To a solution of 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-2-oxoindoline-5-carboxylic acid (40 mg, 0.115 mmol) in DCM (1 mL) were added HATU (52.5 mg, 0.138 mmol) and DIEA (0.060 mL, 0.346 mmol). The mixture was stirred at 50° C. for 10 min, then 3-amino-3-methylthietane 1,1-dioxide (18.68 mg, 0.138 mmol) was added at 50° C. under N₂ atmosphere. The mixture was stirred at 50° C. for 3 h, then poured into H₂O and extracted with DCM. The combined organic layers were washed with brine, dried over Na₂SO₄ (s), filtered and concentrated under reduced pressure. The crude material was purified by flash silica gel chromatography using a 0-100% EtOAc/hexanes gradient to afford the title compound. LC/MS=465 [M+1]. ¹H NMR (500 MHz, METHANOL-d₄) δ 7.94 (d, J=1.53 Hz, 1H), 7.82 (dd, J1=8.32 Hz, J2=1.75 Hz, 1H), 7.65 (t, J=8.01 Hz, 1H), 7.26-7.40 (m, 3H), 6.76-7.13 (m, 2H), 4.61 (br d, J=14.50 Hz, 2H), 4.21-4.29 (m, 2H), 1.86 (s, 3H), 1.54 (s, 6H). Human DGAT2 IC₅₀=29.3 nM

Example 4a and 4b: 1-(3-(difluoromethoxy)phenyl)-3-ethyl-3-methyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide and 1-(3-(difluoromethoxy)phenyl)-3-ethyl-3-methyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide



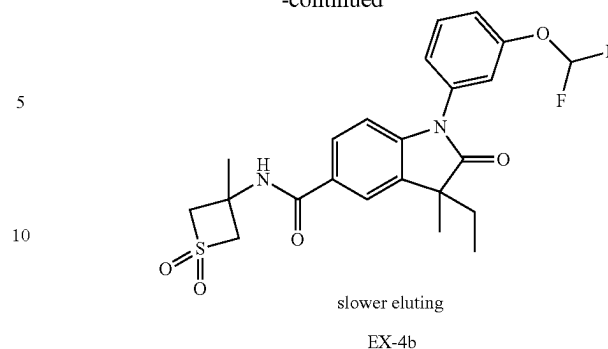
111

-continued



112

-continued



Step A: 3-ethyl-3-methylindolin-2-one

To a mixture of 3-methylindolin-2-one (300 mg, 2.038 mmol) and N,N,N',N'-tetramethylethylenediamine (711 mg, 6.12 mmol) in THF (9 mL) was added n-butyllithium (2.5 M in hexane, 1.794 mL, 4.48 mmol) at -78°C ., and the reaction mixture was stirred at -78°C . for 0.5 h under N_2 . Then, a solution of the iodoethane (477 mg, 3.06 mmol) in THF (3 mL) was added to the mixture at -78°C . under N_2 . The reaction was slowly warmed up to -20°C . and stirred for 1 h. The mixture was poured into H_2O and extracted with DCM ($\times 3$). The combined organic layers were washed with brine, dried over Na_2SO_4 , filtered and concentrated under reduced pressure. The residue was purified by column chromatography on silica (0-100% EtOAc/hexanes) to afford the title compound. LC/MS=176 [M+1].

Step B: 5-bromo-3-ethyl-3-methylindolin-2-one

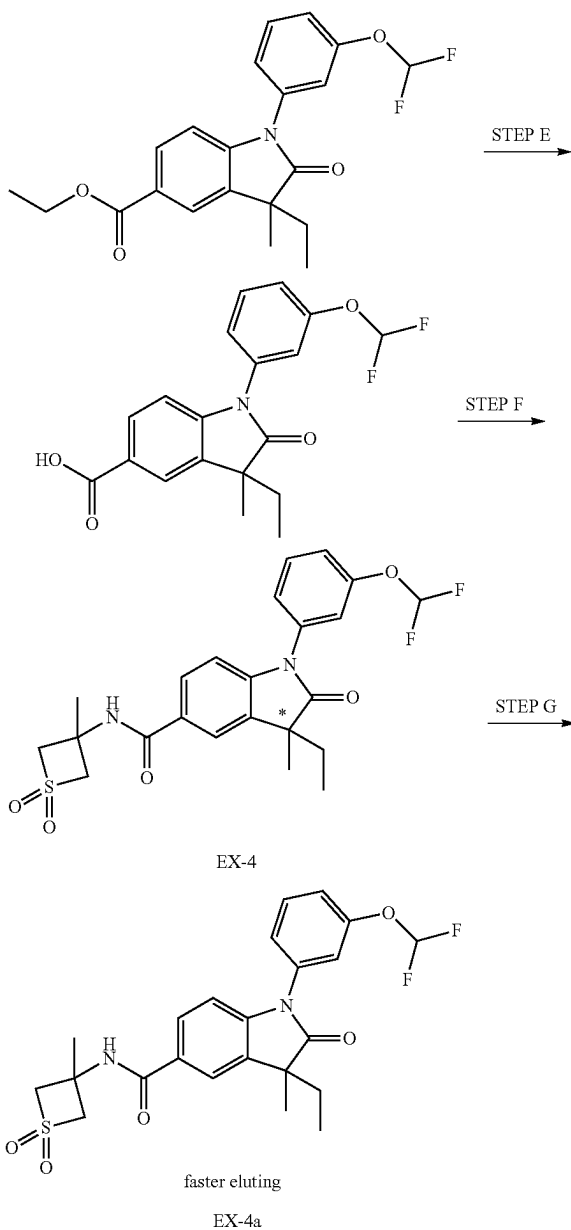
To a solution of 3-ethyl-3-methylindolin-2-one (100 mg, 0.571 mmol) in HOAc (10 mL) was added Br_2 (0.035 mL, 0.685 mmol) at 0°C . The reaction mixture was stirred at 25°C . for 12 h under a N_2 balloon. The reaction mixture was quenched with saturated aqueous Na_2SO_3 solution and diluted with water and extracted with DCM ($\times 3$). The combined organic layers were washed with brine, dried over Na_2SO_4 , then filtered and concentrated under reduced pressure. The residue was purified by column chromatography on silica (0-100% EtOAc/hexanes) to afford the title compound. LC/MS=254 and 256 [M+1].

Step C: 5-bromo-1-(3-(difluoromethoxy)phenyl)-3-ethyl-3-methylindolin-2-one

To a solution of 5-bromo-3-ethyl-3-methylindolin-2-one (50 mg, 0.197 mmol) in acetonitrile (1 mL) were added K_2CO_3 (54.4 mg, 0.394 mmol), N,N'-dimethylethylenediamine (6.94 mg, 0.079 mmol), 1-bromo-3-(difluoromethoxy)benzene (0.193 mL, 1.377 mmol) and copper (I) iodide (7.49 mg, 0.039 mmol) in glove box. The reaction mixture was stirred at 90°C . for 13 h, then poured into H_2O and extracted with DCM ($\times 3$). The combined organic layers were washed with brine, dried over Na_2SO_4 , filtered and concentrated under reduced pressure. The residue was purified by column chromatography on silica (0-100% EtOAc/hexanes) to afford the title compound. LC/MS=396 and 398 [M+1].

Step D: ethyl 1-(3-(difluoromethoxy)phenyl)-3-ethyl-3-methyl-2-oxoindoline-5-carboxylate

To a solution of 5-bromo-1-(3-(difluoromethoxy)phenyl)-3-ethyl-3-methylindolin-2-one (50 mg, 0.126 mmol) in



113

EtOH (5 mL) were added potassium acetate (37.2 mg, 0.379 mmol) and PdCl₂ (dppf) (9.23 mg, 0.013 mmol). The mixture was stirred at 80° C. for 13 h under a CO balloon. LCMS showed the starting material was consumed completely. The mixture was poured into H₂O, and extracted with DCM (×3). The combined organic layers were washed with brine, dried over Na₂SO₄, filtered and concentrated under reduced pressure. The residue was purified by column chromatography on silica (0-100% EtOAc/hexanes) to afford the title compound. LC/MS=390 [M+1].

Step E: 1-(3-(difluoromethoxy)phenyl)-3-ethyl-3-methyl-2-oxoindoline-5-carboxylic acid

To a solution of ethyl 1-(3-(difluoromethoxy)phenyl)-3-ethyl-3-methyl-2-oxoindoline-5-carboxylate (40 mg, 0.103 mmol) in MeOH (2 mL) and water (0.2 mL) was added lithium hydroxide monohydrate (8.62 mg, 0.205 mmol). The mixture was stirred at 25° C. for 24 h, concentrated under reduced pressure and dissolved in H₂O. HCl (1N in water) was added to the mixture until pH=4. Then the mixture was extracted with DCM (×3). The combined organic layers were washed with brine, dried over Na₂SO₄, filtered and concentrated under reduced pressure to afford the title compound. LC/MS=362 [M+1].

Step F: 1-(3-(difluoromethoxy)phenyl)-3-ethyl-3-methyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide

To a solution of 1-(3-(difluoromethoxy)phenyl)-3-ethyl-3-methyl-2-oxoindoline-5-carboxylic acid (37 mg, 0.102 mmol) in DMF (2 mL) were added DIEA (0.089 mL, 0.512 mmol) and HATU (58.4 mg, 0.154 mmol). The mixture was stirred at 25° C. for 0.5 h. Then, 3-amino-3-methylthietane

114

1,1-dioxide (18.00 mg, 0.133 mmol) was added, and the mixture was stirred at 25° C. for 1 h. The crude mixture was purified by mass triggered reverse phase HPLC (ACN/water with 0.1% TFA modifier) to afford the title compound. LC/MS=479 [M+1].

Step G: 1-(3-(difluoromethoxy)phenyl)-3-ethyl-3-methyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide and 1-(3-(difluoromethoxy)phenyl)-3-ethyl-3-methyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide

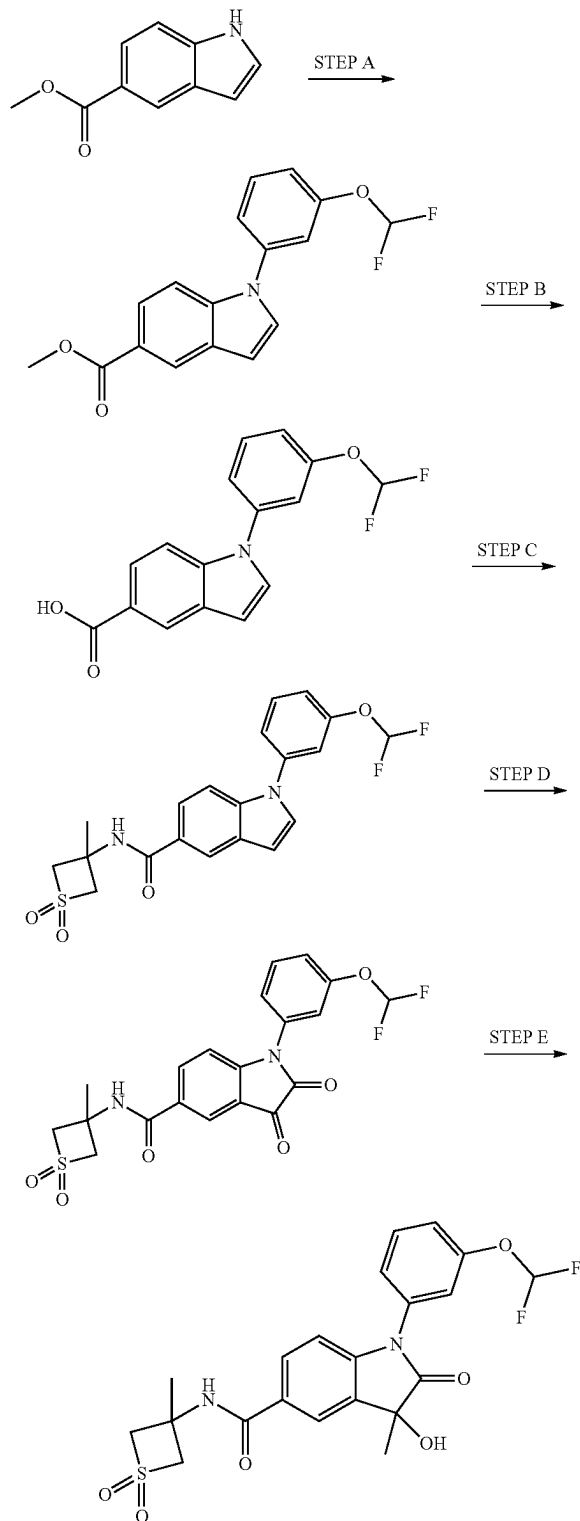
The mixture of the two stereoisomers 1-(3-(difluoromethoxy)phenyl)-3-ethyl-3-methyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide (50 mg, 0.104 mmol) was purified by chiral SFC (IC-H column, 40%/60% ethanol/CO₂) to afford isomer EX-4a (faster eluting). LC/MS=479 [M+1]. ¹H NMR (400 MHz, CHLOROFORM-d₁) δ 7.71 (d, J=1.6 Hz, 1H), 7.61 (dd, J=1.6, 8.0 Hz, 1H), 7.50-7.57 (m, 1H), 7.25-7.28 (m, 1H), 7.16-7.22 (m, 2H), 6.89 (d, J=8.4 Hz, 1H), 6.37-6.78 (m, 2H), 4.60 (d, J=14.8 Hz, 2H), 4.20 (d, J=14.8 Hz, 2H), 2.00-2.16 (m, 1H), 1.83-1.94 (m, 4H), 1.49 (s, 3H), 0.69 (t, J=7.2 Hz, 3H). Human DGAT2 IC₅₀=1103 nM. Isomer EX-4b (slower eluting). LC/MS=479 [M+1]. ¹H NMR (400 MHz, CHLOROFORM-d₁) δ 7.71 (d, J=1.6 Hz, 1H), 7.61 (dd, J=1.6, 8.4 Hz, 1H), 7.50-7.57 (m, 1H), 7.25-7.28 (m, 1H), 7.17-7.23 (m, 2H), 6.89 (d, J=8.4 Hz, 1H), 6.38-6.77 (m, 2H), 4.60 (br d, J=14.8 Hz, 2H), 4.20 (d, J=14.8 Hz, 2H), 2.01-2.13 (m, 1H), 1.84-1.96 (m, 4H), 1.50 (s, 3H), 0.69 (t, J=7.2 Hz, 3H). Human DGAT2 IC₅₀=104 nM.

By using procedures similar to those described in Example 4 with appropriate reagents, the following compounds were synthesized. These compounds were characterized by LC/MS.

Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
5a		1-(3-(difluoromethoxy)phenyl)-3-ethyl-3-methyl-N-((S)-3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide and 1-(3-(difluoromethoxy)phenyl)-3-ethyl-3-methyl-N-((S)-3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide (faster eluting on IC-H column, 40%/60% ethanol/CO ₂)	493	2219
5b		1-(3-(difluoromethoxy)phenyl)-3-ethyl-3-methyl-N-((S)-3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide and 1-(3-(difluoromethoxy)phenyl)-3-ethyl-3-methyl-N-((S)-3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide (slower eluting on IC-H column, 40%/60% ethanol/CO ₂)	493	274

115

Example 6a and 6b: 1-(3-(difluoromethoxy)phenyl)-3-hydroxy-3-methyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide



EX-6a and 6b

116

Step A: methyl 1-(3-(difluoromethoxy)phenyl)-1H-indole-5-carboxylate

To a solution of methyl 1H-indole-5-carboxylate (500 mg, 2.85 mmol) in DMSO (5 mL) was added 1-bromo-3-(difluoromethoxy)benzene (1910 mg, 8.56 mmol), copper (I) iodide (54.4 mg, 0.285 mmol), K_2CO_3 (789 mg, 5.71 mmol) and (1R,2R)-N,N'-dimethyl-1,2-cyclohexanediamine (81 mg, 0.571 mmol) in glove box. The reaction mixture was stirred at 110° C. for 12 h. TLC showed that the desired product was formed. The mixture was poured into H_2O and extracted with DCM ($\times 3$). The combined organic layers were washed with brine, dried over Na_2SO_4 , then filtered and concentrated under reduced pressure. The residue was purified by flash column chromatography on silica (0-100% EtOAc/hexanes) to afford the title compound. LC/MS=318 [M+1].

Step B: 1-(3-(difluoromethoxy)phenyl)-1H-indole-5-carboxylic acid

Lithium hydroxide monohydrate (800 mg, 19.06 mmol) was added to a mixture of methyl 1-(3-(difluoromethoxy)phenyl)-1H-indole-5-carboxylate (1 g, 3.15 mmol) in MeOH (10 mL), THF (3.00 mL) and water (1 mL). The mixture was stirred at 25° C. for 13 h. LCMS showed that the desired product was formed. HCl (1N in water) was added to the mixture until pH=4, and then extracted with DCM ($\times 3$). The combined organic layers were washed with brine, dried over Na_2SO_4 , then filtered and concentrated under reduced pressure to afford the title compound.

Step C: 1-(3-(difluoromethoxy)phenyl)-N-(3-methyl-1,1-dioxidothietan-3-yl)-1H-indole-5-carboxamide

3-amino-3-methylthietane 1,1-dioxide dihydrochloride (0.91 g, 4.37 mmol) was added to a mixture of 1-(3-(difluoromethoxy)phenyl)-1H-indole-5-carboxylic acid (0.95 g, 3.13 mmol), HATU (1.78 g, 4.68 mmol) and DIEA (2.2 mL, 12.60 mmol) in DCM (20 mL). The mixture was stirred at 25° C. for 0.5 h. LCMS showed that the desired product was formed. The mixture was poured into H_2O , and then extracted with DCM ($\times 3$). The combined organic layers were washed with brine, dried over Na_2SO_4 , then filtered and concentrated under reduced pressure. The residue was purified by flash column chromatography on silica (0-100% EtOAc/hexanes) to afford the title compound. LC MS=421 [M+1].

Step D: 1-(3-(difluoromethoxy)phenyl)-N-(3-methyl-1,1-dioxidothietan-3-yl)-2,3-dioxoindoline-5-carboxamide

To a solution of 1-(3-(difluoromethoxy)phenyl)-N-(3-methyl-1,1-dioxidothietan-3-yl)-1H-indole-5-carboxamide (90₀ mg, 2.141 mmol) in acetone (8 mL)/acetic Acid (6 mL)/water (3 mL) was added chromium(VI) oxide (856 mg, 8.56 mmol) at 25° C. The mixture was stirred for 3 h at room temperature. LCMS showed that the desired product was formed. The reaction was poured into sat. NH_4Cl , and then extracted with DCM ($\times 3$). The combined organic layers were washed with brine, dried over Na_2SO_4 , then filtered and concentrated under reduced pressure. The residue was purified by column chromatography on silica (0-100% EtOAc/hexanes) to afford the title compound. LC. MS=451 [M+1].

117

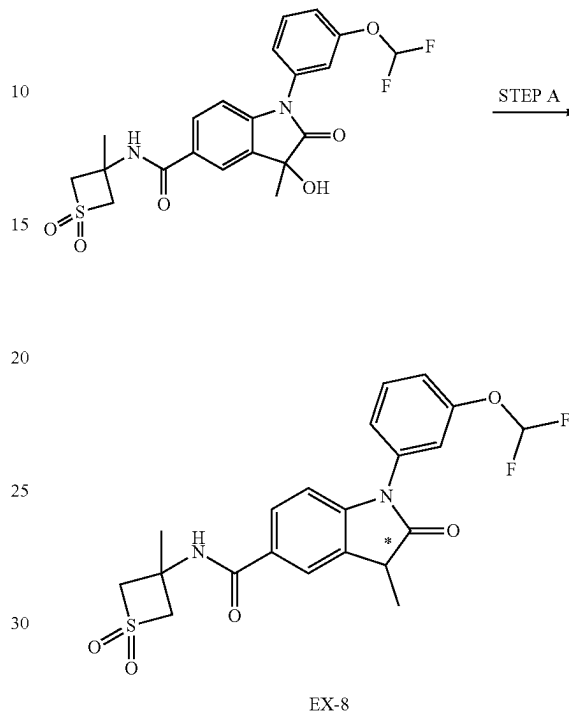
Step E: 1-(3-(difluoromethoxy)phenyl)-3-hydroxy-3-methyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide

A mixture of 1-(3-(difluoromethoxy)phenyl)-N-(3-methyl-1,1-dioxidothietan-3-yl)-2,3-dioxoindoline-5-carboxamide (0.8 g, 1.776 mmol) and lithium bromide (0.31 g, 3.57 mmol) in THF (15 mL) was cooled to -70°C . and treated with methylmagnesium bromide (1.8 mL, 3.60 mmol) and stirred at -70°C . for 2 h under N_2 . LCMS showed that the desired product was formed. The reaction was poured into sat. NH_4Cl , and then extracted with DCM ($\times 3$). The combined organic layers were washed with brine, dried over Na_2SO_4 , then filtered and concentrated under reduced pressure. The residue was purified by mass triggered reverse phase HPLC (ACN/water with 0.1% TFA modifier) to afford the title compound as the TFA salt. LC/MS=467 [M+1]. The mixture of the two stereoisomers was purified by chiral SFC (Column DAICEL CHIRALCEL OD (250 mm $\times$ 30 mm $\times$ 10 μm), Condition 0.1% $\text{NH}_3\text{H}_2\text{O}$ EtOH). EX-6a (faster eluting): LC/MS=467 [M+1]. ^1H NMR (400 MHz, METHANOL- d_4) δ 8.00 (d, J=1.47 Hz, 1H), 7.84 (dd, J=1.47, 8.31 Hz, 1H), 7.58-7.67 (m, 1H), 7.25-7.36 (m, 3H), 6.72-7.13 (m, 2H), 4.57-4.98 (m, 2H), 4.22-4.65 (m, 2H), 1.82 (s, 3H), 1.66 (s, 3H). Human DGAT2 IC_{50} =3933 nM. EX-6b (slower eluting): LC/MS=467 [M+1]. ^1H NMR (400 MHz, METHANOL- d_4) δ 8.00 (d, J=1.96 Hz, 1H), 7.84 (dd, J=1.71, 8.56 Hz, 1H), 7.58-7.66 (m, 1H), 7.24-7.36 (m, 3H), 6.72-7.13 (m, 2H), 4.57-4.86 (m, 2H), 4.22-4.53 (m, 2H), 1.82 (s, 3H), 1.66 (s, 3H). Human DGAT2 IC_{50} =290 nM.

By using procedures similar to those described in Example 6 with appropriate reagents, the following compounds were synthesized. These compounds were characterized by LC/MS.

118

Example 8: 1-(3-(difluoromethoxy)phenyl)-3-hydroxy-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide



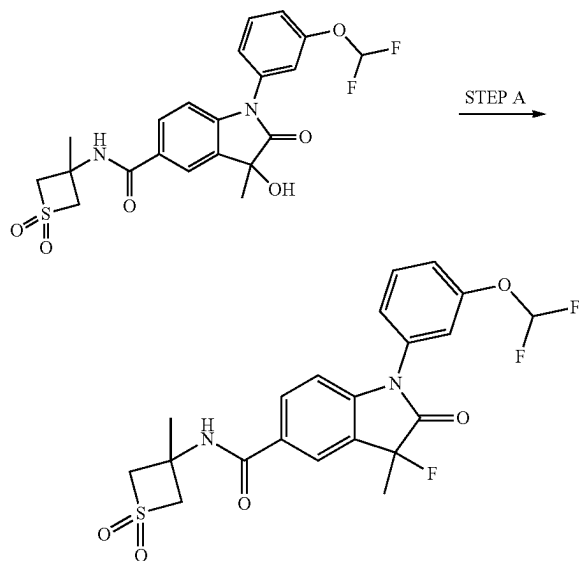
Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC_{50} (nM)
7a		1-(3-(difluoromethoxy)phenyl)-3-hydroxy-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxo-3-(trifluoromethyl)indoline-5-carboxamide (DAICEL CHIRALCEL OD Column, 0.1% $\text{NH}_3\text{H}_2\text{O}$ MeOH, slower eluting)	521	484
7b		1-(3-(difluoromethoxy)phenyl)-3-hydroxy-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxo-3-(trifluoromethyl)indoline-5-carboxamide (DAICEL CHIRALCEL OD, Condition 0.1% $\text{NH}_3\text{H}_2\text{O}$ MeOH, faster eluting)	521	2950

119

Step A: 1-(3-(difluoromethoxy)phenyl)-3-methyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide

Triethylsilane (1.3 g, 11.18 mmol) was added to a solution of 1-(3-(difluoromethoxy)phenyl)-3-hydroxy-3-methyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide (130 mg, 0.279 mmol) in TFA (3 mL) under N₂ atmosphere. The resulting mixture was stirred at 80° C. for 13 h. LCMS showed that the desired product was formed. The reaction was poured into sat. NH₄Cl, and then extracted with DCM (x3). The combined organic layers were washed with brine, dried over Na₂SO₄, then filtered and concentrated under reduced pressure. The residue was purified by mass triggered reverse phase HPLC (ACN/water with 0.1% TFA modifier) to afford the title compound as the TFA salt. LC/MS=451 [M+1]. ¹H NMR (500 MHz, METHANOL-d₄) δ 7.92 (s, 1H), 7.81 (br d, J=8.24 Hz, 1H), 7.64 (t, J=8.09 Hz, 1H), 7.35 (br d, J=7.93 Hz, 1H), 7.26-7.32 (m, 2H), 6.78-7.11 (m, 2H), 4.58-4.60 (m, 2H), 4.23-4.27 (m, 2H), 4.12 (q, J=7.12 Hz, 1H), 3.80 (q, J=7.58 Hz, 1H), 1.85 (s, 3H), 1.61 (d, J=7.63 Hz, 3H). Human DGAT2 IC₅₀=2730 nM

Example 9a and 9b: 1-(3-(difluoromethoxy)phenyl)-3-fluoro-3-methyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide



EX-9a and EX-9b

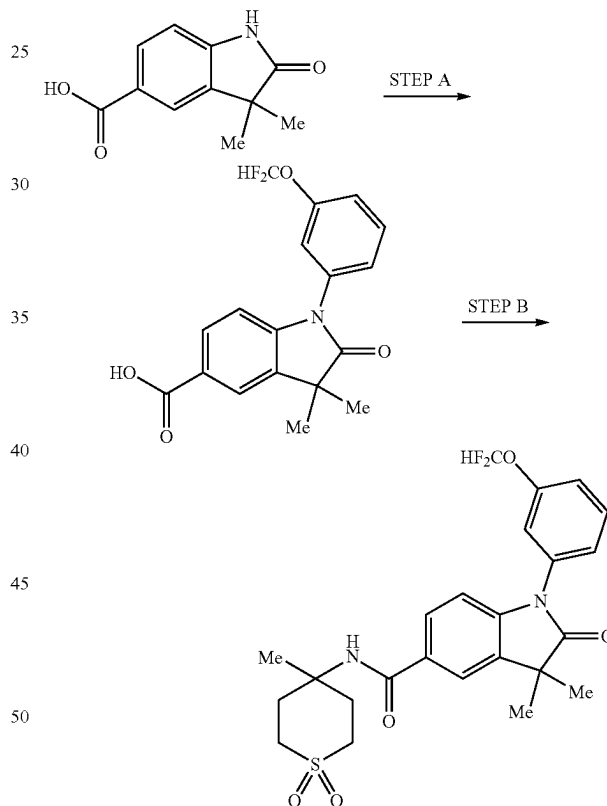
Step A: 1-(3-(difluoromethoxy)phenyl)-3-fluoro-3-methyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide

To a solution of 1-(3-(difluoromethoxy)phenyl)-3-hydroxy-3-methyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide (100 mg, 0.214 mmol) in DCM (7 mL) was added DAST (0.115 mL, 0.870 mmol) at 0° C., the reaction mixture was stirred at 0° C. for 1 h under N₂. LCMS showed that the desired product was formed. The reaction was poured into sat. NaHCO₃, and then extracted with DCM (x3). The combined organic layers were washed with brine, dried over Na₂SO₄, then filtered and concen-

120

trated under reduced pressure. The residue was purified by column chromatography on silica (0-100% EtOAc/hexanes) to afford the title compound. LC/MS=469 [M+1]. The racemic mixtures were purified by chiral SFC (IC-H column, 40%/60% ethanol/CO₂). EX-9a (faster eluting): LC/MS=469 [M+1]. ¹H NMR (400 MHz, METHANOL-d₄) δ 8.11 (t, J=1.76 Hz, 1H), 7.93 (td, J=1.71, 8.31 Hz, 1H), 7.58-7.72 (m, 1H), 7.33-7.38 (m, 1H), 7.27-7.32 (m, 2H), 6.73-7.14 (m, 2H), 4.58 (br d, J=14.48 Hz, 2H), 4.17-4.27 (m, 2H), 1.85-1.93 (m, 3H), 1.83 (s, 3H). Hu DGAT2 IC₅₀=339 nM. EX-9b (slower eluting): LC/MS=469 [M+1]. ¹H NMR (400 MHz, METHANOL-d₄) δ 8.11 (t, J=1.76 Hz, 1H), 7.93 (td, J=1.76, 8.22 Hz, 1H), 7.60-7.68 (m, 1H), 7.35 (dd, J=0.98, 8.80 Hz, 1H), 7.27-7.32 (m, 2H), 6.73-7.14 (m, 2H), 4.58 (br d, J=14.48 Hz, 2H), 4.18-4.26 (m, 2H), 1.85-1.93 (m, 3H), 1.83 (s, 3H). Hu DGAT2 IC₅₀=169 nM.

Example 10: 1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxoindoline-5-carboxamide



EX-10

Step A: 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-2-oxoindoline-5-carboxylic acid

To a stirred solution of 3,3-dimethyl-2-oxoindoline-5-carboxylic acid (5.0 g, 24.4 mmol), 1-(difluoromethoxy)-3-iodobenzene (9.9 g, 36.5 mmol), (1R,2R)-N1,N2-dimethylcyclohexane-1,2-diamine (0.69 g, 4.9 mmol) and potassium carbonate (10.1 g, 73.1 mmol) in acetonitrile (48.7 mL) which was sparged with nitrogen for 10 minutes was added CuI (0.46 g, 2.44 mmol). The heavy-wall pressure vessel

121

was sealed and heated to 95° C. for 18 hours. The reaction was diluted with water, acidified with HCl (1M aq. soln.) and stirred for 45 minutes. The resultant brown solution was extracted with DCM. The combined organic fractions were dried over Na₂SO₄ (s) and dry absorbed on SiO₂. The crude material was purified by flash silica gel chromatography using a 0-100% EtOAc/hexanes gradient to afford the title compound LC/MS=348 [M+1].

Step B: 1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-indoline-5-carboxamide

To a stirred solution of 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-2-oxoindoline-5-carboxylic acid (50 mg, 0.14 mmol), 4-amino-4-methyltetrahydro-2H-thiopyran 1,1-dioxide (23.5 mg, 0.14 mmol), and Hunig's Base (75 µl, 0.43

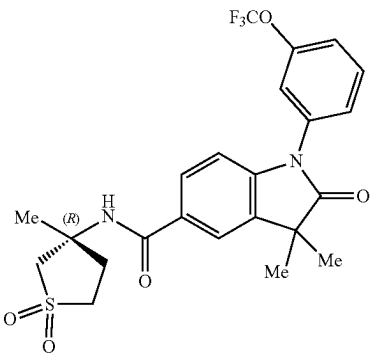
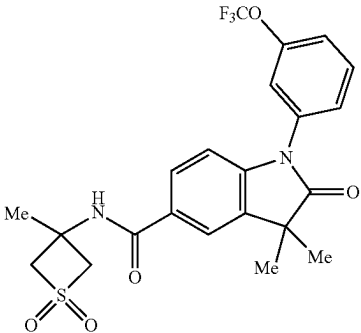
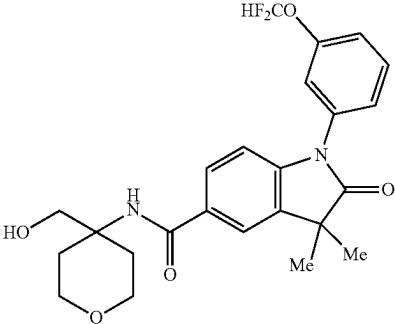
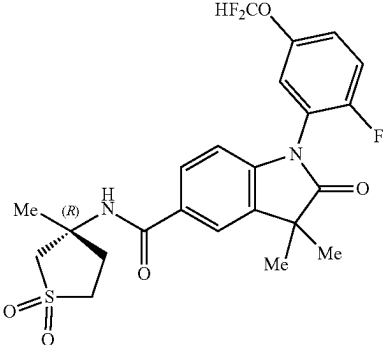
122

mmol) in DMF (1440 µl) was added HATU (57.5 mg, 0.15 mmol). The reaction mixture was allowed to stir at RT for 16 hours, diluted with EtOAc, filtered through a plug of SiO₂, and concentrated in vacuo. The crude residue was purified by mass triggered reverse phase HPLC (ACN/water with 0.1% formic acid modifier) to afford the title compound. LC/MS=493 [M+1]. ¹H NMR (500 MHz, Chloroform-d) δ 7.75 (d, J=1.7 Hz, 1H), 7.60-7.54 (m, 2H), 7.34-7.31 (m, 1H), 7.24 (d, J=8.0 Hz, 2H), 6.94 (d, J=8.2 Hz, 1H), 6.60 (t, J=73.2 Hz, 1H), 5.70 (s, 1H), 3.23-3.15 (m, 2H), 3.03 (d, J=14.2 Hz, 2H), 2.82 (d, J=15.0 Hz, 2H), 2.42-2.33 (m, 2H), 1.62 (s, 3H), 1.55 (s, 6H). Human DGAT2 IC₅₀=39 nM

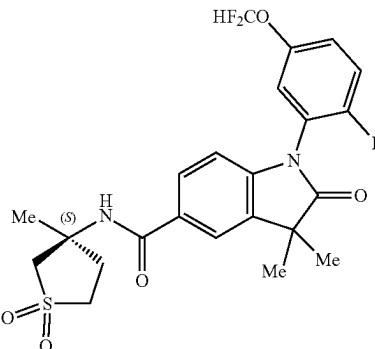
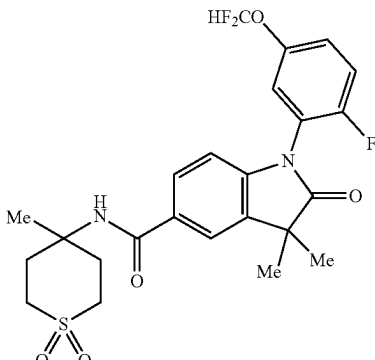
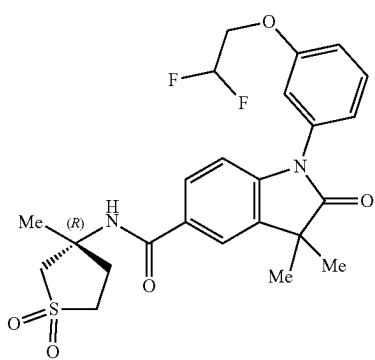
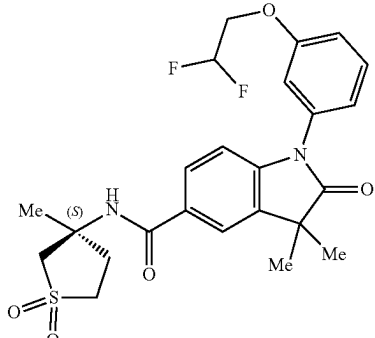
By using procedures similar to those described in Example 10 with appropriate reagents, the following compounds were synthesized. These compounds were characterized by LC/MS.

Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
11		1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-[(3R)-(3-methyl-1,1-dioxo-thiolan-3-yl)]-2-oxo-indoline-5-carboxamide	479	42
12		1-[3-(difluoromethoxy)phenyl]-N-[(1S,2R)-2-hydroxycyclopentyl]-3,3-dimethyl-2-oxo-indoline-5-carboxamide	431	507.6
13		3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-1-[3-(trifluoromethoxy)phenyl]indoline-5-carboxamide	497	229

-continued

Example	Structure	Name	Human	
			LCMS [M + 1]	DGAT2 IC ₅₀ (nM)
14		3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-1-[3-(trifluoromethoxy)phenyl]indoline-5-carboxamide	497	98
15		3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-1-[3-(trifluoromethoxy)phenyl]indoline-5-carboxamide	483	105
16		1-[3-(difluoromethoxy)phenyl]-N-[4-(hydroxymethyl)tetrahydropyran-4-yl]-3,3-dimethyl-2-oxoindoline-5-carboxamide	461	1452
17		1-[5-(difluoromethoxy)-2-fluoro-phenyl]-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxoindoline-5-carboxamide	497	13

-continued

Example	Structure	Name	Human	
			LCMS [M + 1]	DGAT2 IC ₅₀ (nM)
18		1-[5-(difluoromethoxy)-2-fluoro-phenyl]-3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide	497	50
19		1-[5-(difluoromethoxy)-2-fluoro-phenyl]-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-indoline-5-carboxamide	511	13
20		1-[3-(2,2-difluoroethoxy)phenyl]-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide	493	12
21		1-[3-(2,2-difluoroethoxy)phenyl]-3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide	493	35

-continued

Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
22		1-[3-(2,2-difluoroethoxy)phenyl]-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-indoline-5-carboxamide	507	16
23		3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-1-[5-(1,1,2,2-tetrafluoroethoxy)-3-pyridyl]indoline-5-carboxamide	516	9.2
24		3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-1-[5-(1,1,2,2-tetrafluoroethoxy)-3-pyridyl]indoline-5-carboxamide	530	14
25		3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-1-[5-(1,1,2,2-tetrafluoroethoxy)-3-pyridyl]indoline-5-carboxamide	530	28

-continued

Example	Structure	Name	Human	
			LCMS [M + 1]	DGAT2 IC ₅₀ (nM)
26		3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-1-[5-(1,1,2,2-tetrafluoroethoxy)-3-pyridyl]indoline-5-carboxamide	544	1.0
27a		1-[3-(difluoromethoxy)phenyl]-N-(3-hydroxy-1,1-dimethyl-butyl)-3,3-dimethyl-2-oxo-indoline-5-carboxamide (faster eluting in OJ-H (2 x 25 cm), 10% iPrOH/CO ₂)	447	1380
27b		1-[3-(difluoromethoxy)phenyl]-N-(3-hydroxy-1,1-dimethyl-butyl)-3,3-dimethyl-2-oxo-indoline-5-carboxamide (slower eluting in OJ-H (2 x 25 cm) 10% iPrOH/CO ₂)	447	99
28		1-(2-ethoxy-5-fluoro-4-pyridyl)-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide	462	52

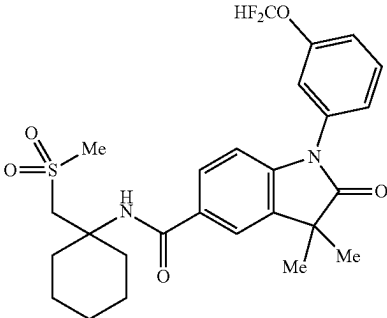
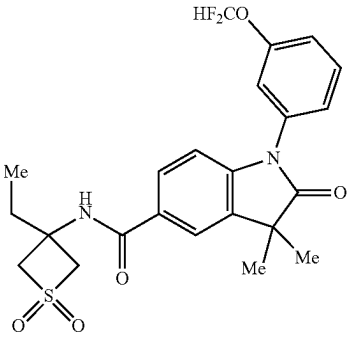
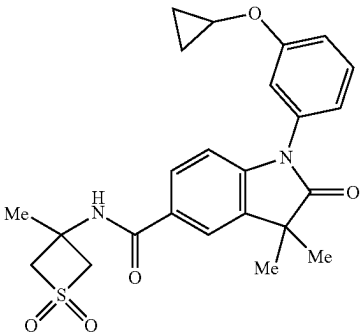
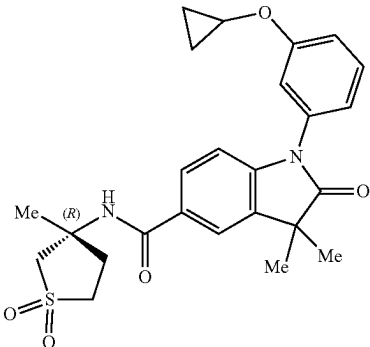
-continued

Example	Structure	Name	Human	
			LCMS [M + 1]	DGAT2 IC ₅₀ (nM)
29		1-(2-ethoxy-5-fluoro-4-pyridyl)-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide	476	56
30		1-(2-ethoxy-5-fluoro-4-pyridyl)-3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide	476	53
31		1-[2-(2,2-difluoroethoxy)pyrimidin-4-yl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide	481	>1000
32		3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-1-(3-quinolyl)indoline-5-carboxamide	450	>1000

-continued

Example	Structure	Name	Human	
			LCMS [M + 1]	DGAT2 IC ₅₀ (nM)
33		3,3-dimethyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-1-(2-methylbenzo[d]oxazol-6-yl)-2-oxoindoline-5-carboxamide	454	2751
34		1-(2-(3,3-difluoroazetidin-1-yl)-3-fluoropyridin-4-yl)-3,3-dimethyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide	509	547.7
35		1-(4-acetamido-3-hydroxyphenyl)-3,3-dimethyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide	472	>10000
36		N-[2-(cyclohexylsulfamoyl)ethyl]-1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-2-oxoindoline-5-carboxamide	536	95.3

-continued

Example	Structure	Name	Human	
			LCMS [M + 1]	DGAT2 IC ₅₀ (nM)
37		1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-[1-(methylsulfonylmethyl)cyclohexyl]-2-oxo-indoline-5-carboxamide	521	74
38		1-[3-(difluoromethoxy)phenyl]-N-(3-ethyl-1,1-dioxo-thietan-3-yl)-3,3-dimethyl-2-oxo-indoline-5-carboxamide	479	17
39		1-[3-(cyclopropoxy)phenyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide	455	15
40		1-[3-(cyclopropoxy)phenyl]-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide	467	22

-continued

Example	Structure	Name	Human	
			LCMS [M + 1]	DGAT2 IC ₅₀ (nM)
41		1-[3-(cyclopropoxy)phenyl]-3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide	467	67
42		1-[3-(cyclopropoxy)phenyl]-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-indoline-5-carboxamide	483	11
43		3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-1-[3-(trifluoromethoxy)phenyl]indoline-5-carboxamide	511	38
44		1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-[(3S)-(3-methyl-1,1-dioxo-thiolan-3-yl)]-2-oxo-indoline-5-carboxamide	479	168

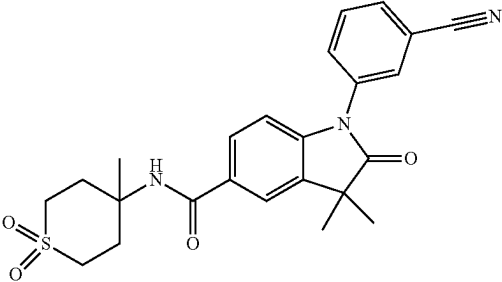
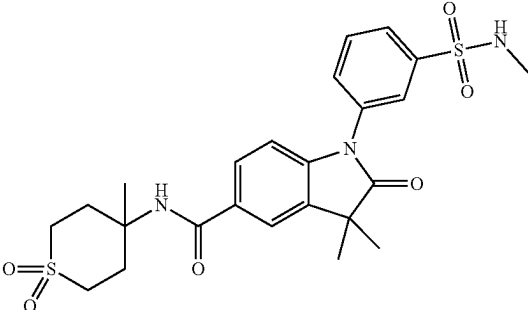
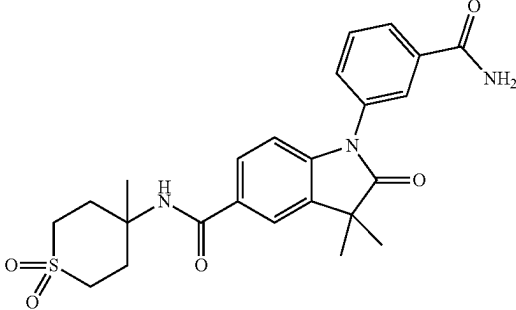
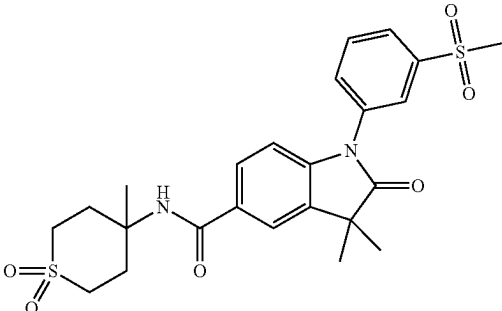
-continued

Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
45		1-[3-(difluoromethoxy)phenyl]-N-(3,3-difluoro-1-methylcyclobutyl)-3,3-dimethyl-2-oxoindoline-5-carboxamide	451	1062
46		(R)-N-(1-(2,2-difluoroethyl)-3-methylpiperidin-3-yl)-1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-2-oxoindoline-5-carboxamide	508	24.2
47		1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-(3-methyl-1-(methylsulfonyl)piperidin-3-yl)-2-oxoindoline-5-carboxamide	522	>10000
48		(R)-N-(1-(acetyl-3-methylpiperidin-3-yl)-1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-2-oxoindoline-5-carboxamide	486	4149
49		1-(3-(difluoromethoxy)phenyl)-N-(4-methoxy-4-(trifluoromethyl)cyclohexyl)-3,3-dimethyl-2-oxoindoline-5-carboxamide	527	407

-continued

Example	Structure	Name	Human	
			LCMS [M + 1]	DGAT2 IC ₅₀ (nM)
50		1-(5-(azetidin-1-yl)-2-fluorophenyl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	519	99
51		1-(3-cyclopropylphenyl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	467	85
52		1-(3-isopropoxyphenyl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	485	123
53		1-(3-fluorophenyl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	445	4051

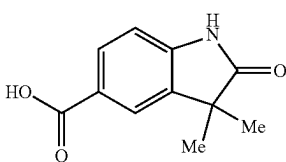
-continued

Example	Structure	Name	Human	
			LCMS [M + 1]	DGAT2 IC ₅₀ (nM)
54		1-(3-cyanophenyl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	452	2551
55		3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-1-(3-(N-methylsulfamoyl)phenyl)-2-oxoindoline-5-carboxamide	520	>9990
56		1-(3-carbamoylphenyl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	470	>9990
57		3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-1-(3-(methylsulfonyl)phenyl)-2-oxoindoline-5-carboxamide	505	>9990

-continued

Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
58		1-(3-(dimethylamino)phenyl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	470	1795
59		1-(3-(difluoromethoxy)phenyl)-N-((3-hydroxytetrahydrofuran-3-yl)methyl)-3,3-dimethyl-2-oxoindoline-5-carboxamide	447	>9990
60		1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-(5-methyl-1,3,4-thiadiazol-2-yl)-2-oxoindoline-5-carboxamide	445	>9990
61		1-(3-(difluoromethoxy)phenyl)-N-((1-ethyl-1H-pyrazol-4-yl)methyl)-3,3-dimethyl-2-oxoindoline-5-carboxamide	455	>9990

Example 62: 1-(4-(2,2-difluoroethoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide



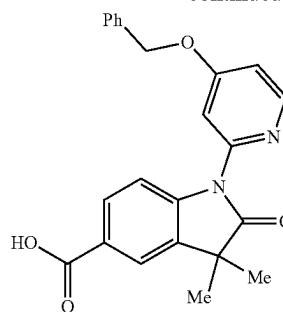
STEP A

55

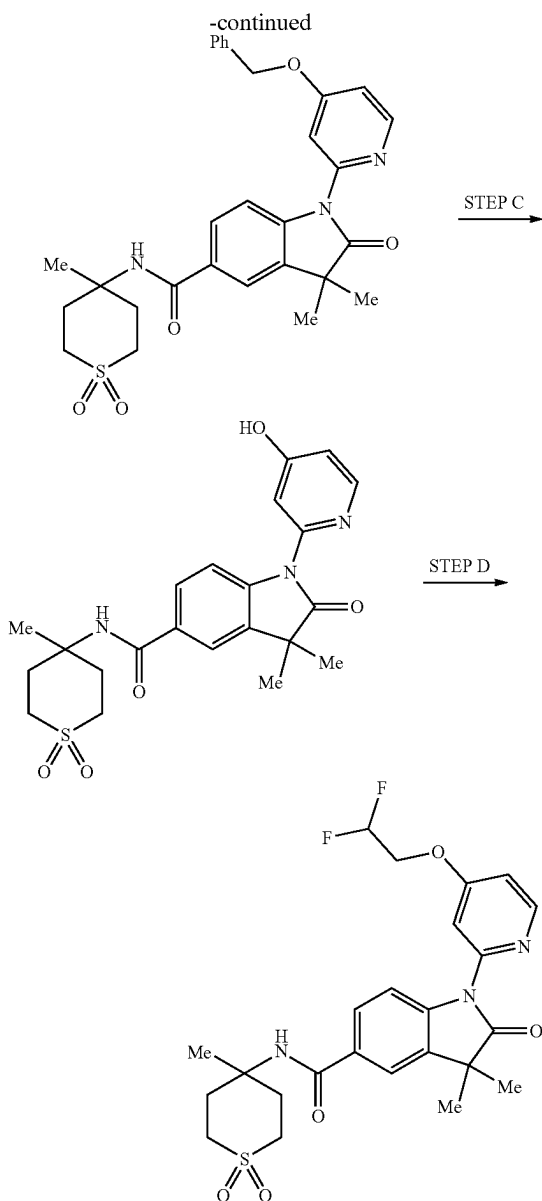
60

65

-continued



STEP B

147

Step A 1-(4-(benzyloxy)pyridin-2-yl)-3,3-dimethyl-2-oxoindoline-5-carboxylic acid

To a stirred mixture of 3,3-dimethyl-2-oxoindoline-5-carboxylic acid (1000 mg, 4.87 mmol), CuI (464 mg, 2.436 mmol) potassium carbonate (2020 mg, 14.62 mmol) 4-(benzyloxy)-2-bromopyridine (3861 mg, 14.62 mmol), and (1R, 2R)-N1,N2-dimethylcyclohexane-1,2-diamine (693 mg, 4.87 mmol) under nitrogen was added acetonitrile (9.7 mL), and the resulting solution was heated to 90° C. for 16 hrs. The reaction was diluted with DCM, poured into HCl (1M aq. soln.) and extracted with DCM. The combined organic fractions were dried over Na₂SO₄, filtered through a plug of silica, and concentrated in vacuo. The crude residue was purified by flash column chromatography on silica (0-100% EtOAc/hexanes) to afford the title compound. LC/MS=389 [M+1].

148

Step B: 1-(4-(benzyloxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide

To a stirred solution of 1-(4-(benzyloxy)pyridin-2-yl)-3,3-dimethyl-2-oxoindoline-5-carboxylic acid (932 mg, 2.4 mmol), DIEA (1.0 mL, 6.00 mmol) and 4-amino-4-methyl-tetrahydro-2H-thiopyran 1,1-dioxide hydrochloride (503 mg, 2.52 mmol) in DMF (24 mL) was added HATU (958 mg, 2.52 mmol). The reaction mixture was stirred for 18 hours. The reaction mixture was diluted with water, and the precipitate collected via filtration. The recovered solid was purified by flash column chromatography on silica (0-70% EtOAc/hexanes) to afford the title compound. LC/MS=534 [M+1].

Step C: 1-(4-hydroxypyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide

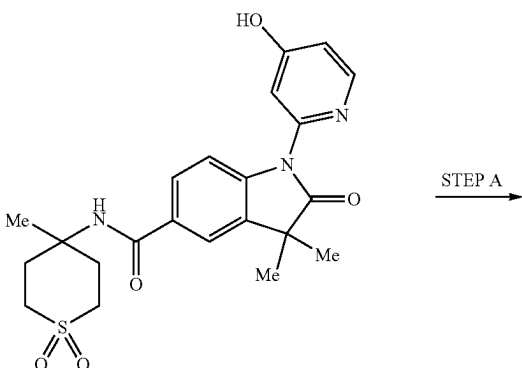
To a stirring solution of 1-(4-(benzyloxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide (0.96 g, 1.80 mmol) in ethanol (18.03 mL) under nitrogen was added triethylsilane (2.88 ml, 18.03 mmol) dropwise via syringe. The reaction mixture was stirred for 3 hrs and was sparged with argon for 5 minutes, diluted with dichloromethane, and filtered though celite. The solvent was removed in vacuo affording the crude title compound which was used without further purification. LC/MS=444 [M+1].

Step D: 1-(4-(2,2-difluoroethoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide

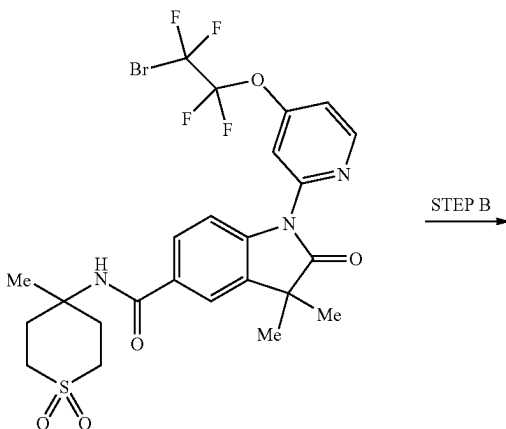
To a stirred solution of 1-(4-hydroxypyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide (155 mg, 0.35 mmol) and cesium carbonate (228 mg, 0.70 mmol) in DMF (1.7 mL) was added 2,2-difluoroethyl trifluoromethanesulfonate (46.0 µL, 0.35 mmol), and the reaction was stirred at 20° C. for 3 hrs. The solvent was removed in vacuo, and the crude reaction mixture was purified by flash column chromatography on silica (0-100% EtOAc/hexanes) to afford the title compound. LC/MS=508 [M+1]. ¹H NMR (500 MHz, DMSO-d₆) δ 8.49 (d, J=5.8 Hz, 1H), 7.95 (d, J=1.4 Hz, 1H), 7.82 (dd, J=8.4, 1.6 Hz, 1H), 7.76 (s, 1H), 7.52 (d, J=8.4 Hz, 1H), 7.40 (d, J=2.3 Hz, 1H), 7.16 (dd, J=5.8, 2.3 Hz, 1H), 6.62-6.32 (m, 1H), 4.53 (td, J=14.7, 3.2 Hz, 2H), 3.18 (t, J=12.0 Hz, 2H), 3.06 (d, J=13.8 Hz, 2H), 2.80 (d, J=14.5 Hz, 2H), 2.02 (t, J=12.2 Hz, 2H), 1.52-1.37 (m, 9H). Human DGAT2 IC₅₀=61 nM.

149

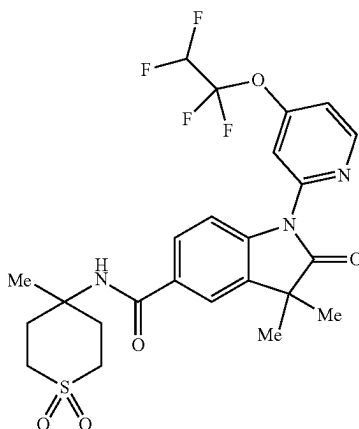
Example 63: 3,3-dimethyl-N-(4-methyl-1,1-dioxido-tetrahydro-2H-thiopyran-4-yl)-2-oxo-1-(4-(1,1,2,2-tetrafluoroethoxy)pyridin-2-yl) indoline-5-carboxamide



STEP A



STEP B



EX-63

150

Step A: 1-(4-(2-bromo-1,1,2,2-tetrafluoroethoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxido-tetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide

5

To a stirred solution of 1-(4-hydroxypyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxido-tetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide (919 mg, 2.07 mmol) and cesium carbonate (1.35 g, 4.14 mmol) in DMSO (10.4 mL) was added 1,2-dibromo-1,1,2,2-tetrafluoroethane (808 mg, 3.11 mmol). The reaction was heated to 60° C. for 16 hrs. After the mixture was cooled, the solvent was removed in vacuo to afford the crude title compound. LC/MS=624 [M+2]

25

Step B: 3,3-dimethyl-N-(4-methyl-1,1-dioxido-tetrahydro-2H-thiopyran-4-yl)-2-oxo-1-(4-(1,1,2,2-tetrafluoroethoxy)pyridin-2-yl) indoline-5-carboxamide

30

Crude 1-(4-(2-bromo-1,1,2,2-tetrafluoroethoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxido-tetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide was dissolved AcOH (10 mL). Zinc (677 mg, 10.36 mmol) was added, and the reaction mixture was heated to 60° C. for ten minutes. The reaction mixture was cooled down, diluted with water, and extracted with 6:1 Et₂O/DCM. The combined organic fractions were washed with NaCl (sat. aq. soln.), dried over MgSO₄, filtered, and concentrated in vacuo. The crude material was purified by flash column chromatography on silica (0-100% EtOAc/hexanes) to afford the title compound (EX-63). LC/MS=544 [M=1] ¹H NMR (500 MHz, DMSO-d₆) δ 8.71 (d, J=5.7 Hz, 1H), 7.96 (d, J=1.4 Hz, 1H), 7.89-7.81 (m, 2H), 7.81-7.72 (m, 2H), 7.43 (d, J=4.4 Hz, 1H), 7.06-6.80 (m, 1H), 3.19 (t, J=12.1 Hz, 2H), 3.06 (d, J=13.9 Hz, 2H), 2.80 (d, J=14.5 Hz, 2H), 2.03 (t, J=12.2 Hz, 2H), 1.51-1.41 (m, 9H). Human DGAT2 IC₅₀=0.8 nM.

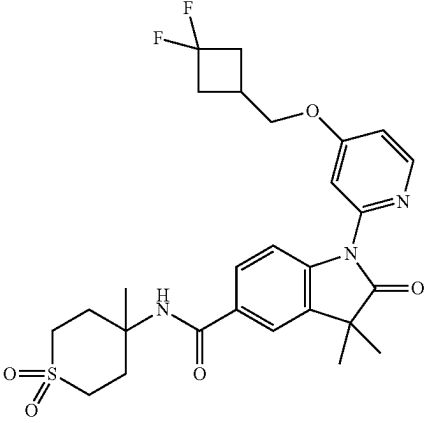
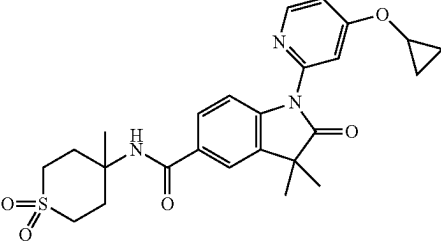
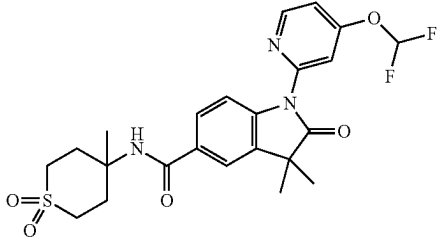
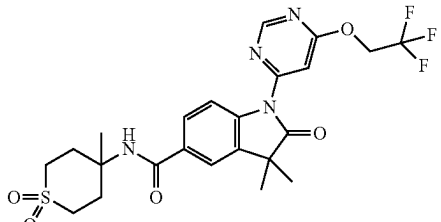
60

By using procedures similar to those described in Example 63 with appropriate reagents, the following compounds were synthesized. These compounds were characterized by LC/MS.

65

Exam- ple	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
64		3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxo-1-(4-(2,2,2-trifluoroethoxy)pyridin-2-yl)indoline-5-carboxamide	526	72
65		1-(4-(2,2-difluoropropoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	522	80
66		3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxo-1-(4-(2,2,3,3-tetrafluoropropoxy)pyridin-2-yl)indoline-5-carboxamide	558	184
67		1-(4-((2,2-difluorocyclopropyl)methoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	534	594

-continued

Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
68		1-((3,3-difluorocyclobutyl)methoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	548	196
69		1-(4-cyclopropoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	484	157
70		1-(4-(difluoromethoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	494	279
71		3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxo-1-(6-(2,2,2-trifluoroethoxy)pyrimidin-4-yl)indoline-5-carboxamide	527	20.4

-continued

Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
72		1-(4-(2,2-difluoroethoxy)-3-fluoropyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	526	821
73		1-(5-(2,2-difluoroethoxy)pyridazin-3-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	509	41
74		1-(4-(2,2-difluoroethoxy)pyrimidin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	509	450
75		1-(6-(2,2-difluoroethoxy)pyrimidin-4-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	509	113

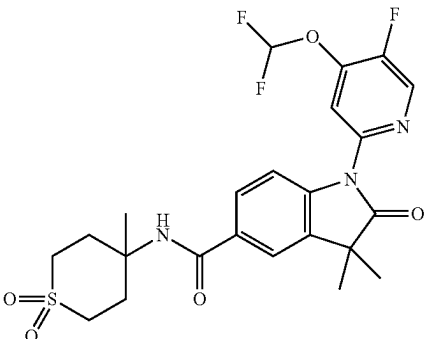
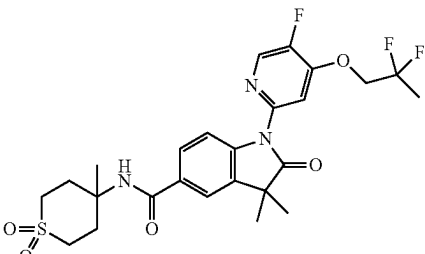
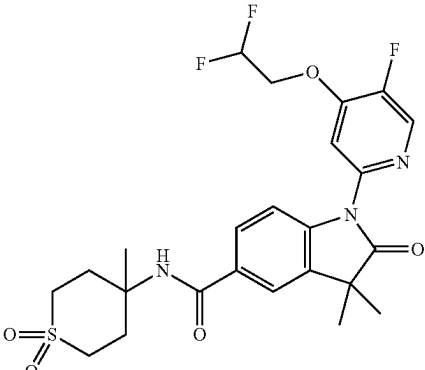
-continued

Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
76		1-(4-isopropoxyphenyl)-3,3-dimethyl-N-(4-methyl-1,1-dioxido-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	486	157
77		1-(4-((1-fluorocyclopropyl)methoxy)phenyl)-3,3-dimethyl-N-(4-methyl-1,1-dioxido-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	516	192
78		1-(4-(cyclopropylmethoxy)phenyl)-3,3-dimethyl-N-(4-methyl-1,1-dioxido-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	498	29
79		1-(4-isobutoxyphenyl)-3,3-dimethyl-N-(4-methyl-1,1-dioxido-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	500	45
80		1-(4-(1,1-difluoroethyl)phenyl)-3,3-dimethyl-N-(4-methyl-1,1-dioxido-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	492	2060

-continued

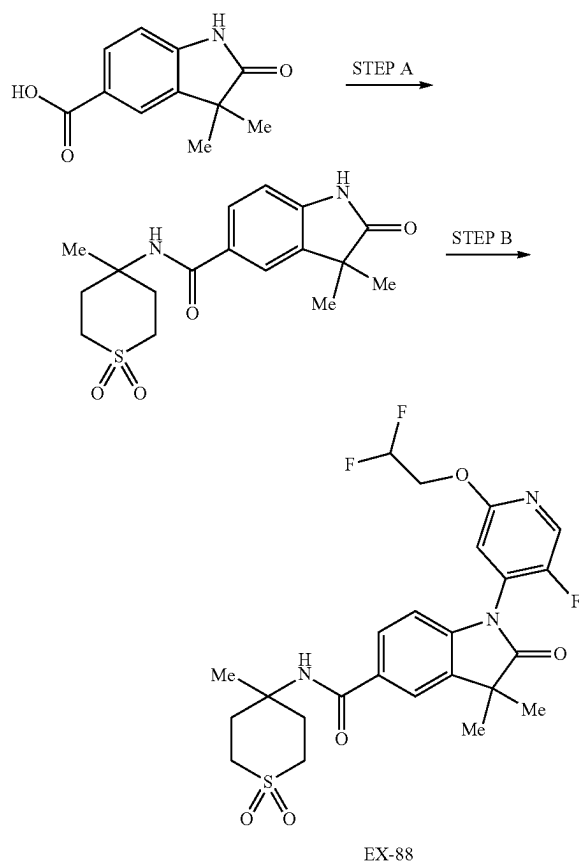
Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
81		1-(5-fluoro-4-(1,1,2,2-tetrafluoroethoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	562	3.2
82		1-(4-(2,2-difluoropropoxy)pyridin-2-yl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	540	24
83		1-(4-(difluoromethoxy)pyridin-2-yl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	512	155
84		1-(4-(2,2-difluoroethoxy)pyridin-2-yl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	526	82

-continued

Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
85		1-(4-(difluoromethoxy)-5-fluoropyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	512	287
86		1-(4-(2,2-difluoropropoxy)-5-fluoropyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	540	280
87		1-(4-(2,2-difluoroethoxy)-5-fluoropyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	526	108

163

Example 88: 1-(2-(2,2-difluoroethoxy)-5-fluoropyridin-4-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide



Step A: 3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide

To a stirred solution of 3,3-dimethyl-2-oxoindoline-5-carboxylic acid (2.5 g, 12.2 mmol), 4-amino-4-methyltetra-

164

hydro-2H-thiopyran 1,1-dioxide hydrochloride (2.4 g, 12.2 mmol), and DIEA (4.24 ml, 24.4 mmol) in DMF (122 ml), was added 2,4,6-tripropyl-1,3,5,2,4,6-trioxatriphosphinane 2,4,6-trioxide (11.6 g, 18.3 mmol), and the reaction mixture was stirred at 20° C. for 48 hrs. The reaction mixture concentrated in vacuo and purified by flash column chromatography on silica (0-100% EtOAc/hexanes) to afford the title compound. LC/MS=350 [M+1].

Step B: 1-(2-(2,2-difluoroethoxy)-5-fluoropyridin-4-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide

To a stirred mixture of 3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide (50 mg, 0.14 mmol), CuI (13.59 mg, 0.071 mmol), potassium carbonate (59.2 mg, 0.43 mmol), 4-bromo-2-(2,2-difluoroethoxy)-5-fluoropyridine (73.1 mg, 0.29 mmol), and (1R,2R)-N1,N2-dimethylcyclohexane-1,2-diamine (20.30 mg, 0.14 mmol) under nitrogen was added acetonitrile (285 µl). The solution was heated to 95° C. for 48 hrs. The reaction mixture was cooled down, diluted with EtOAc, filtered through a plug of silica, and concentrated in vacuo. The crude material was purified by flash column chromatography on silica (0-100% EtOAc/hexanes) to afford the title compound (EX-88). LC/MS=526 [M+1]. ¹H NMR (500 MHz, DMSO-d₆) δ 8.49 (d, J=1.2 Hz, 1H), 7.97 (d, J=1.5 Hz, 1H), 7.82 (dd, J=8.3, 1.6 Hz, 1H), 7.78 (s, 1H), 7.31 (d, J=4.8 Hz, 1H), 6.90 (dd, J=8.2, 1.6 Hz, 1H), 6.43 (tt, J=54.6, 3.4 Hz, 1H), 4.63 (t, J=14.6 Hz, 2H), 3.17 (t, J=11.8 Hz, 2H), 3.05 (d, J=14.1 Hz, 2H), 2.79 (d, J=14.6 Hz, 2H), 2.02 (t, J=11.0 Hz, 2H), 1.52-1.37 (m, 9H). Human DGAT2 IC₅₀=3.4 nM.

By using procedures similar to those described in Example 88 with appropriate reagents, the following compounds were synthesized. These compounds were characterized by LC/MS.

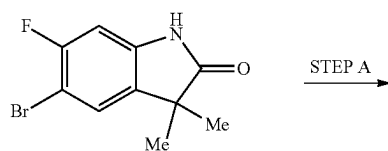
Example	Structure	Name	DGAT2	
			LCMS [M + 1]	IC ₅₀ (nM)
89		1-(2-ethoxy-5-fluoropyridin-4-yl)-3,3-dimethyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide	462	91

-continued

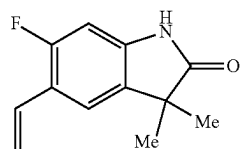
Exam- ple	Structure	Name	DGAT2	
			LCMS [M + 1]	IC50 (nM)
90		1-(2-(difluoromethoxy)-5-fluoropyridin-4-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	512	136
91		1-(2-(2,2-difluoropropoxy)-5-fluoropyridin-4-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	540	5.6
92		1-(5-fluoro-2-isopropoxy-pyridin-4-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	504	378
93		1-(2-(2,2-difluoroethoxy)-5-fluoropyridin-4-yl)-3,3-dimethyl-N-((methylsulfonyl)methyl)cyclobutyl)-2-oxoindoline-5-carboxamide	526	11

167

Example 94: 1-(2-(2,2-difluoroethoxy)-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide

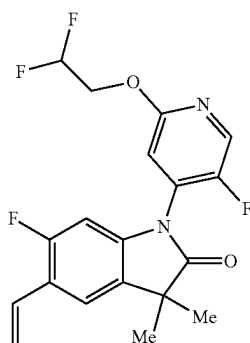


STEP A

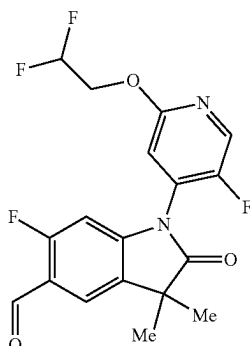


INT-7

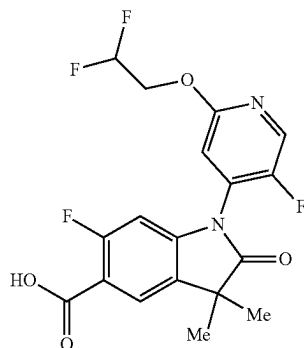
STEP B



STEP C



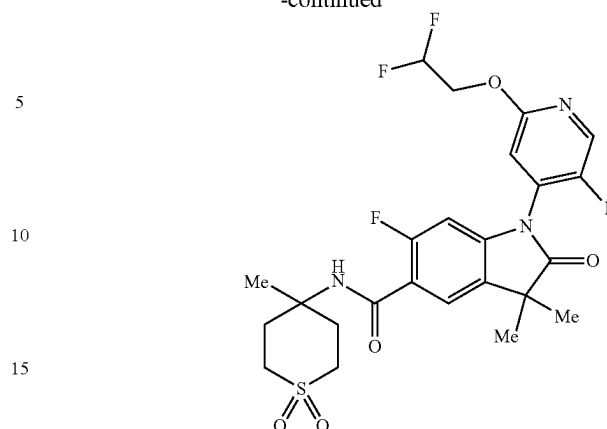
STEP D



STEP E

168

-continued



EX-94

Step A: 6-fluoro-3,3-dimethyl-5-vinylindolin-2-one

To a vial were added 5-bromo-6-fluoro-3,3-dimethylindolin-2-one (2.50 g, 9.69 mmol), potassium vinyltrifluoroborate $\text{CH}_2\text{CHBF}_3\text{K}$ (2.60 g, 19.37 mmol), PdCl_2 (dppf) (0.71 g, 0.97 mmol), triethylamine (4.05 ml, 29.1 mmol). The vial was evacuated and backfilled with nitrogen three times. Ethanol (64.6 ml) and water (6.5 mL) were added sequentially, and the reaction mixture was heated to 90° C. for 6 hrs. The reaction was cooled down, diluted with DCM, dried over MgSO_4 , filtered, and concentrated in vacuo. The crude residue was purified by flash column chromatography on silica (0-50% EtOAc in hexanes) to afford the title compound. LC/MS=206 [M+1].

Step B: 1-(2-(2,2-difluoroethoxy)-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-5-vinylindolin-2-one

To a stirred mixture of 6-fluoro-3,3-dimethyl-5-vinylindolin-2-one (200 mg, 0.98 mmol), CuI (93 mg, 0.49 mmol), potassium carbonate (404 mg, 2.92 mmol), 4-bromo-2-(2,2-difluoroethoxy)-5-fluoropyridine (499 mg, 1.95 mmol) and (1R,2R)-N1,N2-dimethylcyclohexane-1,2-diamine (139 mg, 0.98 mmol) under nitrogen was added acetonitrile (1.9 ml). The reaction mixture was heated to 95° C. for 48 hrs, then diluted with EtOAc, filtered through a plug of silica, and concentrated in vacuo. The crude residue was purified by flash column chromatography on silica (0-100% EtOAc in hexanes) to afford the title compound. LC/MS=381 [M+1].

Step C: 1-(2-(2,2-difluoroethoxy)-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-2-oxoindoline-5-carbaldehyde

To a stirred solution of 1-(2-(2,2-difluoroethoxy)-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-5-vinylindolin-2-one (50 mg, 0.131 mmol), 2,6-dimethylpyridine (30.5 μl , 0.263 mmol) and $\text{H}_4\text{K}_2\text{O}_6\text{Os}$ (1.21 mg, 3.29 μmol) in 1,4-dioxane (657 μl) and water (657 μl) was added NaIO_4 (112 mg, 0.526 mmol) at 0° C. The reaction mixture was stirred at 20° C. for 3 hrs. The reaction mixture was diluted with NaCl (sat. aq. soln.) and extracted with ethyl acetate. The combined organic fractions were dried over MgSO_4 , filtered, and concentrated in vacuo. The crude material was purified by

169

flash column chromatography on silica (0-50% EtOAc in hexanes) to afford the title compound. LC/MS=382 [M+1].

Step D: 1-(2-(2,2-difluoroethoxy)-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-2-oxoindoline-5-carboxylic acid

To a stirred solution of 1-(2-(2,2-difluoroethoxy)-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-2-oxoindoline-5-carbaldehyde (0.049 g, 0.127 mmol), 2-methylbut-2-ene (0.14 ml, 1.27 mmol), and sodium dihydrogen phosphate (0.10 g, 0.83 mmol) in a mixture of t-BuOH (0.85 ml), acetone (0.85 ml), and water (0.85 ml) was added sodium chlorite (0.034 g, 0.38 mmol) at 0° C. The reaction mixture was stirred for 2 hrs. The crude mixture was extracted with DCM. The combined organic fractions were dried over Na₂SO₄, filtered, and evaporated in vacuo. The crude material was used directly in subsequent reactions. LC/MS=399 [M+1].

Step E: 1-(2-(2,2-difluoroethoxy)-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxido-tetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide

To a stirred solution of 1-(2-(2,2-difluoroethoxy)-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-2-oxoindoline-5-car-

170

boxylic acid (44 mg, 0.110 mmol), 4-amino-4-methyltetrahydro-2H-thiopyran 1,1-dioxide hydrochloride (22.06 mg, 0.110 mmol), and DIEA (57.9 μ l, 0.33 mmol) in DCM (1.1 ml) was added HATU (44.1 mg, 0.12 mmol). The reaction mixture was stirred at RT for 16 hrs, then diluted with EtOAc, filtered through a plug of SiO₂, and concentrated in vacuo. The crude material was purified by flash column chromatography on silica (0-100% EtOAc/hexanes) to afford the title compound (EX-94). LC/MS=544 [M+1]. ¹H NMR (500 MHz, DMSO-d₆) δ 8.48 (d, J=1.2 Hz, 1H), 8.10 (s, 1H), 7.73 (d, J=6.7 Hz, 1H), 7.30 (d, J=4.8 Hz, 1H), 6.90 (d, J=9.0 Hz, 1H), 6.43 (tt, J=54.5, 3.4 Hz, 1H), 4.63 (s, 2H), 3.19-3.01 (m, 4H), 1.99 (t, J=11.9 Hz, 2H), 1.44 (s, 9H). The signal for 2H were obscured. Human DGAT2 IC₅₀=0.6 nM.

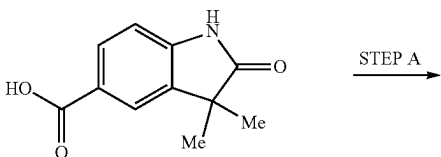
By using procedures similar to those described in Example 94 with appropriate reagents, the following compounds were synthesized. These compounds were characterized by LC/MS.

Example	Structure	Name	DGAT2	
			LCMS [M + 1]	IC50 (nM)
95		1-(2-ethoxy-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxido-tetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	508	1.9
96		1-(2-(difluoromethoxy)-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxido-tetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	530	132
97		1-(2-(difluoromethoxy)-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide	502	231

-continued

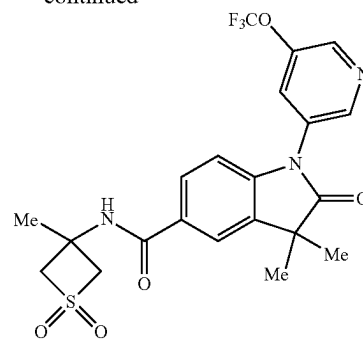
Example	Structure	Name	LCMS [M + 1]	DGAT2 IC50 (nM)
98		1-(2-(3,3-difluoroazetidin-1-yl)-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide	555	83.8
99		N-(4-(difluoromethyl)-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-1-(2-ethoxy-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-2-oxoindoline-5-carboxamide	544	73

Example 100: 3,3-dimethyl-N-(3-methyl-1,1-dioxo-
thietan-3-yl)-2-oxo-1-[5-(trifluoromethoxy)-3-
pyridyl]indoline-5-carboxamide

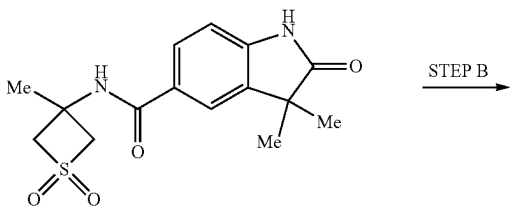


STEP A

-continued



EX-100



STEP B

Step A: 3,3-dimethyl-N-(3-methyl-1,1-dioxidothi-
etan-3-yl)-2-oxoindoline-5-carboxamide

To a stirred solution of 3,3-dimethyl-2-oxoindoline-5-
carboxylic acid (1.00 g, 4.87 mmol) and 3-amino-3-meth-
ylthietane 1,1-dioxide hydrochloride (1.26 g, 7.31 mmol) in
DMF (48 mL), was added propanephosphonic acid anhy-
dride (3.10 g, 4.87 mmol). The reaction mixture was allowed
to stir at 20° C. for 48 hours, then diluted with HCl (1M aq.
soln.) and extracted with DCM. The combined organic
fractions were washed with HCl (1M aq. soln.) and NaCl

173

(sat. aq. soln.), dried over MgSO_4 (s), filtered, and concentrated in vacuo to afford the title compound. LC/MS=323 [M+1].

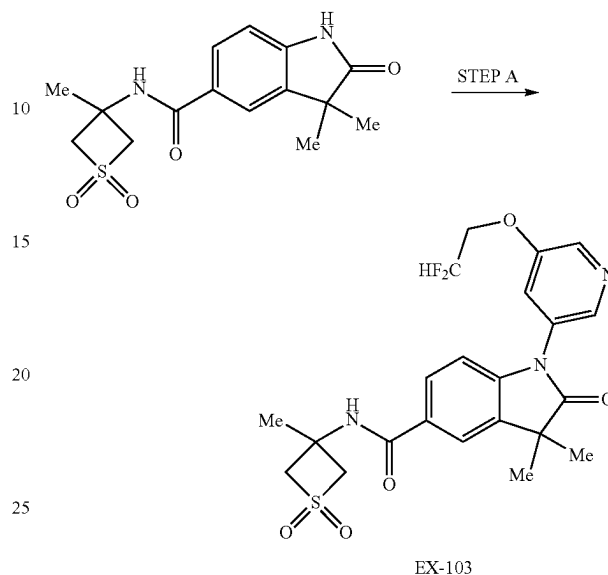
Step B: 3,3-dimethyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxo-1-(5-(trifluoromethoxy)pyridin-3-yl) indoline-5-carboxamide

To a stirred solution of 3,3-dimethyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide (50 mg, 0.16 mmol), (5-(trifluoromethoxy)pyridin-3-yl) boronic acid (64.2 mg, 0.31 mmol), and $\text{Cu}(\text{OAc})_2$ (56.3 mg, 0.31 mmol) in DMF (310 μl) was added pyridine (125 μl , 1.55 mmol). The vial was heated to 90° C. under air overnight. The reaction mixture was diluted with EtOAc, filtered through a plug of SiO_2 , and concentrated in vacuo. The crude residue was purified by mass triggered reverse phase HPLC (ACN/water with 0.1% formic acid modifier) to afford the title compound. LC/MS=484 [M+1]. ^1H NMR (500 MHz, Methanol- d_4) δ 8.77 (d, J=1.8 Hz, 1H), 8.68 (s, 1H), 8.07 (s, 1H), 7.93 (d, J=1.5 Hz, 1H), 7.82 (dd, J=8.3, 1.7 Hz, 1H), 7.01 (d, J=8.3 Hz, 1H), 4.58 (d, J=14.5 Hz, 2H), 4.23 (d, J=14.7 Hz, 2H), 1.83 (s, 3H), 1.54 (s, 6H). Human DGAT2 IC_{50} =372 nM.

By using procedures similar to those described in Example 100 with appropriate reagents, the following compounds were synthesized. These compounds were characterized by LC/MS.

174

Example 103: 1-[5-(2,2-difluoroethoxy)-3-pyridyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide



To a stirred mixture of 3,3-dimethyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide (50 mg, 0.16 mmol), CuI (2.95 mg, 0.016 mmol) and potassium carbonate (42.9 mg, 0.310 mmol) under nitrogen were added 3-bromo-5-(2,2-difluoroethoxy)pyridine (73.8 mg, 0.310

Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC_{50} (nM)
101		1-[5-(difluoromethoxy)-3-pyridyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide	466	314
102		1-(5-ethoxy-3-pyridyl)-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide	444	111

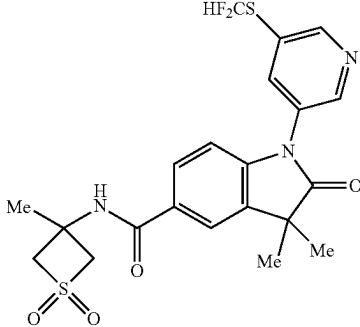
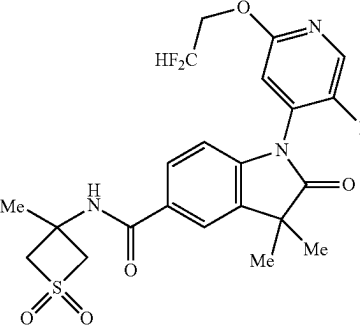
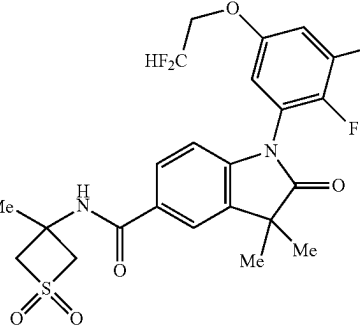
175

mmol) and (1R,2R)-N1,N2-dimethylcyclohexane-1,2-diamine (4.41 mg, 0.031 mmol) as a solution in acetonitrile (310 μ l) via syringe. The resultant solution was heated to 95° C. for 16 hours, then diluted with EtOAc, filtered through a plug of SiO₂, and concentrated in vacuo. The crude residue was purified by mass triggered reverse phase HPLC (ACN/ water with 0.1% formic acid modifier) to afford the title compound. LC/MS=480 [M+1]. ¹H NMR (500 MHz, Chloroform-d) δ 8.44 (s, 2H), 7.80 (d, J=1.7 Hz, 1H), 7.64 (dd,

176

J=8.3, 1.8 Hz, 1H), 7.41 (t, J=2.3 Hz, 1H), 6.97 (d, J=8.3 Hz, 1H), 6.39 (s, 1H), 6.16 (tt, J=54.7, 3.9 Hz, 1H), 4.61 (d, J=14.5 Hz, 2H), 4.33 (td, J=12.8, 4.0 Hz, 2H), 4.24 (d, J=14.7 Hz, 2H), 1.95 (s, 3H), 1.56 (s, 6H). Human DGAT2 IC₅₀=50 nM

By using procedures similar to those described in Example 103 with appropriate reagents, the following compounds were synthesized. These compounds were characterized by LC/MS.

Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
104		1-[5-(difluoromethylsulfanyl)-3-pyridyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide	482	197
105		1-[2-(2,2-difluoroethoxy)-5-fluoro-4-pyridyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide	498	22
106		1-[5-(2,2-difluoroethoxy)-2,3-difluorophenyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide	515	4.8

-continued

Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
107		1-[5-(2,2-difluoroethoxy)-2-fluorophenyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide	497	11
108		1-[6-(2,2-difluoroethoxy)pyrazin-2-yl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide	481	627.8
109		1-[6-(2,2-difluoroethoxy)pyridazin-4-yl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide	481	>10000
110		1-[3-(2,2-difluoroethoxy)phenyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide	479	54

-continued

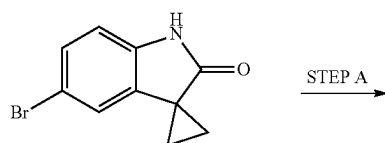
Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
111		1-[5-(difluoromethylsulfanyl)-2-fluorophenyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide	499	35
112		1-[5-(difluoromethoxy)-2-fluorophenyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide	483	7.8
113		1-[5-(1,1-difluoroethyl)-3-pyridyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide	464	820.7
114		1-[5-(1,1-difluoroethoxy)-3-pyridyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide	480	12

-continued

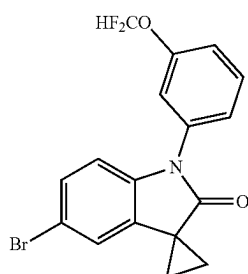
Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
115		1-[5-(2,2-difluoroethoxy)-2-methyl-3-pyridyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide	494	213
116		1-(5-(difluoromethoxy)pyridin-3-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxo-indoline-5-carboxamide	494	34
117		3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxo-1-(5-(trifluoromethoxy)pyridin-3-yl)indoline-5-carboxamide	512	124
118		1-(5-(1,1-difluoro-2-hydroxyethoxy)pyridin-3-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxo-indoline-5-carboxamide	524	26

183

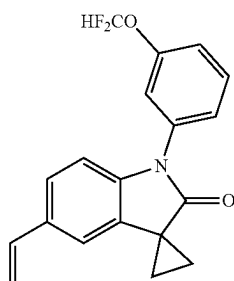
Example 119: (R)-1'-(3-(difluoromethoxy)phenyl)-N-(3-methyl-1,1-dioxidotetrahydrothiophen-3-yl)-2'-oxospiro[cyclopropane-1,3'-indoline]-5'-carboxamide



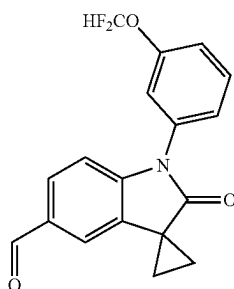
STEP A



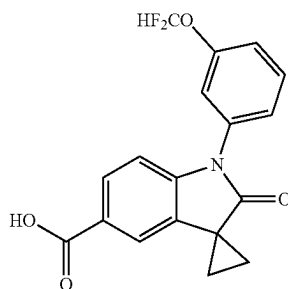
STEP B



STEP C



STEP C



STEP D

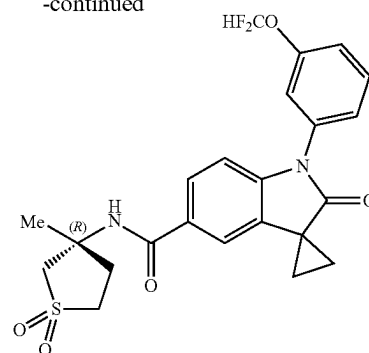
184

-continued

5

10

15



EX-119

Step A: 5'-bromo-1'-(3-(difluoromethoxy)phenyl)spiro[cyclopropane-1,3'-indolin]-2'-one

20

25

30

35

To a stirred solution of 5'-bromospiro[cyclopropane-1,3'-indolin]-2'-one (0.50 g, 2.100 mmol), (3-(difluoromethoxy)phenyl) boronic acid (0.79 g, 4.20 mmol), and Cu(OAc)₂ (0.38 g, 2.10 mmol) in DMF (4.2 mL) was added pyridine (0.85 mL, 10.5 mmol). The reaction mixture was stirred at 75° C. for 60 hours under air, then poured into water and acidified with HCl (1M Aq. Soln.). The mixture was extracted with DCM, washed with NaHCO₃ (sat. aq. soln.) and NaCl (sat. aq. soln.), dried over MgSO₄ (s), filtered, and concentrated in vacuo. The crude residue was subjected to flash silica gel column chromatography using a 0-40% EtOAc/hexanes gradient to afford the title compound. LC/MS=381 [M+1].

Step B: 1'-(3-(difluoromethoxy)phenyl)-5'-vinylspiro[cyclopropane-1,3'-indolin]-2'-one

40

45

50

55

At 20° C. a 40 mL screw cap vial equipped with a magnetic stir bar was charged with 5'-bromo-1'-(3-(difluoromethoxy)phenyl)spiro[cyclopropane-1,3'-indolin]-2'-one (0.22 g, 0.58 mmol), trifluoro (vinyl)-14-borane, potassium salt (0.078 g, 0.581 mmol), PdCl₂ (dppf) (0.043 g, 0.058 mmol), Na₂CO₃ (0.19 g, 1.74 mmol). The vial was evacuated and backfilled with nitrogen three times. Dioxane (4.8 mL) and water (1.0 mL) were added sequentially, and the reaction mixture was heated to 90° C. for 6 hours. LCMS analysis indicated high conversion to the desired product. The reaction was diluted with DCM, dried over MgSO₄ (s), filtered, and concentrated in vacuo. The crude residue was subjected to flash silica gel column chromatography using a 0-50% EtOAc/hexanes gradient to afford the title compound. LC/MS=330 [M+1].

Step C: 1'-(3-(difluoromethoxy)phenyl)-2'-oxospiro[cyclopropane-1,3'-indoline]-5'-carbaldehyde

60

65

To a stirred solution of 1'-(3-(difluoromethoxy)phenyl)-5'-vinylspiro[cyclopropane-1,3'-indolin]-2'-one (127 mg, 0.39 mmol), 2,6-dimethylpyridine (90 µL, 0.78 mmol) and K₂[OsO₂(OH)₄] (3.57 mg, 9.70 µmol) in dioxane (1.9 mL) and water (1.9 mL) was added sodium periodate (332 mg, 1.55 mmol) at 0° C. The reaction mixture was stirred at 20° C. for 3 hours. LCMS analysis indicated the desired product was formed in high conversion. The reaction mixture was diluted with NaCl (sat. aq. soln.) and extracted with EtOAc. The combined organic fractions were dried over MgSO₄ (s),

185

filtered, and concentrated in vacuo. The crude material was subjected to flash silica gel column chromatography using a 0-50% EtOAc/hexanes gradient to afford the title compound. LC MS=330 [M+1].

Step D: 1'-(3-(difluoromethoxy)phenyl)-2'-oxospiro[cyclopropane-1,3'-indoline]-5'-carboxylic acid

To a stirred solution of 1'-(3-(difluoromethoxy)phenyl)-2'-oxospiro[cyclopropane-1,3'-indoline]-5'-carbaldehyde (97 mg, 0.30 mmol), 2-methylbut-2-ene (0.31 mL, 2.95 mmol), and NaH₂PO₄ (0.12 g, 0.96 mmol) in a mixture of t-BuOH (2.0 mL) and water (2.0 mL) was added NaClO₂ (0.04 g, 0.44 mmol) at 0° C. The reaction mixture was stirred for 16 hours. The crude mixture was extracted with DCM. The combined organic fractions were dried over Na₂SO₄ (s), filtered, and evaporated in vacuo to give the title compound. LC/MS=346 [M+1].

Step E: (R)-1'-(3-(difluoromethoxy)phenyl)-N-(3-methyl-1,1-dioxido-tetrahydrothiophen-3-yl)-2'-oxospiro[cyclopropane-1,3'-indoline]-5'-carboxamide

To a stirred solution of 1'-(3-(difluoromethoxy)phenyl)-2'-oxospiro[cyclopropane-1,3'-indoline]-5'-carboxylic acid

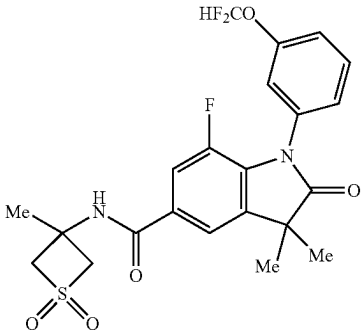
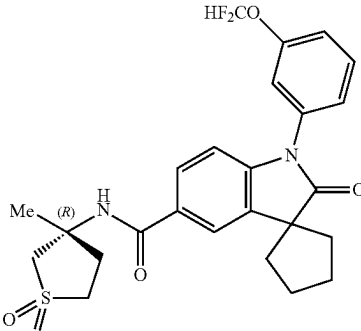
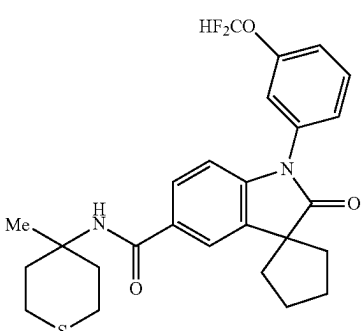
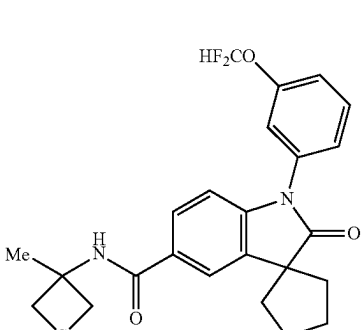
186

(33.8 mg, 0.098 mmol), (R)-3-amino-3-methyltetrahydrothiophene 1,1-dioxide hydrochloride (18.20 mg, 0.098 mmol), and Hunig's Base (51.3 µL, 0.29 mmol) in DMF (980 µL) was added HATU (39.1 mg, 0.10 mmol). The reaction mixture was allowed to stir at RT for 16 hours, then was diluted with EtOAc, filtered through a plug of SiO₂, and concentrated in vacuo. The crude residue was purified by mass triggered reverse phase HPLC (ACN/water with 0.1% formic acid modifier) to afford the title compound. LC/MS=477 [M+1]. ¹H NMR (500 MHz, Chloroform-d) δ 7.60-7.54 (m, 2H), 7.41 (d, J=1.6 Hz, 1H), 7.38-7.33 (m, 1H), 7.28 (m, 1H), 7.23 (d, J=8.3 Hz, 1H), 6.98 (d, J=8.2 Hz, 1H), 6.60 (t, J=73.3 Hz, 1H), 6.28 (s, 1H), 3.65 (d, J=13.5 Hz, 1H), 3.50-3.39 (m, 1H), 3.35-3.25 (m, 1H), 3.17 (d, J=13.8 Hz, 1H), 3.15-3.06 (m, 1H), 2.27 (ddd, J=13.8, 10.3, 8.3 Hz, 1H), 1.94 (q, J=4.0 Hz, 2H), 1.79 (s, 3H), 1.75 (q, J=4.1 Hz, 2H). Human DGAT2 IC₅₀=118 nM

By using procedures similar to those described in Example 119 with appropriate reagents, the following compounds were synthesized. These compounds were characterized by LC/MS.

Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
120		1-[3-(difluoromethoxy)phenyl]-7-fluoro-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide	497	53
121		1-[3-(difluoromethoxy)phenyl]-7-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-indoline-5-carboxamide	511	74

-continued

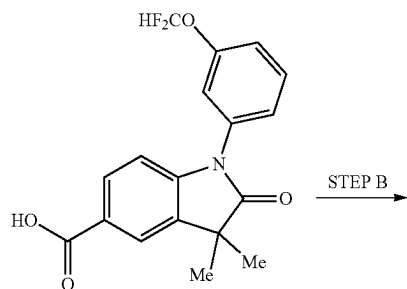
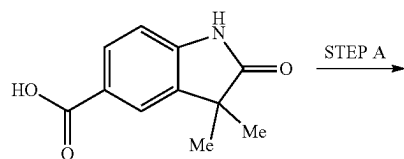
Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
122		1-[3-(difluoromethoxy)phenyl]-7-fluoro-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide	483	19
123		1'-[3-(difluoromethoxy)phenyl]-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2'-oxo-spiro[cyclopentane-1,3'-indoline]-5'-carboxamide	505	64
124		1'-[3-(difluoromethoxy)phenyl]-N-(4-methyl-1,1-dioxo-thian-4-yl)-2'-oxo-spiro[cyclopentane-1,3'-indoline]-5'-carboxamide	519	31
125		1'-[3-(difluoromethoxy)phenyl]-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2'-oxo-spiro[cyclopentane-1,3'-indoline]-5'-carboxamide	491	68

-continued

Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
126		1'-[3-(difluoromethoxy)phenyl]-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2'-oxo-spiro[cyclopropane-1,3'-indoline]-5'-carboxamide	463	461
127		1'-[3-(difluoromethoxy)phenyl]-N-(4-methyl-1,1-dioxo-thian-4-yl)-2'-oxo-spiro[cyclopropane-1,3'-indoline]-5'-carboxamide	491	234

35

Example 128: 1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-(5-methyl-1,1-dioxo-thiolan-3-yl)-2-oxo-indoline-5-carboxamide



-continued

40

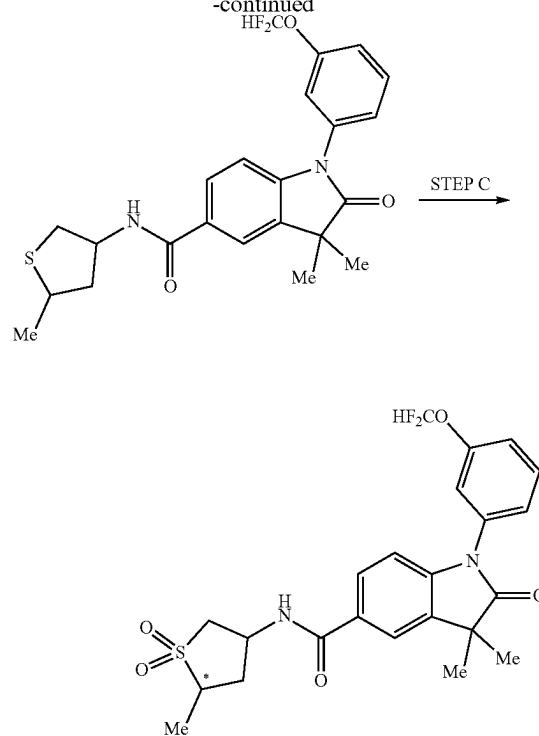
45

50

55

60

65



EX-128

191

Step A: 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-2-oxoindoline-5-carboxylic acid

To a stirred mixture of 3,3-dimethyl-2-oxoindoline-5-carboxylic acid (1.00 g, 4.87 mmol), CuI (0.093 g, 0.49 mmol), and K₂CO₃ (2.02 g, 14.62 mmol) were added 1-(difluoromethoxy)-3-iodobenzene (1.97 g, 7.31 mmol) and (1R,2R)-N1,N2-dimethylcyclohexane-1,2-diamine (0.14 g, 0.98 mmol) as a solution in acetonitrile (9.8 mL) and the resultant solution was heated to 90° C. for 16 hours. The reaction mixture was diluted with DCM, poured into HCl (1M aq. soln.) and extracted with DCM. The combined organic fractions were dried over Na₂SO₄ (s), filtered through a plug of SiO₂, and concentrated in vacuo. The crude residue was subjected to flash silica gel column chromatography using a 0-80% EtOAc/hexanes gradient to afford the title compound. LC/MS=348 [M+1].

Step B: 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-(5-methyltetrahydrothiophen-3-yl)-2-oxoindoline-5-carboxamide

To a stirred solution of 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-2-oxoindoline-5-carboxylic acid (0.10 g, 0.29 mmol), 5-methyltetrahydrothiophen-3-amine (0.034 g, 0.29 mmol), and Hunig's Base (0.15 mL, 0.86 mmol) in DCM (2.9 mL) was added HATU (0.115 g, 0.30 mmol). The reaction mixture was allowed to stir at RT for 16 hours, then was diluted with EtOAc, filtered through a plug of SiO₂, and concentrated in vacuo to give the crude title compound. LC/MS=447 [M+1].

Step C: 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-(5-methyl-1,1-dioxidotetrahydrothiophen-3-yl)-2-oxoindoline-5-carboxamide

To a stirred solution of 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-(5-methyltetrahydrothiophen-3-yl)-2-oxoindoline-5-carboxamide (129 mg, 0.29 mmol) in DCM (2.9 mL) was added m-CPBA (177 mg, 0.72 mmol). The reaction mixture was allowed to stir at RT for 0.5 hours, then was diluted with EtOAc, filtered through a plug of SiO₂, and concentrated in vacuo. The crude residue was purified by mass triggered reverse phase HPLC (ACN/water with 0.1% formic acid modifier) to afford the title compound. LC/MS=479 [M+1].

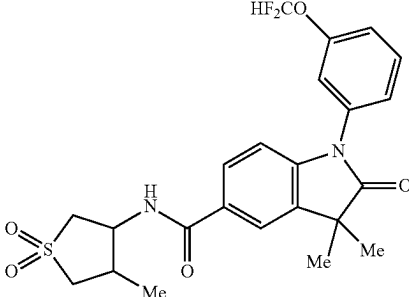
192

Step D: 1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-(5-methyl-1,1-dioxo-thiolan-3-yl)-2-oxoindoline-5-carboxamide

The mixture of 4 stereoisomers of 1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-(5-methyl-1,1-dioxo-thiolan-3-yl)-2-oxoindoline-5-carboxamide was purified by chiral SFC (IC column, 20% ethanol/CO₂) to afford isomers EX-128a and EX-128b as a single peak (fast eluting).

The mixture of EX-128a and EX-128b was subsequently purified by chiral SFC (AD-H column 15% ethanol/CO₂) to afford isomer EX-128a (fast eluting). LC/MS=479 [M+1]. ¹H NMR (500 MHz, Chloroform-d) δ 7.78 (d, J=1.6 Hz, 1H), 7.63 (dd, J=8.3, 1.7 Hz, 1H), 7.57 (t, J=8.0 Hz, 1H), 7.31 (d, J=8.4 Hz, 1H), 7.23 (d, J=8.2 Hz, 2H), 6.94 (d, J=8.2 Hz, 1H), 6.75 (s, 1H), 6.60 (t, J=7.3 Hz, 1H), 4.98 (s, 1H), 3.47-3.34 (m, 2H), 3.23 (d, J=13.7 Hz, 1H), 2.78-2.67 (m, 1H), 2.18 (ddd, J=14.5, 9.1, 5.9 Hz, 1H), 1.55 (s, 6H), 1.45 (d, J=7.0 Hz, 3H). Human DGAT2 IC₅₀=90 nM. EX-128b (slow eluting). LC/MS=479 [M+1]. ¹H NMR (500 MHz, Chloroform-d) δ 7.77 (d, J=1.6 Hz, 1H), 7.62 (dd, J=8.3, 1.7 Hz, 1H), 7.57 (t, J=8.3 Hz, 1H), 7.34-7.30 (m, 1H), 7.23 (d, J=7.7 Hz, 2H), 6.94 (d, J=8.3 Hz, 1H), 6.77-6.43 (m, 2H), 4.94 (dt, J=16.0, 7.9 Hz, 1H), 3.64 (dd, J=13.9, 8.4 Hz, 1H), 3.33-3.20 (m, 1H), 3.07 (dd, J=13.9, 5.6 Hz, 1H), 2.82 (dt, J=14.4, 7.5 Hz, 1H), 1.91 (ddd, J=13.4, 11.3, 8.3 Hz, 1H), 1.54 (s, 6H), 1.47 (d, J=6.7 Hz, 3H). Human DGAT2 IC₅₀=404.2 nM. EX-128c (middle eluting). LC/MS=479 [M+1]. ¹H NMR (500 MHz, Chloroform-d) δ 7.78 (d, J=1.6 Hz, 1H), 7.63 (dd, J=8.3, 1.8 Hz, 1H), 7.57 (t, J=8.2 Hz, 1H), 7.31 (d, J=8.5 Hz, 1H), 7.23 (d, J=8.1 Hz, 2H), 6.94 (d, J=8.2 Hz, 1H), 6.79-6.44 (m, 2H), 5.00-4.95 (m, 1H), 3.47-3.33 (m, 2H), 3.23 (d, J=14.0 Hz, 1H), 2.78-2.67 (m, 1H), 2.18 (ddd, J=14.5, 9.1, 5.9 Hz, 1H), 1.55 (s, 6H), 1.45 (d, J=7.0 Hz, 3H). Human DGAT2 IC₅₀=1935 nM. EX-128d (slow eluting). LC/MS=479 [M+1]. ¹H NMR (500 MHz, Chloroform-d) δ 7.77 (d, J=1.5 Hz, 1H), 7.62 (dd, J=8.3, 1.8 Hz, 1H), 7.57 (t, J=8.3 Hz, 1H), 7.31 (d, J=8.5 Hz, 1H), 7.23 (d, J=7.7 Hz, 2H), 6.94 (d, J=8.3 Hz, 1H), 6.80-6.43 (m, 2H), 4.95 (dq, J=15.8, 7.9 Hz, 1H), 3.64 (dd, J=13.9, 8.4 Hz, 1H), 3.32-3.17 (m, 1H), 3.07 (dd, J=13.9, 5.7 Hz, 1H), 2.83 (dt, J=14.5, 7.5 Hz, 1H), 1.91 (ddd, J=13.5, 11.3, 8.4 Hz, 1H), 1.54 (s, 6H), 1.48 (d, J=6.7 Hz, 3H). Human DGAT2 IC₅₀>10000 nM.

By using procedures similar to those described in Example 128 with appropriate reagents, the following compounds were synthesized. These compounds were characterized by LC/MS.

Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
129a		1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thiolan-3-yl)-2-oxoindoline-5-carboxamide (Peak 1 on OJ-H (2 × 25 cm), 20% MeOH/CO ₂)	479	>10000

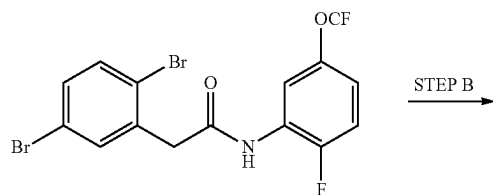
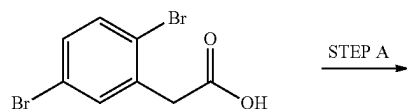
-continued

Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
129b		1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thiolan-3-yl)-2-oxo-indoline-5-carboxamide (Peak 2 on OJ-H (2 × 25 cm), 20% MeOH/CO ₂)	479	>10000
129c		1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thiolan-3-yl)-2-oxo-indoline-5-carboxamide (Peak 3 on OJ-H (2 × 25 cm), 20% MeOH/CO ₂)	479	85
129d		1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thiolan-3-yl)-2-oxo-indoline-5-carboxamide (Peak 4 on OJ-H (2 × 25 cm), 20% MeOH/CO ₂)	479	36
130a		1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-(2-methylsulfonylcyclopentyl)-2-oxo-indoline-5-carboxamide (Peak 1 on AS-H (2 × 25 cm), 10% MeOH/CO ₂)	493	>10000
130b		1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-(2-methylsulfonylcyclopentyl)-2-oxo-indoline-5-carboxamide (Peak 2 on AS-H (2 × 25 cm), 10% MeOH/CO ₂)	493	225.7

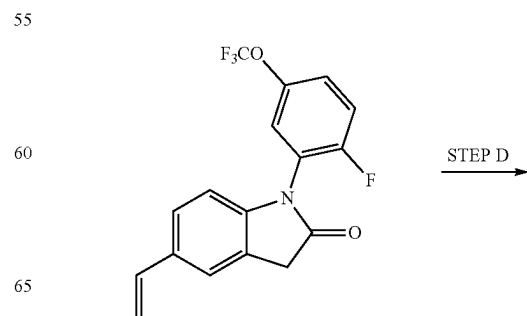
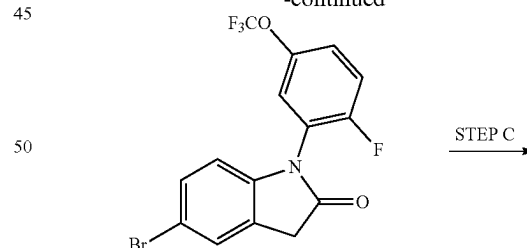
-continued

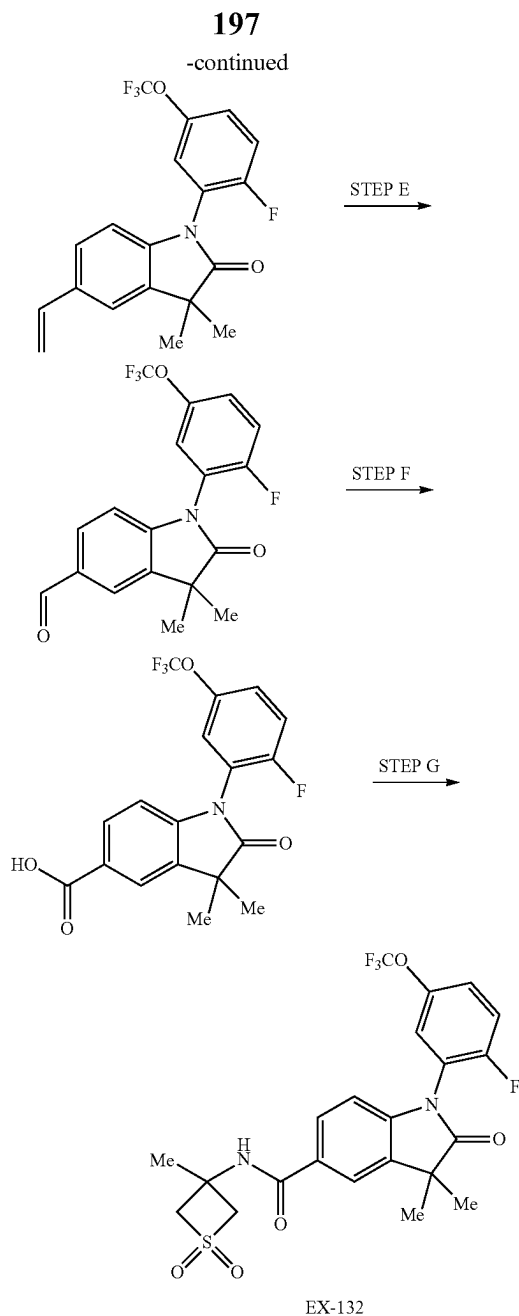
Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
130c		1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-(2-methylsulfonylcyclopentyl)-2-oxo-indoline-5-carboxamide (Peak 3 on AS-H (2 x 25 cm), 10% MeOH/CO ₂)	493	24
130d		1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-(2-methylsulfonylcyclopentyl)-2-oxo-indoline-5-carboxamide (Peak 4 on AS-H (2 x 25 cm), 10% MeOH/CO ₂)	493	536.7
131		1-[3-(difluoromethoxy)phenyl]-N-(1,1-dimethyl-2-methylsulfonyl-ethyl)-3,3-dimethyl-2-oxo-indoline-5-carboxamide	481	260

Example 132: 1-[2-fluoro-5-(trifluoromethoxy)phenyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide



-continued





Step A: 2-(2,5-dibromophenyl)-N-(2-fluoro-5-(trifluoromethoxy)phenyl) acetamide

To a stirred solution of 2-(2,5-dibromophenyl) acetic acid (1.00 g, 3.40 mmol) in pyridine (34.0 mL) was added HATU (1.36 g, 3.57 mmol), and the reaction mixture was allowed to stir at 50° C. for 30 minutes. 2-fluoro-5-(trifluoromethoxy) aniline (0.66 g, 3.40 mmol) was added via syringe, and the reaction mixture was allowed to stir at 50° C. for 16 h. The solvent was removed in vacuo. The residue was taken up in toluene, and the solvent was again removed in vacuo. The crude material was dry absorbed on SiO₂ and subjected to flash silica gel column chromatography using a 0-40% EtOAc/hexanes gradient to afford the title compound. LC/MS=470 [M+1].

198

Step B: 5-bromo-1-(2-fluoro-5-(trifluoromethoxy)phenyl) indolin-2-one

To a stirred mixture of 2-(2,5-dibromophenyl)-N-(2-fluoro-5-(trifluoromethoxy)phenyl) acetamide (1.55 g, 3.29 mmol), CuI (0.063 g, 0.33 mmol), and K₂CO₃ (0.91 g, 6.58 mmol) was added (1R,2R)-N1,N2-dimethylcyclohexane-1,2-diamine (0.094 g, 0.66 mmol) as a solution in acetonitrile (6.6 mL). The resultant reaction mixture was heated to 50° C. for 16 hours, then diluted with DCM, poured into HCl (1M aq. soln.) and extracted with DCM. The combined organic fractions were dried over Na₂SO₄ (s), filtered through a plug of SiO₂, and concentrated in vacuo. The crude residue was subjected to flash silica gel column chromatography using a 0-50% EtOAc/hexanes gradient to afford the title compound. LC/MS=390 [M+1].

Step C: 1-(2-fluoro-5-(trifluoromethoxy)phenyl)-5-vinylindolin-2-one

To a stirred solution of 5-bromo-1-(2-fluoro-5-(trifluoromethoxy)phenyl) indolin-2-one (1.05 g, 2.69 mmol), potassium vinyltrifluoroborate (0.43 g, 3.23 mmol), PdCl₂ (dppf) (0.20 g, 0.27 mmol) and K₂CO₃ (1.12 g, 8.08 mmol) in dioxane (22.5 mL) was added water (4.5 mL) under nitrogen. The reaction mixture was heated to 90° C. for 16 hours, then was diluted with DCM, dried over MgSO₄ (s), filtered, and concentrated in vacuo. The crude residue was subjected to flash silica gel column chromatography using a 0-50% EtOAc/hexanes gradient to afford the title compound. LC MS=338 [M+1].

Step D: 1-(2-fluoro-5-(trifluoromethoxy)phenyl)-3,3-dimethyl-5-vinylindolin-2-one

To a stirred solution of 1-(2-fluoro-5-(trifluoromethoxy)phenyl)-5-vinylindolin-2-one (648 mg, 1.92 mmol) and Cs₂CO₃ (1.88 g, 5.76 mmol) in DMF (7.7 mL) was added MeI (359 µl, 5.76 mmol) dropwise via syringe. The reaction mixture was stirred at 20° C. for 1 hour, then was quenched with water and extracted with DCM. The combined organic fractions were washed with NaCl (sat. aq. soln.), dried over MgSO₄ (s), filtered, and concentrated in vacuo. The crude residue was subjected to flash silica gel column chromatography using a 0-30% EtOAc/hexanes gradient to afford the title compound. LC/MS=366 [M+1].

Step E: 1-(2-fluoro-5-(trifluoromethoxy)phenyl)-3,3-dimethyl-2-oxoindoline-5-carbaldehyde

To a stirred solution of 1-(2-fluoro-5-(trifluoromethoxy)phenyl)-3,3-dimethyl-5-vinylindolin-2-one (648 mg, 1.77 mmol), 2,6-dimethylpyridine (411 µl, 3.55 mmol) and K₂[OsO₂ (OH)₄] (16.34 mg, 0.044 mmol) in dioxane (8.9 mL) and water (8.9 mL) was added NaIO₄ (1.51 g, 7.10 mmol) at 0° C. The reaction mixture was stirred at 20° C. for 3 hours, then diluted with NaCl (sat. aq. soln.) and extracted with EtOAc. The combined organic fractions were dried over MgSO₄ (s), filtered, and concentrated in vacuo. The crude material was subjected to flash silica gel column chromatography using a 0-50% EtOAc/hexanes gradient to afford the title compound. LC/MS=368 [M+1].

Step F: 1-(2-fluoro-5-(trifluoromethoxy)phenyl)-3,3-dimethyl-2-oxoindoline-5-carboxylic acid

To a stirred solution of 1-(2-fluoro-5-(trifluoromethoxy)phenyl)-3,3-dimethyl-2-oxoindoline-5-carbaldehyde (0.43

199

g, 1.17 mmol), 2-methylbut-2-ene (1.237 mL, 11.68 mmol), and NaH_2PO_4 (0.91 g, 7.59 mmol) in a mixture of t-BuOH (7.8 mL), acetone (7.8 mL), and water (7.8 mL) was added NaClO_2 (0.32 g, 3.50 mmol) at 0° C., and the reaction mixture was stirred at 20° C. for 16 hours. The crude mixture was extracted with DCM. The combined organic fractions were dried over Na_2SO_4 (s), filtered, and evaporated in vacuo to afford the crude title compound. LC/MS=384 [M+1].

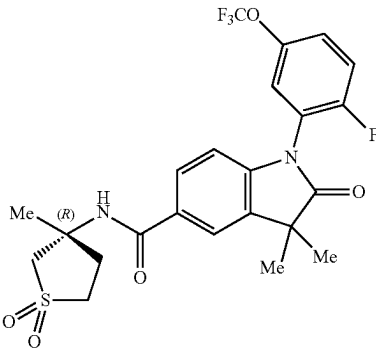
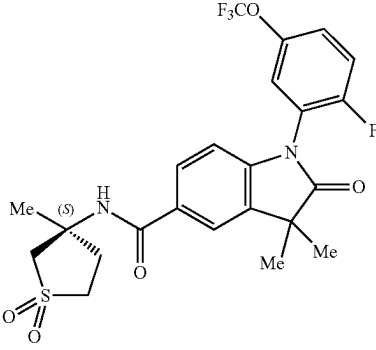
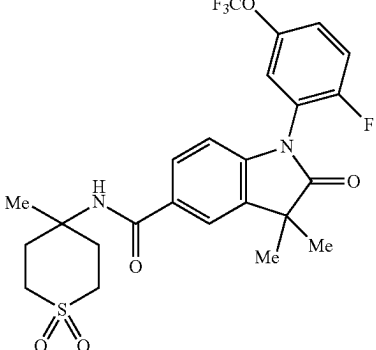
Step G: 1-[2-fluoro-5-(trifluoromethoxy)phenyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide

To a stirred solution of 1-(2-fluoro-5-(trifluoromethoxy)phenyl)-3,3-dimethyl-2-oxoindoline-5-carboxylic acid (20.0 mg, 0.052 mmol), 3-amino-3-methylthietane 1,1-dioxide

200

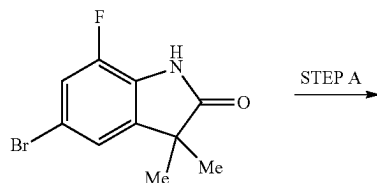
hydrochloride (8.96 mg, 0.052 mmol), and Hunig's Base (27.3 μL , 0.16 mmol) in DMF (522 μL) was added HATU (20.83 mg, 0.055 mmol), and the reaction mixture was allowed to stir at RT for 16 hours. The reaction mixture was diluted with EtOAc, filtered through a plug of SiO_2 , and concentrated in vacuo. The crude residue was purified by mass triggered reverse phase HPLC (ACN/water with 0.1% formic acid modifier) to afford the title compound. LC/MS=501 [M+1]. ^1H NMR (500 MHz, Chloroform- d) δ 7.78 (d, $J=1.6$ Hz, 1H), 7.64 (dd, $J=8.2, 1.7$ Hz, 1H), 7.37 (d, $J=5.3$ Hz, 3H), 6.77-6.70 (m, 1H), 6.37 (s, 1H), 4.61 (d, $J=14.3$ Hz, 2H), 4.23 (d, $J=14.5$ Hz, 2H), 1.95 (s, 3H), 1.57 (s, 6H). Human DGAT2 $\text{IC}_{50}=51$ nM

By using procedures similar to those described in Example 132 with appropriate reagents, the following compounds were synthesized. These compounds were characterized by LC/MS.

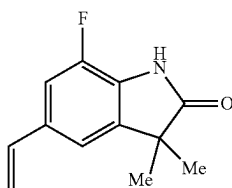
Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC_{50} (nM)
133		1-[2-fluoro-5-(trifluoromethoxy)phenyl]-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide	515	32
134		1-[2-fluoro-5-(trifluoromethoxy)phenyl]-3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide	515	72
135		1-[2-fluoro-5-(trifluoromethoxy)phenyl]-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-indoline-5-carboxamide	529	29

201

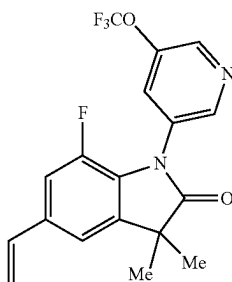
Example 136: 7-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-1-[5-(trifluoromethoxy)-3-pyridyl]indoline-5-carboxamide



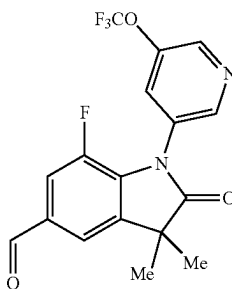
STEP A



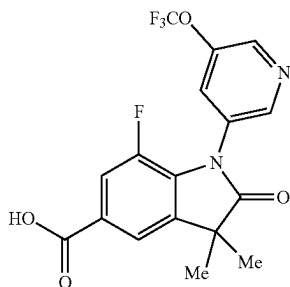
STEP B



STEP C



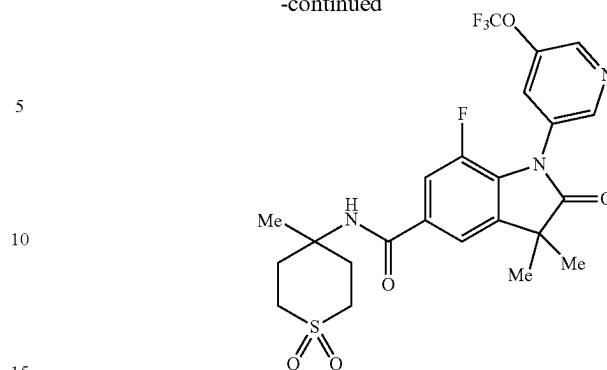
STEP D



STEP E

202

-continued



EX-136

Step A: 7-fluoro-3,3-dimethyl-5-vinylindolin-2-one

To a stirred solution of 5-bromo-7-fluoro-3,3-dimethylindolin-2-one (2.02 g, 7.83 mmol), trifluoro (vinyl)-14-borane, potassium salt (1.26 g, 9.39 mmol), PdCl₂ (dppf) (0.57 g, 0.78 mmol), Na₂CO₃ (2.49 g, 23.48 mmol) in dioxane (65.2 mL) was added water (13.0 mL) under a nitrogen atmosphere. The reaction mixture was heated to 90° C. for 6 hours, then diluted with DCM, dried over MgSO₄ (s), filtered, and concentrated in vacuo. The crude residue was subjected to flash silica gel column chromatography using a 0-50% EtOAc/hexanes gradient to afford the title compound. LC/MS=206 [M+1].

Step B: 7-fluoro-3,3-dimethyl-1-(5-(trifluoromethoxy)pyridin-3-yl)-5-vinylindolin-2-one

To a stirred mixture of 7-fluoro-3,3-dimethyl-5-vinylindolin-2-one (250 mg, 1.22 mmol), CuI (23.20 mg, 0.12 mmol), and K₂CO₃ (337 mg, 2.44 mmol) were added 3-bromo-5-(trifluoromethoxy)pyridine (590 mg, 2.44 mmol) and (1R,2R)-N1,N2-dimethylcyclohexane-1,2-diamine (34.7 mg, 0.24 mmol) as a solution in acetonitrile (2.4 mL). The reaction mixture was heated to 95° C. for 16 hours, then diluted with DCM, poured into HCl (1M aq. soln.) and extracted with DCM. The combined organic fractions were dried over Na₂SO₄ (s), filtered through a plug of SiO₂, and concentrated in vacuo. The crude residue was subjected to flash silica gel column chromatography using an EtOAc/hexanes gradient to afford the title compound. LC/MS=367 [M+1]

Step C: 7-fluoro-3,3-dimethyl-2-oxo-1-(5-(trifluoromethoxy)pyridin-3-yl) indoline-5-carbaldehyde

To a stirred solution of 7-fluoro-3,3-dimethyl-1-(5-(trifluoromethoxy)pyridin-3-yl)-5-vinylindolin-2-one (344 mg, 0.94 mmol), 2,6-dimethylpyridine (218 µL, 1.88 mmol) and K₂[OsO₂ (OH)₄] (8.65 mg, 0.023 mmol) in dioxane (4.7 mL) and water (4.7 mL) was added sodium periodate (803 mg, 3.76 mmol) at 0° C. The reaction mixture was stirred at 20° C. for 3 hours, then diluted with NaCl (sat. aq. soln.) and extracted with EtOAc. The combined organic fractions were dried over MgSO₄ (s), filtered, and concentrated in vacuo. The crude material was subjected to flash silica gel column chromatography using a 0-50% EtOAc/hexanes gradient to afford the title compound. LC/MS=369 [M+1].

203

Step D: 7-fluoro-3,3-dimethyl-2-oxo-1-(5-(trifluoromethoxy)pyridin-3-yl) indoline-5-carboxylic acid

To a stirred solution of 7-fluoro-3,3-dimethyl-2-oxo-1-(5-(trifluoromethoxy)pyridin-3-yl) indoline-5-carbaldehyde (0.32 g, 0.86 mmol), 2-methylbut-2-ene (0.91 mL, 8.58 mmol), and NaH_2PO_4 (0.67 g, 5.58 mmol) in a mixture of *t*-BuOH (5.72 mL), acetone (5.7 mL), and water (5.7 mL) was added NaClO_2 (0.23 g, 2.57 mmol) at 0° C. The reaction mixture was stirred for 16 hours at 0° C. The crude mixture was extracted with DCM. The combined organic fractions were dried over Na_2SO_4 (s), filtered, and evaporated in vacuo to provide the title compound. LCMS=385 [M+1].

Step E: 7-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxo-1-(5-(trifluoromethoxy)pyridin-3-yl) indoline-5-carboxamide

To a stirred solution of 7-fluoro-3,3-dimethyl-2-oxo-1-(5-(trifluoromethoxy)pyridin-3-yl) indoline-5-carboxylic acid

204

(30 mg, 0.078 mmol), 4-amino-4-methyltetrahydro-2H-thiopyran 1,1-dioxide hydrochloride (15.59 mg, 0.078 mmol), and Hunig's Base (40.9 μL , 0.23 mmol) in DMF (781 μL) was added HATU (31.2 mg, 0.082 mmol). The reaction mixture was allowed to stir at 20° C. for 16 hours. The crude residue was purified by mass triggered reverse phase HPLC (ACN/water with 0.1% formic acid modifier) to afford the title compound. LC/MS=530 [M+1]. ^1H NMR (500 MHz, Chloroform-*d*) δ 8.70-8.60 (m, 2H), 7.68 (s, 1H), 7.56 (d, $J=1.5$ Hz, 1H), 7.40 (dd, $J=11.4, 1.4$ Hz, 1H), 5.68 (s, 1H), 3.23-3.13 (m, 2H), 3.05 (d, $J=14.5$ Hz, 2H), 2.81 (d, $J=14.8$ Hz, 2H), 2.46-2.34 (m, 2H), 1.59 (s, 6H). Human DGAT2 $\text{IC}_{50}=40$ nM.

By using procedures similar to those described in Example 136 with appropriate reagents, the following compounds were synthesized. These compounds were characterized by LC/MS.

Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC_{50} (nM)
137		1-[5-(difluoromethoxy)-2-fluoro-phenyl]-7-fluoro-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide	501	0.51
138		1-[5-(difluoromethoxy)-2-fluoro-phenyl]-7-fluoro-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide	515	1.2
139		1-[5-(difluoromethoxy)-2-fluoro-phenyl]-7-fluoro-3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide	515	4.4

-continued

Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
140		1-[5-(difluoromethoxy)-2-fluoro-phenyl]-7-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-indoline-5-carboxamide	529	2.1
141		1-(2-ethoxy-5-fluoro-4-pyridyl)-7-fluoro-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide	480	10.1
142		1-(2-ethoxy-5-fluoro-4-pyridyl)-7-fluoro-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide	494	4.6
143		1-(2-ethoxy-5-fluoro-4-pyridyl)-7-fluoro-3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide	494	9.3

-continued

Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
144		1-(2-ethoxy-5-fluoro-4-pyridyl)-7-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-indoline-5-carboxamide	508	2.3
145		1-[5-(difluoromethoxy)-2-fluoro-phenyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-pyrrolo[2,3-b]pyridine-5-carboxamide	484	91
146		1-[5-(difluoromethoxy)-2-fluoro-phenyl]-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-pyrrolo[2,3-b]pyridine-5-carboxamide	498	351
147		1-[5-(difluoromethoxy)-2-fluoro-phenyl]-3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-pyrrolo[2,3-b]pyridine-5-carboxamide	498	227

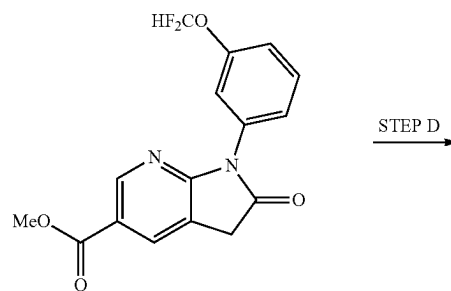
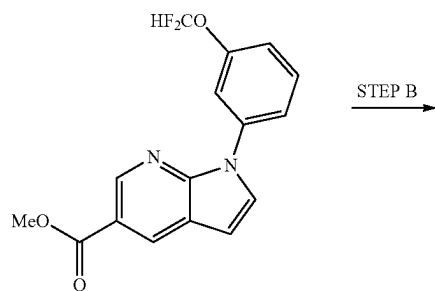
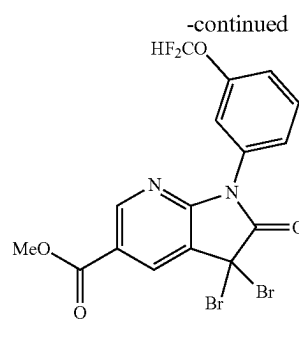
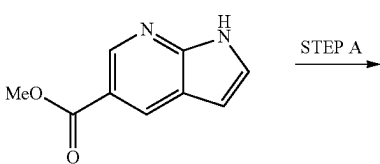
-continued

Exam- ple	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
148		1-[5-(difluoromethoxy)-2-fluoro-phenyl]-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-pyrrolo[2,3-b]pyridine-5-carboxamide	512	90
149		7-fluoro-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-1-[5-(trifluoromethoxy)-3-pyridyl]indoline-5-carboxamide	502	111
150		7-fluoro-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-1-[5-(trifluoromethoxy)-3-pyridyl]indoline-5-carboxamide	516	122
151		7-fluoro-3,3-dimethyl-N-(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-1-[5-(trifluoromethoxy)-3-pyridyl]indoline-5-carboxamide	516	454

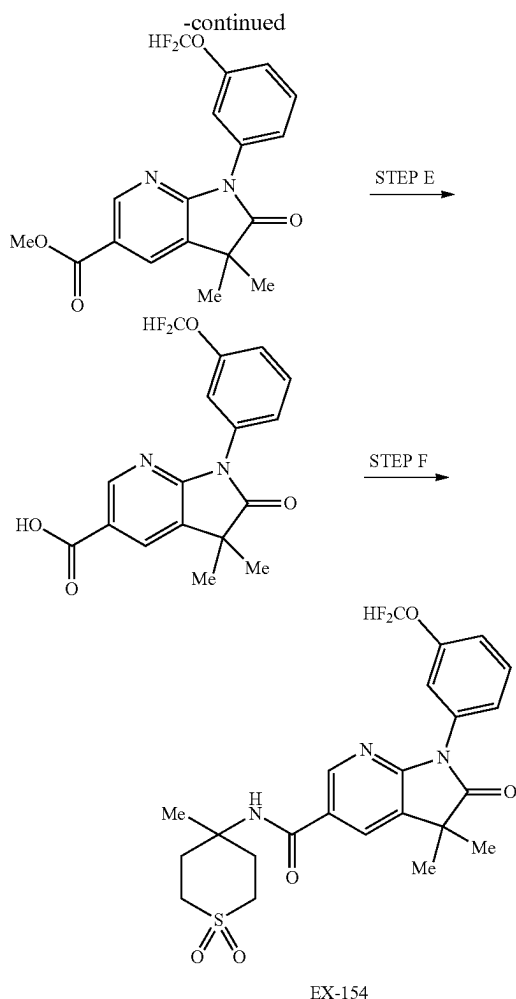
-continued

Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
152		1-(3-(difluoromethoxy)phenyl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxindoline-5-carboxamide	511	16.9
153		7-cyano-1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxindoline-5-carboxamide	490	6.8

Example 154: 1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-pyrrolo[2,3-b]pyridine-5-carboxamide



213



Step A: methyl 1-(3-(difluoromethoxy)phenyl)-1H-pyrrolo[2,3-b]pyridine-5-carboxylate

To a stirred mixture of methyl 1H-pyrrolo[2,3-b]pyridine-5-carboxylate (1.00 g, 5.68 mmol), CuI (0.11 g, 0.57 mmol), and K_3PO_4 (2.53 g, 11.92 mmol) was added 1-(difluoromethoxy)-3-iodobenzene (2.30 g, 8.51 mmol) and (1R, 2R)-N1,N2-dimethylcyclohexane-1,2-diamine (0.161 g, 1.14 mmol) as a solution in toluene (5.7 mL). The resultant reaction mixture was heated to 95° C. for 16 hours, then diluted with EtOAc, filtered through a plug of SiO_2 , and concentrated in vacuo. The crude residue was subjected to flash silica gel column chromatography using a 0-30% EtOAc/hexanes gradient to afford the title compound. LC/MS=319 [M+1].

Step B: methyl 3,3-dibromo-1-(3-(difluoromethoxy)phenyl)-2-oxo-2,3-dihydro-1H-pyrrolo[2,3-b]pyridine-5-carboxylate

To a stirred solution of methyl 1-(3-(difluoromethoxy)phenyl)-1H-pyrrolo[2,3-b]pyridine-5-carboxylate (1.60 g, 5.03 mmol) in t-BuOH (40.2 mL) and dioxane (10.05 mL) was added Pyridinium tribromide (5.14 g, 16.09 mmol) portion wise over 30 min at 20° C. After 5 hours the reaction mixture was diluted with water and extracted with DCM. The combined organic fractions were washed with $NaHCO_3$ (sat. aq. soln.) and NaCl (sat. aq. soln.), dried over

214

$MgSO_4$ (s), filtered, and concentrated in vacuo to afford the crude title compound. LC/MS=491 [M+1].

Step C: methyl 1-(3-(difluoromethoxy)phenyl)-2-oxo-2,3-dihydro-1H-pyrrolo[2,3-b]pyridine-5-carboxylate

To a stirred suspension of methyl 3,3-dibromo-1-(3-(difluoromethoxy)phenyl)-2-oxo-2,3-dihydro-1H-pyrrolo[2,3-b]pyridine-5-carboxylate (2.46 g, 5.00 mmol) in acetic acid (25 mL) was added zinc (3.27 g, 50.0 mmol) portion wise over 30 min. The suspension became solid and the mixture was sonicated for 3 h. A second portion of zinc (3.27 g, 50.0 mmol) was added and the reaction mixture was sonicated for an additional hour. The reaction mixture was diluted with DCM, filtered, and concentrated in vacuo. The crude residue was subjected to flash silica gel column chromatography using a 0-50% EtOAc/hexanes gradient to afford the title compound. LC/MS=335 [M+1].

Step D: methyl 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-2-oxo-2,3-dihydro-1H-pyrrolo[2,3-b]pyridine-5-carboxylate

To a stirred solution of methyl 1-(3-(difluoromethoxy)phenyl)-2-oxo-2,3-dihydro-1H-pyrrolo[2,3-b]pyridine-5-carboxylate (300 mg, 0.90 mmol) and Cs_2CO_3 (877 mg, 2.69 mmol) in DMF (3.6 mL) was added MeI (168 μ L, 2.69 mmol) dropwise via syringe. The reaction mixture was stirred at 20° C. for 1 hour, then quenched with water and extracted with DCM. The combined organic fractions were washed with NaCl (sat. aq. soln.), dried over $MgSO_4$ (s), filtered, and concentrated in vacuo. The crude residue was subjected to flash silica gel column chromatography using a 0-40% EtOAc/hexanes gradient to afford the title compound. LC/MS=363 [M+1].

Step E: 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-2-oxo-2,3-dihydro-1H-pyrrolo[2,3-b]pyridine-5-carboxylic acid

To a stirred solution of methyl 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-2-oxo-2,3-dihydro-1H-pyrrolo[2,3-b]pyridine-5-carboxylate (0.33 g, 0.90 mmol) and LiOH (0.22 g, 9.00 mmol) in THF (6.1 mL) and methanol (3.0 mL) was added water (1.5 mL). The reaction mixture was allowed to stir at 20° C. for 3 hours, then acidified with 1M HCl, diluted with water, and extracted with DCM. The combined organic fractions were dried over $MgSO_4$ (s), filtered, and concentrated in vacuo to afford the title compound. LC/MS=349 [M+1].

Step F: 1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-pyrrolo[2,3-b]pyridine-5-carboxamide

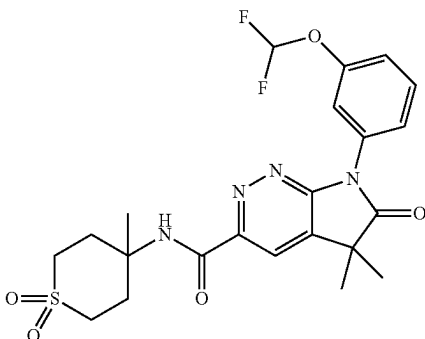
To a stirred solution of 1-(3-(difluoromethoxy)phenyl)-3,3-dimethyl-2-oxo-2,3-dihydro-1H-pyrrolo[2,3-b]pyridine-5-carboxylic acid (25 mg, 0.072 mmol), 4-amino-4-methyl-tetrahydro-2H-thiopyran 1,1-dioxide (11.72 mg, 0.072 mmol), and Hunig's Base (37.6 μ L, 0.22 mmol) in DMF (718 μ L) was added HATU (28.7 mg, 0.075 mmol). The reaction mixture was stirred at RT for 16 hours, then diluted with EtOAc, filtered through a plug of SiO_2 , and concentrated in vacuo. The crude residue was purified by mass triggered reverse phase HPLC (ACN/water with 0.1% formic acid modifier) to afford the title compound. LC/MS=494 [M+1]. 1H NMR (500 MHz, Chloroform-d) δ 8.51 (d, J=2.1 Hz, 1H), 8.00 (d, J=2.1 Hz, 1H), 7.55 (t, J=8.1 Hz, 1H), 7.51-7.46 (m, 1H), 7.42 (s, 1H), 7.21 (d, J=8.3 Hz, 1H), 6.60 (t, J=73.5 Hz, 1H), 5.71 (s, 1H), 3.22-3.12 (m, 2H), 3.04 (d,

J=14.5 Hz, 2H), 2.80 (d, J=15.1 Hz, 2H), 2.45-2.32 (m, 2H), 1.62 (s, 3H), 1.57 (d, J=3.6 Hz, 6H). Human DGAT2 IC₅₀=161 nM.

By using procedures similar to those described in Example 154 with appropriate reagents, the following compounds were synthesized. These compounds were characterized by LC/MS.

Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
155		1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-pyrrolo[2,3-b]pyridine-5-carboxamide	466	888.3
156		1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-pyrrolo[2,3-b]pyridine-5-carboxamide	480	465.1
157		1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-pyrrolo[2,3-b]pyridine-5-carboxamide	480	176.8
158		1-[3-(difluoromethoxy)phenyl]-3,3-dimethyl-N-(4-methyl-1,1-dioxido-tetrahydro-2H-thiopyran-4-yl)-2-oxo-2,3-dihydro-1H-pyrrolo[2,3-c]pyridine-5-carboxamide	494	126.5

-continued

Example	Structure	Name	LCMS [M + 1]	Human DGAT2 IC ₅₀ (nM)
159		7-(3-(difluoromethoxy)phenyl)-5,5-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-6-oxo-6,7-dihydro-5H-pyrrolo[2,3-c]pyridazine-3-carboxamide	495	1036

ASSAYS

Insect Cell Expression and Membrane Preparation

Sf-9 insect cells were maintained in Grace's insect cell culture medium with 10% heated-inactivated fetal bovine serum, 1% Pluronic F-68 and 0.14 µg/ml Kanamycine sulfate at 27° C. in a shaker incubator. After infection with untagged baculovirus expressing human DGAT2 (hDGAT2) at multiplicity of infection (MOI) 3 for 48 hours, cells were harvested. Cell pellets were suspended in buffer containing 10 mM Tris-HCl pH 7.5, 1 mM EDTA, 250 mM sucrose and Complete Protease Inhibitor Cocktail (Sigma Aldrich), and sonicated on ice. Cell debris were removed by centrifugation at 2000×g for 15 minutes. Membrane fractions were isolated by ultracentrifugation (100,000×g), resuspended in the same buffer, and frozen (−80° C.) for later use. The protein concentration was determined with the Pierce™ BCA Protein Assay Kit (Thermo Fisher Scientific). Expression of protein levels was analyzed by immunoblotting with rabbit anti-DGAT2 antibody (Abcam, ab102831) and donkey anti-rabbit IgG H&L Alexa Fluor® 647 (Abcam, ab150075) followed by detection using Typhoon FLA9000 (GE Healthcare).

LC/MS/MS Analysis Method

LC/MS/MS analyses were performed using Thermal Fisher's LX4-TSQ Vantage system. This system consists of an Agilent binary high-performance liquid chromatography (HPLC) pump and a TSQ Vantage triple quadrupole MS/MS instrument. For each sample, 2 µL samples from the top organic layer of in-plate liquid-liquid extraction were injected onto a Thermo Betabasic C4 column (2.1 mm×20 mm, 5 µm particle size). The samples were then eluted using the following conditions; mobile phase: Isopropanol:acetonitrile/10 mM ammonium formate=50/35/15 (v/v/v), flow rate: 0.8 mL/min, temperature: 25° C. Data was acquired in positive mode using a heated electrospray ionization (HESI) interface. The operational parameters for the TSQ Vantage MS/MS instrument were a spray voltage of 3000 V, capillary temperature of 280° C., vaporizer temperature 400° C., sheath gas 45 arbitrary unit, Aux gas 10 arbitrary units, S-lens 165 and collision gas 1.0 mTorr. Standard reference material (SRM) chromatograms of ¹³C₁₈-triolein (Q1:920.8>Q3:621.3) and internal standard ¹³C₂₁-triolein (Q1:923.8>Q3:617.3) were collected for 33 sec. The peak area was integrated by Xcalibur Quan soft-

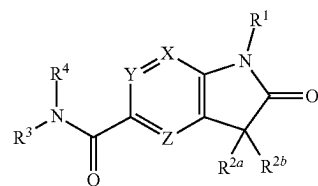
ware. The ratio between the ¹³C₁₈-triolein generated in the reaction and spiked in internal standard ¹³C₂₁-triolein was used to generate percentage inhibition and IC₅₀ values. Compound percentage inhibition was calculated by the following formula: Inhibition % = 1 - [(compound response - low control) / (high control - low control)] × 100%. Potent compounds were titrated and IC₅₀ were calculated by 4 parameter sigmoidal curve fitting formula.

DGAT2 Enzymatic Activity Assay

DGAT2 activity was determined by measuring the amount of enzymatic product ¹³C₁₈-triolein (¹³C-1,2,3-Tri (cis-9-octadecenoyl)glycerol) using the membrane prep mentioned above. The assay was carried out in ABgene 384-well assay plates in a final volume of 25 µL at rt. The assay mixture contained the following: assay buffer (100 mM Tris-Cl, pH 7.0, 20 mM MgCl₂, 5% ethanol), 25 µM of diolein, 5 µM of ¹³C oleoyl-CoA and 8 ng/µL of DGAT2 membrane.

What is claimed is:

1. A compound of Formula I:



or a pharmaceutically acceptable salt thereof wherein:

X, Y, and Z are independently selected from N and C(R⁵);
R¹ is a 5- or 6-membered heteroaryl containing 1, 2, 3 or 4 heteroatoms independently selected from N, O, and S, wherein the heteroaryl is unsubstituted or substituted with 1, 2, or 3 R⁶;

R^{2a} and R^{2b} are independently selected from

- (1) hydrogen,
- (2) halogen,
- (3) hydroxy,
- (4) (C₁₋₆)alkyl,
- (5) (C₁₋₆)haloalkyl,
- (6) R^{2a} and R^{2b} join to form spiro(C₃₋₈)cycloalkyl wherein the spiro(C₃₋₈)cycloalkyl is unsubstituted or

219

mono or di-substituted with substituents independently selected from C₁₋₃alkyl, halogen, and OH, and

- (7) R^{2a} and R^{2b} join to form spiro 4- to 8-membered heterocyclyl containing 1 or 2 heteroatoms independently selected from N, O, and S wherein the heterocyclyl is unsubstituted or substituted by 1, 2, or 3 R⁷;

R³ is

- (1) 4- to 7-membered heterocyclyl containing 1, 2 or 3 heteroatoms independently selected from N, O and S,
- (2) 5- or 6-membered heteroaryl containing 1, 2 or 3 heteroatoms independently selected from N, O, and S,
- (3) —(C₁₋₆)alkyl-heteroaryl, wherein the heteroaryl is a 5- or 6-membered heteroaryl containing 1, 2, or 3 heteroatoms independently selected from N, O and S,
- (4) —(C₁₋₆)alkyl-aryl,
- (5) —(C₁₋₆)alkyl-heterocyclyl, wherein the heterocyclyl is a 3- to 6-membered ring containing 1 or 2 heteroatoms independently selected from N, O and S,
- (6) (C₁₋₆)alkyl,
- (7) (C₃₋₆)cycloalkyl,
- (8) (C₁₋₆)hydroxyalkyl,
- (9) —(C₁₋₆)alkyl-S(O)₂—NR^{8a}R^{8b}, or
- (10) —(C₁₋₆)alkyl-S(O)₂—(C₁₋₃)alkyl,

wherein each cycloalkyl, or heterocyclyl is unsubstituted or substituted with 1, 2, or 3 R⁹, and wherein each alkyl, aryl, or heteroaryl is unsubstituted or substituted with 1, 2, or 3 R¹⁰;

R⁴ is

- (1) hydrogen, or
- (2) (C₁₋₃)alkyl,

or R³ and R⁴ combine together with the nitrogen atom to which they are attached to form a mono- or bicyclic heterocyclyl ring containing 1 or 2 heteroatoms independently selected from N, O and S, wherein the heterocyclyl ring is unsubstituted or substituted by 1, 2, or 3 R¹¹,

when present, each R⁵ is independently selected from

- (1) hydrogen,
- (2) (C₁₋₃)alkyl,
- (3) (C₁₋₃)haloalkyl,
- (4) cyano, and
- (5) halogen,

when present, each R⁶ is independently selected from

- (1) cyano,
- (2) halogen,
- (3) —OC₁₋₆alkyl
- (4) (C₃₋₆)cycloalkyl, unsubstituted or substituted with halogen, and OH,
- (5) —(C=O)NH₂,
- (6) (C₃₋₆)cycloalkyloxy wherein the cycloalkyl is unsubstituted or substituted with halogen or OH,
- (7) hydroxy,
- (8) —NR¹¹R¹¹,
- (9) —NH(C=O)(C₁₋₆)alkyl,
- (10) (C₂₋₆) cyclic amine, unsubstituted or substituted with one or two halogen substituents,
- (11) —(C₁₋₆)haloalkyl
- (12) —O(C₁₋₆)haloalkyl unsubstituted or substituted with OH,

220

- (13) —O(C₀₋₃)alkyl-(C₃₋₆)cycloalkyl optionally unsubstituted or substituted with one or two halogen substituents,

- (14) —SO₂(C₁₋₆)alkyl,
- (15) —SO₂NH(C₁₋₆)alkyl,
- (16) —SC₁₋₆alkyl,
- (17) —SC₁₋₆haloalkyl, and
- (18) (C₁₋₆)alkyl,

when present, each R⁷ is independently selected from

- (1) (C₁₋₃)alkyl,
- (2) halogen,
- (3) (C₁₋₃)alkoxy-,
- (4) —(C₁₋₃)haloalkyl, and
- (5) hydroxy,

when present, R^{8a} and R^{8b} are independently selected from

- (1) hydrogen,
- (2) (C₁₋₃)alkyl, and
- (3) (C₃₋₇)cycloalkyl;

when present, each R⁹ is independently selected from

- (1) (C₁₋₃)alkyl,
- (2) —(C₁₋₃)haloalkyl,
- (3) OXO,
- (4) (C₃₋₆)cycloalkyl,
- (5) —N(R¹¹)₂,
- (6) hydroxy,
- (7) (C₁₋₃)alkoxy,
- (8) cyano,
- (9) halogen,
- (10) —SO₂(C₁₋₆)alkyl,
- (11) —(C₁₋₆)alkyl SO₂(C₁₋₆)alkyl,
- (12) —C(O)(C₁₋₃)alkyl, and
- (13) —O(C₁₋₃)alkyl,

when present, R¹⁰ is independently selected from

- (1) hydrogen,
- (2) (C₁₋₃)alkyl,
- (3) (C₁₋₃)alkoxy-,
- (4) hydroxy,
- (5) halogen,
- (6) (C₁₋₃)alkyl-S—,
- (7) —(C₁₋₃)haloalkyl, and
- (8) N(R¹¹)₂, and

R¹¹, when present, is independently

- (1) hydrogen, or
- (2) (C₁₋₃)alkyl.

2. The compound of claim 1, or a pharmaceutically acceptable salt thereof, wherein R¹ is a 6 membered heteroaryl containing one or two nitrogen atoms substituted with one or two substituents independently selected from halogen, hydroxy, C₁₋₃alkyl, C₁₋₃haloalkyl, C₃₋₆cycloalkyl, —OC₁₋₆alkyl, —OC₁₋₃haloalkyl unsubstituted or substituted with OH, —OC₀₋₃alkyl-C₃₋₆cycloalkyl unsubstituted or substituted with 1-2 halogens, azetidiny unsubstituted or substituted with 1-2 halogens, CN, —SC₁₋₃alkyl, and —SC₁₋₃haloalkyl.

3. The compound of claim 2, or a pharmaceutically acceptable salt thereof, wherein R¹ is a 6 membered heteroaryl containing one or two nitrogen atoms substituted with one or two substituents independently selected from: C₁₋₃alkyl; C₁₋₃haloalkyl-OC₁₋₆alkyl; —OC₁₋₃haloalkyl unsubstituted or substituted with —OH; —O-cyclopropyl unsubstituted or substituted with halogen; —O—C₁₋₃alkyl-C₃₋₆cycloalkyl unsubstituted or substituted with one or two fluoro substituents; —SC₁₋₃alkyl; —SC₁₋₃haloalkyl; azetidiny unsubstituted or substituted with one or two fluoro substituents; and halogen.

221

4. The compound of claim 3 or a pharmaceutically acceptable salt thereof, wherein R^1 is a pyridinyl, substituted with one or two substituents independently selected from F; $-\text{CH}_3$; $-\text{CF}_2\text{CH}_3$; $-\text{OCH}(\text{CH}_3)_2$; $-\text{OCHF}_2$; $-\text{OCH}_2\text{CH}_3$; $-\text{OCF}_2\text{CH}_3$; $-\text{OCF}_2\text{CHF}_2$; $-\text{OCH}_2\text{CH}(\text{CH}_3)_2$; $-\text{OCH}_2\text{CHF}_2$; $-\text{OCH}_2\text{CF}_3$; $-\text{OCF}_3$; O-cyclopropyl; $-\text{O}-\text{CH}_2$ -cyclopropyl unsubstituted or substituted with 1 or 2 F; $-\text{O}-\text{CH}_2$ -cyclobutyl unsubstituted or substituted with 2 F; $-\text{OCH}_2\text{CF}_2\text{CH}_3$; $-\text{OCH}_2\text{CF}_2\text{CHF}_2$; $-\text{OCF}_2\text{CH}_2\text{OH}$; azetidiny; 3,3-difluoroazetidiny; and $-\text{SCHF}_2$.

5. The compound of claim 1, or a pharmaceutically acceptable salt thereof, wherein R^{2a} and R^{2b} are selected from one of the following:

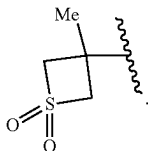
- (1) R^{2a} and R^{2b} are hydrogen,
- (2) R^{2a} is hydrogen and R^{2b} is (C_{1-6}) alkyl, or R^{2a} is (C_{1-6}) alkyl and R^{2b} is hydrogen,
- (3) R^{2a} is halogen and R^{2b} is (C_{1-6}) alkyl, or R^{2a} is (C_{1-6}) alkyl and R^{2b} is halogen,
- (4) R^{2a} is hydroxy and R^{2b} is (C_{1-6}) alkyl, or R^{2a} is (C_{1-6}) alkyl and R^{2b} is hydroxy,
- (5) R^{2a} is hydroxy and R^{2b} is (C_{1-6}) haloalkyl, or R^{2a} is (C_{1-6}) haloalkyl and R^{2b} is hydroxy,
- (6) R^{2a} and R^{2b} are each independently selected from (C_{1-6}) alkyl,
- (7) R^{2a} and R^{2b} join to form spiro 4- to 8-membered heterocyclyl containing 1 or 2 heteroatoms independently selected from N, O and S wherein the spiroheterocyclyl is unsubstituted or substituted by 1, 2, or 3 R^7 , or
- (8) R^{2a} and R^{2b} join to form spiro (C_{3-8}) cycloalkyl unsubstituted or optionally mono-substituted, or di-substituted with C_{1-63} alkyl, halogen, or OH.

6. The compound of claim 5, or a pharmaceutically acceptable salt thereof, wherein R^{2a} and R^{2b} are independently selected from H, and C_{1-3} alkyl, or together form a spiro C_{3-8} cycloalkyl.

7. The compound of claim 1, or a pharmaceutically acceptable salt thereof, wherein R^3 is

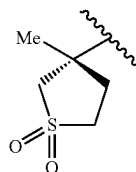
- (1) 4- to 6-membered heterocyclyl containing 1, 2 or 3 heteroatoms independently selected from N, O and S,
- (2) (C_{3-6}) cycloalkyl,
- (3) (C_{1-6}) hydroxyalkyl,
- (4) $-(\text{C}_{1-6})$ alkyl- $\text{S}(\text{O})_2-\text{NR}^{8a}\text{R}^{8b}$,
- (5) $-(\text{C}_{1-6})$ alkyl- $\text{S}(\text{O})_2-(\text{C}_{1-3})$ alkyl,
- (6) $-(\text{C}_{1-6})$ alkyl-heterocyclyl, wherein the heterocyclyl is a 5-membered heterocyclyl containing one oxygen,
- (7) 5- to 6-membered heteroaryl containing 1, 2, or 3 heteroatoms independently selected from N, O and S, or
- (8) $-(\text{C}_{1-6})$ alkyl-heteroaryl, wherein the heteroaryl is a 5-membered heteroaryl containing 2 N heteroatoms, wherein each cycloalkyl, or heterocyclyl is unsubstituted or substituted with 1, 2, or 3 R^9 , and wherein each alkyl or heteroaryl is unsubstituted or substituted with 1, 2, or 3 R^{10} .

8. The compound of claim 1, or a pharmaceutically acceptable salt thereof, wherein R^3 is

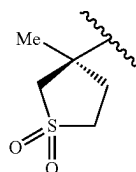


222

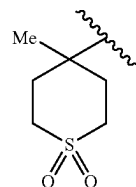
9. The compound of claim 1, or a pharmaceutically acceptable salt thereof, wherein R^3 is



10. The compound of claim 1, or a pharmaceutically acceptable salt thereof, wherein R^3 is



11. The compound of claim 1, or a pharmaceutically acceptable salt thereof, wherein R^3 is



12. The compound of claim 1, or a pharmaceutically acceptable salt thereof, wherein R^4 is H.

13. The compound of claim 1, or a pharmaceutically acceptable salt thereof, wherein R^5 is H, F, Cl, or CN.

14. The compound of claim 1, or a pharmaceutically acceptable salt thereof, wherein R^6 is halogen; hydroxy; $-\text{CN}$; $-\text{C}_{1-6}$ alkyl; $-\text{C}_{1-6}$ haloalkyl; $-\text{C}_{3-6}$ cycloalkyl unsubstituted or substituted with halogen or OH; $-\text{OC}_{1-6}$ alkyl; $-\text{OC}_{1-6}$ haloalkyl unsubstituted or substituted with OH; $-\text{O}(\text{C}_{1-3})$ alkyl- (C_{3-6}) cycloalkyl unsubstituted or substituted with one or two halogen substituents; $-\text{OC}_{3-6}$ cycloalkyl; $-\text{SO}_2\text{NH}(\text{C}_{1-6})$ alkyl; $-\text{S}(\text{O})_2\text{C}_{1-6}$ alkyl; $-\text{S}-\text{C}_{1-6}$ alkyl; $-\text{SC}_{1-6}$ haloalkyl; $-\text{NH}(\text{C}-\text{O})\text{C}_{1-3}$ alkyl; (C_{2-6}) cyclic amine unsubstituted or substituted with one or two halogen substituents; or $-\text{N}((\text{C}_{1-3})\text{alkyl})_2$.

15. The compound of claim 14, or a pharmaceutically acceptable salt thereof, wherein R^6 is F; $-\text{CN}$; $-\text{OH}$; $-\text{CH}_3$; $-\text{CF}_2\text{CH}_3$; $-\text{OCF}_2\text{CH}_3$; $-\text{OCHF}_2$; $-\text{OCF}_3$; $-\text{OCH}_2\text{CF}_3$; $-\text{OCH}_2\text{CH}_3$; $-\text{OCH}(\text{CH}_3)_2$; $-\text{OCF}_2\text{CHF}_2$; $-\text{OCH}_2\text{CHF}_2$; $-\text{OCHF}_2\text{CH}_3$; $-\text{OCH}_2\text{CF}_2\text{CH}_3$; $\text{OCH}_2\text{CH}(\text{CH}_3)_2$; $\text{OCH}_2\text{CF}_2\text{CHF}_2$; -cyclopropyl; -O-cyclopropyl; OCH_2 -cyclopropyl unsubstituted or substituted with 1 or 2 F; $-\text{OCH}_2$ -cyclobutyl unsubstituted or substituted with 1 or 2 F; azetidine unsubstituted or substituted with 1 or 2 F; $-\text{SCHF}_2$; $-\text{S}(\text{O})_2\text{CH}_3$; $-\text{S}(\text{O})_2\text{NHCH}_3$; $-\text{N}(\text{CH}_3)_2$; $-\text{OCF}_2\text{CH}_2\text{OH}$; or $-\text{NH}(\text{C}=\text{O})\text{C}_{1-3}$ alkyl.

16. The compound of claim 1, or a pharmaceutically acceptable salt thereof, wherein R^{8a} and R^{8b} are independently selected from hydrogen and cyclohexyl.

223

17. The compound of claim 1, or a pharmaceutically acceptable salt thereof, wherein R^9 is $=O$, $-OH$, halogen, $OC_{1-3}alkyl$, $C_{1-3}alkyl$, $C_{1-3}haloalkyl$, $-SO_2(C_{1-3})alkyl$, $-C_{(1-3)}alkylSO_2(C_{1-3})alkyl$, or $C(O)C_{1-3}alkyl$.

18. The compound of claim 1, or a pharmaceutically acceptable salt thereof, wherein R^{10} is hydrogen, $-OH$, $-CH_3$, $-CH_2CH_3$, F, or CF_3 .

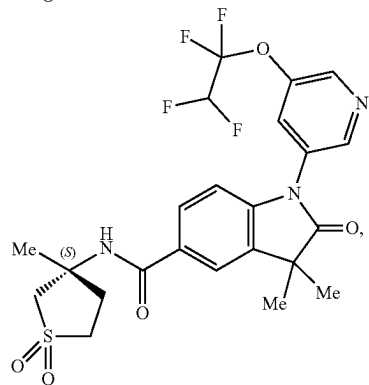
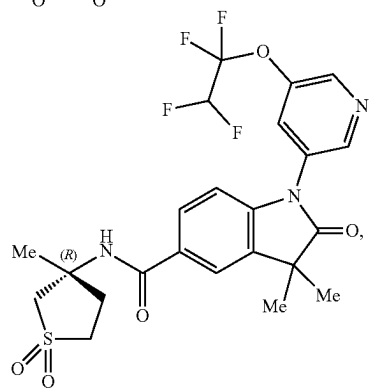
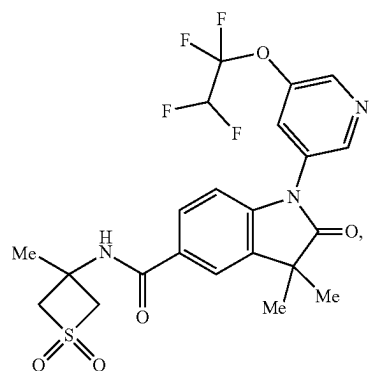
19. The compound of claim 1, or a pharmaceutically acceptable salt thereof, wherein X is N, Y is $C(R^5)$, and Z is $C(R^5)$.

20. The compound of claim 1, or a pharmaceutically acceptable salt thereof, wherein X is $C(R^5)$, Y is N, and Z is $C(R^5)$.

21. The compound of claim 1, or a pharmaceutically acceptable salt thereof, wherein X, Y, and Z are each $C(R^5)$.

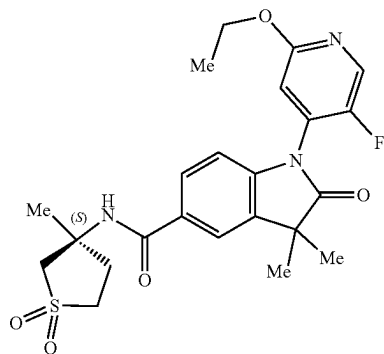
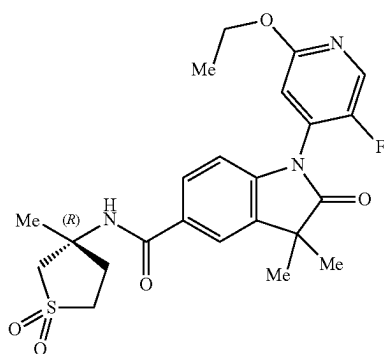
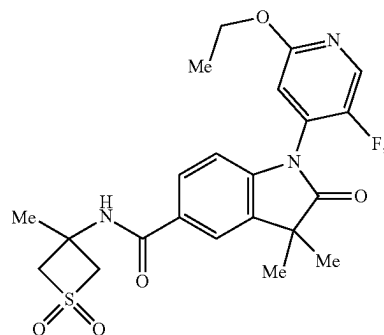
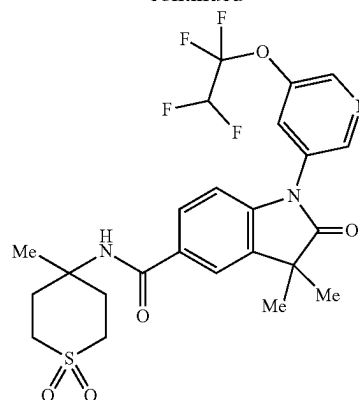
22. The compound of claim 1, or a pharmaceutically acceptable salt thereof, wherein X is N, Y is N and Z is $C(R^5)$.

23. The compound of claim 1, which is:



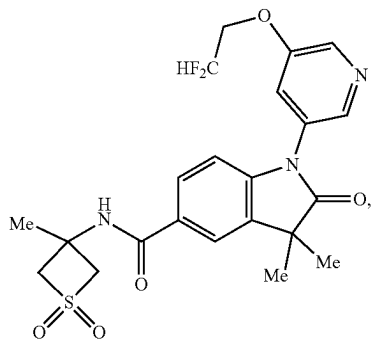
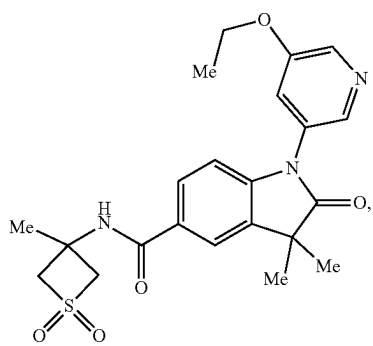
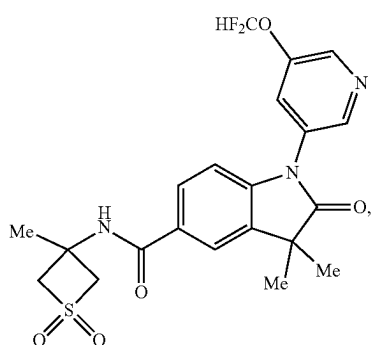
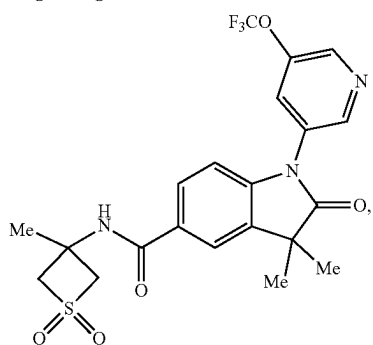
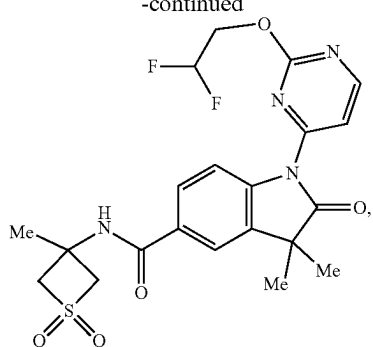
224

-continued

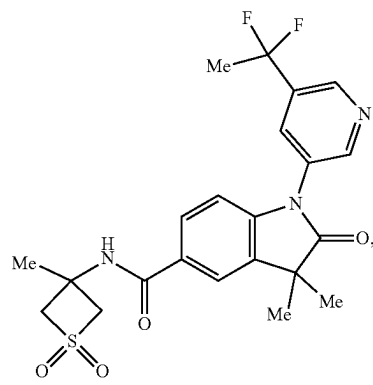
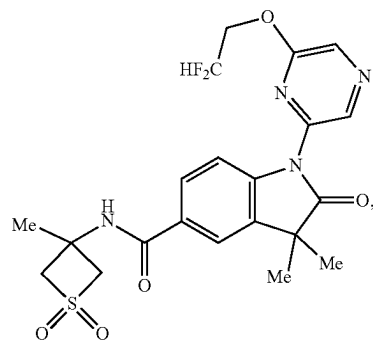
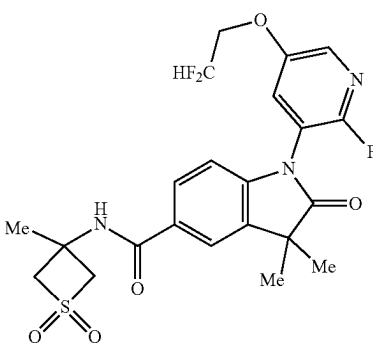
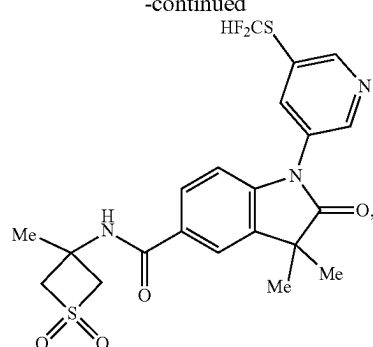


225

-continued

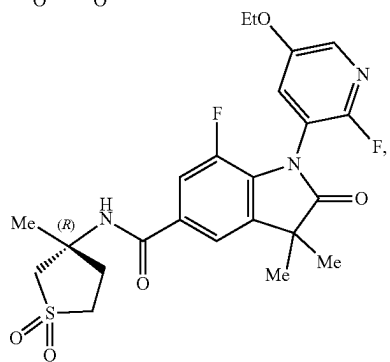
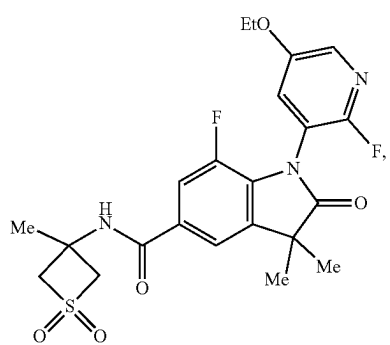
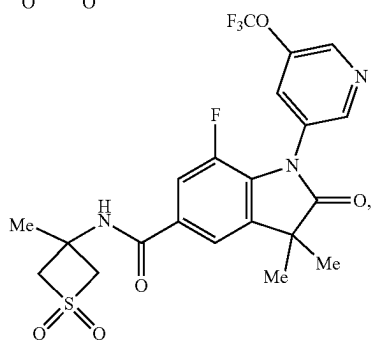
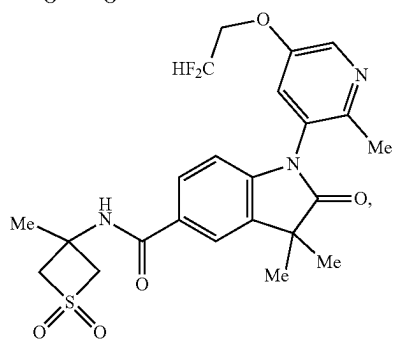
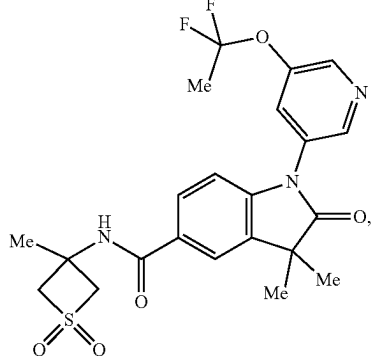
**226**

-continued



227

-continued

**228**

-continued

5

10

15

20

25

30

35

40

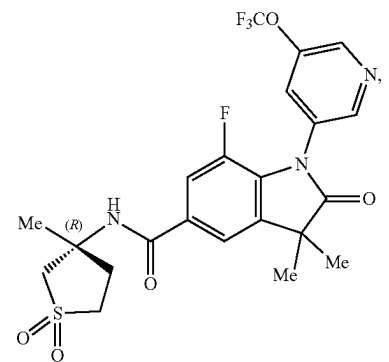
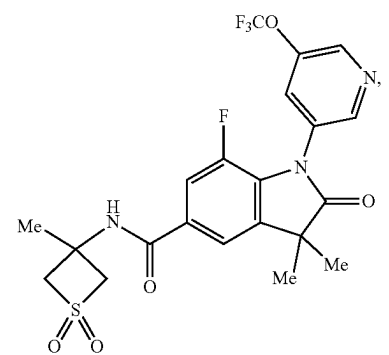
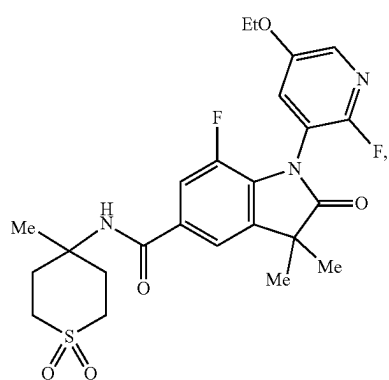
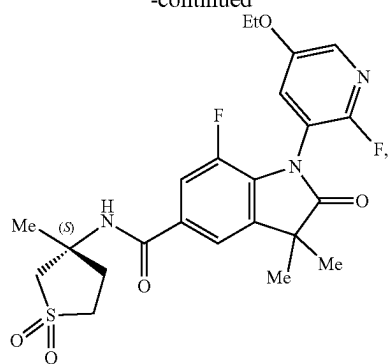
45

50

55

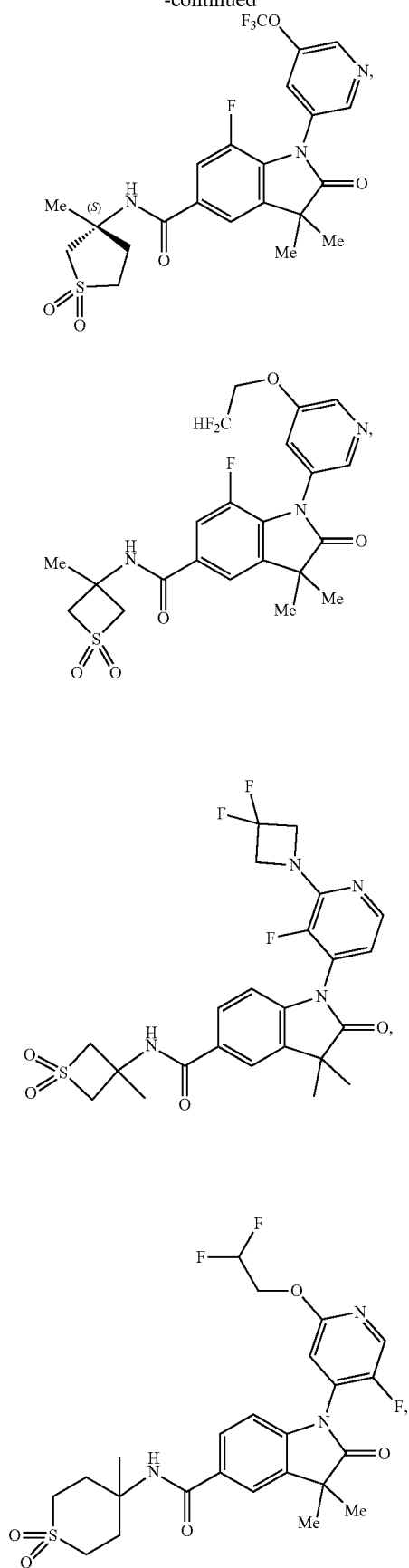
60

65

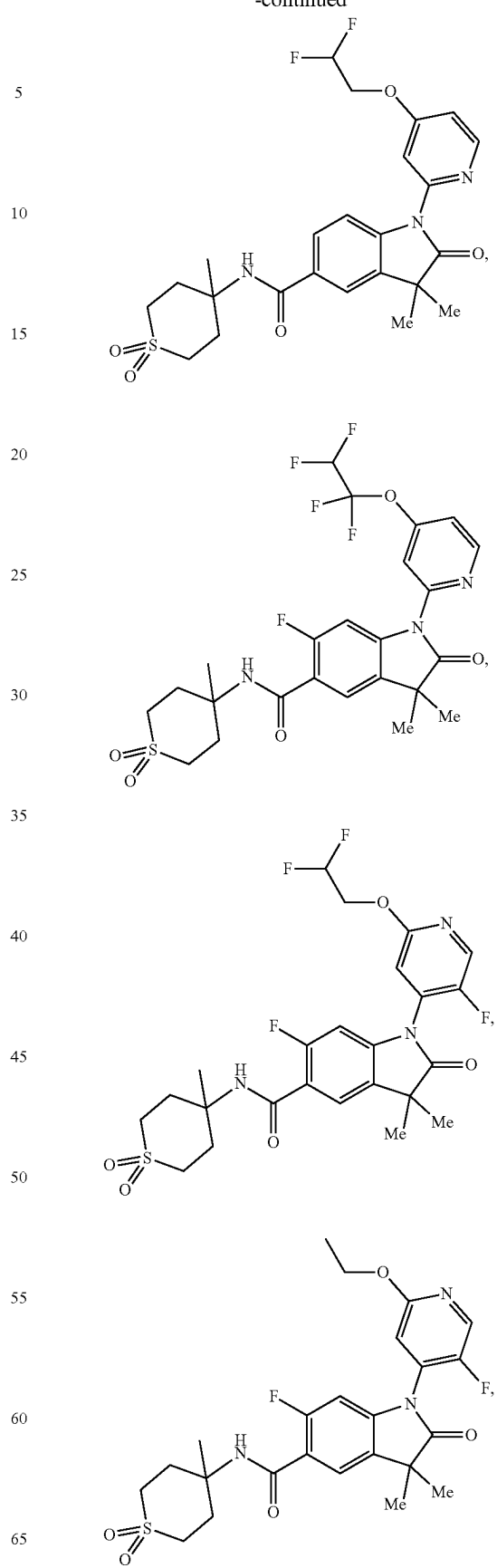


229

-continued

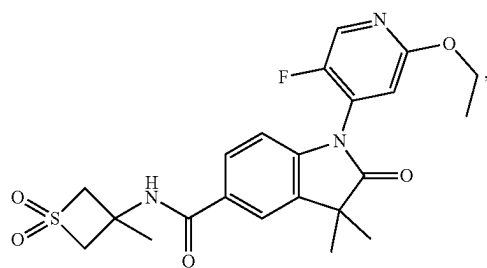
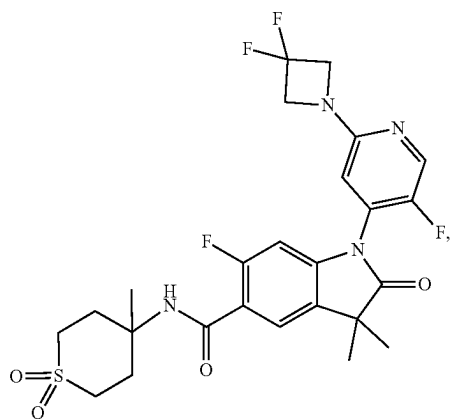
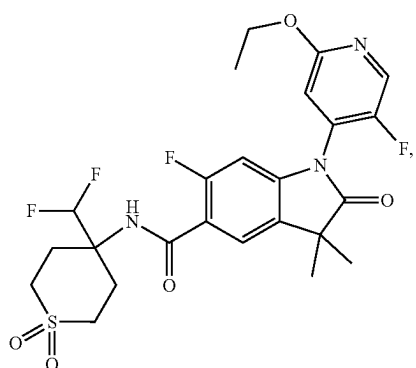
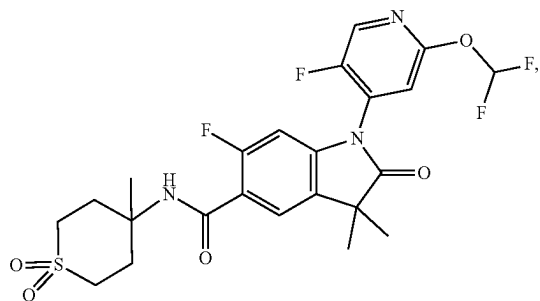
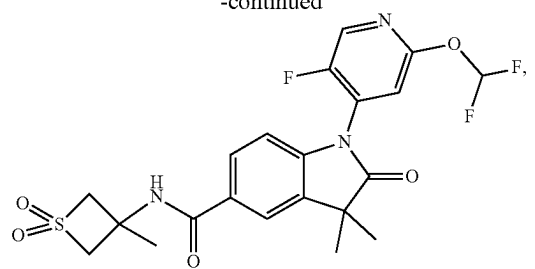
**230**

-continued

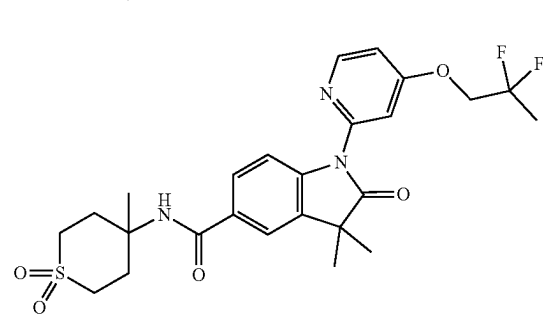
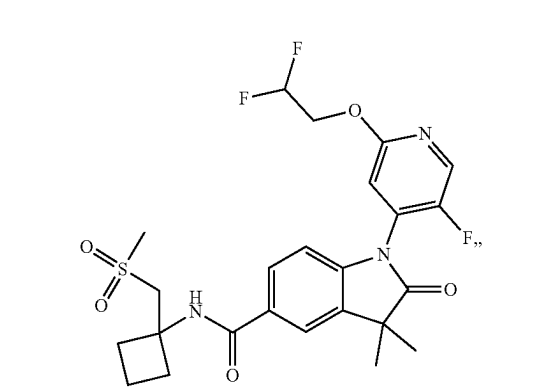
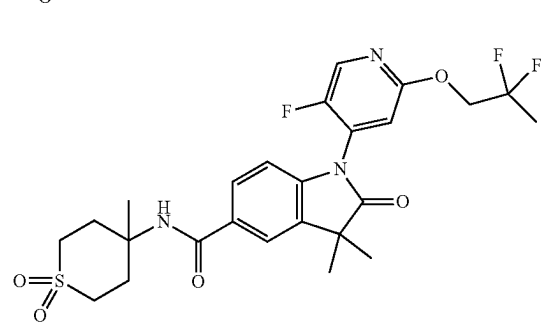
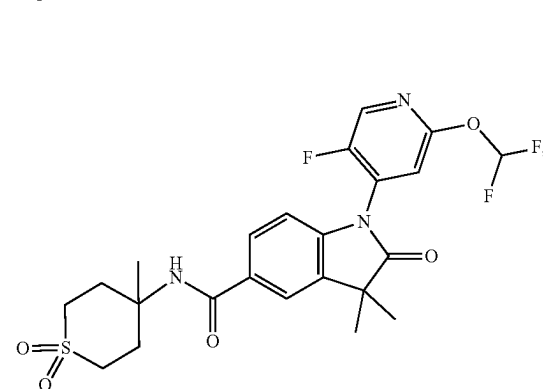
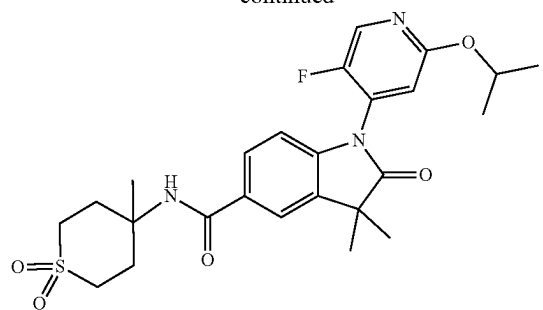


231

-continued

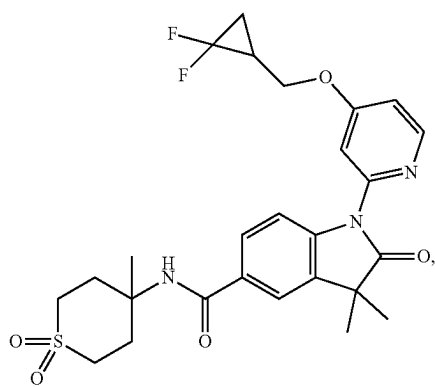
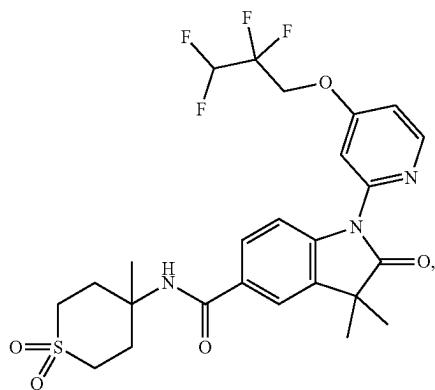
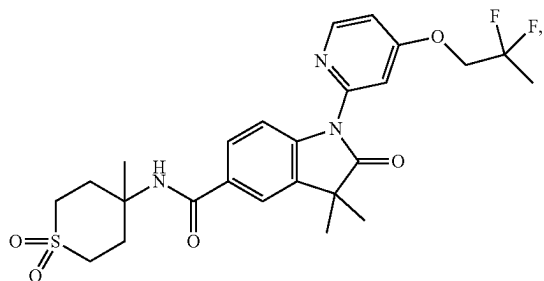
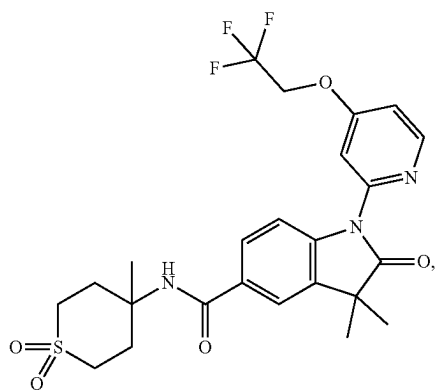
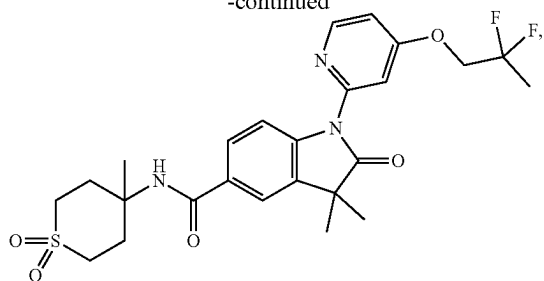
**232**

-continued

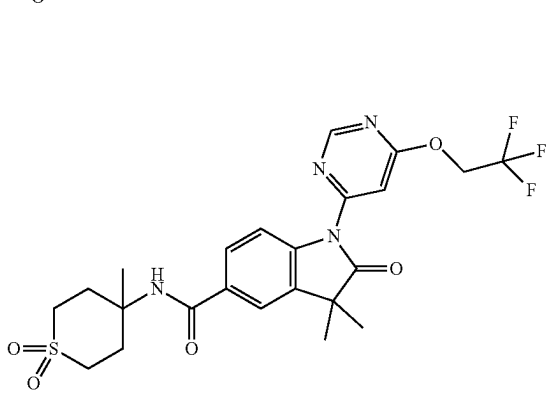
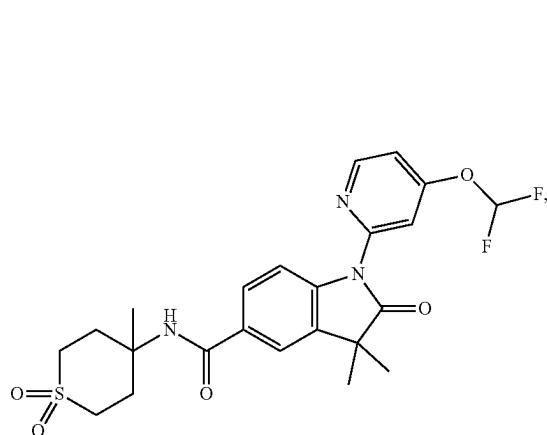
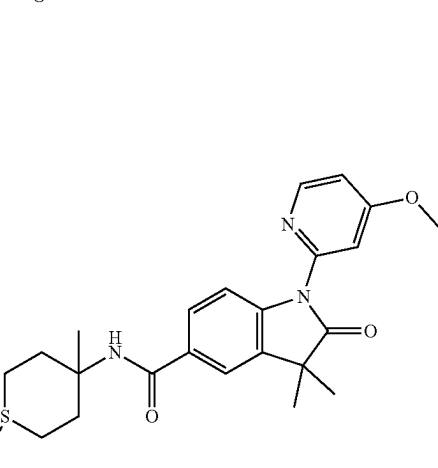
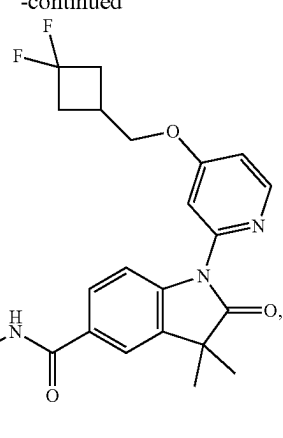


233

-continued

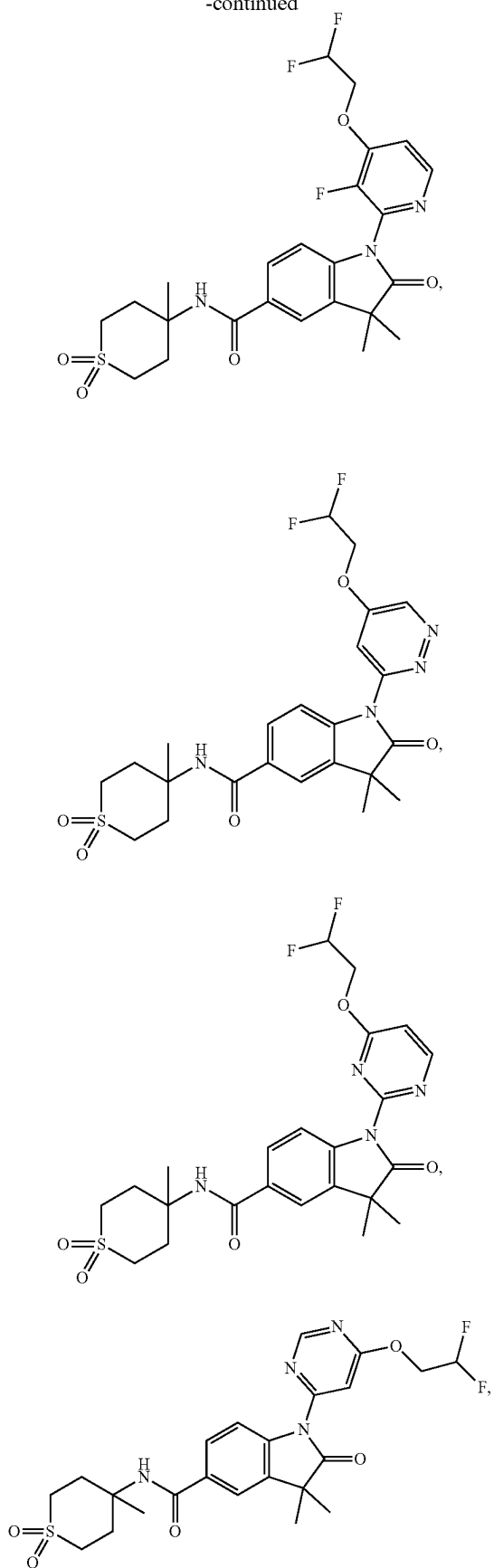
**234**

-continued

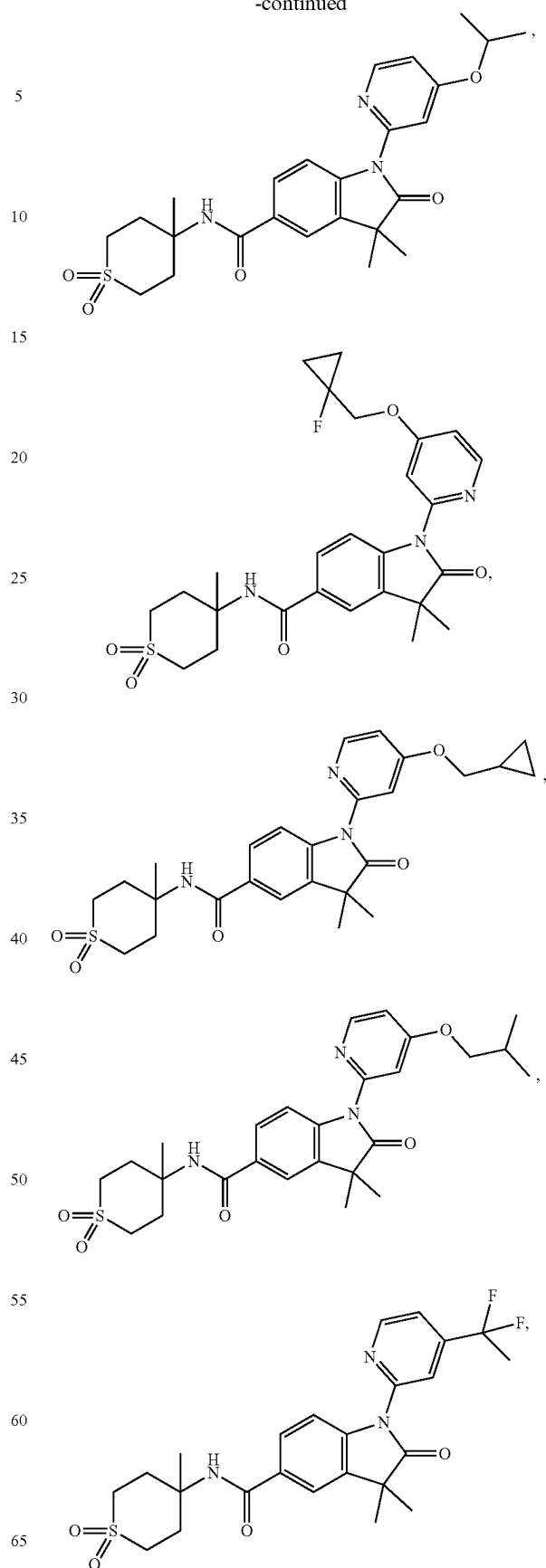


235

-continued

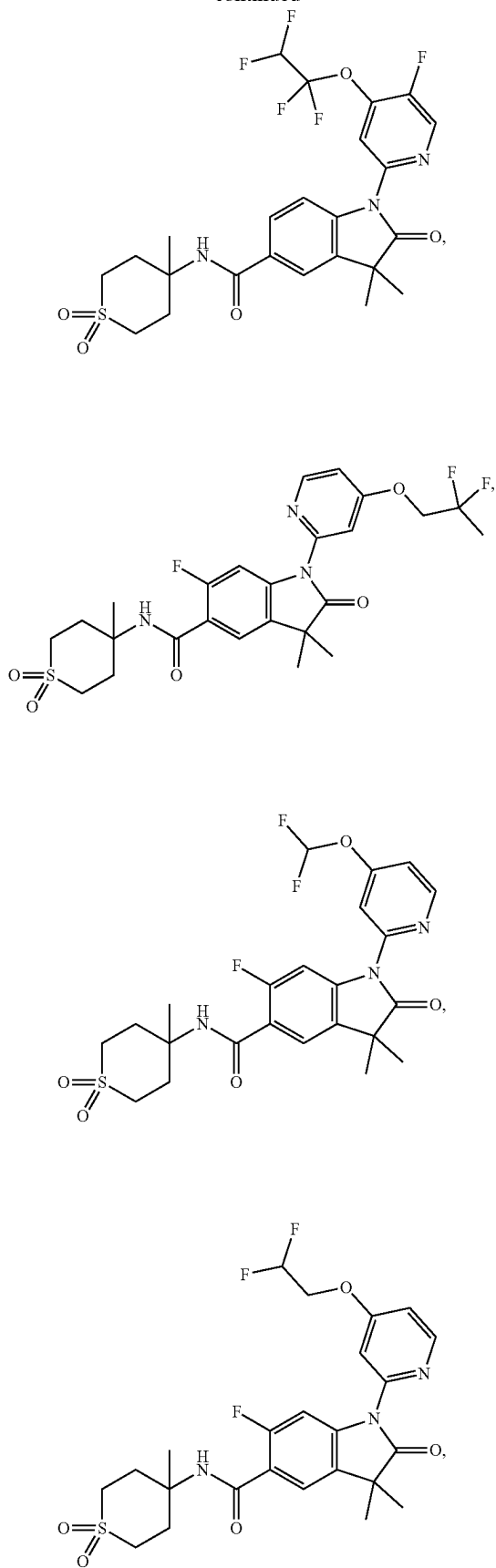
**236**

-continued

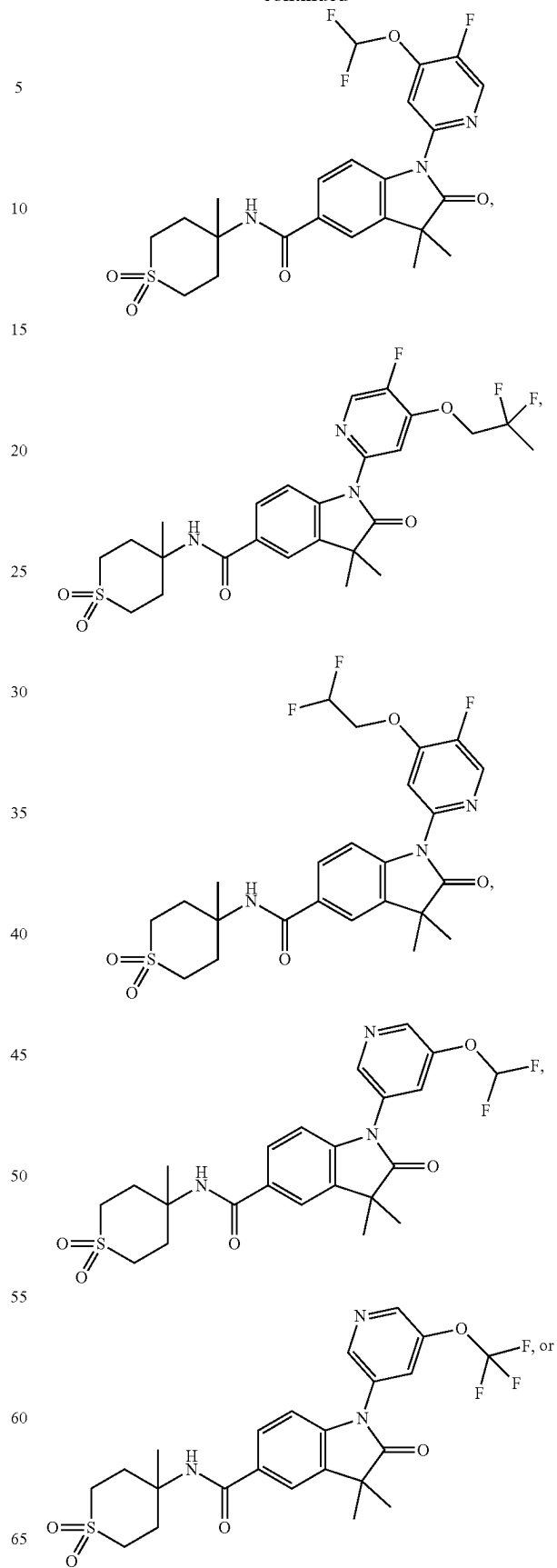


237

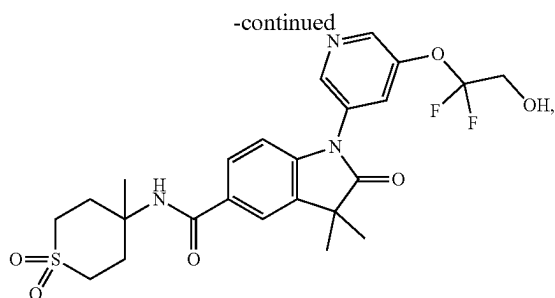
-continued

**238**

-continued



239



or a pharmaceutically acceptable salt thereof.

24. The compound of claim 1 which is:

- 3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-1-[5-(1,1,2,2-tetrafluoroethoxy)-3-pyridyl]indoline-5-carboxamide,
- 3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-1-[5-(1,1,2,2-tetrafluoroethoxy)-3-pyridyl]indoline-5-carboxamide,
- 3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-1-[5-(1,1,2,2-tetrafluoroethoxy)-3-pyridyl]indoline-5-carboxamide,
- 3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-1-[5-(1,1,2,2-tetrafluoroethoxy)-3-pyridyl]indoline-5-carboxamide,
- 1-(2-ethoxy-5-fluoro-4-pyridyl)-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
- 1-(2-ethoxy-5-fluoro-4-pyridyl)-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide,
- 1-(2-ethoxy-5-fluoro-4-pyridyl)-3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide,
- 1-[2-(2,2-difluoroethoxy)pyrimidin-4-yl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
- 3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-1-[5-(trifluoromethoxy)-3-pyridyl]indoline-5-carboxamide,
- 1-[5-(difluoromethoxy)-3-pyridyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
- 1-(5-ethoxy-3-pyridyl)-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
- 1-[5-(2,2-difluoroethoxy)-3-pyridyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
- 1-[5-(difluoromethylsulfanyl)-3-pyridyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
- 1-[2-(2,2-difluoroethoxy)-5-fluoro-4-pyridyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
- 1-[6-(2,2-difluoroethoxy) pyrazin-2-yl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
- 1-[5-(1,1-difluoroethyl)-3-pyridyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
- 1-[5-(1,1-difluoroethoxy)-3-pyridyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,

240

- 1-[5-(2,2-difluoroethoxy)-2-methyl-3-pyridyl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
- 7-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-1-[5-(trifluoromethoxy)-3-pyridyl]indoline-5-carboxamide,
- 1-(2-ethoxy-5-fluoro-4-pyridyl)-7-fluoro-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
- 1-(2-ethoxy-5-fluoro-4-pyridyl)-7-fluoro-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide,
- 1-(2-ethoxy-5-fluoro-4-pyridyl)-7-fluoro-3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-indoline-5-carboxamide,
- 1-(2-ethoxy-5-fluoro-4-pyridyl)-7-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxo-thian-4-yl)-2-oxo-indoline-5-carboxamide,
- 7-fluoro-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-1-[5-(trifluoromethoxy)-3-pyridyl]indoline-5-carboxamide,
- 7-fluoro-3,3-dimethyl-N-[(3R)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-1-[5-(trifluoromethoxy)-3-pyridyl]indoline-5-carboxamide,
- 7-fluoro-3,3-dimethyl-N-[(3S)-3-methyl-1,1-dioxo-thiolan-3-yl]-2-oxo-1-[5-(trifluoromethoxy)-3-pyridyl]indoline-5-carboxamide,
- 1-[6-(2,2-difluoroethoxy)pyridazin-4-yl]-3,3-dimethyl-N-(3-methyl-1,1-dioxo-thietan-3-yl)-2-oxo-indoline-5-carboxamide,
- 1-(2-(3,3-difluoroazetidin-1-yl)-3-fluoropyridin-4-yl)-3,3-dimethyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide,
- 1-(2-(2,2-difluoroethoxy)-5-fluoropyridin-4-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
- 1-(4-(2,2-difluoroethoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
- 3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxo-1-(4-(1,1,2,2-tetrafluoroethoxy)pyridin-2-yl) indoline-5-carboxamide,
- 1-(2-(2,2-difluoroethoxy)-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
- 1-(2-ethoxy-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
- 1-(2-ethoxy-5-fluoropyridin-4-yl)-3,3-dimethyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide,
- 3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxo-1-(4-(2,2,2-trifluoroethoxy)pyridin-2-yl) indoline-5-carboxamide,
- 1-(4-(2,2-difluoropropoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
- 3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxo-1-(4-(2,2,3,3-tetrafluoropropoxy)pyridin-2-yl) indoline-5-carboxamide,
- 1-(4-((2,2-difluorocyclopropyl) methoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
- 1-(4-((3,3-difluorocyclobutyl) methoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,

241

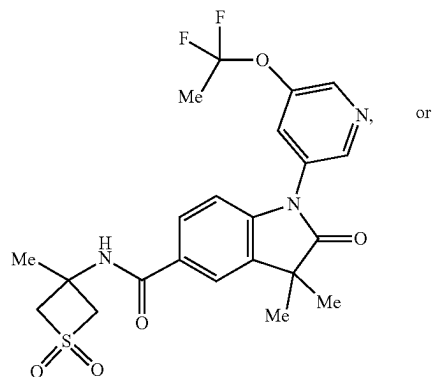
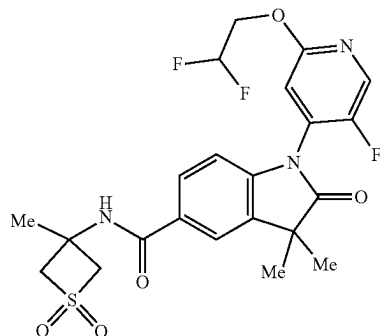
- 1-(4-cyclopropoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-(difluoromethoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(2-(difluoromethoxy)-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxo-1-(6-(2,2,2-trifluoroethoxy)pyrimidin-4-yl) indoline-5-carboxamide,
 1-(5-(difluoromethoxy)pyridin-3-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxo-1-(5-(trifluoromethoxy)pyridin-3-yl) indoline-5-carboxamide,
 1-(2-(difluoromethoxy)-5-fluoropyridin-4-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(2-(difluoromethoxy)-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-N-(3-methyl-1,1-dioxidothietan-3-yl)-2-oxoindoline-5-carboxamide,
 1-(2-(2,2-difluoroethoxy)-5-fluoropyridin-4-yl)-3,3-dimethyl-N-(1-((methylsulfonyl)methyl)cyclobutyl)-2-oxoindoline-5-carboxamide,
 1-(4-(2,2-difluoroethoxy)-3-fluoropyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(2-(2,2-difluoropropoxy)-5-fluoropyridin-4-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(5-(2,2-difluoroethoxy)pyridazin-3-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-(2,2-difluoroethoxy)pyrimidin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(6-(2,2-difluoroethoxy)pyrimidin-4-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-isopropoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-(1-fluorocyclopropyl) methoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-(cyclopropylmethoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-isobutoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-(1,1-difluoroethyl) pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(5-fluoro-4-(1,1,2,2-tetrafluoroethoxy)pyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(5-(1,1-difluoro-2-hydroxyethoxy)pyridin-3-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(2-(3,3-difluoroazetidin-1-yl)-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,

242

- 1-(4-(2,2-difluoropropoxy)pyridin-2-yl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-(difluoromethoxy)pyridin-2-yl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-(2,2-difluoroethoxy)pyridin-2-yl)-6-fluoro-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-(difluoromethoxy)-5-fluoropyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-(2,2-difluoropropoxy)-5-fluoropyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,
 1-(4-(2,2-difluoroethoxy)-5-fluoropyridin-2-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide, N-(4-(difluoromethyl)-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-1-(2-ethoxy-5-fluoropyridin-4-yl)-6-fluoro-3,3-dimethyl-2-oxoindoline-5-carboxamide, or
 1-(5-fluoro-2-isopropoxy)pyridin-4-yl)-3,3-dimethyl-N-(4-methyl-1,1-dioxidotetrahydro-2H-thiopyran-4-yl)-2-oxoindoline-5-carboxamide,

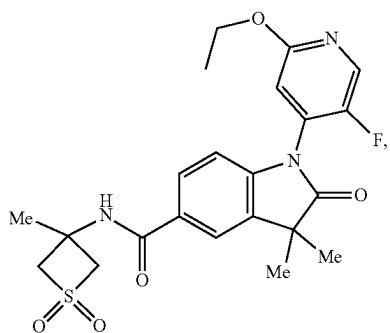
or a pharmaceutically acceptable salt thereof.

25. A compound which is:

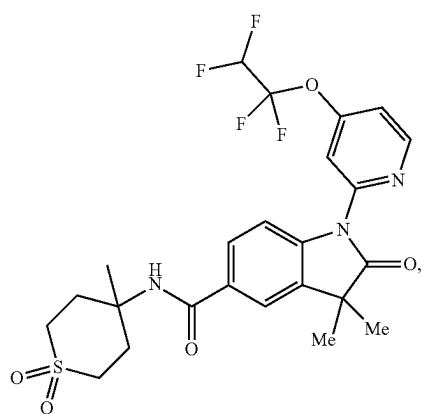
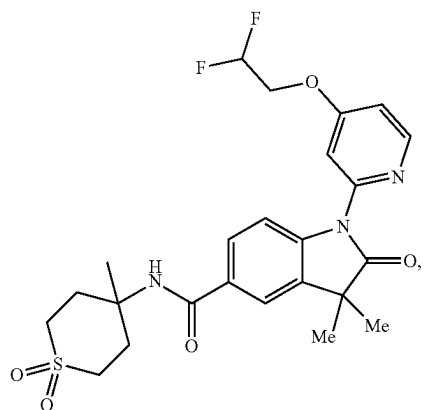
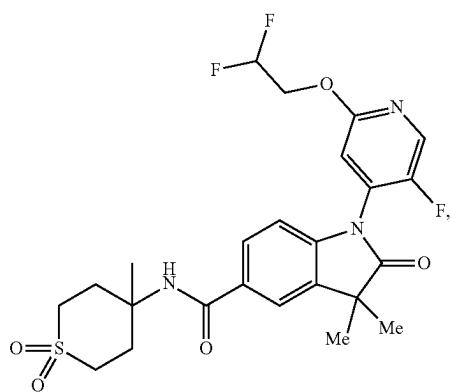


243

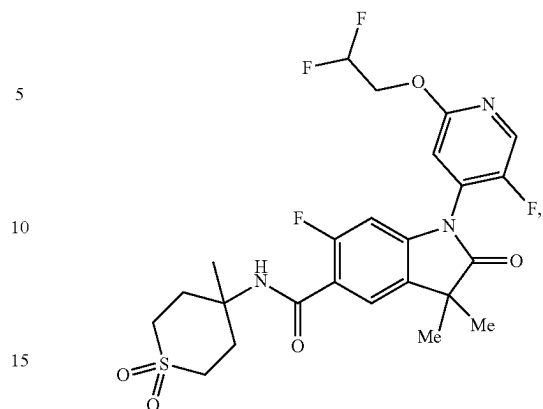
-continued



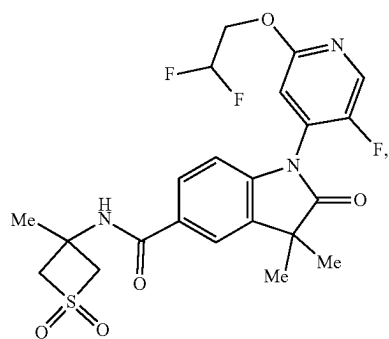
or a pharmaceutically acceptable salt thereof.

26. A compound which is:**244**

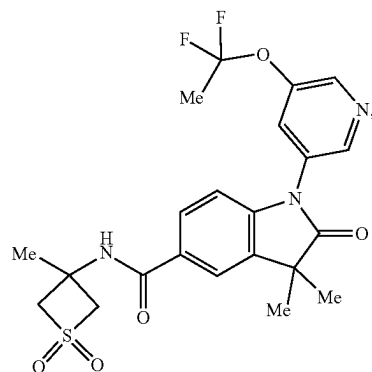
-continued



or a pharmaceutically acceptable salt thereof.

27. The compound of claim **25** which is

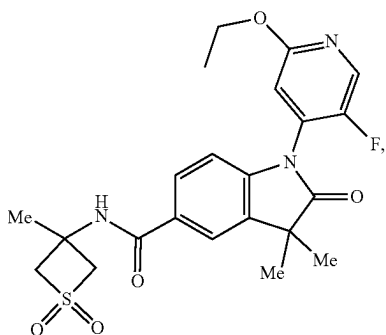
or a pharmaceutically acceptable salt thereof.

28. The compound of claim **25** which is

or a pharmaceutically acceptable salt thereof.

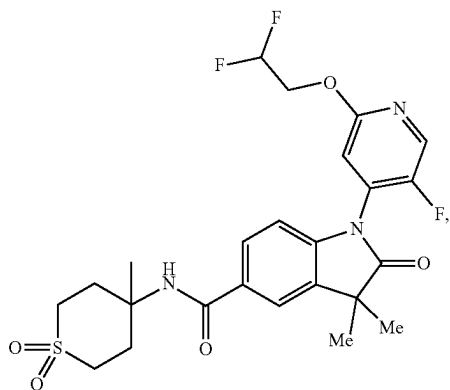
245

29. The compound of claim 25 which is



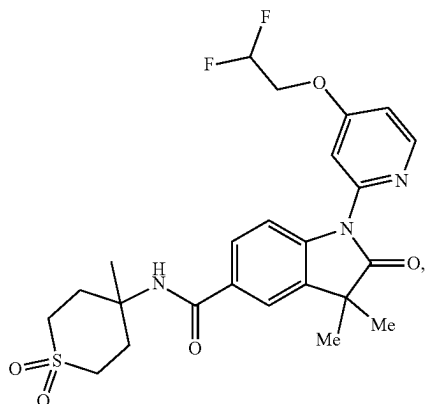
or a pharmaceutically acceptable salt thereof.

30. The compound of claim 26 which is



or a pharmaceutically acceptable salt thereof.

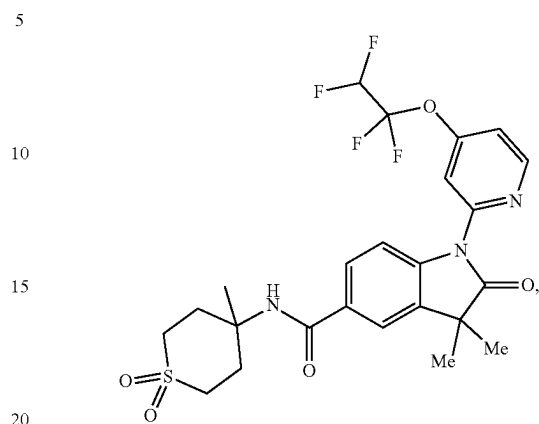
31. The compound of claim 26 which is



or a pharmaceutically acceptable salt thereof.

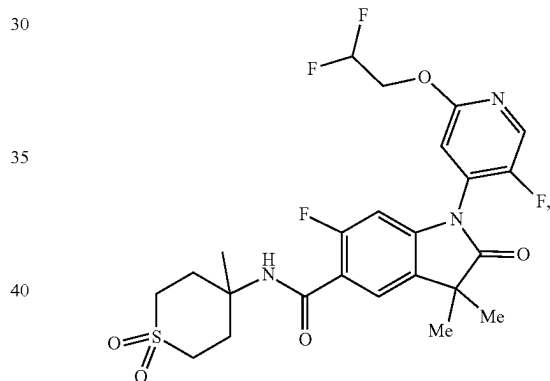
246

32. The compound of claim 26 which is



or a pharmaceutically acceptable salt thereof.

33. The compound of claim 26 which is



or a pharmaceutically acceptable salt thereof.

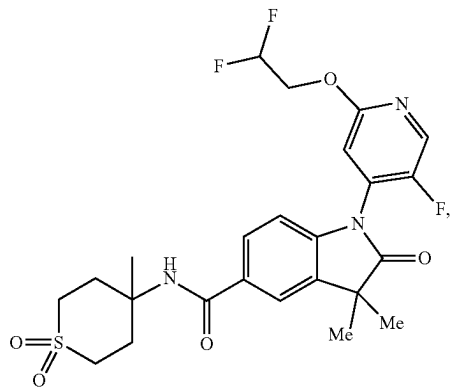
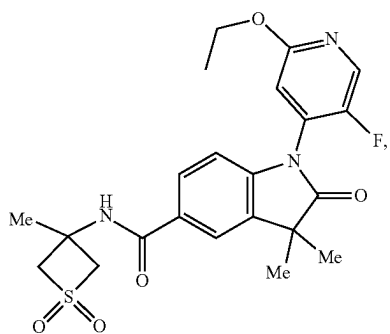
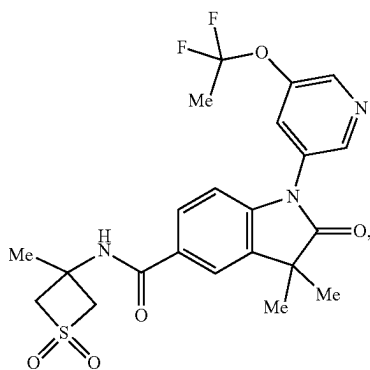
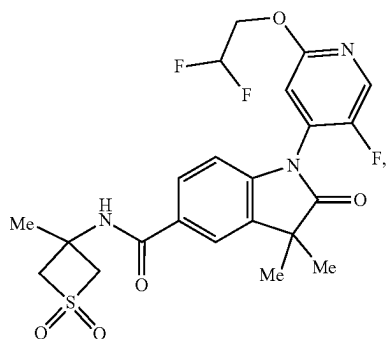
34. A composition for treating a condition selected from hepatic steatosis, nonalcoholic steatohepatitis (NASH), fibrosis, type-2 diabetes mellitus, obesity, hyperlipidemia, hypercholesterolemia, atherosclerosis, cognitive decline, dementia, cardiorenal diseases and heart failure comprising a compound of claim 1, or a pharmaceutically acceptable salt thereof, and a pharmaceutically carrier.

35. A composition comprising a pharmaceutically acceptable carrier and a compound according to claim 1, or a pharmaceutically acceptable salt thereof.

36. A method for treating a condition selected from hepatic steatosis, nonalcoholic steatohepatitis (NASH), fibrosis, type-2 diabetes mellitus, obesity, hyperlipidemia, hypercholesterolemia, atherosclerosis, cognitive decline, dementia, cardiorenal diseases and heart failure comprising administering to a patient in need thereof of a therapeutically effective amount of a compound of claim 1 or a pharmaceutically acceptable salt thereof.

247

37. A compound selected from:

**248**

-continued

5

10

15

20

25

30

35

40

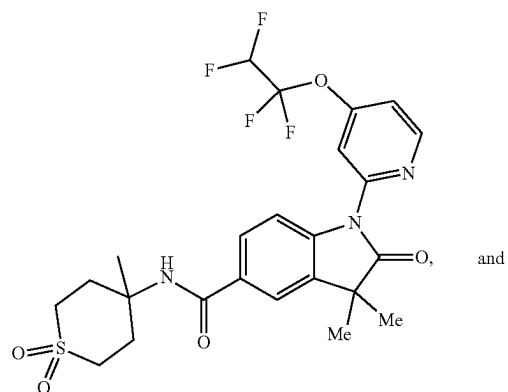
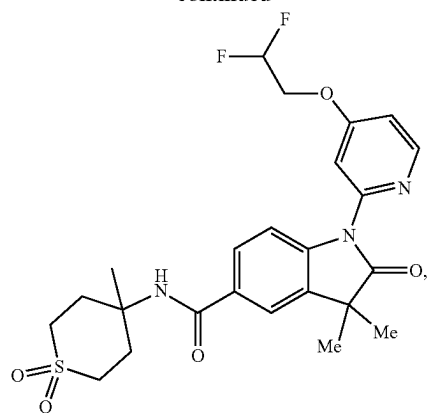
45

50

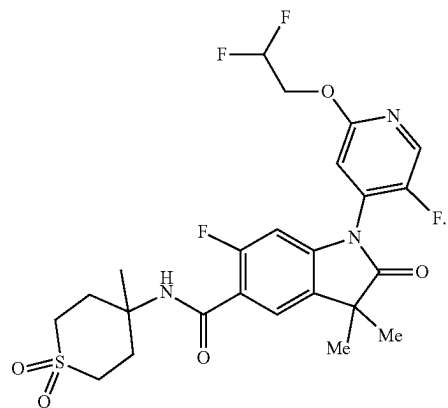
55

60

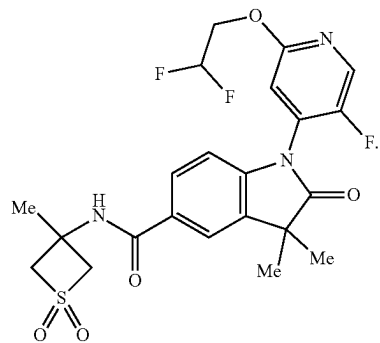
65

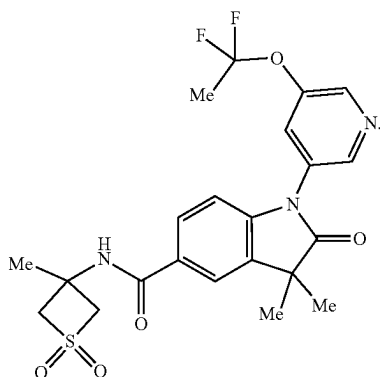
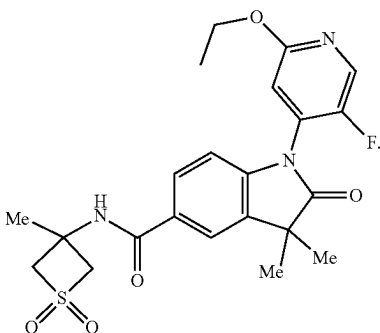
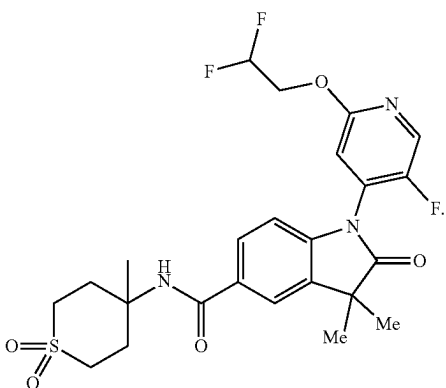
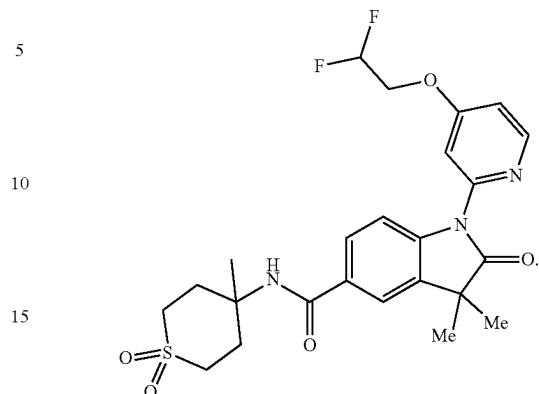
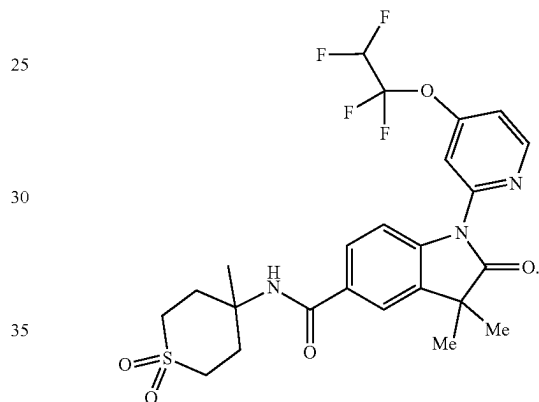
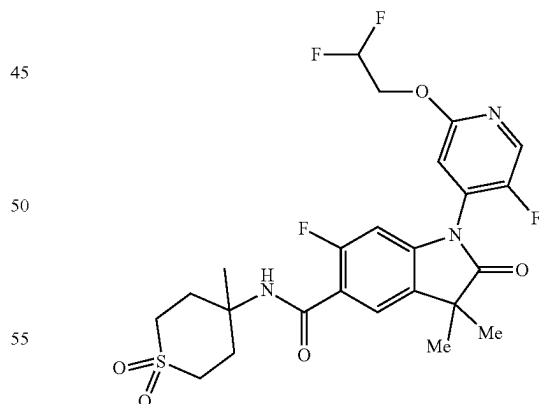


and



38. The compound of claim 37 which is



249**39.** The compound of claim **37** which is**40.** The compound of claim **37** which is**41.** The compound of claim **37** which is**250****42.** The compound of claim **37** which is**43.** The compound of claim **37** which is**44.** The compound of claim **37** which is

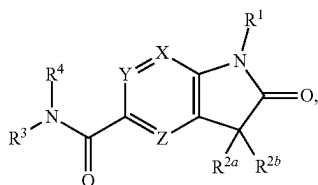
45. A method for treating a condition selected from hepatic steatosis, nonalcoholic steatohepatitis (NASH), fibrosis, type-2 diabetes mellitus, obesity, hyperlipidemia, hypercholesterolemia, atherosclerosis, cognitive decline, dementia, cardiorenal diseases and heart failure comprising administering to a patient in need thereof a therapeutically effective amount of a compound of claim **25** or a pharmaceutically acceptable salt thereof.

251

46. A method for treating a condition selected from hepatic steatosis, nonalcoholic steatohepatitis (NASH), fibrosis, type-2 diabetes mellitus, obesity, hyperlipidemia, hypercholesterolemia, atherosclerosis, cognitive decline, dementia, cardiorenal diseases and heart failure comprising administering to a patient in need thereof a therapeutically effective amount of a compound of claim 26 or a pharmaceutically acceptable salt thereof.

47. A method for treating a condition selected from hepatic steatosis, nonalcoholic steatohepatitis (NASH), fibrosis, type-2 diabetes mellitus, obesity, hyperlipidemia, hypercholesterolemia, atherosclerosis, cognitive decline, dementia, cardiorenal diseases and heart failure comprising administering to a patient in need thereof a therapeutically effective amount of a compound of claim 37.

48. A compound of Formula I:

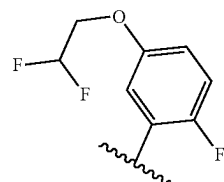


or a pharmaceutically acceptable salt thereof wherein:

252

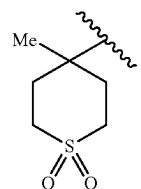
X, Y, and Z are each C(R⁵);

R¹ is



R^{2a} and R^{2b} are each methyl;

R³ is



R⁴ is hydrogen; and
each R⁵ is hydrogen or fluoro.

* * * * *